

# ***Quantity One***<sup>®</sup>

*User Guide for Version 4.6.0*  
*Windows and Macintosh*

# Quantity One User Guide

---

Bio-Rad Technical Service Department

Phone: (800) 424-6723  
(510) 741-2612

Fax: (510) 741-5802

E-mail: LSG.TechServ.US@Bio-Rad.com

## **Notice:**

No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopy, recording, or any information storage or retrieval system, without permission in writing from Bio-Rad.

**Quantity One** is a registered trademark of Bio-Rad Laboratories. **The Discovery Series** is a trademark of Bio-Rad Laboratories. All other trademarks and registered trademarks are of their respective companies.

WASTE text engine © 1993–1997 Marco Piovaneli

## **Limitations of Liability:**

Bio-Rad is not responsible for the misinterpretation of results obtained by following the instructions in this guide. Whenever possible, you should contact the Technical Services Department at Bio-Rad to discuss your results. As with all scientific studies, we recommend that you repeat your experiment at least once before making any significant conclusions for presentation or publication.

Copyright © 2005 by Bio-Rad Laboratories. All rights reserved.

# Table of Contents

|                                               |            |
|-----------------------------------------------|------------|
| <b>1. Introduction .....</b>                  | <b>1-1</b> |
| 1.1. Overview of Quantity One .....           | 1-1        |
| 1.2. Digital Data and Signal Intensity .....  | 1-2        |
| 1.3. Gel Quality .....                        | 1-3        |
| 1.4. Quantity One Workflow .....              | 1-4        |
| 1.5. Computer Requirements .....              | 1-5        |
| 1.6. Installation .....                       | 1-6        |
| 1.7. Software License .....                   | 1-9        |
| 1.8. Downloading from the Internet .....      | 1-13       |
| 1.9. Quantity One Basic .....                 | 1-14       |
| 1.10. Contacting Bio-Rad .....                | 1-14       |
| <b>2. General Operation .....</b>             | <b>2-1</b> |
| 2.1. Menus and Toolbars .....                 | 2-1        |
| 2.2. File Commands .....                      | 2-6        |
| 2.3. Imaging Device Acquisition Windows ..... | 2-14       |
| 2.4. Exit .....                               | 2-15       |
| 2.5. Preferences .....                        | 2-15       |
| 2.6. User Settings .....                      | 2-23       |
| <b>3. Viewing and Editing Images .....</b>    | <b>3-1</b> |
| 3.1. Magnifying and Positioning Tools .....   | 3-1        |
| 3.2. Density Tools .....                      | 3-5        |
| 3.3. Showing and Hiding Overlays .....        | 3-6        |
| 3.4. Multi-Channel Viewer .....               | 3-7        |
| 3.5. 3D Viewer .....                          | 3-10       |
| 3.6. Image Stack Tool .....                   | 3-13       |

# Quantity One User Guide

---

|                                                     |            |
|-----------------------------------------------------|------------|
| 3.7. Colors .....                                   | 3-14       |
| 3.8. Transform .....                                | 3-17       |
| 3.9. Resizing and Reorienting Images .....          | 3-23       |
| 3.10. Whole-Image Background Subtraction .....      | 3-29       |
| 3.11. Filtering Images .....                        | 3-33       |
| 3.12. Invert Data .....                             | 3-39       |
| 3.13. Text Overlays .....                           | 3-39       |
| 3.14. Erasing All Analysis from an Image .....      | 3-42       |
| 3.15. Sort and Recalculate .....                    | 3-42       |
| 3.16. Automation Manager .....                      | 3-42       |
| <b>4. Lanes .....</b>                               | <b>4-1</b> |
| 4.1. Defining Lanes .....                           | 4-1        |
| 4.2. Lane-Based Background Subtraction .....        | 4-9        |
| 4.3. Compare Lanes .....                            | 4-13       |
| 4.4. Lane-based Arrays .....                        | 4-16       |
| <b>5. Bands .....</b>                               | <b>5-1</b> |
| 5.1. How Bands Are Identified and Quantified .....  | 5-2        |
| 5.2. Band Detection .....                           | 5-3        |
| 5.3. Identifying and Editing Individual Bands ..... | 5-10       |
| 5.4. Plotting Traces of Bands .....                 | 5-12       |
| 5.5. Band Attributes .....                          | 5-13       |
| 5.6. Band Information .....                         | 5-16       |
| 5.7. Gauss-Modeling Bands .....                     | 5-18       |
| 5.8. Irregularly Shaped Bands in Lanes .....        | 5-21       |
| <b>6. Standards and Band Matching .....</b>         | <b>6-1</b> |
| 6.1. Standards .....                                | 6-1        |
| 6.2. Band Matching .....                            | 6-13       |
| 6.3. Quantity Standards .....                       | 6-24       |

|                                                        |             |
|--------------------------------------------------------|-------------|
| <b>7. Volume Tools .....</b>                           | <b>7-1</b>  |
| 7.1. Creating a Volume .....                           | 7-1         |
| 7.2. Volume Labels .....                               | 7-5         |
| 7.3. Volume Features .....                             | 7-7         |
| 7.4. Moving, Copying, and Deleting Volumes .....       | 7-8         |
| 7.5. Volume Standards .....                            | 7-8         |
| 7.6. Volume Background Subtraction .....               | 7-10        |
| 7.7. Volume Arrays .....                               | 7-12        |
| <b>8. Colony Counting .....</b>                        | <b>8-1</b>  |
| 8.1. Defining the Counting Region .....                | 8-2         |
| 8.2. Counting the Colonies .....                       | 8-3         |
| 8.3. Displaying the Results .....                      | 8-4         |
| 8.4. Making and Erasing Individual Colonies .....      | 8-5         |
| 8.5. Using the Histogram to Distinguish Colonies ..... | 8-6         |
| 8.6. Ignoring a Region of the Dish .....               | 8-8         |
| 8.7. Saving/Resetting the Count .....                  | 8-9         |
| 8.8. Saving to a Spreadsheet .....                     | 8-10        |
| <b>9 ReadyAgarose™ 96 Plus .....</b>                   | <b>9-1</b>  |
| 9.1. ReadyAgarose Wizard .....                         | 9-1         |
| <b>10. EXQuest Spot Cutter .....</b>                   | <b>10-1</b> |
| 10.1. Calibration and setup .....                      | 10-2        |
| 10.2. Manual Excision .....                            | 10-6        |
| 10.3. Lane/Volume Excision .....                       | 10-18       |
| <b>11. Differential Display and VNTRs .....</b>        | <b>11-1</b> |
| 11.1. Differential Display .....                       | 11-1        |
| 11.2. Variable Number Tandem Repeats .....             | 11-3        |

|                                           |             |
|-------------------------------------------|-------------|
| <b>12. Reports .....</b>                  | <b>12-1</b> |
| 12.1. Report Window .....                 | 12-1        |
| 12.2. Lane and Match Reports .....        | 12-5        |
| 12.3. Band Types Report .....             | 12-6        |
| 12.4. 1-D Analysis Report .....           | 12-8        |
| 12.5. Similarity Comparison Reports ..... | 12-9        |
| 12.6. Volume Analysis Report .....        | 12-20       |
| 12.7. Volume Regression Curve .....       | 12-23       |
| 12.8. VNTR Report .....                   | 12-25       |
| <b>13. Printing and Exporting .....</b>   | <b>13-1</b> |
| 13.1. Print Image .....                   | 13-1        |
| 13.2. Page Setup .....                    | 13-1        |
| 13.3. Print Settings .....                | 13-1        |
| 13.4. Image Report .....                  | 13-3        |
| 13.5. Video Print .....                   | 13-4        |
| 13.6. Export to TIFF Image .....          | 13-5        |
| 13.7. Exporting a JPEG Image .....        | 13-8        |
| <b>Appendix A.</b>                        |             |
| <b>Gel Doc EQ .....</b>                   | <b>A-1</b>  |
| <b>Appendix B.</b>                        |             |
| <b>Gel Doc XR .....</b>                   | <b>B-1</b>  |
| <b>Appendix C.</b>                        |             |
| <b>ChemiDoc EQ .....</b>                  | <b>C-1</b>  |
| <b>Appendix D.</b>                        |             |
| <b>ChemiDoc XRS .....</b>                 | <b>D-1</b>  |

|                                                                                |            |
|--------------------------------------------------------------------------------|------------|
| <b>Appendix E.</b>                                                             |            |
| <b>GS-800 Imaging Densitometer .....</b>                                       | <b>E-1</b> |
| <br><b>Appendix F.</b>                                                         |            |
| <b>Personal Molecular Imager FX .....</b>                                      | <b>F-1</b> |
| <br><b>Appendix G.</b>                                                         |            |
| <b>Molecular Imager FX Family (FX Pro, FX Pro Plus and Molecular FX) .....</b> | <b>G-1</b> |
| <br><b>Appendix H.</b>                                                         |            |
| <b>VersaDoc .....</b>                                                          | <b>H-1</b> |
| <br><b>Appendix I.</b>                                                         |            |
| <b>Other Features .....</b>                                                    | <b>I-1</b> |
| <br><b>Appendix J.</b>                                                         |            |
| <b>21 CFR Part 11 .....</b>                                                    | <b>J-1</b> |





# Preface

## 1. About This Document

This user guide is designed to be used as a reference in your everyday use of Quantity One<sup>®</sup> Software. It provides detailed information about the tools and commands of Quantity One for the Windows and Macintosh platforms. Any platform differences in procedures and commands are noted in the text.

This guide assumes that you have a working knowledge of your computer operating system and its conventions, including how to use a mouse and standard menus and commands, and how to open, save, and close files. For help with any of these techniques, see the documentation that came with your computer.

This guide uses certain text conventions to describe specific commands and functions.

| Example                                     | Indicates                                                                                                    |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| File > Open                                 | Choosing the Open command under the File menu.                                                               |
| Dragging                                    | Positioning the cursor on an object and holding down the left mouse button while you move the mouse.         |
| Ctrl+s                                      | Holding down the Control key while typing the letter s.                                                      |
| Right-click/<br>Left-click/<br>Double-click | Clicking the right mouse button/<br>Clicking the left mouse button/<br>Clicking the left mouse button twice. |

Some of the illustrations of menus and dialog boxes found in this manual are taken from the Windows version of the software, and some are taken from the Macintosh version. Both versions of a menu or dialog box will be shown only when there is a significant difference between the two.

## 2. Bio-Rad Listens

The staff at Bio-Rad are receptive to your suggestions. Many of the new features and enhancements in this version of Quantity One are a direct result of conversations with our customers. Please let us know what you would like to see in the next version of Quantity One by faxing, calling, or e-mailing our Technical Services staff. You can also use Solobug (installed with Quantity One) to make software feature requests.

# 1. Introduction

## 1.1 Overview of Quantity One

Quantity One is a powerful, flexible software package for imaging and analyzing 1-D electrophoresis gels, dot blots, arrays, and colonies.

The software is supported on Windows and Macintosh operating systems and has a graphical interface with standard pull-down menus, toolbars, and keyboard commands.

Quantity One can image and analyze a wide variety of biological data, including radioactive, chemiluminescent, fluorescent, and color-stained samples acquired from densitometers, phosphor imagers, fluorescent imagers, and gel documentation systems.

An image of a sample is captured using the controls in the imaging device window and displayed on your computer monitor. Image processing and analysis operations are performed using commands from the menus and toolbars.

Images can be magnified, annotated, rotated, and resized. They can be printed using standard and video printers.

All data in the image can be quickly and accurately quantitated using the Volume tools.

The lane-based functions can be used to determine molecular weights, isoelectric points, VNTRs, presence/absence and up/down regulation of bands, and other values. The software can measure total and average quantities, determine relative and actual amounts of protein, and count colonies in a Petri dish.

The software can cope with distortions in the shape of lanes and bands. Lanes can be adjusted along their lengths to compensate for any curvature or smiling of gels.

Image files can be shared among all The Discovery Series™ software. Images can also be easily converted into TIFF format for compatibility with other Macintosh and Windows applications.

# 1.2 Digital Data and Signal Intensity

The Bio-Rad imaging devices supported by Quantity One are light and/or radiation detectors that convert signals from biological samples into digital data. Quantity One then displays the digital data on your computer screen, in the form of gray scale or color images.

A data object as displayed on the computer is composed of tiny individual screen pixels. Each pixel has an X and Y coordinate, and a value Z. The X and Y coordinates are the pixel's horizontal and vertical positions on the image, and the Z value is the signal intensity of the pixel.

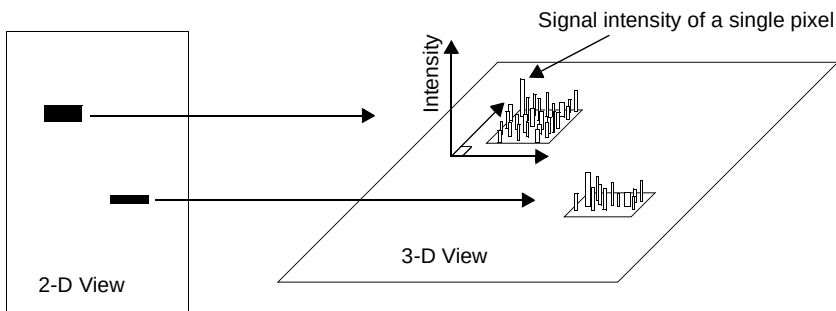


Fig. 1-1. Representation of the pixels in two digitally imaged bands in a gel.

For a data object to be visible and quantifiable, the intensity of its clustered pixels must be higher than the intensity of the pixels that make up the background of the image. The total intensity of a data object is the sum of the intensities of all the pixels that make up the object. The mean intensity of a data object is the total intensity divided by the number of pixels in the object.

The units of signal intensity are Optical Density (O.D.) in the case of the GS-700™ imaging densitometer, GS-710™ calibrated imaging densitometer, and GS-800™ calibrated densitometer, the Gel Doc EQ™ system, ChemiDoc EQ™ system, ChemiDoc XRS™ system with a white light source, or the Fluor-S™ MultiImager system, Fluor-S™ MAX MultiImager system, Fluor-S™ MAX2 MultiImager system and VersaDoc™ imaging systems with white light illumination. Signal intensity is expressed in counts when using the Personal Molecular Imager™ system or the Molecular Imager FX™ system, Molecular Imager FX Pro™ fluorescent imager,

Molecular Imager FX Pro Plus™ multiimager system, or in the case of the Gel Doc EQ, ChemiDoc EQ, ChemiDoc XRS, Fluor-S, Fluor-S MAX, or VersaDoc when using the UV light source.

### 1.3 Gel Quality

Quantity One is very tolerant of an assortment of electrophoretic artifacts. Lanes do not have to be perfectly straight or parallel. Bands do not have to be perfectly resolved.

However, for accurate lane-based quantitation, bands should be reasonably flat and horizontal. Lane-based quantitation involves calculating the average intensity of pixels across the band width and integrating over the band height. For the automatic band finder to function optimally, bands should be well-resolved.

Dots that appear as halos, rings, or craters, or that are of unequal diameter, may be incorrectly quantified using the automatic functions.

### 1.4 Quantity One Workflow

The following steps are involved in using Quantity One.

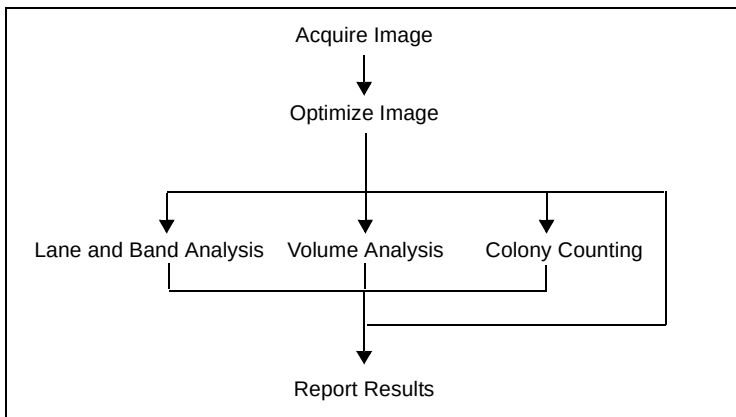


Fig. 1-2. Quantity One workflow.

#### 1.4.a Acquire Image

Before you can use Quantity One to analyze a biological image, you need to capture the image and save it as an image file. This may be done with one of the several Bio-Rad imaging instruments supported by this software: the Molecular Imager FX and Personal Molecular Imager systems; the GS-800 Imaging Densitometer; the Gel Doc EQ, ChemiDoc EQ, and ChemiDoc XRS Gel documentation systems; and the VersaDoc.

The resulting images can be stored in files on a computer hard disk, network file server, or removable disks.

#### 1.4.b Optimize Image

Once you have acquired an image of your sample, you may need to reduce noise or background density in the image. Quantity One has a variety of functions to minimize image background while maintaining data integrity.

### 1.4.c Analyze Image

Once a “clean” image is available, you can use Quantity One to gather and analyze your biological data. In the case of 1-D gels, the software has tools for identifying lanes and defining, quantifying, and calculating the values of bands. Volume tools allow you to easily measure and compare the quantities of bands, spots, or arrays. The colony counting controls allow you to count the number of colonies in a Petri dish, as well as perform batch analysis.

Qualitative and quantitative data can be displayed in tabular and graphical formats.

### 1.4.d Report Results

When your analysis is complete, you can print your results in the form of simple images, images with overlays, reports, tables, and graphs. You can export your images and data to other applications for further analysis.

## 1.5 Computer Requirements

This software is supported on Windows XP and Windows 2000, or on a Macintosh PowerPC running Mac OS 9.2.2 or Mac OS 10.2.6.

The computer memory requirements are mainly determined by the file size of the images you will scan and analyze. High-resolution image files can be very large. For this reason, we recommend that you archive images on a network file server or high-capacity removable disk.

### PC

The following is the **recommended** system configuration for installing and running on a PC:

|                   |                                           |
|-------------------|-------------------------------------------|
| Operating system: | Windows 2000<br>Windows XP                |
| Processor:        | Pentium 333 MHz or faster                 |
| RAM:              | ≥ 256 MB for all Bio-Rad imaging systems. |

## Quantity One User Guide

---

|                  |                                                            |
|------------------|------------------------------------------------------------|
| Hard disk space: | ≥ 3 GB                                                     |
| Monitor:         | 17" monitor, 1024 x 768 resolution (required), True color. |
| USB port:        | Required for the Hardware Security Key (HSK).              |
| Printer:         | Optional.                                                  |

### Macintosh

The following is the **recommended** system configuration for installing and running on a Macintosh:

|                   |                                                                    |
|-------------------|--------------------------------------------------------------------|
| Operating system: | MacOS 9.2.2, MacOS 10.2.6.                                         |
| Processor/Model:  | PowerPC G3 processor or faster.                                    |
| RAM:              | ≥ 256 MB for all Bio-Rad imaging systems.                          |
| Hard disk space:  | ≥ 3 GB                                                             |
| Monitor:          | 17" monitor, 1024 x 768 resolution (required), Millions of colors. |
| USB port:         | Required for the Hardware Security Key (HSK).                      |
| Printer:          | Optional.                                                          |

**Note:** The default amount of memory assigned to this program on the Macintosh is 128 MB. If the total RAM in your Macintosh is 128 MB or less, you should reduce the amount of memory assigned to the program to 10 MB less than your total RAM. With the application icon selected, go to File > Get Info in your Finder to reduce the memory requirements for the application. See your Macintosh computer documentation for details.

## 1.6 Installation

Refer to the Installation Guide for detailed information regarding installation of The Discovery Series.



### 1.6.a Hardware Security Key (HSK)

The Discovery Series software is password-protected using a Hardware Security Key (HSK), which is included in your software package. You must attach the Hardware Security Key to your computer before you can run the software.



Fig. 1-3. Hardware Security Key

Before proceeding with installation, plug the HSK into any available USB port on your computer.

The code for the HSK is EYYCY, which is printed on the key itself. Use this code to identify the HSK that belongs to The Discovery Series software.

**Note:** Initial installation of a network server requires the Hardware Security Key included in the software package. Installation of an additional Network Client User to a Network License Server System does not require an HSK. Please refer to the Network License Installation Guide that ships with Network License.

### 1.6.b Installing The Discovery Series for Windows

**Note:** You must be a member of the Administrators group to install Discovery Series software.

Insert The Discovery Series CD-ROM. The installation wizard will start automatically. (If the CD does not auto-start, click Start in your taskbar then click Run. In the Open field type d:\setup.exe where d is the letter of your CD-ROM drive.) Select the software application you want to install. On each panel, click Next when you are ready to proceed. You must accept the license agreement to continue with installation.

## Quantity One User Guide

---

The installer program guides you through the installation. The installer creates a default directory under Program Files on your computer called Bio-Rad\The Discovery Series (to select a different directory, click Browse). The application program will be placed in the Bin folder inside The Discovery Series folder. An additional folder for storing sample images is also located in The Discovery Series folder. User profiles will be created and stored in the Documents and Settings folder for each user.

The installer places a shortcut to the application and user guide on your desktop and creates a The Discovery Series folder in Programs on your Windows Start menu.

After installation, you must reboot your computer before using an imaging device.

**Note:** If you are installing in a Windows 2000 environment, you must start the application before allowing any other user access to the application.

### Uninstalling The Discovery Series from Windows

If you need to uninstall The Discovery Series for any reason, go to Add/Remove Programs in the Control Panel. Highlight the application you want to remove and click Remove.

**Note:** Uninstalling The Discovery Series software does not remove any system files in the Documents and Settings folder located in Application Data. If you want to remove these files as well, you must do so manually.

### 1.6.c Installing the Discovery Series on a Macintosh

The Mac install process involves installation of the application, dongle drivers, and Roper support. Insert the Discovery Series CD-ROM into your Macintosh. The TDS-Mac folder opens on your desktop, displaying the installers for The Discovery Series applications. Double-click on the installer for your application.



Fig. 1-4. Installation program icon (Macintosh).

Insert the Discovery Series CD-ROM. The TDS-Mac folder opens on your desktop, displaying the installers for The Discovery Series applications. Double-click the installer for your application. You must accept the license agreement to proceed with installation. The ReadMe contains important information about the HSK. Click Continue.

The installation wizard lists the types of installations in the pull down list. If you want to install the application and documentation only, select Easy Install from the list.

If you want to install drivers as well, select Custom from the pull down list. (Remember to also select the application in custom install.)

**Note:** If the computer is going to be used to acquire images from Gel Doc EQ, ChemiDoc EQ, ChemiDoc XRS, or the VersaDoc system, then select the driver for your particular imaging device now.

Install Location identifies where the application will be installed. If you want to specify a different location, click the pull-down button.

Click Install to proceed. Once installation is complete, click Quit. To return to the installer, click Continue.

### Uninstalling The Discovery Series from a Macintosh

Insert the Discovery Series CD-ROM. The TDS-Mac folder opens on your desktop, displaying the installers for The Discovery Series applications. Double-click the installer for your application. In the installer screen, select Uninstall from the pull down list.

## 1.7 Software License

When the software opens for the first time, you will see a *Software License* screen that shows the current status of your software license.

With a new HSK or network license, you receive a 30-day temporary license (“Your license will expire on \_\_\_\_\_”). The temporary license is designed to give you time to purchase the software, if you have not already done so.

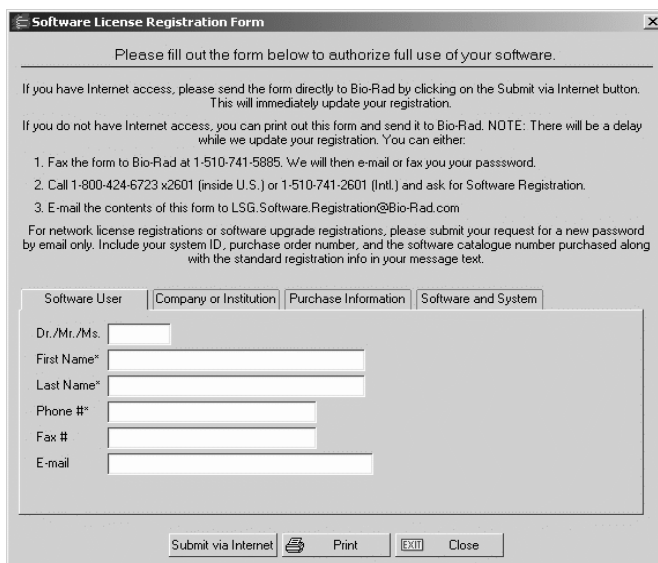


Fig. 1-5. Temporary license screen.

During the 30-day period, the *Software License* screen will appear every time you open the software. To use the software during this period, click on the **Run** button.

Network license holders can click on the **Check License** button at any time during the 30-day period to activate their full network license. (If your network license is not activated when you click on **Check License**, notify your network administrator.)

HSK users have 30 days to purchase the software and obtain a purchase order number and software serial number from Bio-Rad. When you have this information, click on the **Check License** or **Registration Form** button in the *Software License* screen to register your software.



The image shows a screenshot of a 'Software License Registration Form' window. The window has a title bar with the text 'Software License Registration Form' and a close button. Below the title bar, there is a section with instructions: 'Please fill out the form below to authorize full use of your software.' followed by two paragraphs explaining how to submit the form (via Internet or by fax/email). Below the instructions, there are four tabs: 'Software User', 'Company or Institution', 'Purchase Information', and 'Software and System'. The 'Software User' tab is currently selected. Under this tab, there are several input fields: 'Dr./Mr./Ms.', 'First Name\*', 'Last Name\*', 'Phone #\*', 'Fax #', and 'E-mail'. At the bottom of the window, there are four buttons: 'Submit via Internet', 'Print', 'EXIT', and 'Close'.

Fig. 1-6. Software License Registration Form.

Fill out the information in the *Software License Registration Form*. Be sure to enter your purchase order number and software serial number under the *Purchase Information* tab when registering.

### 1.7.a Registering by Internet

If you have Internet access from your computer, click on the Submit via Internet button to send the *Software Registration Form* directly to Bio-Rad.

Your information will be submitted, and a temporary password will be generated automatically and sent back to your computer. Simply continue to run the application as before.

Bio-Rad will confirm your purchase information and generate a permanent license. After 2–3 days, click on **Check License** in the *Software License* screen again to update to a permanent password. (The *Software License* screen will not appear automatically after the temporary password has been generated; the software will

## Quantity One User Guide

---

simply open normally. Go to the *Help* menu and select **Register** to open the *Software License* screen.)

### 1.7.b Registering by Fax or E-mail

If you do not have Internet access, click on the **Print** button in the *Software License Registration Form* and fax the form to Bio-Rad at the number listed on the form. Alternatively, you can enter the contents of the form into an e-mail and send it to Bio-Rad at the address listed in the *Registration Form*.

Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

### 1.7.c Entering a Password

If you fax or e-mail your registration information, you will receive a password from Bio-Rad. You must enter this password manually.

To enter your password, click on **Enter Password** in the *Software License* screen. If you are not currently in the *Software License* screen, select **Register** from the *Help* menu.

**Enter Password**

To register, please contact Bio-Rad during US/Pacific business hours:  
Phone: 1-800-424-6723 x2601 (in the U.S.)  
1-510-741-2601 (Intl.)  
Ask for Software Registration.  
Fax: 1-510-741-5885  
e-mail: LSG.Software.Registration@Bio-Rad.com

Please provide the information in the completed registration form.

For network license registrations or software upgrade registrations, please submit your request for a new password by email only. Include your system ID, purchase order number, and the software catalogue number purchased along with the standard registration info in your message text.

Current license  
Single system license.  
License expires on 24-Aug-2002.

System ID  
CMXXXLXNE3

Password  
Quantity One  OK Not OK

Enter Cancel

Fig. 1-7. Enter Password screen.

In the *Enter Password* screen, type in your password in the field.

Once you have typed in the correct password, the **OK** light next to the password field will change to green and the **Enter** button will activate. Click on **Enter** to run the program.

## 1.8 Downloading from the Internet

You can download a trial version of the software from Bio-Rad's Web site. Go to The Discovery Series download page at [www.bio-rad.com/softwaredownloads](http://www.bio-rad.com/softwaredownloads) and select from the list of applications. Follow the instructions to download the installer onto your computer, then run the installer.

After installation, double-click on the application icon to run the program. The software will open and the *Software License* screen will be displayed.

**Note:** If you attempt to start the downloaded program and receive an “Unable to obtain authorization” message, you will need a Hardware Security Key to run the program. Contact Bio-Rad to obtain a key.



Fig. 1-8. Free Trial screen.

In the *Software License* screen, click on the **Free Trial** button. This will open the *Software License Registration Form*. Enter the required information (you will not

have a purchase order number or software serial number, and can leave these fields blank) and click on **Submit Via Internet**.

A free trial password will be automatically downloaded to your computer. This password will allow you to use the software for 30 days.

If you decide to purchase the software during that period, contact Bio-Rad to receive a software package and a Hardware Security Key. You can then complete the registration process as described in the previous sections.

## 1.9 Quantity One Basic

Quantity One can be run in Basic mode. Quantity One Basic does not require a software license. The program can be installed and used simultaneously on unlimited numbers of computers. Quantity One Basic is a limited version of the flexible and powerful Quantity One.

The following functionality is active in Basic Mode: Image acquisition with Bio-Rad imaging devices, Transform, Crop, Flip, Rotate, Text Tool, Volume Rectangle Tool, Volume Circle Tool, Density Tools, ReadyAgarose 96 PLUS, Manual Excision, Print, Export to TIFF, and Save.

## 1.10 Contacting Bio-Rad

Bio-Rad technical service hours are from 8:00 a.m. to 4:00 p.m., Pacific Standard Time in the U.S.

Phone: 800-424-6723  
510-741-2612

Fax: 510-741-5802

E-mail: LSG.TechServ.US@Bio-Rad.com

For software registration:

Phone: 800-424-6723 (in the U.S.)



+1-510-741-6996 (outside the U.S.)



## 2. General Operation

This chapter describes the graphical interface of Quantity One, how to access the various commands, how to open and save images, how to set preferences, and how to perform other basic file commands.

### 2.1 Menus and Toolbars

#### 2.1.a Menu Bar

Quantity One has a standard menu bar with pull-down menus that contain all the major features and functions available in the software.

- *File*—Opening and saving files, imaging device controls, printing, exporting.
- *Edit*—Preferences, other settings.
- *View*—Image magnification and viewing tools, tools for viewing image data.
- *Image*—Image transform, advanced crop, image processing and modification.
- *Lane*—Lane-finding tools.
- *Band*—Band-finding and band-modeling tools.
- *Match*—Tools for calculating molecular weights and other values from standards, tools for comparing lanes and bands in lanes.
- *Volume*—Band quantity and array data tools.
- *Analysis*—Colony counting, Differential Display, VNTR analysis.
- *Reports*—Band and lane analysis reports, Phylogenetic Tree, Similarity Matrix.
- *Window*—Commands for arranging multiple image windows.
- *Help*—Quick Guides, on-line Help, software registration.

Below the menu bar is the main toolbar, containing some of the most commonly used commands. Next to the main toolbar are the status boxes, which provide information about cursor selection and toolbar buttons.

### 2.1.b Main Toolbar

The main toolbar appears below the menu. It includes buttons for the main file commands (**Open, Save, Print**) and essential viewing tools (**Zoom Box, Grab**, etc.), as well as buttons that open the secondary toolbars and the most useful Quick Guides (*Printing, Volumes, Molecular Weight, and Colony Counting*).



Fig. 2-1. Main toolbar.

### Tool Help

If you hold the cursor over a toolbar icon, the name of the command will pop up below the icon. This utility is called Tool Help. Tool Help appears on a time delay basis that can be specified in the *Preferences* dialog box (see section 2.5, Preferences). You can also specify how long the Tool Help will remain displayed.

### 2.1.c Status Boxes

There are two status boxes, which appear to the right of the main toolbar.

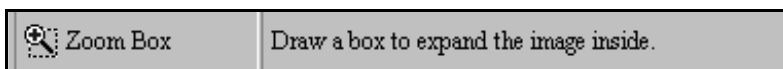


Fig. 2-2. Status boxes.

The first box displays any function that is assigned to the mouse. If you select a command such as **Zoom Box**, the name and icon of that command will appear in this status box and remain there until another mouse function is selected or the mouse is deassigned.

The second status box is designed to supplement Tool Help (see above). It provides additional information about the toolbar buttons. If you hold your cursor over a

button, a short explanation about that command will be displayed in this second status box.

### 2.1.d Secondary Toolbars

Secondary toolbars contain groups of related functions. You can open these toolbars from the main toolbar or from the *View > Toolbars* submenu.

The secondary toolbars can be toggled between vertical, horizontal, and expanded formats by clicking on the resize button on the toolbar itself.

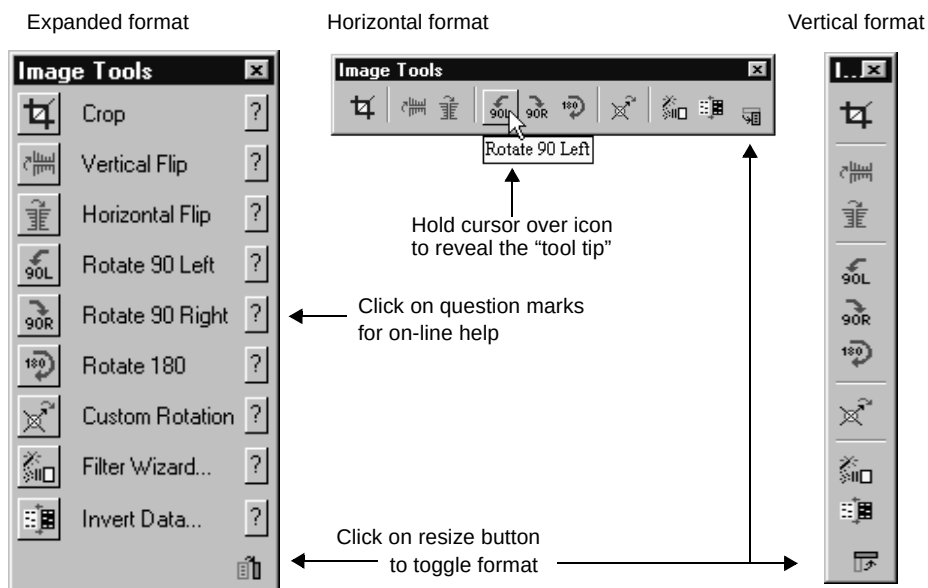


Fig. 2-3. Secondary toolbar formats and features.

The expanded toolbar format shows the name of each of the commands. Click on the ? icon next to the name to display on-line Help for that command.

### 2.1.e Quick Guides

The Quick Guides are designed to guide you through the major applications of the software. They are listed under the *Help* menu; four of these are also available on the main toolbar.

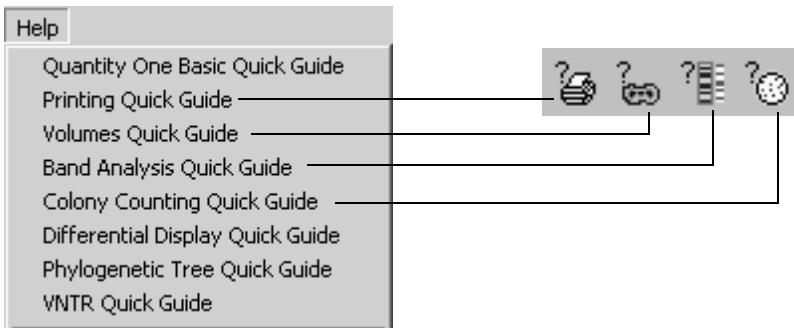


Fig. 2-4. Quick Guides listed on Help menu and main toolbar.

The Quick Guides are similar in design to the secondary toolbars, but are application-specific. Each Quick Guide contains all of the functions for a particular application, from opening the image to outputting data.

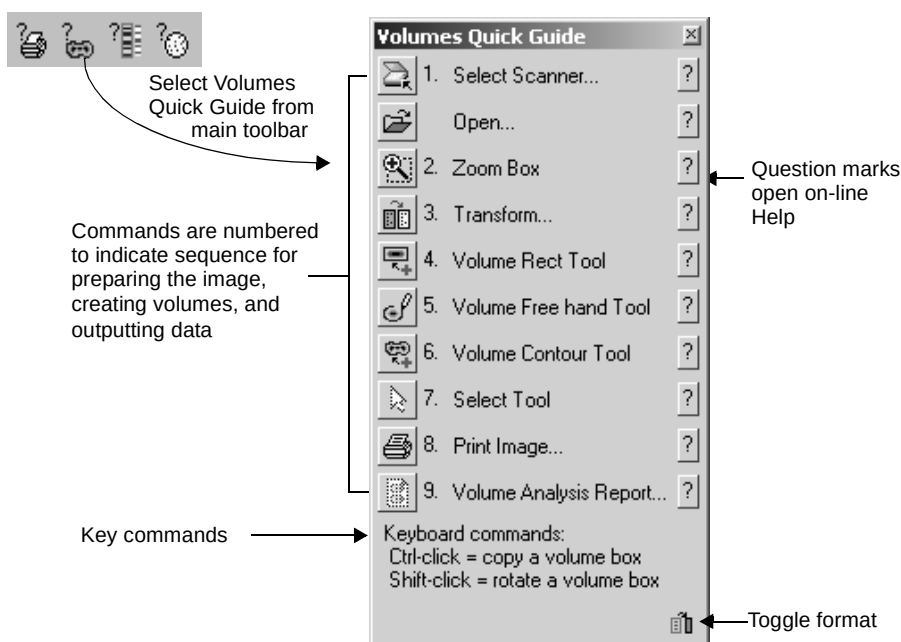


Fig. 2-5. Example of a Quick Guide: Volumes

In their expanded format, the Quick Guide commands are numbered as well as named. The numbers provide a suggested order of operation; however, not every command is required for every application.

As with the secondary toolbars, you can click on the ? next to the name of a function to display the Help text.

### 2.1.f Right-Click Context Menu

With an image open, right-click anywhere on the image to display a context menu of common commands.

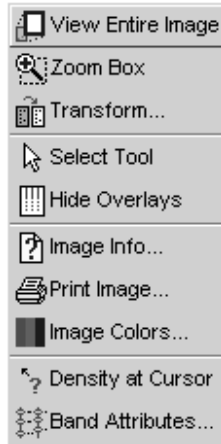


Fig. 2-6. Selecting Zoom Box from the right-click context menu.

You can select commands from this menu as you would from a standard menu.

### 2.1.g Keyboard Commands

Many commands and functions can be performed using keyboard keys (e.g., press the F1 key for **View Entire Image**; press Ctrl+S for **Save**). Select **Keyboard Layout** from the *Help* menu to display a list of keys and key combinations and their associated commands.

The pull-down menus also list the shortcut keys for the menu commands.

## 2.2 File Commands

The basic file commands and functions are located on the *File* menu.



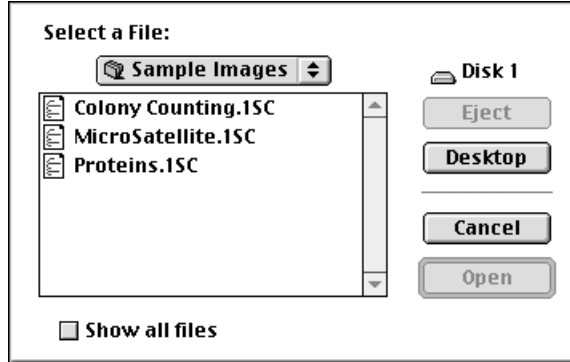
|                         |        |
|-------------------------|--------|
| Open...                 | Ctrl+O |
| Save                    | Ctrl+S |
| Save As...              |        |
| Save All                | Ctrl+Y |
|                         |        |
| Close                   | Alt+F4 |
| Close All               |        |
|                         |        |
| Change Version...       |        |
| Revert to Saved...      |        |
| Image Info...           | Ctrl+I |
| Reduce File Size...     |        |
|                         |        |
| Gel Doc EQ...           |        |
| ChemiDoc EQ...          |        |
| ChemiDoc XRS...         |        |
| GS-700...               |        |
| GS-710...               |        |
| GS-800...               |        |
| Fluor-S...              |        |
| Fluor-S MAX...          |        |
| Fluor-S MAX2...         |        |
| VersaDoc...             |        |
| Personal FX...          |        |
| FX...                   |        |
|                         |        |
| Print                   | ▶      |
| Export to TIFF Image... |        |
| Export to JPEG Image... |        |
|                         |        |
| Exit                    | Ctrl+Q |

Fig. 2-7. File menu.

### 2.2.a Opening Images

To open a saved image, select **Open** from the *File* menu or click on **Open** button on the main toolbar. This opens the standard *Open* dialog box for your operating system.

Macintosh version:



Windows version:

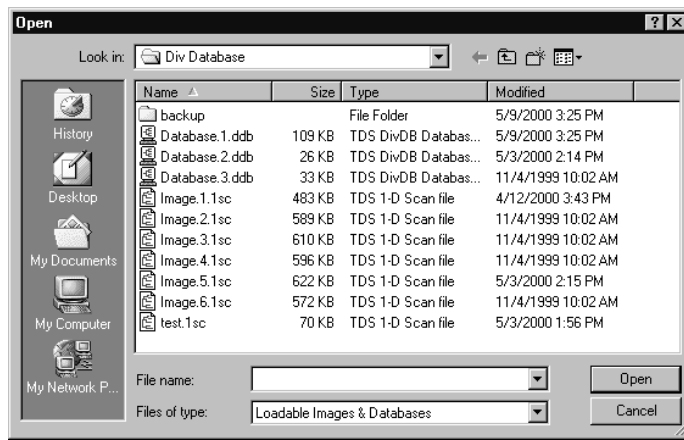


Fig. 2-8. Open dialog box.

In the dialog box, open a file by double-clicking on the file name. To open multiple files, first select them using Ctrl-click or Shift-click key combinations, and then click on the **Open** button.

**Note:** Due to operating system limitations, it is possible to have image files open in two or more locations over a network simultaneously. To safeguard your image files keep them in a protected folder or on your local machine.

You can also open images from other The Discovery Series software (PDQuest, Diversity Database, DNACode).

**Note:** It is possible to move image data between The Discovery Series software applications on different platforms (Windows and Macintosh).

The application comes with a selection of sample images. In Windows, these are located in The Discovery Series/Sample Images/1D directory. On the Macintosh, they are stored in the Sample Images folder in the Quantity One folder.

### Opening TIFF Images

The **Open** command can also be used to import TIFF images created using other software applications.

There are many types of TIFF formats that exist on the market. Not all are supported by The Discovery Series. There are two broad categories of TIFF files that are supported:

1. 8-bit Grayscale. Most scanners have an option between line art, full color, and grayscale formats. Select grayscale for use with The Discovery Series software. In a grayscale format, each pixel is assigned a value from 0 to 255, with each value corresponding to a particular shade of gray.
2. 16-bit Grayscale. Bio-Rad's Molecular Imager FX and Personal Molecular Imager and Fluor-S, and VersaDoc imaging systems use 16-bit pixel values to describe intensity of scale. Molecular Dynamics™ and Fuji™ imagers also use 16-bit pixel values. The Discovery Series understands these formats and can interpret images from both Bio-Rad and Molecular Dynamics storage phosphor systems.

**Note:** The program can import 8- and 16-bit TIFF images from both Macintosh and PC platforms.

TIFF files that are *not* supported include:

1. 1-bit Line Art. This format is generally used for scanning text for optical character recognition or line drawings. Each pixel in an image is read as either black or white. Because the software needs to read continuous gradations to perform gel analysis, this on-off pixel format is not used.
2. 24-bit Full Color or 256 Indexed Color. These formats are frequently used for retouching photographs and are currently unsupported in The Discovery Series,

although most scanners that are capable of producing 24-bit and indexed color images will be able to produce grayscale scans as well.

3. **Compressed Files.** The software does not read compressed TIFF images. Since most programs offer compression as a selectable option, files intended for compatibility with The Discovery Series should be formatted with the compression option turned off.

### 2.2.b Saving Images

To save a new image or an old image with changes, select **Save** from the *File* menu. In Windows, new images will be given a **.1sc** extension when they are first saved.

**Save As** can be used to save a new image, rename an old image, or save a copy of an image to a different directory. The standard *Save As* dialog box for your operating system will open.

To save all open images, select **Save All** from the *File* menu or click on the button on the main toolbar.

### 2.2.c Closing Images

To close an image, select **Close** from the *File* menu. To close all open images, select **Close All**. You will be prompted to save any changes before closing.

### 2.2.d Revert to Saved

To reload the last saved version of an image, select **Revert to Saved** from the *File* menu. Because any changes you made since last saving the file will be lost, you will be prompted to confirm the command.

### 2.2.e Image Info

**Image Info** on the *File* menu opens a dialog box containing general information about the selected image, including scan date, scan area, number of pixels in the image, data range, and the size of the file. Type any description or comments about the image in the *Description* field.

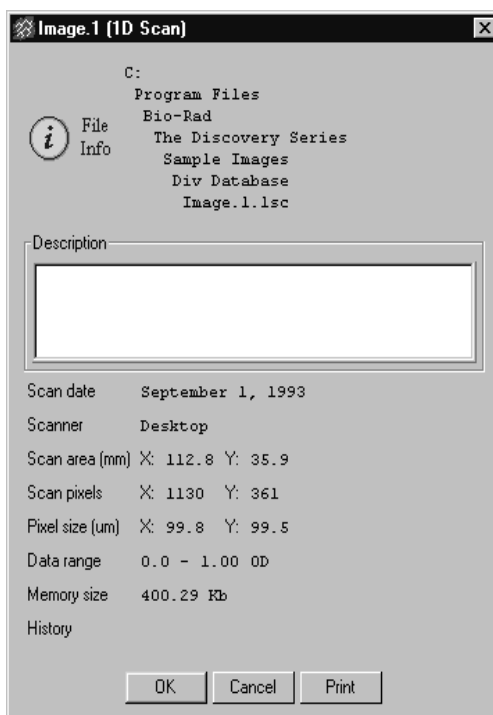


Fig. 2-9. Image Info box.

History lists the changes made to the image including the date. If you have Security Mode active, the name of the user who made the change is also listed (See Section 2.5, Preferences for information on Security Mode).

To print the file info, click on the **Print** button in the dialog box.

### Changing the Image Dimensions

You can change the dimensions of certain images using the *Image Info* dialog box. This feature is only available for images captured by a camera or imported TIFF images in which the dimensions are not already specified.

## Quantity One User Guide

For these types of images, the *Image Info* dialog box will include fields for changing the image dimensions.

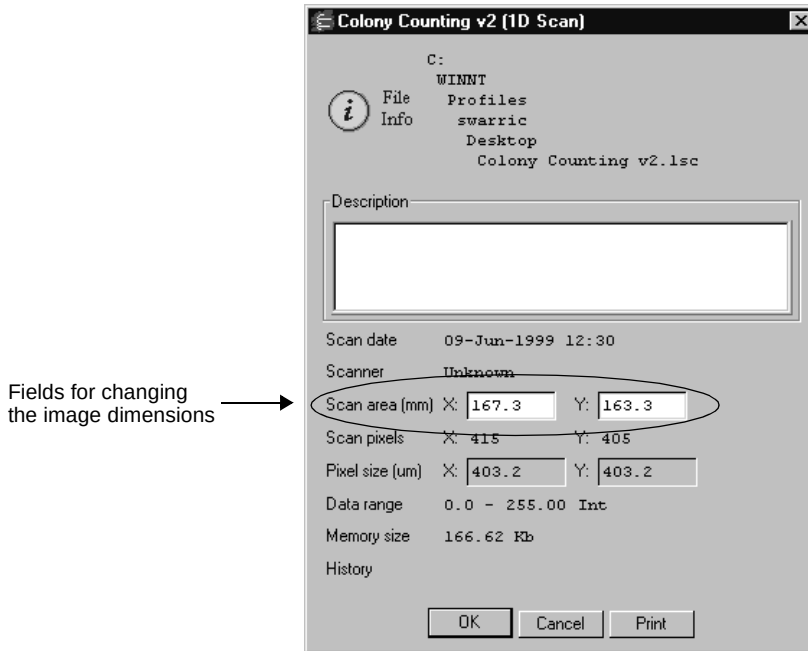


Fig. 2-10. Image Info dialog box with fields for changing the image dimensions.

Enter the new image dimensions (in millimeters) in the appropriate fields. Note that the pixel size in the image (in micrometers) will change to retain the same number of pixels in the image.

### 2.2.f Reduce File Size

High-resolution image files can be very large, which can lead to problems with opening and saving. To reduce the file size of an image, you can reduce the image resolution by reducing the number of pixels in the image. (You can also trim unneeded parts of an image to reduce its memory size. See section 3.9.a, Cropping Images.)

This function is comparable to scanning at a lower resolution, in that you are increasing the size of the pixels in the image, thereby reducing the total number of pixels and thus the file size.

**Note:** In most cases, reducing the resolution of an image will not affect quantitation. In general, as long as the pixel size remains less than 10 percent of the size of the objects in your image, changing the pixel size will not affect quantitation.

Select **Reduce File Size** from the *File* menu to open the *Reduce File Size* dialog box. The dialog box lists the size of the pixels in the image (**Pixel Size: X by Y microns**), the number of pixels in the image (**Pixel Count: X by Y pixels**), and the memory size of the image.

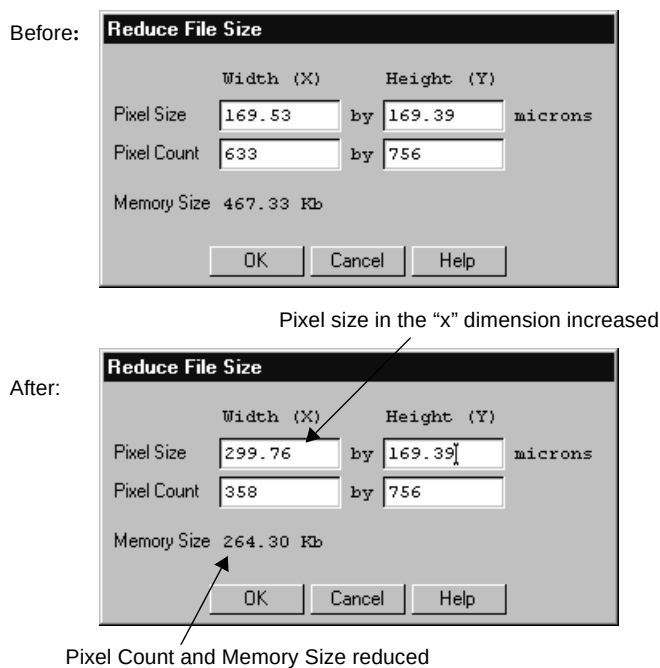


Fig. 2-11. Reduce File Size dialog box, before and after pixel size increase.

Lower the resolution by entering *lower* values in the *Pixel Count* fields or *higher* values in the *Pixel Size* fields (see the figure for an example).

**Note:** Since with most 1-D gels you are more concerned with resolving bands in the vertical direction than the horizontal direction, you may want to reduce the file size by making rectangular pixels. That is, keep the pixel size in the “y” dimension the same, while increasing the size in the “x” dimension.

When you are finished, click on the **OK** button.

A pop-up box will give you the option of reducing the file size of the displayed image or making a copy of the image and then reducing the copy's size.

*Reducing the file size is an irreversible process.* For that reason, we suggest that you first experiment with a copy of the image. Then, when you are satisfied with the reduced image, delete the original.

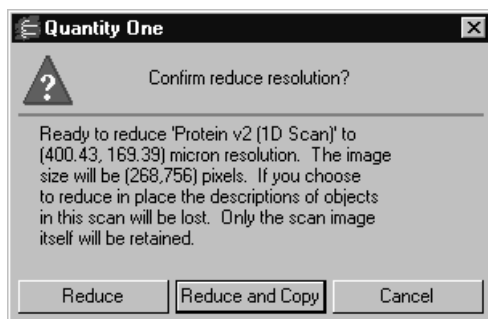


Fig. 2-12. Confirm Reduce File Size pop-up box.

## 2.3 Imaging Device Acquisition Windows

The *File* menu contains a list of Bio-Rad imaging devices supported by Quantity One. These are:

1. Gel Doc EQ
2. ChemiDoc EQ
3. ChemiDoc XRS
4. GS-800 Calibrated Densitometer



5. VersaDoc
6. Personal Molecular Imager FX
7. Molecular Imager FX

To open the acquisition window for an imaging device, select the name of that device from the *File* menu.

See the individual chapters on the imaging devices for more details.

## 2.4 Exit

To close Quantity One, select **Exit** from the *File* menu. You will be prompted to save your changes to any open files.

## 2.5 Preferences

You can customize basic features of Quantity One—such as menu options, display settings, and toolbars—using the *Preferences* dialog box. Select **Preferences** from the *Edit* menu to open this dialog.

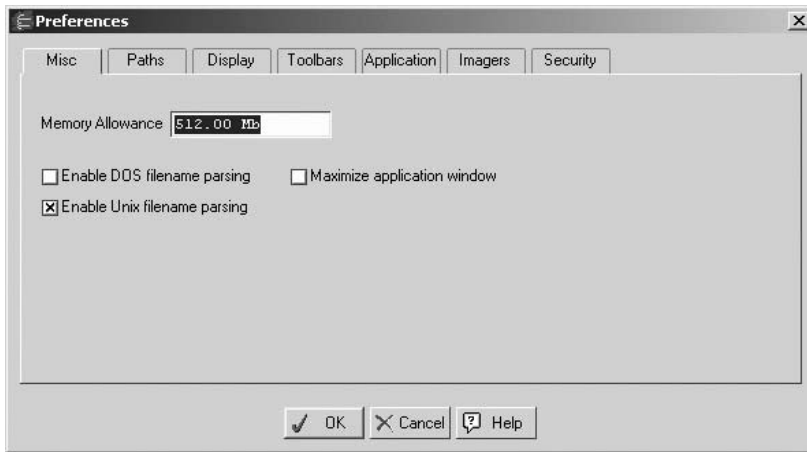


Fig. 2-13. Preferences dialog box.

Click on the appropriate tab to access groups of related preferences. After you have selected your preferences, click on **OK** to implement them.

### 2.5.a Misc.

Click on the *Misc* tab to access the following preferences.

#### Memory Allowance

To specify the amount of virtual memory allocated for the application at start-up, enter a value (in megabytes) in the **Memory Allowance** field. The default value of 512 megabytes is recommended. If you receive a warning message that the amount of virtual memory is set too high, you can enter a smaller value in this field. However, this should be considered a temporary fix, and you should consider expanding your hard drive.

#### Enable DOS Filename Parsing

If this checkbox is selected, for 8-character file names ending in two digits, the final two digits are interpreted as version and exposure numbers. For example, the file name IMAGE-11.1sc would be parsed as IMAGE ver 1 xpo 1.1sc. This is designed to

enable backwards compatibility for users with DOS image files. You should only check this box if you are using these image files.

### Enable UNIX Filename Parsing

This is similar to DOS file name parsing. Windows and Macintosh users are unlikely to run into difficulties with UNIX parsing; therefore, this setting is checked by default.

### Maximize Application Window

In the Windows version, select the **Maximize Application Window** checkbox to automatically maximize the application window when Quantity One first opens. If this is unchecked, the menu and status bars will appear across the top of the screen and any toolbars will appear “floating” on the screen.

## 2.5.b Paths

Click on the *Paths* tab to access the following preference.

### Temporary File Location

Temporary image files are normally stored in the TMP directory of your The Discovery Series folder. The full path is listed in the field. To change the location of your temporary files, type in a new path or click **Browse** and select a new directory. To return to the default TMP directory, click **Default**.

## 2.5.c Display

Click on the *Display* tab to access the following preferences.

### Zoom %

**Zoom %** determines the percentage by which an image zooms in or out when you use the **Zoom In** and **Zoom Out** functions. This percentage is based on the size of the image.

### Pan %

**Pan %** determines the percentage by which the image moves side to side or up and down when you use the arrow keys. This percentage is based on the size of the image.

### Band Style

Bands in your gel image can be marked with brackets that define the top and bottom boundaries of the band, or they can be marked with a dash at the center of the band. Indicate your preference by clicking on the **Brackets** or **Lines** button. (This setting can be temporarily changed in the *Band Attributes* dialog box. However, all newly opened images will use the preferences setting.)

### Auto “Imitate Zoom”

When this checkbox is selected, the magnifying and image positioning commands used in one window will be applied to all open windows. This is useful, for example, if you want to compare the same band of group of bands in different gels; magnify the band(s) in one gel, and the same area will be magnified in all the other gel images.

Note that the images must be approximately the same size.

### AutoScale Annotations

Select AutoScale Annotations if you want to display annotations in the image as a percentage of the image window not the image. This allows you to zoom in on an area of the image but have displayed annotations remain the same size in the image window.

### Disable hardware Acceleration for 3D Viewer

If you are having problems using the 3D Viewer, install the latest drivers for your graphics card. If, after updating your drivers, you are still having problems, select Disable hardware acceleration for 3D viewer. This allows Quantity One to emulate a graphics driver to render a 3D image.

**Note:** Disabling hardware acceleration reduces performance for the 3D Viewer.

### 2.5.d Toolbar

Click on the Toolbar tab to access the following preferences, which determine the behavior and positioning of the secondary toolbars and Quick Guides.

#### Show Default Quick Guide at Startup

If this checkbox is selected, the *Volumes Quick Guide* will open automatically when you open the program. If you are in Basic mode, the Basic Quick Guide opens instead.

#### Align Quick Guide with Document

If this checkbox is selected, the Quick Guides will open flush with the edge of your documents. Otherwise, they will appear flush with the edge of the screen.

#### Guides Always on Top

If this checkbox is selected, Quick Guides will always appear on top of images and never be obscured by them.

#### Quick Guide Placement and Toolbar Placement

These checkboxes determine on which side of the screen the Quick Guides and toolbars will first open.

#### Placement Behavior

This setting determines whether a Quick Guide or toolbar will always pop up in the same place and format (**Always Auto**), or whether they will pop up in the last location they were moved to and the last format selected (**Save Prior**).

#### Toolbar Orientation

These option buttons specify whether toolbars will first appear in a vertical, horizontal, or expanded format when you open the program.

### Tool Help Delay and Persistence

Specify the amount of time the cursor must remain over a toolbar icon before the Tool Help appears by entering a value (in seconds) in the **Tool Help Delay** field.

Specify the amount of time that the Tool Help will remain on the screen after you move the cursor off a button by entering a value (in seconds) in the **Tool Help Persistence** field.

### 2.5.e Application

Click on the Application tab to access the following preferences.

#### Relative Quantity Calculation

The **Relative Quantity Calculation** option allows you to define how the relative quantities of defined bands in lanes will be determined for all reports, histograms, and band information functions: either as a percentage of the signal intensity of an entire lane or as a percentage of the signal intensity of the defined bands in a lane.

Selecting **% of Lane** means that the total intensity in the lane (including bands and the intensity between bands) will equal 100 percent and the intensity of a band in that lane will be reported as a fraction thereof.

Selecting **% of Bands in Lane** means that the sum of the intensity of the defined bands in a lane will equal 100 percent, and the intensity of an individual band will be reported as a fraction of that sum.

If you create, adjust, or remove bands in a lane with **Relative Quantity** defined as **% of Bands in Lane**, the relative quantities of the remaining bands will be updated.

#### Relative Front Calculation

The **Relative Front Calculation** option lets you select the method for calculating the relative positions of bands in lanes. This affects the calculation of both Relative Front and Normalized Rf values.

Relative front is calculated by either:

1. Dividing the distance a band has traveled down a lane by the length of the lane (**Follow Lane**). This is useful if your gel image is curved or slanted.
2. Dividing the vertical distance a band has traveled from the top of a lane by the vertical distance from the top of the lane to the bottom (**Vertical**).

**Note:** “Lane” and “band” refer here to lanes and bands as defined by overlays on the gel image. For example, the top of a lane refers to the beginning of the lane line created in Quantity One, not necessarily the actual gel lane.

Note that if a lane is straight and vertical, both calculation methods will give the same result.

### 2.5.f Devices

This tab contains preferences for imaging devices and the EXQuest Spot Cutter.

Specify the imaging devices that you want to appear on the File menu. By default, all supported imaging devices are selected; clear the checkboxes of the imagers that you do not want to include on the File menu.

If you have connected the EXQuest Spot Cutter, select it under Spot Cutter. Click Setup spot cutter to run the spot cutter calibration wizard.

**Note:** If you are changing the spot cutter settings, you need to restart the application before you can perform calibration.

### 2.5.g Security

The Security tab of the Preferences dialog box contains the security options for CFR mode and the option for enabling Secure mode.

#### CFR Mode

Setting the Security preferences for CFR mode requires the user have administrator privileges. To locate the Security Preferences, select Preferences from the Edit menu and click the Security tab.

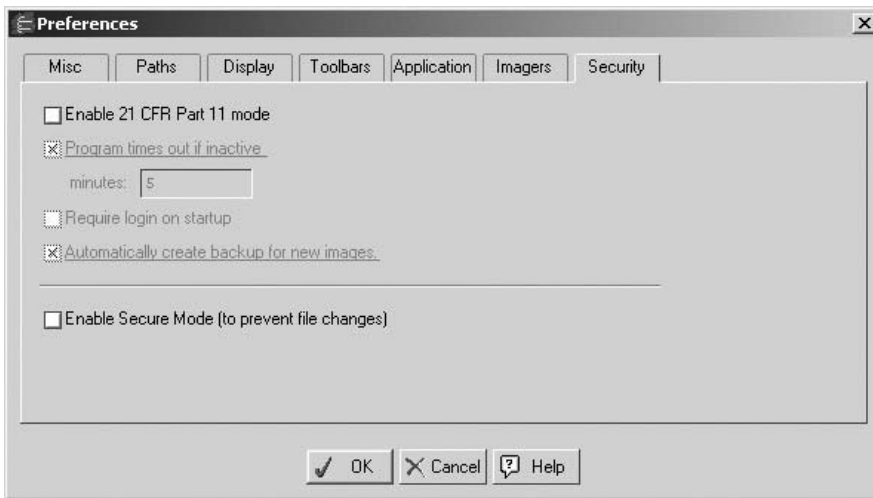


Fig. 2-14. Security Preferences

The following options are available on the Security tab of the Preferences dialog box:

**Enable 21 CFR Part 11 mode** - If this box is unchecked, all other options are inactive. To activate the other options and CFR mode check this box.

**Program times out if inactive** - When this box is checked, the user must enter their password to resume using Quantity One. The minutes field indicates the amount of time that must elapse before the application times out.

**Require login on startup** - Check this box to require the user to enter the user's password when the application opens. This password is the same password as the one used for logging on to Windows. This is an added security measure to prevent unauthorized users from starting Quantity One on a machine where they are not logged in.

**Automatically create backup for new images** - This option, when selected, creates a backup file when you save a newly acquired or imported image. Locate the backup file in the same folder as the original.

See Appendix J, 21 CFR Part 11, for further information.



### Secure Mode

Secure mode is not associated with CFR. However, you can enable Secure mode when CFR is enabled. Secure mode prevents any user from making changes to raw image data on the local machine. If a user opens any image on a machine where Secure mode is enabled, any commands that change the data, such as Subtract background, Filter Wizard, etc. as well as all analysis commands will be inactive (grayed out).

To activate Secure mode, select Secure mode. Once checked, you will be prompted for a new Secure mode password which will be required for disabling Secure Mode.

**Note:** Secure mode only affects images opened on the local machine. Images stored on a network can still be opened and modified on other machines where Secure mode is disabled.

## 2.6 User Settings

If Quantity One is on a workstation with multiple users, each user can have his or her own set of preferences and settings.

In multiple-user situations, the preferences and settings are associated with individual user names. Under Windows, your user name is the name you use to log onto the computer. On a Macintosh, your user name is the **Owner Name** on the *File Sharing* control panel.

If you do not log onto your Windows PC or do not have a Owner Name on your Macintosh, then you do not have a user name and your preferences and settings will be saved in a generic file.



# 3. Viewing and Editing Images

This chapter describes the viewing tools for magnifying and optimizing images. This chapter also describes the tools for cropping, flipping, and rotating images, reducing background intensity and filtering noise, and adding text overlays to images.

These tools are located on the *View*, *Image*, *Window*, and *Edit* menus.

**Note:** The following chapters contain instructions for analyzing X-ray films, wet and dry gels, blots, and photographs. For the sake of simplicity, these are all referred to as “gels.”

## 3.1 Magnifying and Positioning Tools

The magnifying and positioning tools are located on the *View* menu and *Window* menu; some of these functions are also found on the main toolbar.

These commands will only change how the image is displayed on the computer screen. *They will not change the underlying data.*

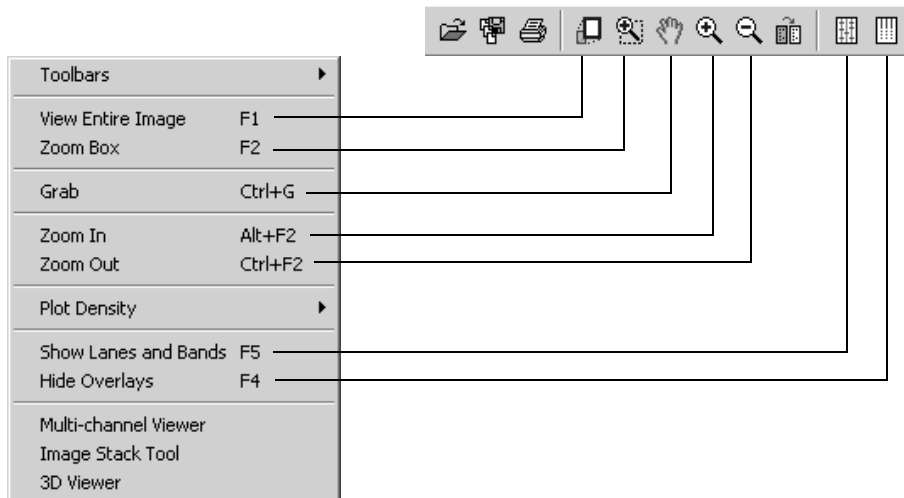


Fig. 3-1. Viewing functions on View menu and main toolbar.

### Zoom Box

Use **Zoom Box** to select a small area of the image to magnify so that it fills the entire image window.

Click on the **Zoom Box** button on the main toolbar or select the command from the View menu. Then drag the cursor on the image to enclose the area you want to magnify, and release the mouse button. The area of the image you selected will be magnified to fill the entire window.

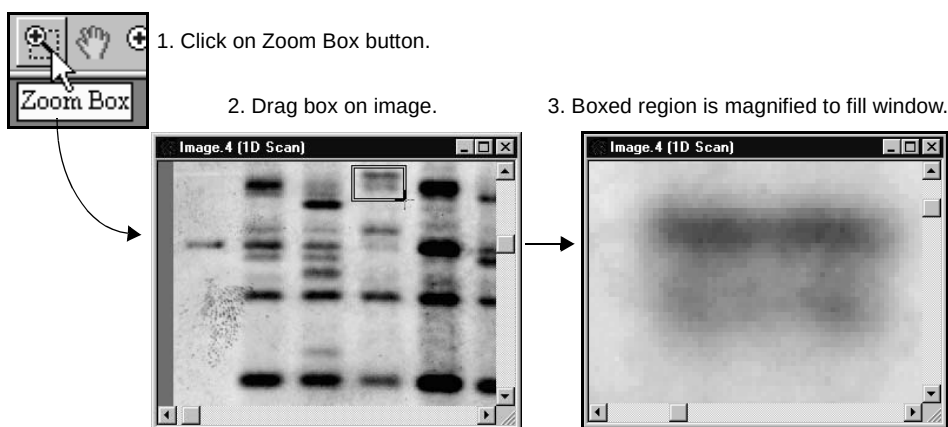


Fig. 3-2. Zoom Box tool.

### Zoom In/Zoom Out

These tools work like standard magnifying tools in other applications.

Click on the **Zoom In** or **Zoom Out** button on the main toolbar (or select from the *View* menu). The cursor will change to a magnifying glass. Click on an area of the image to zoom in or out a defined amount, determined by the setting in the *Preferences* dialog (see section 2.5, *Preferences*).

### Grab

This tool allows you to change the position of the image in the image window. Select **Grab** from the main toolbar or *View* menu. The cursor will change to a “hand” symbol. Drag the cursor on the image to move the image in any direction.

### Arrow Keys

You can also move the image inside the image window by using the **Arrow** keys on the keyboard. Click on an arrow button to shift the image incrementally within the window. The amount the image shifts is determined by the **Pan %** setting in the *Preferences* dialog (see section 2.5, *Preferences*).

### View Entire Image

If you have magnified part of an image or moved part of an image out of view, select **View Entire Image** from the main toolbar or *View* menu to return to the original, full view of the image.

### Centering an Image

You can center the image window on any point in an image quickly and easily using the **F3** key command. This is useful if you are comparing the same region on two gel images and want to center both image windows on the same point.

Position the cursor on the point on the image that you want at the center of the image window, then press the **F3** key. The image will shift so that point is at the center of the image window.

### Imitate Zoom

To magnify the same area on multiple images at the same time, use the **Imitate Zoom** command on the *Window* menu.

First, adjust the magnification in one of the images. Then, with that image window still selected, select **Imitate Zoom**. The zoom factor and region of the selected image will be applied to all the images.

**Note:** **Imitate Zoom** only works on images with similar dimensions.

### Tiling Windows

If you have more than one image open, the **Tile** commands on the *Window* menu allow you to arrange the images neatly on the screen.

Select **Tile** to resize all the windows and arrange them on the screen left to right and top to bottom.

Select **Tile Vertical** to resize all windows and arrange them side-by-side on the screen.

Select **Tile Horizontal** to resize all windows and stack them top-to-bottom on the screen.

## 3.2 Density Tools

The density tools on the *View > Plot Density* submenu and the *Density Tools* toolbar are designed to provide a quick measure of the data in a gel image.

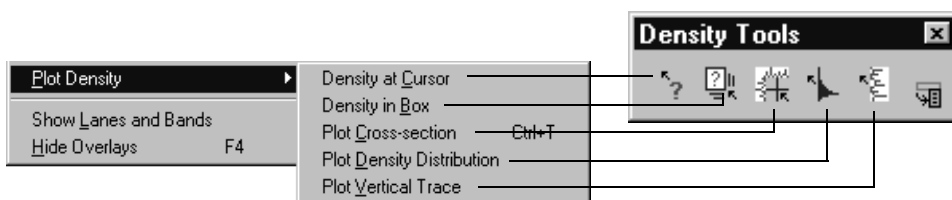


Fig. 3-3. Density tools on the menu and toolbar.

**Note:** The density traces will be slightly different than the traces for functions such as **Plot Lane** or **Plot Band**, because the sampling width is only one image pixel.

### Density at Cursor

Select **Density at Cursor** and click on a band or spot to display the intensity of that point on the image. It also shows the average intensity for a 3 x 3 pixel box centered on that point.

### Density in Box

Select **Density in Box** and drag a box on the image to display the average and total intensity within the boxed region.

### Plot Density Distribution

Select **Plot Density Distribution** to display a histogram of the signal intensity distribution for the part of the image displayed in the image window. The average intensity is marked in yellow on the histogram.

The histogram will appear along the right side of the image. Magnify the image to display the data for a smaller region.

### Plot Cross-section

Select **Plot Cross-section** and click or drag on the image to display an intensity trace of a cross-section of the gel at that point. The horizontal trace is displayed along the top of the image, and the vertical trace is displayed along the side of the image.

The intensity at the point you clicked on is displayed, as is the maximum intensity along the lines of the cross-section.

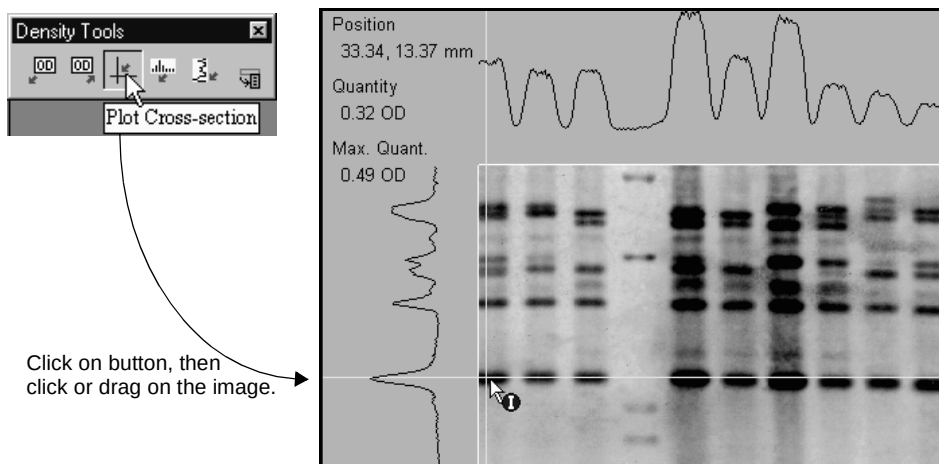


Fig. 3-4. Plot Cross-section tool.

### Plot Vertical Trace

Select **Plot Vertical Trace** and click or drag on the image to plot an intensity trace of a vertical cross-section of the image centered on that point.

## 3.3 Showing and Hiding Overlays

To conceal all plots, traces, info boxes, and overlays on an image, select **Hide Overlays** from the main toolbar or View menu.



**Note:** Click once on **Hide Overlays** to conceal the overlays. Click twice to deassign any function that has been assigned to the mouse.

To redisplay the lane and band overlays, select **Show Lanes and Bands** from the *View* menu or main toolbar.

### 3.4 Multi-Channel Viewer

The *Multi-Channel Viewer* can display different types and levels of fluorescence in a gel that has been imaged at different wavelengths. You can merge the data from up to three different images of the same gel.

**Note:** The gel images being compared must be exactly the same size. When changing image filters, be careful not to move the gel. If the images are not exactly the same size, you can use the **Crop** tool (see Section 3.9.a, Cropping Images) to resize them.

With at least one image open, select **Multi-Channel Viewer** from the *View* menu. The first open image will be displayed in the viewer window using the **Red channel**, and the image name will be displayed in the field at the top of the viewer.

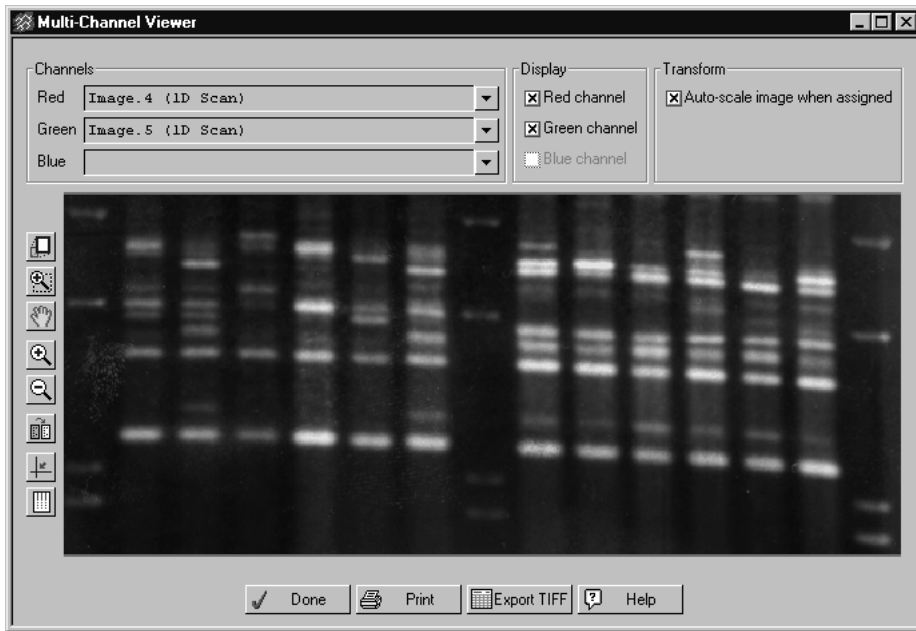


Fig. 3-5. Multi-Channel Viewer.

**Note:** The color channel used to display the image in the viewer has no relation to the filter used to capture the image. The red, green, and blue channels are only used to distinguish different images.

To add another image to the viewer, make sure the image is open and click on the pull-down button next to the **Green** or **Blue** name field. Select the image name from the pull-down list. Add a third image using the same procedure.



Fig. 3-6. Selecting images to display in the viewer.

To reassign the different images to different channels, use the pull-down buttons to the right of the name fields. Select **<clear>** from the pull-down list to remove an image from that channel of the viewer.

### Viewing Options

To remove a particular color channel from the display, click in the checkbox associated with that channel to deselect it.

Select the **Auto-Scale Image When Assigned** checkbox to automatically adjust the brightness and contrast of each loaded image based on the data range in the image. This invokes the **Auto-scale** command from the *Transform* dialog (see section 3.8, *Transform*) when an image is first opened in the viewer. Note that this setting affects only how the image is displayed in the viewer, not the actual data.

**Note:** If you deselect this checkbox, any images currently displayed will remain auto-scaled. Click on the **Transform** button in the viewer and click on the **Reset** button in the *Transform* dialog to undo auto-scaling.

Buttons for various viewing tools are included in the *Multi-Channel Viewer*. Tools such as **Zoom Box** and **Grab** will change the display of all the images in the viewer at once.

Click on the **Transform** button to open the *Transform* dialog. In the dialog, you can adjust the display of each channel independently by selecting the appropriate channel option button. Similarly, the **Plot Cross-section** command will report the intensity of each channel separately.

### Exporting and Printing

Click on the **Export** button to export a 24-bit TIFF image of the merged view. This will open a version of the *Export to TIFF* dialog (see section 13.6, *Export to TIFF Image*). Note that you cannot export data from the *Multi-Channel Viewer*—only the current view of the image (designated as **Publishing Mode** in the *Export* dialog). The colors in the viewer will be preserved in the exported TIFF image.

To print a copy of the merged view to a color or grayscale printer, click on the **Print** button.

## 3.5 3D Viewer

The *3D Viewer* allows you to see a three-dimensional rendering of a portion of your image. This is important for such instances as determining whether a selected band is actually two or more separate bands.

To see a 3D rendering of a portion of your image, select 3D Viewer from the View menu. Your cursor turns into a crosshair. Click and drag your cursor over the image area you would like to view creating a box.

**Note:** viewing a large area of your image may reduce performance.

- To reposition the box, position your cursor at the center of the box. The cursor appearance will change to a multidirectional arrow symbol. You can then drag the box to a new position.
- To resize the box, position your cursor on a box corner. The cursor appearance will change to a bi-directional arrow. You can then drag that corner in or out, resizing the box.
- To redraw the box, position your cursor outside the box and click once. The box disappears, and you can then draw a new box.

To view the selected area, position your cursor inside the box slightly off-center. The cursor appearance will change to an arrow. Click once to open the 3D Viewer.

**Note:** If you are having problems using the 3D Viewer, install the latest drivers for your graphics card. If, after updating your drivers, you are still having problems, go to the Display tab of the Preferences dialog box and select Disable hardware

acceleration for 3D viewer. This allows Quantity One to emulate a graphics driver to render a 3D image. See Section 2.5.c, Display, for further information.

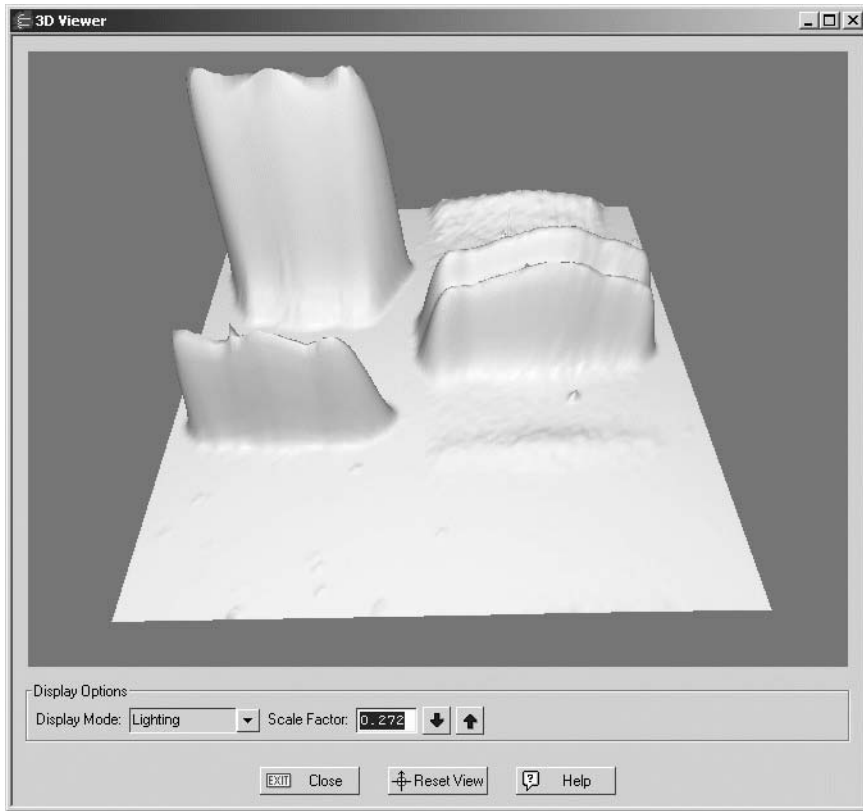


Fig. 3-7. 3D Viewer

### 3.5.a Positioning the Image

#### Windows

For Windows operating systems, use your mouse to reposition and rotate the image.

## Quantity One User Guide

---

- **Rotate the image** - Left click and drag to rotate the image.
- **Reposition the image** - Right click and drag to reposition the image.
- **Zoom in/out** - To zoom in or out, Click the center mouse button or roll the wheel. If you do not have a three button mouse or a mouse with a wheel, hold down the shift key and left click and drag to zoom in or out.

### Macintosh

For Macintosh operating systems, use mouse and keyboard combinations to reposition and rotate the image.

- **Rotate the image** - Click and drag to rotate the image.
- **Reposition the image** - Ctrl>click and drag to reposition the image.
- **Zoom in/out** - Shift>click and drag to zoom in or out.

### 3.5.b Display Mode

The 3D Viewer window allows you to view the image in three different modes; wire frame, lighting, and textured.

- Wire-frame shows the image in a transparent frame view.
- Lighting shows the image with different areas of light and shadow depending on the angle of view. Use the slider bar to adjust the intensity of the lighting.
- Texture gives the image texture.

Use the Scale function to scale the image. This is useful for viewing shallow spots in the 3D Viewer.

If you lose the image because you moved it too far past the window border, or rotated it and disoriented the view, click Reset View to return the image to the original view.

**Note:** Reset view does not change the scale factor. To reset the scale factor, close the 3D Viewer and click the box again to re-open the 3D Viewer with the original scale factor.

## 3.6 Image Stack Tool

Use the *Image Stack Tool* to scroll through a series of related gel images. You can easily compare bands that appear, disappear, or change size in different gels run under different conditions.

**Note:** The images should be close to the same size, with bands in the same relative positions. You can use the **Crop** tool to resize images.

With all the images open, select **Image Stack Tool** from the *View* menu. The *Image Stack Tool* window will open.

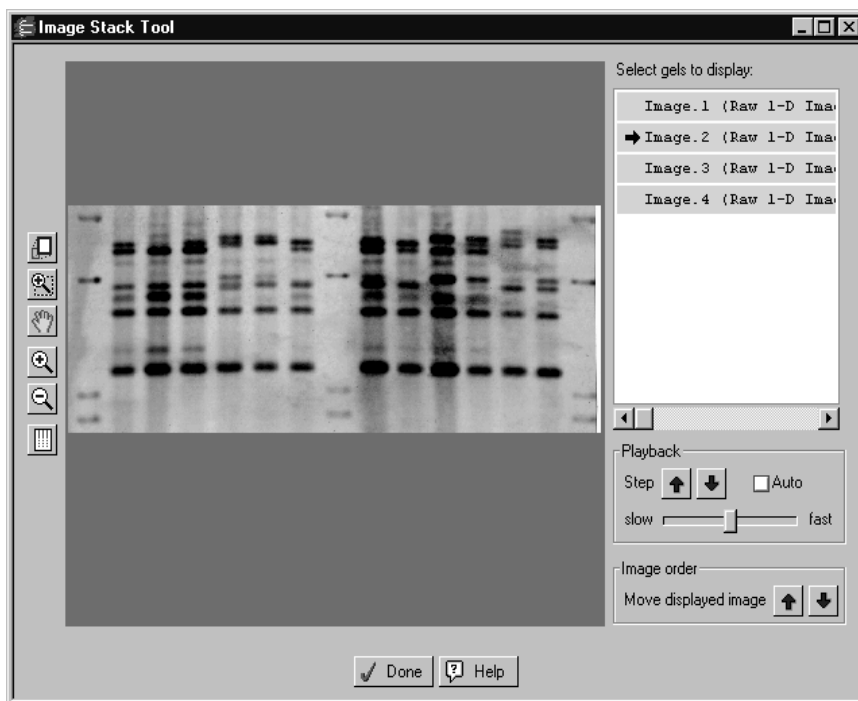


Fig. 3-8. Image Stack Tool.

## Quantity One User Guide

---

In the *Image Stack Tool* window, all open gels are listed in the field to the right of the display window. To select an image to display, click on a gel name. The name will appear highlighted with an arrow and the image will appear in the window.

Click on another gel name to display that image.

Buttons for various viewing tools are aligned next to the *Image Stack Tool* window. These commands will change the display of all the images in the stacker at once (e.g., magnifying one image will magnify the same relative area in all the images).

Using the controls below the list of names, you can reorder the images and/or scroll through them in the stacker.

### Reordering Images

To reorder the images in the stacker, first select an image name in the list, and then click on the **Move displayed image** arrow buttons to move it up or down in the list.

### Image Playback

Using the controls under **Playback**, you can scroll through the images in the stacker.

First, highlight some or all of the gel names using **Shift-click** or **Ctrl-click** key commands. With multiple images selected, the **Step** arrow buttons become active. Click on the arrow buttons to scroll through the list of selected gels.

Alternatively, click on the **Auto** checkbox next to the arrow buttons to begin automatically scrolling through the list. You can adjust the auto-scroll speed using the **Slow/Fast** slider.

## 3.7 Colors

Select **Colors** from the *Edit* menu to open a dialog in which you can adjust the colors of the image, as well as windows, buttons, overlays, and other features of the display.



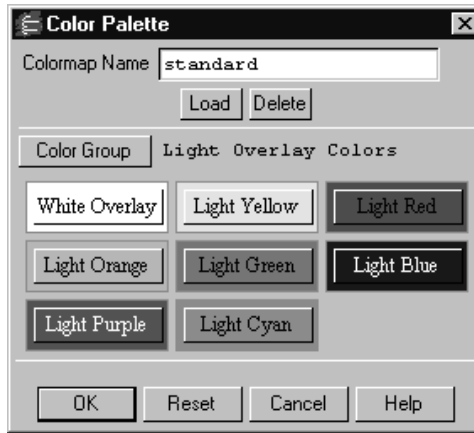


Fig. 3-9. Colors dialog.

### Selecting a Color Group

In the *Colors* dialog, click on the **Color Group** button to select the colors of a particular group of objects (e.g., pop-up boxes, image colors, etc.).

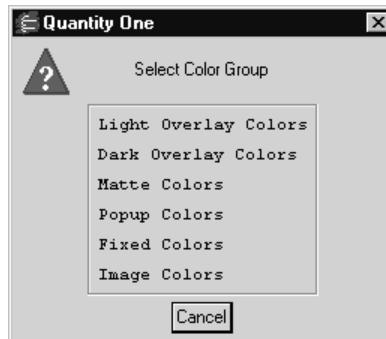


Fig. 3-10. List of Color Groups.

Click on a color group in the list to select it.

### Changing a Color

After you have selected the color group to change, click on the specific color button. In the *Color Edit* dialog, adjust the RGB values of the color you selected.

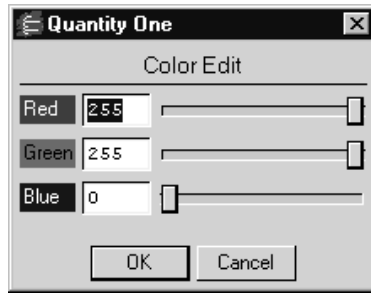


Fig. 3-11. Color Edit dialog.

### Saving/Selecting a Defined Set of Colors

After you have changed the colors within color groups, you can save these settings for future use on other images. The **Colormap Name** field displays the name of a defined set of colors and color groups. There are several predefined colormaps, or you can create your own.

To select a predefined colormap, click on the **Load** button.

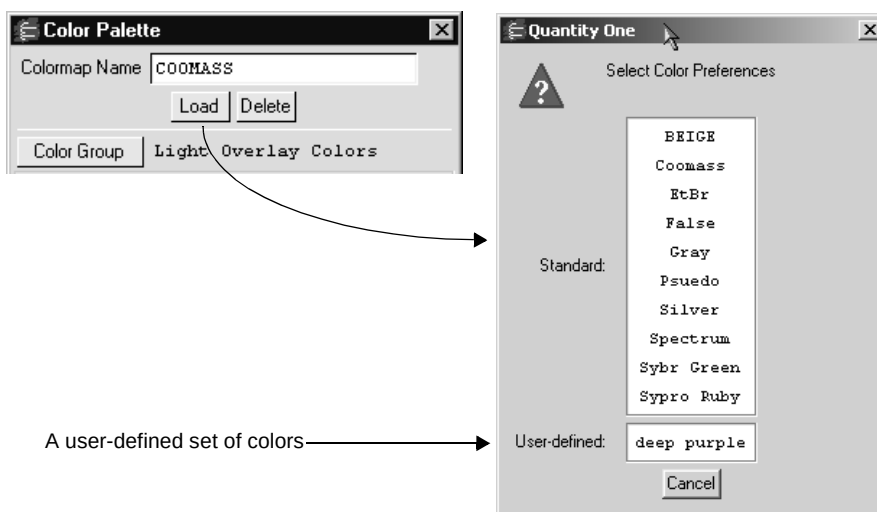


Fig. 3-12. Selecting a Colormap.

From the list displayed, click on the set of colors you want to apply.

To create your own colormap, adjust the colors within the color groups as described above and type in a new colormap name. Click on **OK** to apply the changes.

To remove a colormap, click on the **Delete** button. Select the colormap to be deleted from the displayed list. A pop-up box will ask you to confirm the deletion.

To return to the **Standard** colormap, click on the **Reset** button. All colors will reset to their default values.

### 3.8 Transform

Use the *Transform* dialog to adjust the image brightness and contrast and optimize the image display. These controls affect the image display only, and *will not change* the underlying data.

With an image open, select **Transform** from the *Image* menu or main toolbar.

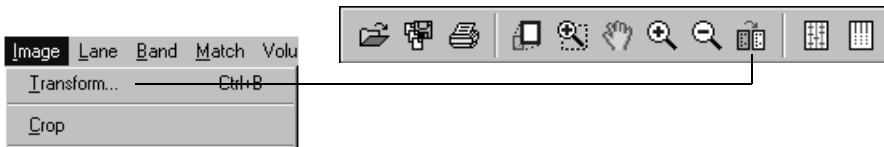


Fig. 3-13. Transform command.

The *Transform* dialog contains a Preview Window, a Frequency Distribution histogram, a Transform Plot, and three main methods of optimizing the image: **Auto-scale**, **High** and **Low** sliders, and a **Gamma** slider. You can use these controls to adjust the way the software transforms raw image data into the visual display.

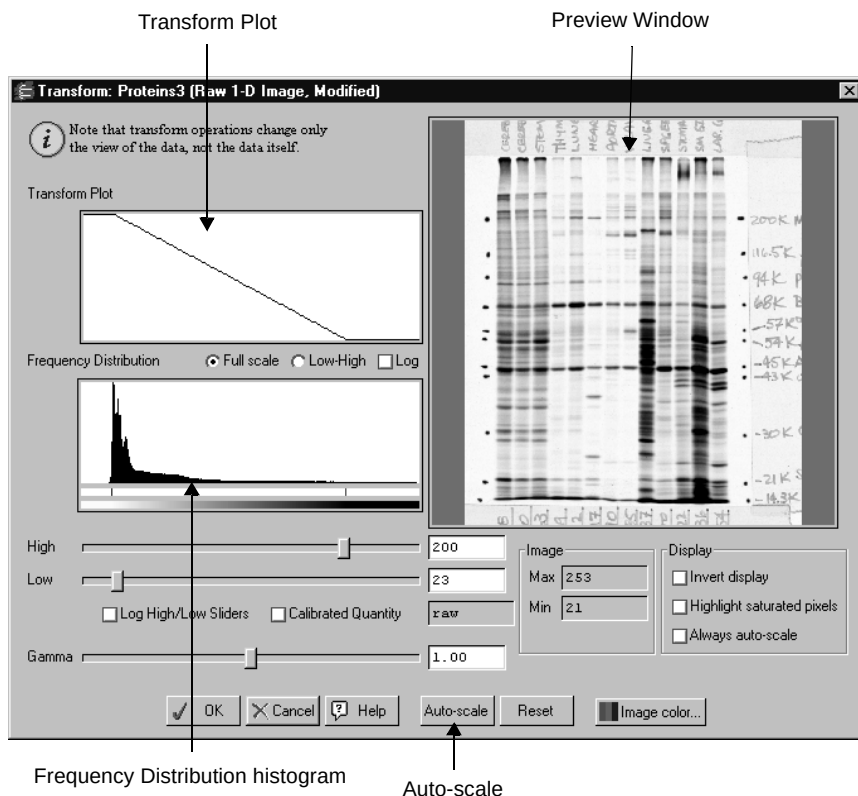


Fig. 3-14. Transform dialog.

### 3.8.a Transform Subwindows

#### Preview Window

The Preview Window shows a smaller view of the same image that is displayed in the main image window. Changes in the controls are automatically reflected in the Preview Window. They are only applied to the main image when you click on **OK**.

You can use viewing tools such as **Zoom Box**, **Grab** and **View Entire Image** in the Preview Window just as you can in the main image window, to focus on particular regions of interest.

### Frequency Distribution Histogram

The Frequency Distribution histogram shows the total data range in the image and the amount of data at each point in the range. In a typical scan, there is a signal spike at the left (“gray”) end of the histogram due to background noise.

### Transform Plot

The Transform Plot is a logarithmic representation of how the raw pixel data are mapped to the pixels of the computer screen.

## 3.8.b Transform Controls

### Auto-scale

Click on the **Auto-scale** button to optimize the image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). This enhances minor variations in the image, making fine details easier to see. You can then “fine-tune” the display using the **High**, **Low**, and **Gamma** sliders described below.

### High/Low Sliders

If **Auto-scale** doesn’t give you the appearance you want, use the **High** and **Low** sliders to redraw the image manually. Drag the **High** slider handle to the left to make weak signals appear darker. Drag the **Low** slider handle to the right to reduce background noise.

As you drag the sliders, the slider markers on the Frequency Distribution histogram will move. Everything to the left of the **Low** marker will be remapped to minimum intensity, while everything to the right of the **High** marker will be remapped to maximum intensity. Using the histogram, you can position the markers at either end of the data range in the image, and use the low slider to cut off the “spike” of background noise.

## Chapter 3. Viewing and Editing Images

You can also type specific **High** and **Low** values in the text boxes next to the sliders. Click anywhere on the slider bars to move the sliders incrementally.

**Log High/Low Sliders** changes the feedback from the slider handles, so that when you drag them, the slider markers move a shorter distance in the histogram. This allows for finer adjustments when the data is in a narrow range.

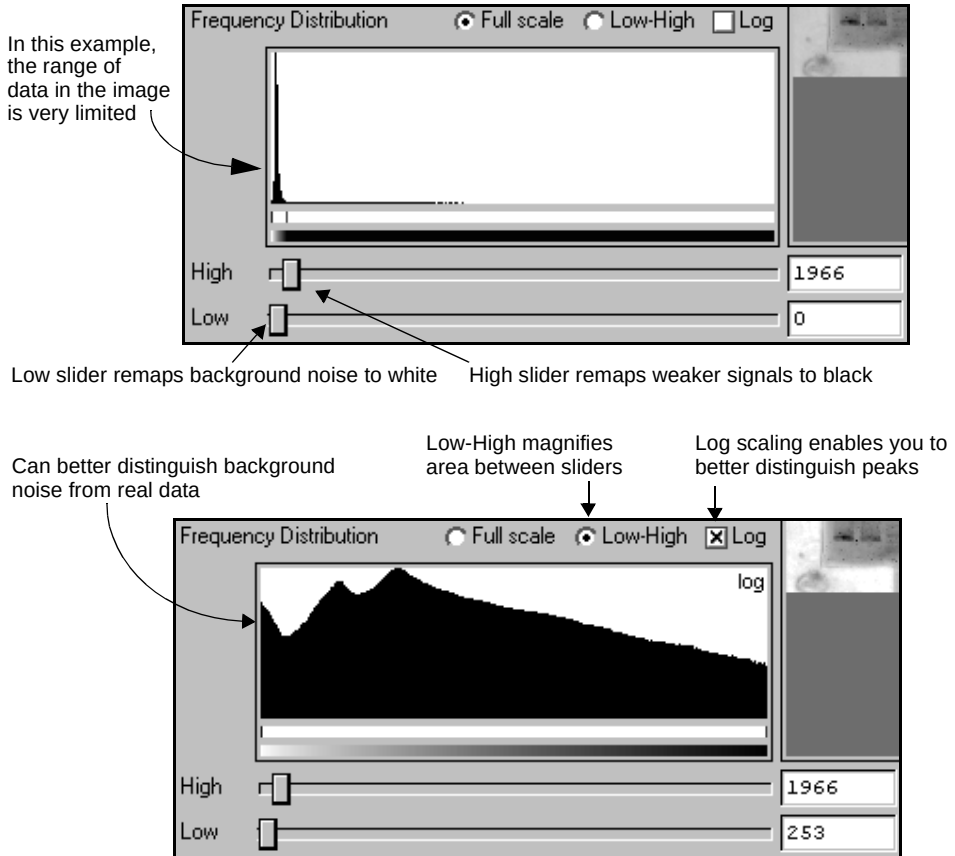


Fig. 3-15. Two views of the Frequency Distribution histogram.

### Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjust the **Gamma** slider handle to expand or compress the contrast range at the dark or light end of the range. This is reflected in the Transform Plot and Preview Window.

### 3.8.c Other Features

#### Full Scale and Low-High

The **Full Scale** and **Low-High** option buttons adjust how the range of data in the image is displayed in the Frequency Distribution histogram and Transform Plot. They do not change how the data is displayed in the image window.

Select **Full Scale** to adjust the displays so that they show the full intensity range of the image.

Select **Low-High** to magnify the range between the **Low** and **High** sliders. This makes it easier to view the data if it does not occupy the full intensity range of the image.

#### Log

The **Log** checkbox changes the way the data is displayed in the histogram so you can better discern subtle changes in signal intensity.

#### Image Max/Min and Units

Image **Max** and **Min** display the range of intensity in the gel image.

The image units are determined by the type of scanner used to create the image. For images measured in O.D.s, you can display the maximum and minimum O.D values in the image by selecting the **Calibrated Quantity** checkbox. If this box is not selected, the maximum and minimum numeric pixel values are displayed.



### Image Color

Click on this button to open a list of color maps, which you can define using the **Colors** command on the *Edit* menu (see section 3.7, Colors). Select a color map from the list to change the image in both the *Transform* window and the image window.

### Invert Display

Select the **Invert Display** checkbox to change light bands on a dark background to dark bands on a light background, and visa versa. The image data will not change—only the display.

### Highlight Saturated Pixels

Select the **Highlight Saturated Pixels** checkbox to highlight areas of saturation in the image in red.

### Always Auto-Scale

Select this checkbox to automatically **Auto-Scale** every new image that you open. The software will examine the data range in every image and optimize it accordingly. This setting disables the other image-optimization controls in the *Transform* dialog.

### Reset

To return to an unmodified view of the image, click on **Reset**.

## 3.9 Resizing and Reorienting Images

The *Image* menu and toolbar contain commands for changing the size and orientation of images.

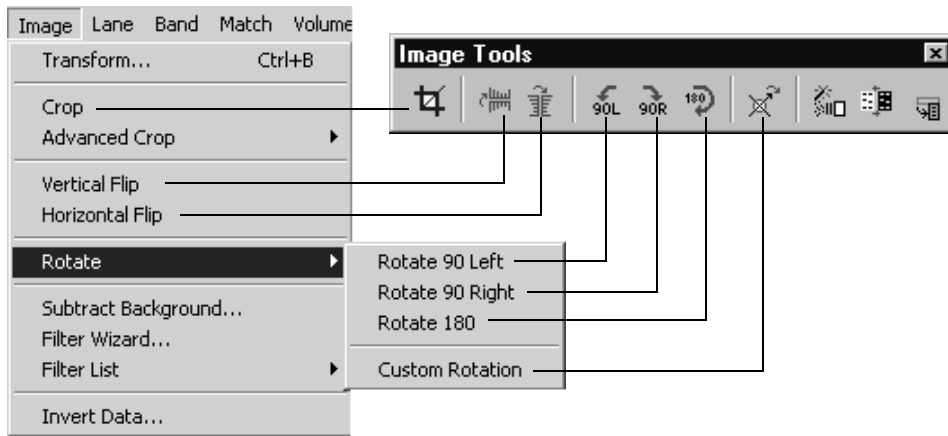


Fig. 3-16. Resizing and reorienting tools.

**Note:** Many of these commands will permanently change the image. You will be prompted to confirm any permanent changes.

### 3.9.a Cropping Images

Use the **Crop** tool to eliminate unwanted parts of an image, such as border space around the gel. You can also use this command to reduce the file size of an image.

Select **Crop** from the *Image* menu or toolbar. The cursor will change to a Crop symbol.

Define the region to be cropped by dragging the cursor across the image, creating a box. Everything outside the box will be deleted.

The dimensions of the crop area (in millimeters and number of pixels) and the file size of the image inside the crop area are listed at the bottom of the crop box.

1. To *reposition* the crop box, position the cursor at the center of the box. The cursor will change to a multidirectional arrow. Then drag the box to a new position.

2. To *resize* the box, position the cursor on a box border line or corner. The cursor will change to a bidirectional arrow. You can then drag the border or corner in or out, resizing the box.
3. To *redraw* the box, position the cursor outside the box. The cursor will change back to the Crop tool, and you can redraw the box.

After you are satisfied with the size of crop box, position the cursor inside the box slightly off-center. The cursor will change to a scissors symbol. Then click to perform the crop.

A pop-up box will prompt you to: (1) crop the original image, (2) crop a copy of the original image, or (3) cancel the operation.

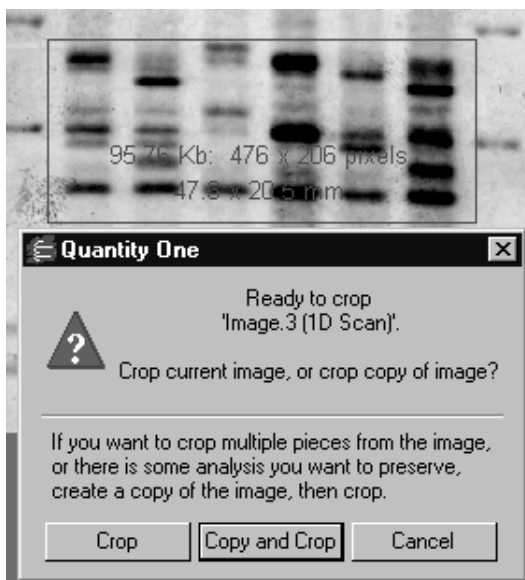


Fig. 3-17. Crop box and pop-up Crop dialog.

If you click on the **Copy and Crop** button, you will be prompted to enter the name and version number of the image copy before cropping.

### Advanced Crop Commands

To ensure that your scans are exactly the same size and shape, you can use the tools on the Image > Advanced Crop submenu to save the crop box from one scan and apply it to others. These tools also allow you to crop a gel of the same scan.

To define and save a crop box and apply it to another scan:

1. Select Define Crop Area from the submenu and drag a crop box on an image. Position the box as described in Cropping Images.
2. Select Place Crosshair from the submenu and click a landmark inside the box that is present in all the gels you want to crop. This could be a spot or some other image detail. The crosshair will make it easier to position the box in the other images so that it encloses the same area.
3. Select Save Crop Settings from the submenu, enter a name for the current crop settings in the pop-up box, and click Apply.
4. Complete the crop action in the current image by positioning your cursor inside the box slightly off-center and clicking to perform the crop, as described in Cropping Images.
5. Open or select the next image you want to crop, select Load Crop Settings from the submenu, and select the name of the settings you saved. The crop box and crosshair will appear on the image.
6. Reposition the crop box so that the crosshair is correctly aligned with the appropriate image object, then complete the crop as described in step 4 above.

To delete any crop settings you have saved, select Delete Crop Settings from the submenu and choose the settings to be deleted from the list.

### 3.9.b Flipping and Rotating Images

Use the image flipping and rotating commands to reorient lanes and bands for proper analysis.

**Note:** These actions will erase any analysis you have performed on the image. You will be prompted to confirm the changes.

### Flipping

To flip the image right-to-left, select **Horizontal Flip** from the *Image* menu or toolbar. To flip the image top-to-bottom, select **Vertical Flip**.

### 90° Rotations

Select **Rotate 90 Left**, **Rotate 90 Right**, or **Rotate 180** from the *Image > Rotate* menu or *Image* toolbar to perform the specified rotation.

### Custom Rotation

Use the **Custom Rotation** command to rotate the image in increments other than 90°. Select **Custom Rotation** from the *Image > Rotate* submenu or *Image* toolbar. A green “plus” sign will appear next to the cursor. Click on the image and a circular overlay with an orange arrow will appear. A pop-up box will indicate the angle of rotation in degrees and radians.

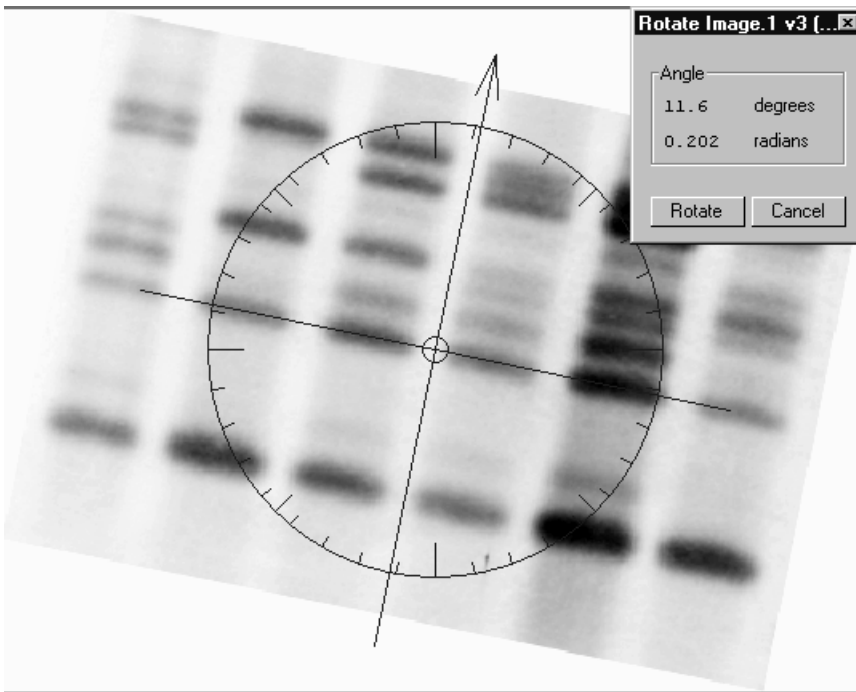


Fig. 3-18. Custom rotation; the arrow points in the direction of the new top of the image.

To perform the rotation, position the cursor on the arrowhead and drag. As you drag, the arrow will rotate and the angle in the box will change. Position the arrow so that it points in the direction of the new top of the image.

**Note:** To center the arrow on a particular point on the image (e.g., to align along a particular lane), position the cursor on the point and press the F3 key. The center of the arrow will shift to the new position.

To complete the rotation, click on the **Rotate** button in the pop-up box. Another window will open containing the rotated image, and you will be prompted to save this image under a new name or version number.

If you are not satisfied with the rotated image, close the window without saving and start over.

**Note:** Because an image is composed of square or rectangular pixels, **Custom Rotation** performs some minor smoothing on the image for rotations other than 90°. Also, any analysis performed on the image will be lost.

### 3.10 Whole-Image Background Subtraction

Image background due to gel opacity, random signal noise, or other factors can interfere with quantitation and data analysis. Quantity One has several tools for subtracting background intensity from gel images. This section describes whole image background subtraction. You can also subtract background from individual lanes (see section 4.2, Lane-Based Background Subtraction) and bands (see section 7.6, Volume Background Subtraction).

Whole-image background subtraction is useful for reducing background resulting from the opacity of the carrier medium (film, gel matrix, or blot matrix) or film fogging.

**Note:** Whole-image background subtraction permanently changes the image. You will be prompted to confirm the change.

Select **Subtract Background** from the *Image* menu.

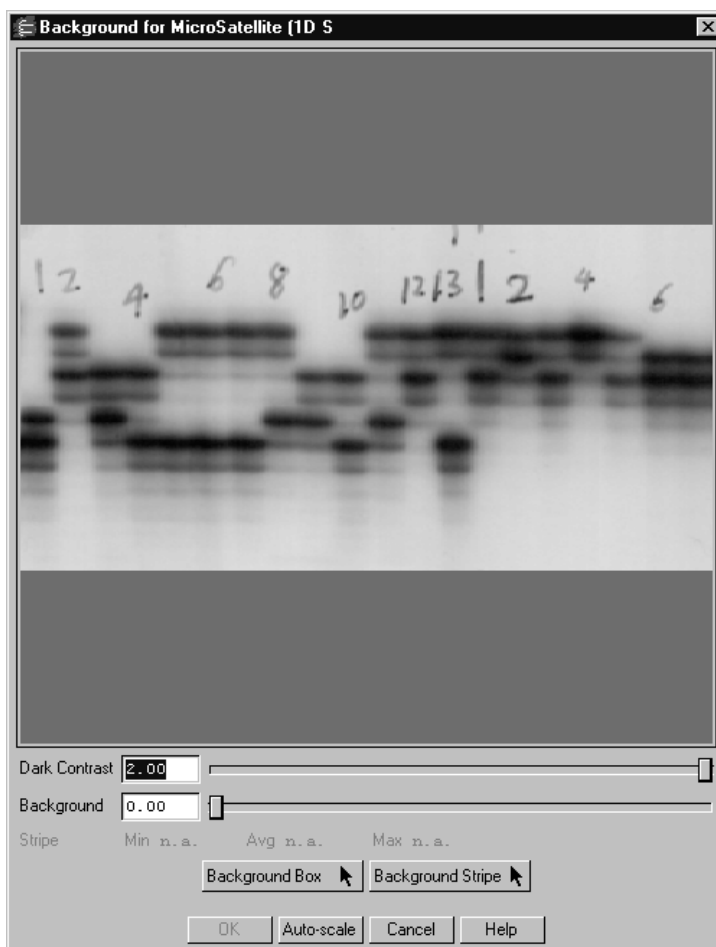


Fig. 3-19. Subtract Background dialog.

The *Subtract Background* dialog has a preview window, which contains a smaller view of the image. Changes in the subtract background controls are reflected in the preview window and are only applied to the main image when you click on **OK**.



### Auto-scale

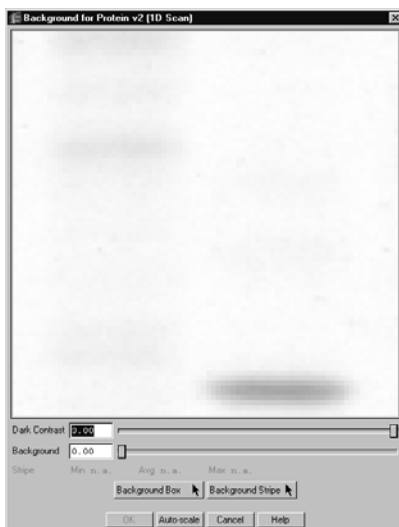
Click on the **Auto-scale** button to automatically adjust the **Dark Contrast** and **Background** settings to optimal levels. You can then manually adjust these settings using the other controls.

### Dark Contrast Slider

Use the **Dark Contrast** slider to reveal the level of background in the image before subtraction. This slider is similar to the **High** slider in the *Transform* dialog. Drag the slider handle to the left to make faint signals appear stronger. Click on the slider bar to move the slider incrementally, or type a value into the field next to the slider.

**Note:** The **Dark Contrast** slider itself does not eliminate background intensity; therefore, the **OK** button will not activate if you only adjust this slider. If you want to adjust the display contrast *without* subtracting background, use the **Transform** command (see section 3.8, Transform).

Before adjustment: the image seems clear of background



After adjustment: background levels are more apparent

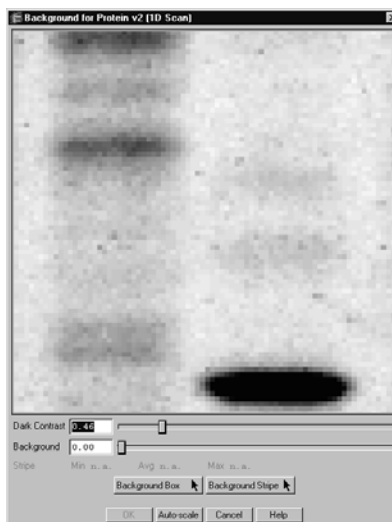


Fig. 3-20. Dark Contrast adjustment reveals true levels of background in the image.

### Background Slider

To manually adjust the background subtraction levels, drag the **Background** slider to the right. You can also move the slider incrementally by clicking on the slider bar, or type a specific value into the field next to the bar.

Objects with signal intensities lower than the subtraction level will be eliminated from the image when you click on **OK**.

### Background Box

Use the **Background Box** function to define a background area in the gel that is representative of the background in the entire image. This method of subtraction is useful for images with uniform backgrounds.

Click on the **Background Box** button, then drag on a background area of the image. The average intensity of the pixels in the box will be used as the background level to be subtracted from the entire image.

### Background Stripe

The **Background Stripe** function is useful for gels in which the background changes from top to bottom (e.g., gradient gels).

Click on the **Background Stripe** button, then drag on a background region to create a rectangular box down the length of the image. The average intensity of each horizontal row of pixels in the stripe will be subtracted from each pixel in that row across the entire gel. This way, if the image has more background at the bottom than at the top, more background will be removed from the lower regions of the image.

**Note:** Make sure that the background stripe runs the entire length of the lanes down the gel. The average of the topmost row in the stripe will be subtracted from all rows above the stripe, and the average of the bottommost row will be subtracted from all rows below.

The minimum and maximum intensities in the stripe are displayed next to the **Min** and **Max** labels in the box. Also, the average intensity value for the entire stripe is displayed next to **Avg**.

### Completing the Subtraction

When you are satisfied with the background subtraction shown in the preview image, click on **OK**. Because whole-image background subtraction is irreversible, you will be prompted to subtract from the original image, subtract from a copy, or cancel the operation.

If you choose **Copy and Subtract**, enter the name and/or version number for the new copy in the pop-up box and click on **OK** to complete the command.

## 3.11 Filtering Images

Filtering is a process that removes small noise features on an image while leaving larger features (e.g., bands) relatively unaffected. A wide range of filters are available for removing different types of noise from images. Depending on the nature of your data, you will probably need to use only one or two of the available filters. However, you should experiment with several different filters before selecting the ones that work best for your images.

The filtering commands are located on the *Image* menu and toolbar.

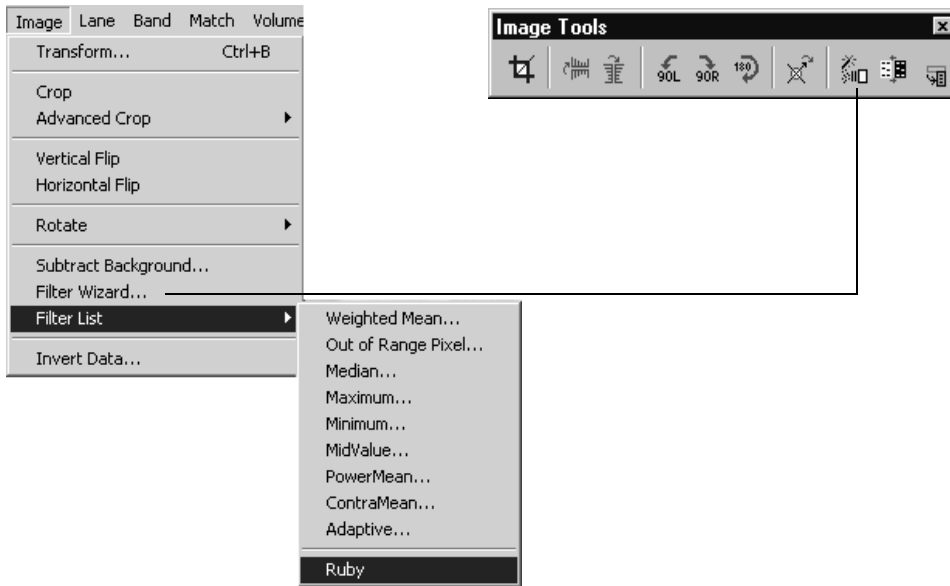


Fig. 3-21. Filtering commands.

**Note:** Since filtering is an irreversible process, you will be asked if you want to create a copy of the original image before you filter. If you are experimenting with various filters, you should create copies of the image and compare them side-by-side. If you filter the original image and save it, *you cannot return to the original, unfiltered state.*

### 3.11.a Filter Wizard

The *Filter Wizard* is designed to guide you through the filter selection process. First, you identify the type of noise in the image. Next, select the size of the filter to use on that noise. Finally, filter the image.

Select **Filter Wizard** from the *Image* menu or toolbar to open the dialog.

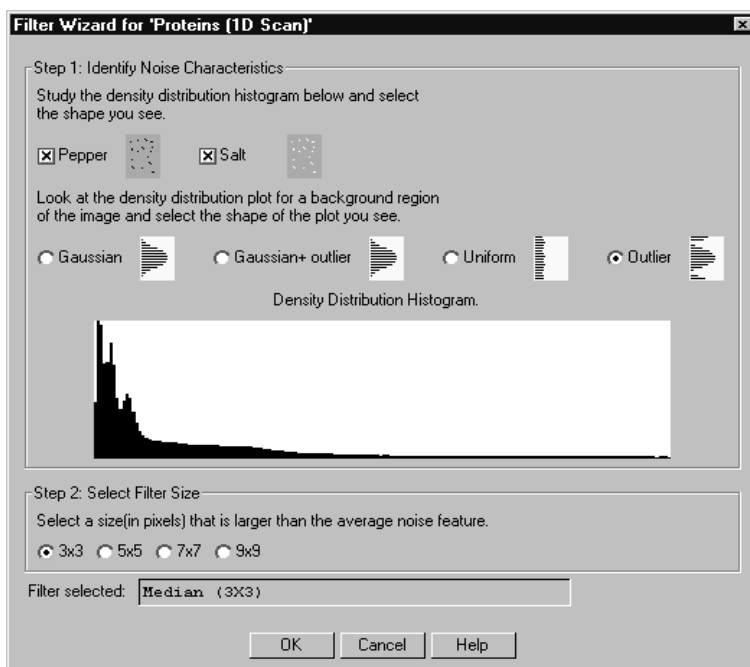


Fig. 3-22. Filter Wizard dialog.

The dialog contains settings for identifying the different types of noise in the image. It also includes a density distribution histogram of the noise in the image to aid in filter selection.

### Step I: Identify Noise Characteristics

The first step is to identify the type of noise in the image. Examine both the image and the density distribution histogram, then select one, both, or neither of the following checkboxes:

- **Salt.** This type of noise appears as specks that are lighter than the surrounding background. The density distribution histogram of this type of noise displays noise peaks at the high end of the range (right end of the plot). This type of noise is common in electronic cameras with malfunctioning pixels. It can also be

caused by dust or lint in the imaging optics or scratches on photographic film. Salt is a type of outlier noise (see below).

- **Pepper.** This type of noise appears as specks that are darker than the surrounding background. The distribution histogram of this type of noise displays noise peaks at the low end of the range (left end of the plot). Its causes are similar to those of salt noise. Pepper is a type of outlier noise (see below).

Next, select one of the following option buttons to describe additional features of the noise.

- **Gaussian.** The distribution histogram of this type of noise has a Gaussian profile, usually at the bottom of the data range. This type of noise is usually an electronic artifact created by cameras and sensors, or by a combination of independent unknown noise sources.
- **Uniform noise.** This type of noise appears in the histogram as a uniform layer of noise across the data range of the image.
- **Outlier noise.** This category of noise includes salt and pepper noise (see above). The distribution histogram of this type of noise displays noise peaks at the high and low ends of the range.

After you have identified the type of noise, go to the next step.

### Step 2: Select Filter Size

Image noise is filtered by means of a filtering window (or kernel), which is measured in pixels. This filtering window slides across the image, processing the pixels within it.

The available filter dimensions range from **3 x 3** to **9 x 9** pixels. To select an appropriate size, magnify a background region of the image so that you can see the individual pixels. The filter size you select should be larger than the average noise feature but smaller than the data features.

**Note:** A smaller filter will alter the image less than a larger filter. Large filters can result in better suppression of noise, but can also blur desirable features in the image.

### Step 3: Begin Filtering

After you have completed the selections, the filter name and size will be displayed at the bottom of the *Filter Wizard* dialog.

To begin filtering, click on the **OK** button. Because filtering is an irreversible process, you will be prompted to filter the original image, filter a copy of the image, or cancel the operation.

If you choose **Copy and Filter**, enter a name and/or version number for the new copy in the pop-up box and click on **OK**.

### 3.11.b Selecting a Filter Directly

If you know the type and size of filter you want, you can select it directly from the *Image > Filter List* submenu. The submenu includes all the available filters.

The types of filters are:

- **Weighted Mean.** This filter is useful for reducing Gaussian noise. It calculates the weighted mean of the pixels within the filtering window and uses it to replace the value of the pixel being processed.
- **Out of Range Pixel.** This filter is useful for suppressing salt-and-pepper noise; its effect on Gaussian noise is minimal. This filter calculates the mean of the pixel values in the filtering window, including the pixel being processed. If the difference between the mean and the individual pixel value is above a certain threshold, then the individual value is replaced by the mean.
- **Median.** Also useful for suppressing salt-and-pepper noise, this filter calculates the median value of the pixels within the filtering window and uses it to replace the value of the pixel being processed. The median filter produces very little blurring if a small-sized window is selected.
- **Maximum.** This filter is useful for eliminating pepper noise in an image (it worsens the effect of salt noise). It replaces the value of the pixel being processed with the maximum value of the pixels within the filtering window.
- **Minimum.** This filter replaces the value of the pixel being processed with the minimum pixel value within the filtering window. This filter is useful for eliminating salt noise in an image (it worsens the effect of pepper).
- **MidValue.** This filter is useful for suppressing uniform noise within an image; however, it worsens the effect of pepper and salt. This filter replaces the value of

the pixel being processed with the mean of the maximum and minimum pixel values within the filtering window.

- **PowerMean.** This filter is useful for suppressing salt and Gaussian noise within an image (it worsens the effect of pepper noise). It replaces the value of the pixel being processed with the power mean of the pixel values within the filtering window.
- **ContraMean.** This filter is useful for suppressing pepper and Gaussian noise within an image (it worsens the effect of salt). It replaces the value of the pixel being processed with the contra-harmonic mean of the pixel values within the filtering window.
- **Adaptive.** This filter is useful for suppressing Gaussian noise and salt and/or pepper within an image. If the image contains a mix of salt and pepper, select this filter.

To begin filtering, select a filter type from the pull-down list. A pop-up box will prompt you to select a filter size.

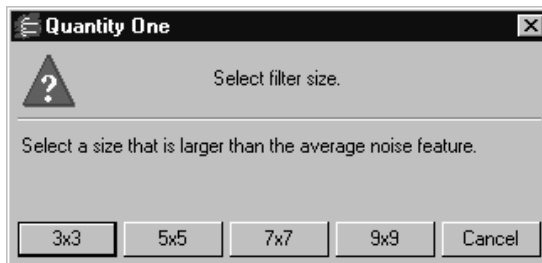


Fig. 3-23. Selecting a filter size.

Click on a button to select a size. (See, Step 2: Select Filter Size for guidance.) Because filtering is an irreversible process, you will be prompted to filter the original image, filter a copy of the image, or cancel the operation.

If you choose **Copy and Filter**, enter a name and/or version number for the new copy in the pop-up box and click on **OK**.



## 3.12 Invert Data

The **Invert** checkbox in the *Transform* dialog (see “Invert Display” on page 23) inverts the display of the image. However, in some cases you may need to invert the actual image data.

If the image has light bands or spots on a dark background (i.e., the signal intensity of the background is greater than the signal intensity of the sample), you need to invert the data before analysis.

Select **Invert Data** from the *Image* menu or toolbar. This function is reversible. You may need to use the **Transform** controls to adjust the appearance of the inverted image.

## 3.13 Text Overlays

To create and display textual notes directly on the image, select **Text Overlay Tools** from the *Edit* menu or main toolbar. This will open the *Text Overlay Tools* toolbar.

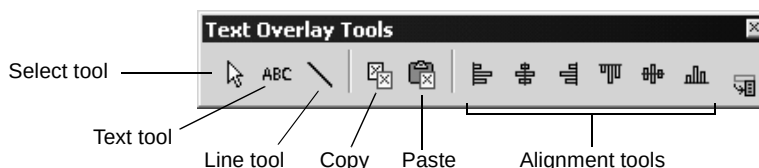


Fig. 3-24. Text Overlay Tools toolbar.

### Creating a Text Overlay

To create a text overlay, click on the **Text Tool**, then click on the image at the spot where you want the text to appear. This opens the *Text Overlay Properties* dialog.

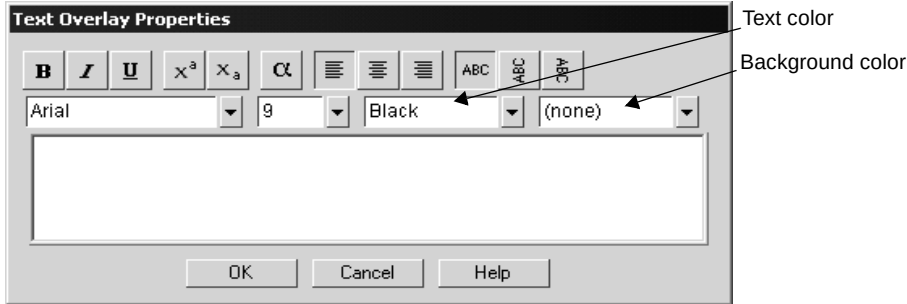


Fig. 3-25. Text Overlay Properties dialog.

To enter text, type in the main field. Use the buttons in the dialog to select the properties of the text, including format, alignment, and justification.

Select the font style, font size, color of the text, and color of the background in the text box using the pull-down lists.

After you have typed the text, click on **OK**. The text will appear on the image at the spot where you originally clicked.

### Editing a Text Overlay

To edit a text overlay, make sure the **Text Tool** or **Select Tool** is selected, and then double-click on the overlay to open the *Text Overlay Properties* dialog. The existing text will be displayed and can be edited.

### Line Tool

You can use the **Line Tool** to draw a line between text and an image feature, or between any two points of interest on the image.

Click on the **Line Tool** button, then drag on the image to create the line.

To resize or adjust a line, make sure the **Line Tool** or **Select Tool** is selected, and then position the cursor on one end of the line (marked by a circle) and drag.

To add arrowheads to a line, make sure the **Line Tool** or **Select Tool** is selected, and then double-click on the middle of the line. A dialog will pop up with options to add arrowheads to one or both ends of the line.

### Moving and Copying Text and Lines

You can move, copy, or delete a single text overlay or line or a group of overlays and lines within an image. You can also copy and paste between images.

First, you must select the object(s). Click on the **Select Tool** button. To select a single overlay or line, click on it. To select multiple objects, either drag a box around them or hold down the **Shift** key and click on them individually. When dragging to select a group of objects, make sure that you completely surround all the objects to be selected.

Each selected overlay/line will have a green border.

- To *move* the selected object(s), position the cursor over the selection and drag.
- To *copy within an image*, hold down the **Ctrl** key while dragging the selected object(s). The copy will be created and dragged to the new position.
- To *delete* the selected object(s), press the **Delete** key.
- To *copy between images*, click on the **Copy to Clipboard** button, then open or select the image you want to copy to and click on the **Paste from Clipboard** button. The copied object(s) will be pasted into the new image in the same relative position they were copied from.

**Note:** If you are pasting into an image with a different pixel size (i.e., resolution), you will receive a message that the placement of the copy may not be exact. Click on **OK** to complete the paste, then position the pasted objects manually.

### Viewing Previously Created Text Overlays/Lines

Previously created text overlays and lines will appear on the image when you open the *Text Overlay Tools* toolbar.

If you have concealed all overlays using the **Hide Overlays** command (section 3.3, Showing and Hiding Overlays), click on any of the buttons on the *Text Overlay Tools* toolbar to redisplay the text.

### 3.14 Erasing All Analysis from an Image

To delete all analysis and overlays from an image (including any lanes, bands, volumes, standards, text overlays, etc.), select **Clear Analysis** from the *Edit* menu. Since this process is irreversible, you will be prompted to confirm the selection.

### 3.15 Sort and Recalculate

To update, renumber, and recalculate all lane and band information, select **Sort and Recalculate** from the *Edit* menu.

### 3.16 Automation Manager

The **Automation Manager** allows you to save objects such as lanes, automated band detection, standard bandsets, volume overlays, and text and line overlays in a template file. These files can then be automatically applied to images either individually or in batches. To open the Automation Manager, select *Analysis>Automation Manager*.

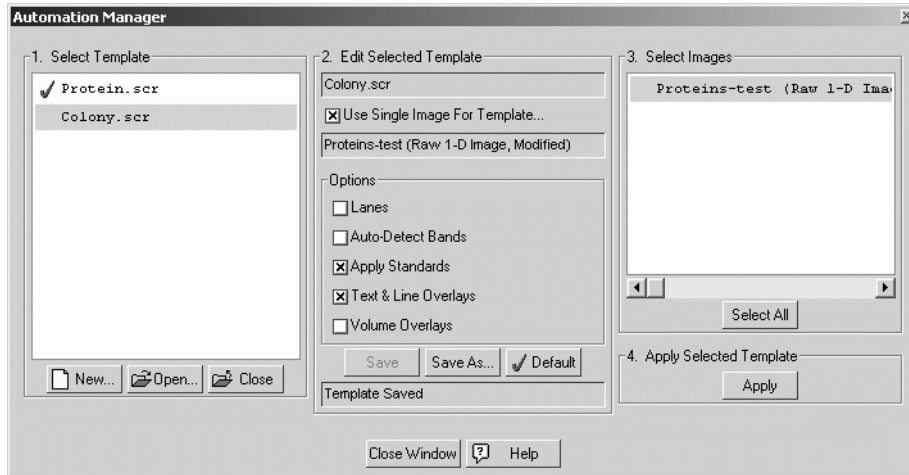


Fig. 3-26. Automation Manager dialog

### 3.16.a Step 1 - Select Template

The Select template field lists all the templates that are currently open. The Automation Manager remembers the files that were open the last time the application was open. To create a new template, click New. A new template appears in the list. See Section 3.16.b, Step 2 - Edit Selected Template for how to modify a new template. To open an existing template not currently in the list, click Open. This opens the standard Open dialog. To remove a template from the list, click Close.

### 3.16.b Step 2 - Edit Selected Template

Step 2, Edit Selected Template, lists the name of the currently selected template and the list of options available to the template. To edit the template, check the option(s) you would like to include and/or uncheck the option(s) you would like to remove from the template.

### Use Single Image for Template

The Automation Manager allows you to specify a single image as the source for all options you select. Checking the *Use Single image for template* box opens the **Select Source Image for this Script item** dialog opens.



Fig. 3-27. Select an image

Make sure the image you select contains the object(s) you wish to apply to the destination images.

If you wish to use options from multiple images, then uncheck the box labeled, **Use single image for template**. As you check each option in step two you will be prompted to select a source image for that option.

**Note:** The source image must be open when the option check box is selected. Once you have finished checking options from a source, the source image can be closed.

When you are satisfied with your changes, click Save, or Save As to save it as a new template. If this is a new template, you must enter a new name for the template when you click Save. To change the name of a template, click Save As and enter a new name.

### Default Automation

The default automation template is applied to the currently active image when you select Analysis>Apply Default Automation. To set a new default template, highlight the desired template in the Automation Manager and click Set as Default.

To run the default automation template on the currently active image, select Apply Default Automation from the Analysis menu.

Once a default template is chosen, The Automation Manager does not need to be open to be applied, nor do you need to select a default each time Quantity One opens as it remembers the default from the last open session of the application.

### 3.16.c Step 3 - Select Images

This portion of the Automation Manager lists the currently open images. Select the images to which you wish to apply the selected template. Use ctrl>click and shift>click to select more than one image, or click Select All to select all the images in the list.

### 3.16.d Step 4 - Apply Selected Template

Click **Apply** to apply the selected template in step 1 to the selected images in step 3.





# 4. Lanes

Before you can use many of the analysis functions, you must first define lanes and bands on the gel image. This chapter describes the tools for defining lanes.

**Note:** If you want to compare bands across lanes (using standards or band matching), the lane lines should be approximately the same length, with their starting points aligned across the top of the image. This is important for calculating the relative mobility of the bands. If gel wells are visible in the image, you should center the start points of the lane lines on the wells and position the ends of the lanes slightly below the last band for best results.

## 4.1 Defining Lanes

You can define lanes individually or as part of a frame. The functions for doing this are on the *Lane* menu and toolbar.

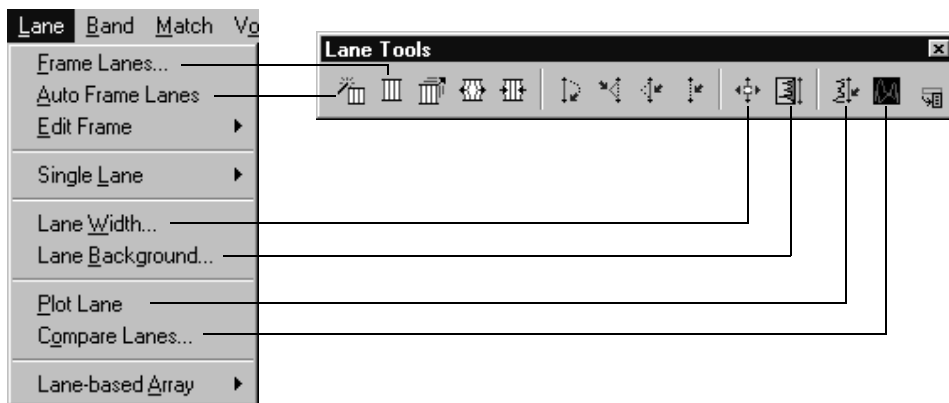


Fig. 4-1. Lane menu and toolbar.

In Quantity One, lanes are defined by red **lane lines** overlaid on a gel. The lane lines can be created individually (see section 4.1.c, Single Lanes), or they can be created as part of a **lane frame**.

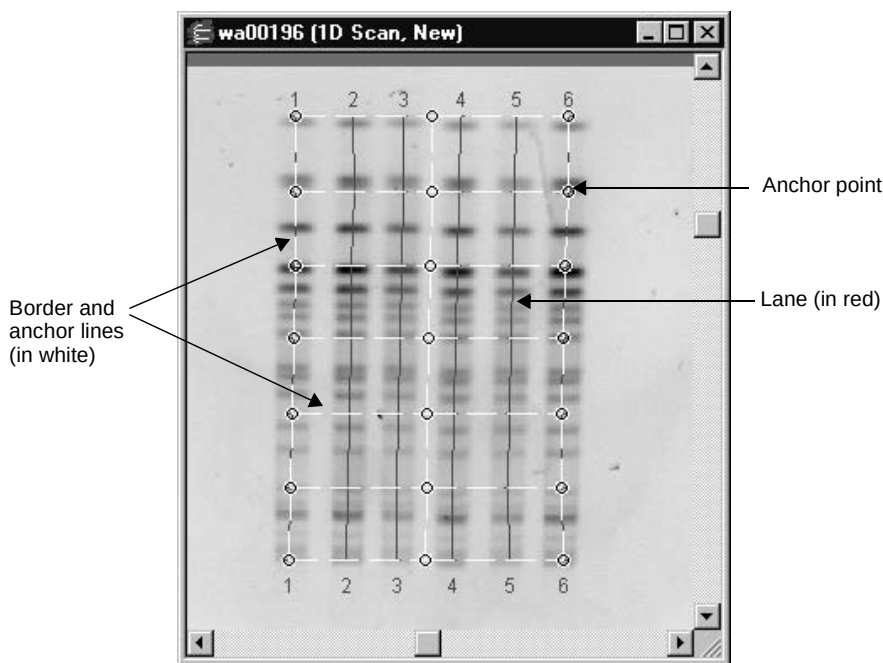


Fig. 4-2. Features of a lane frame.

### 4.1.a Lane Frames

The fastest way to define all the lanes on a gel is to create a lane frame using the **Auto Frame Lanes** command. If this command doesn't work well on your images, you can create and place a lane frame manually.

#### Auto Frame Lanes

**Note:** **Auto Frame Lanes** works best on gels with large numbers of clearly defined lanes and bands. Also, the lanes should be reasonably vertical and contain approximately the same amount of sample.

Select **Auto Frame Lanes** from the *Lane* menu or toolbar. The program will automatically detect the lanes and place a frame over them.

The lane frame contains individual lane lines numbered sequentially from left to right. The border and anchor lines of the frame are marked with dashed white lines, the lanes are solid red lines, and each anchor point (interior and corner) is marked with a circle.

The top and bottom of the frame are parallel with the top and bottom of the image. However, the interior anchor points and lines will “bend” the frame to follow the actual lanes in the gel, compensating for any curving or distortion in the gel.

If **Auto Frame Lanes** detects too few or too many lanes, you can add or delete lanes using the single lane commands (see section 4.1.c, Single Lanes).

If **Auto Frame Lanes** does not work on the image, you will be prompted to create a lane frame manually. To delete the lane frame, select **Clear Analysis** from the *Edit* menu.

### Manual Frame Lanes

If **Auto Frame Lanes** does not work with your images, you can frame the lanes manually.

Select **Frame Lanes** from the *Lane* menu or toolbar. In the dialog, enter the number of lanes in the gel and click on **OK**.

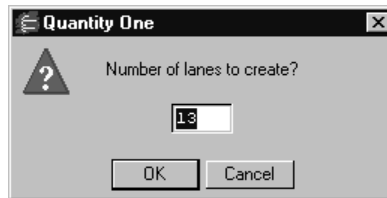


Fig. 4-3. Frame Lanes dialog.

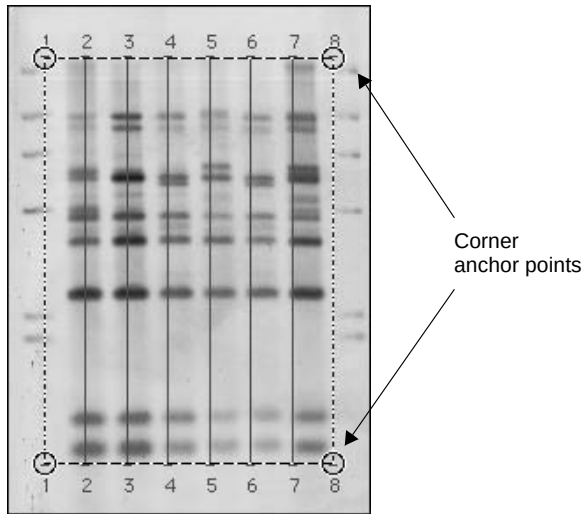


Fig. 4-4. Lane frame created using the Frame Lanes command.

The lane frame will be marked by corner anchor points, with no interior anchors. You can edit the frame as described below.

### 4.1.b Editing the Frame

If the frame is too large or small, or does not follow the lanes on the image, you can adjust it using the frame editing commands.

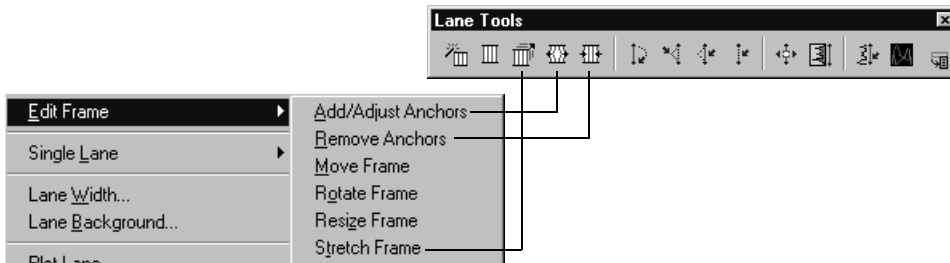


Fig. 4-5. Edit Frame tools.

### Adjusting the Entire Frame

The following commands are located on the *Lane > Edit Frame* submenu:

- To *stretch* the frame in one direction (e.g., to encompass additional bands at the top or bottom of the image), select **Stretch Frame** from the submenu or *Lane* toolbar and drag an anchor point in or out. The opposite anchor point will remain fixed while the frame expands or contracts.
- To *move* the entire frame to a new position, select **Move Frame** from the submenu and drag an anchor point. The entire frame will move.
- To *rotate* the frame, select **Rotate Frame** from the submenu and drag an anchor point. The entire frame will rotate.
- To *resize* the entire frame, select **Resize Frame** from the submenu and drag an anchor point in or out. The frame will expand or contract from the center.

### Adding and Adjusting Frame Anchors

To adjust a corner anchor point, select **Add/Adjust Anchors** from the *Edit Frame* submenu or toolbar and drag the anchor. This will move the anchor point and attached frame lines.

If the gel lanes are not straight, you can create additional anchor points along the frame to change the shape of individual lines.

Still using **Add/Adjust Anchors**, click anywhere on the frame lines. This creates interior anchor points both where you clicked and on the other side of the frame, connected by a frame line. Then drag the anchor points to bend the frame.

Add and adjust as many anchor points as you need to bend the lane lines to follow the lanes in the gel.

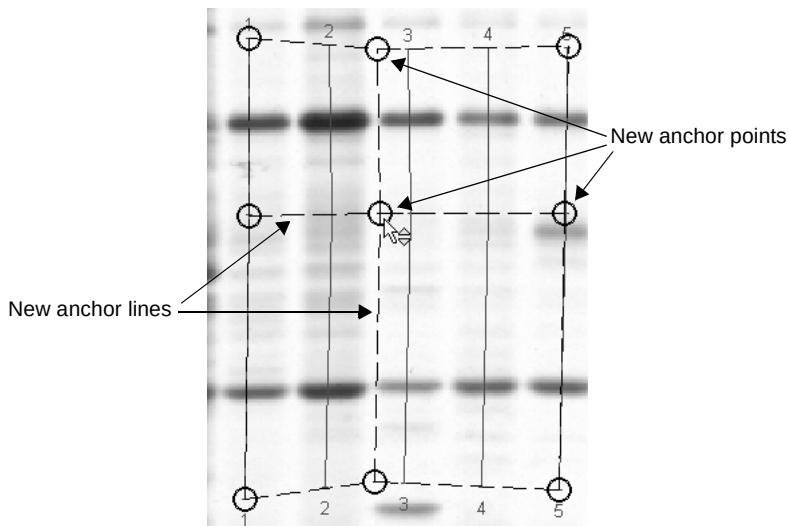


Fig. 4-6. Adjusting the anchor points of the lane frame.

### Removing/Unadjusting Anchors

To remove an anchor point, select **Unadjust Anchors** from the *Edit Frame* submenu or toolbar and click on the anchor. The anchor will disappear and the adjusted lanes will straighten out.

#### 4.1.c Single Lanes

You can define individual lanes using the single lane tools. These are located on the *Lane > Single Lane* submenu or on the *Lane* toolbar.

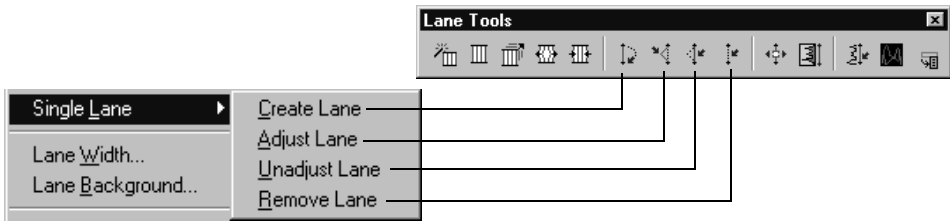


Fig. 4-7. Single Lane tools.

**Note:** You can use the single lane commands on any lane within a frame; however, the lane will be detached from the frame.

To mark an individual lane, select **Create Lane** and drag a line from the top to the bottom of the gel lane. The lane line will be marked in red. Repeat this procedure to manually mark all the lanes you want to analyze in the gel.

**Note:** If the lane numbering gets out of sequence, select **Sort and Recalculate** from the *Edit* menu to renumber the lanes.

### Adjusting Single Lanes

You can adjust the position of any lane line. Select **Adjust Lane** from the *Lane > Single Lane* submenu or the *Lane* toolbar and either drag one of the existing anchor points or click anywhere on the lane to create a new anchor point and drag it into position.

To undo any lane adjustments, select **Unadjust Lane** from the submenu or toolbar and click on an anchor point to remove it. If you remove the anchor points at either end of the lane line, you will delete the entire line.

### Deleting Lanes

You can delete both single lane lines and lines from a lane frame.

Select **Remove Lane** from the *Single Lane* submenu or toolbar and click on the lane. You will be prompted to confirm the deletion.

**Note:** If you delete a lane from a group of lanes or a frame, select **Sort and Recalculate** from the *Edit* menu to renumber the remaining lanes.

### 4.1.d Lane Width

Lane width is important for band quantitation. Only the region of the band within the lane sampling width is quantitated, so the defined lanes should be slightly *wider* than the actual lanes in the gel.

You can adjust the sampling width of all the lanes using the *Detect Bands* dialog (see section 5.2.a, Detection Parameters). See section 5.1, How Bands Are Identified and Quantified, for a full discussion of the effect of sampling width on band quantitation.

To adjust the width of a single lane, select **Lane Width** from the *Lane* menu or toolbar and click on the lane line. The *Sample Width* dialog will display the current width of the lane.

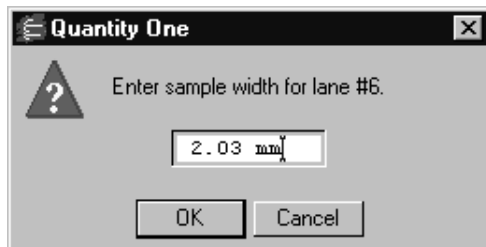


Fig. 4-8. Sample Width dialog.

Enter a new width in millimeters and click on the **OK** button.

### 4.1.e Lane Profile

After you have defined a lane, you can review the intensity profile of the lane. A lane profile provides a quick visualization of the intensity of your sample data, and is also useful for determining the level of background in the gel.

Select **Plot Lane** from the *Lane* menu or toolbar, then click on a lane. A lane profile graph will be displayed.



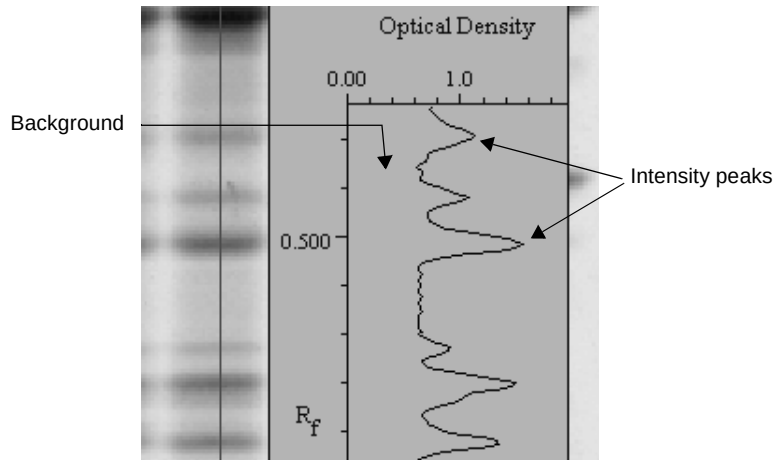


Fig. 4-9. Profile of a defined lane.

In the lane profile, bands are represented by peaks and background intensity is represented by the baseline region below the peaks. The profile is generated by calculating the average intensity of each horizontal row of pixels across the specified width of the lane.

To close the lane profile, click on **Hide Overlays** on the main toolbar.

## 4.2 Lane-Based Background Subtraction

After defining lanes, we strongly recommend that you perform lane-based background subtraction. This is the best method for removing background intensity from lanes, and is required for Gaussian modeling of bands.

Lane-based background subtraction uses a “rolling disk” method of subtraction, named for a hypothetical disk that rolls along underneath the lane profile, removing different intensity levels along the length of the lane.

The size of the disk determines how much background will be subtracted. A large disk will follow the profile trace less closely, touching fewer points along the trace

and removing less background. A smaller disk will more closely follow the profile trace, removing more background.

A disk radius that is too large will result in poor removal of background. A disk radius that is too small may subtract actual data.

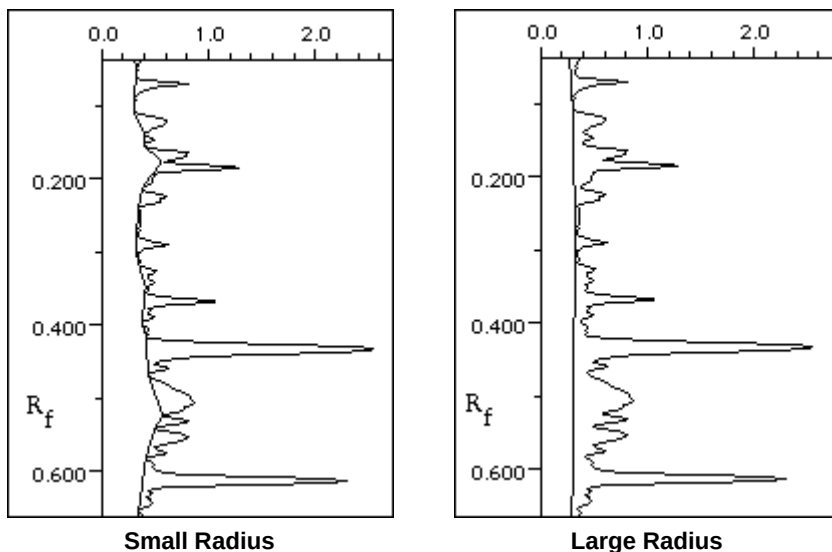


Fig. 4-10. Examples of the background trace for small and large rolling disks. The small disk follows the profile trace more closely, resulting in more background subtraction.

Select **Lane Background** from the *Lane* menu or toolbar and click on a lane. The lane will be highlighted, the lane profile will be displayed, and the *Lane Background Subtraction* dialog will open.

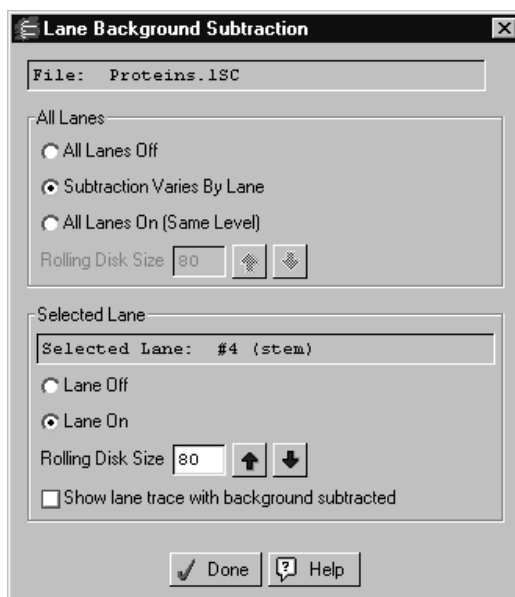


Fig. 4-11. Lane Background Subtraction dialog.

In the dialog, you can set the same subtraction level for all lanes or specify an individual subtraction level for the selected lane. Any changes you make will be automatically applied to the image. To close the dialog, click on **Done**.

### 4.2.a Profile Trace

When you make changes in the dialog, note that the profile trace of the lane also changes. In the standard view, the original “raw” trace of the line is shown in black, and the orange line represents the background beneath the peaks of the trace as defined by the rolling disk.

As you change the size of the rolling disk, the orange line changes, following the contours of the profile more or less closely as shown in Fig. 4-10.

To display the trace with the background removed, select **Show lane trace with background subtracted**. The orange line in the trace disappears along with the

background, and you can visually compare the relative intensities of the bands in the lane.

### 4.2.b All Lanes Off

To turn off lane-based background subtraction for all the lanes in the image, select the **All Lanes Off** button.

### 4.2.c All Lanes On (Same Level)

Select this option to set the same subtraction level for all lanes. The **Rolling Disk Size** field in the *All Lanes* section of the dialog displays the rolling disk radius in number of pixels. Enter a new value in the field, or use the arrows to change the value in 10 percent increments.

Typical rolling disk sizes range from 50 to 150. As you change the size, study the level of background subtraction in the lane trace.

### 4.2.d Subtraction Varies by Lane

You can set a different subtraction level for each lane in the gel using the controls in the *Selected Lane* area of the dialog. Any changes you make with these controls will only be applied to the selected lane; to select a different lane, click on it with the **Lane Background** command assigned.

Click on the **Lane Off** button to turn off subtraction for the selected lane. Select **Lane On** to turn on subtraction for the lane.

To adjust the subtraction level for the lane, change the value in the **Rolling Disk Size** field under *Selected Lane*. This field displays the rolling disk radius in number of pixels. Enter a new value in the field, or use the arrows to change the value in 10 percent increments.

Typical rolling disk sizes range from 50 to 150. As you change the size, study the level of background subtraction in the lane trace.

When you make changes to the selected lane, the **Subtraction Varies by Lane** button is selected. To turn off individual lane subtraction, choose either **All Lanes Off** or **All Lanes On**.

## 4.3 Compare Lanes

The *Compare Lanes* graph allows you to superimpose the intensity profiles of any number of lanes from any number of open images.

Select **Compare Lanes** from the *Lane* menu or toolbar, then click on the first lane you want to display. The *Compare Lanes* window will open.

**Note:** The image must have defined lanes for this command to work.

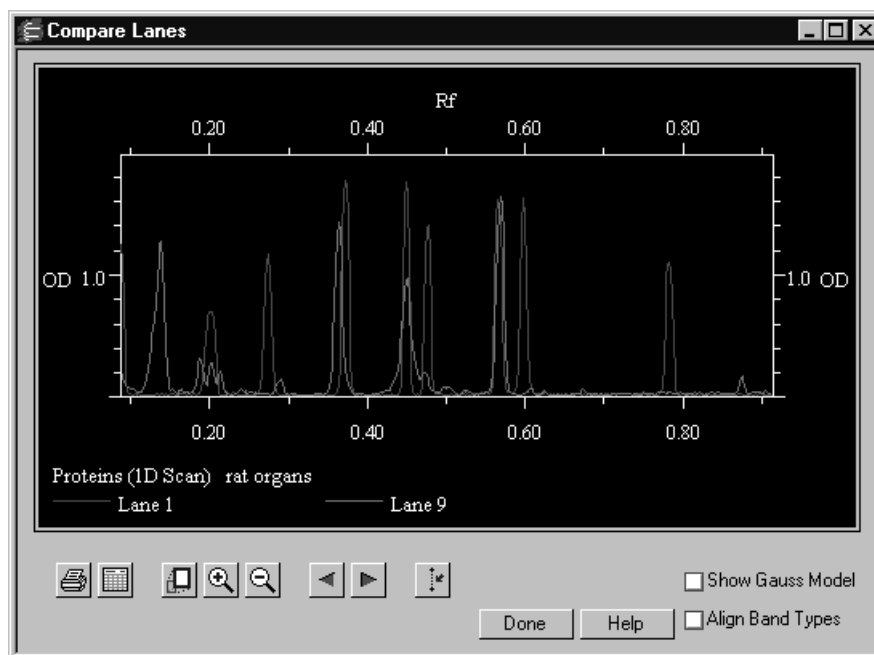


Fig. 4-12. The Compare Lanes dialog.

The X axis of the graph is the Rf value and the Y axis is the pixel intensity value at each point along the lane. *Compare Lanes* automatically “best fits” lanes within the display window to maximize the range of intensity values included in the graph. Rf values are displayed from 0.0 to 1.0.

### Adding and Removing Lanes from the Graph

To add a lane to the graph, click directly on the lane in the image. The plot of the lane will appear in the graph.

Each lane you add will be displayed in one of eight colors, and will be identified by color in a legend underneath the graph. If you add more than eight lanes, the colors will repeat, but each lane will still be identified underneath the trace display. There is no limit to the number of lanes that can be displayed simultaneously.

To remove a lane profile, click on the **Remove Lane** button. A pop-up box will prompt you to select the lane to remove. If you remove a lane, the colors of the remaining lanes will change. Check the lane legend for an updated color code.

### Magnifying the Graph

Use the **Zoom In** and **Zoom Out** buttons in the dialog to magnify regions of interest in the profiles. Alternatively, drag the cursor horizontally across the graph and release the mouse button to magnify the defined range.

**Note:** The magnifying functions in **Compare Lanes** only magnify the profile in the direction of the X axis. Therefore, the profile will appear to “stretch” without increasing in height.

The **Full View** button returns the graph to its default display.

If you have magnified part of the graph, the **Left arrow** and **Right arrow** scroll buttons can be used to pan left or right along the graph.

### Show Gaussian Modeling

The **Show Gauss Model** checkbox is active if any of the selected lanes includes Gaussian modeling (see section 5.7, Gauss-Modeling Bands). If you select this checkbox, the Gaussian-fitted profiles will be superimposed on the regular lane profiles. The Gaussian profiles are displayed in white.

### Align Band Types

The **Align Band Types** checkbox is active if any of the selected lanes includes defined band types (see section 6.2, Band Matching).

If this checkbox is selected, the profiles of all bands that have been identified as the same band type will be stretched and superimposed on one another, so their peaks align. This is useful if the same band appears as peaks in slightly different positions in different lanes, and you want to align the peaks to confirm that they are all the same band type.

**Note:** This command only changes the lane profiles as they are displayed in the *Compare Lanes* dialog, and will not affect image data in any way.

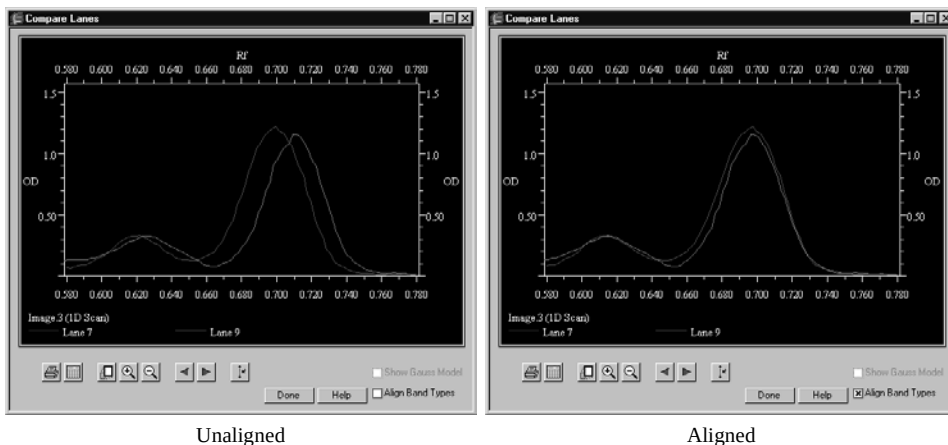


Fig. 4-13. Two band types as they appear in two lanes, before alignment and after.

Note that this function will not align band types from different band sets (e.g., Band Type 1 in Band Set A and Band Type 1 in Band Set B will not be aligned). However, the same band types from different images will be aligned.

**Note:** The Rf values in the X axis will no longer be accurate if **Align Band Types** is selected, since some band profiles will be stretched and their peaks shifted.

### Printing and Exporting

Click on the **Print** button to print a copy of the *Compare Lanes* display.

Click on the **Export** button to export the data points in the graph to a spreadsheet. This will open the *Compare Lanes Export* dialog.

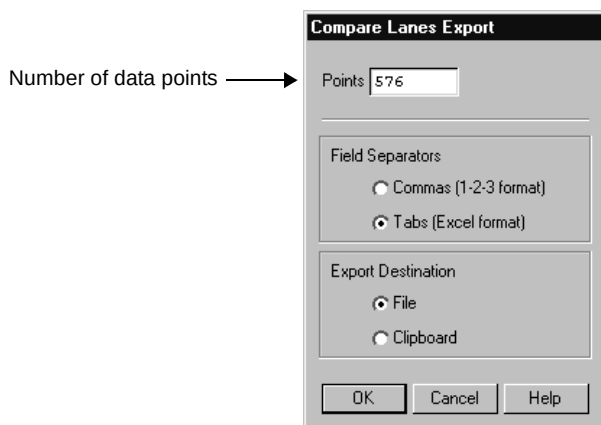


Fig. 4-14. Compare Lanes Export dialog.

This *Export* dialog includes a field for the number of data points to be taken along the length of each lane. The default value in this field is the maximum number of data points that are available for the lanes you are comparing.

Select the export format (tab or comma delimited) and destination (file or clipboard), then click on **OK**.

**Note:** The exported data will be different depending on whether you have checked **Align Band Types**, **Show Gauss Model**, or neither. If the **Show Gauss Model** checkbox is selected, each lane that has been Gaussian fitted will have two columns of data: one for the Gaussian-fitted profile and one for the regular profile. If the **Align Band Types** checkbox is selected, the exported values will reflect the stretched and shifted profiles of those lanes that have been aligned.

## 4.4 Lane-based Arrays

The lane-based array functions allow you to create a lane frame for the cells in an array. You can then specify the cell dimensions and quantitate them using the **Quantity Standards** function (see section 6.3, Quantity Standards).



**Note:** You can quantitate arrays outside of lanes using volume arrays (see section 7.7, Volume Arrays).

The first step in defining an array is specifying the number of columns and rows and creating an array frame.

Got to the *Lane* menu, open the *Lane-based Array* submenu, and select **Frame Array**.



Fig. 4-15. Lane-based Array tools.

Enter the number of columns in the array and click on **OK**.

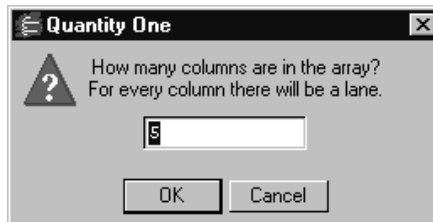


Fig. 4-16. Setting number of array columns.

In the next box, enter the number of rows and click on **OK**.

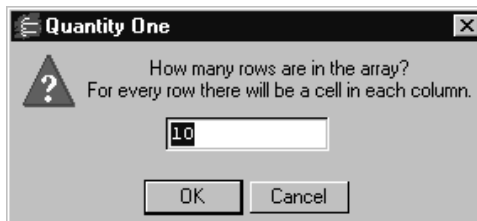


Fig. 4-17. Setting the number of array rows.

## Quantity One User Guide

---

The array matrix will appear on the image. Each column will be marked by a red line, and each cell will be marked by top and bottom brackets.

**Note:** If the cells appear marked by lines instead of brackets, select **Band Attributes** from the *Band* menu and select **Brackets** in the dialog.

When you first create the array matrix, it will probably not be centered on the columns and cells in the actual image. In the next step, you will adjust the position of the matrix.

### Adjusting the Array Matrix

The **Add/Adjust Anchors** tool will be automatically assigned to the mouse after you create the frame (otherwise, select it from the *Lane > Edit Frame* submenu). Position the cursor on the corner points of the frame and drag them into position so that the red lines run down the middle of the array columns and the top and bottom brackets are centered on the array cells (see section 4.1.b, Editing the Frame for guidance on adjusting frames).

If necessary, the **Adjust Lane** command (see “Adjusting Single Lanes” on page 7) and **Adjust Band** command (see section 5.3.b, Adjusting Bands) can be used to adjust the placement of columns and cells within the frame.

### Reducing Background in the Array

After you have positioned the array, you should reduce lane background using the **Lane Background** command (see page 4.2, Lane-Based Background Subtraction). Lane background will affect quantitation of the cells.

### Setting Array Cell Height and Width

Now you should adjust the cell brackets so that they completely enclose the cells in the array.

Select **Array Cell Height** from the *Lane > Lane-based Arrays* submenu and enter the height in millimeters of the array cells.

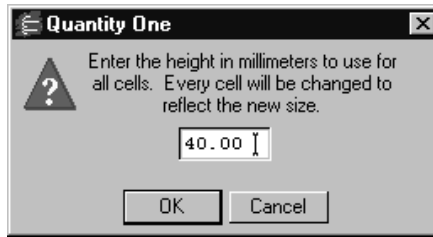


Fig. 4-18. Setting the array cell height.

When you click on **OK**, the cell brackets will adjust to the specified height. If you aren't sure of the exact height, you can experiment with different values.

Select **Array Cell Width** and enter the width in millimeters of all the cells in the array.

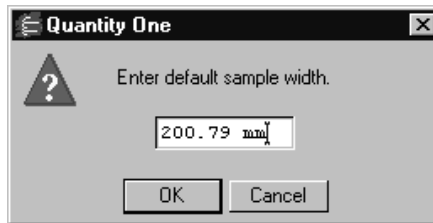


Fig. 4-19. Setting the array cell width.

When you click on **OK**, the cell brackets will adjust to the specified width. If you aren't sure of the exact width, you can experiment with different values.

## Analyzing Array Data

When the brackets fully enclose each cell in the array, you are ready to analyze the data. You can display various measures of cell quantity on the image using the **Band Attributes** command on the *Band* menu. With the *Band Attributes* dialog open, select from **Peak Density**, **Average Density**, **Trace Quantity**, **Relative Quantity**, and other measures. You can also report these values by selecting **Lane Reports** from the *Reports* menu.

## Quantity One User Guide

---

To use known quantities to calculate unknowns, you can use the **Quantity Standards** function (see section 6.3, Quantity Standards).

# 5. Bands

After you have defined the lanes on the gel image, you can automatically identify and quantitate bands using a set of adjustable parameters.

**Note:** You can quantitate bands, arrays, or other objects outside of lanes using volumes. See Chapter 7 for details.

The tools for band detection are located on the *Band* menu and on the *Band Tools* toolbar.

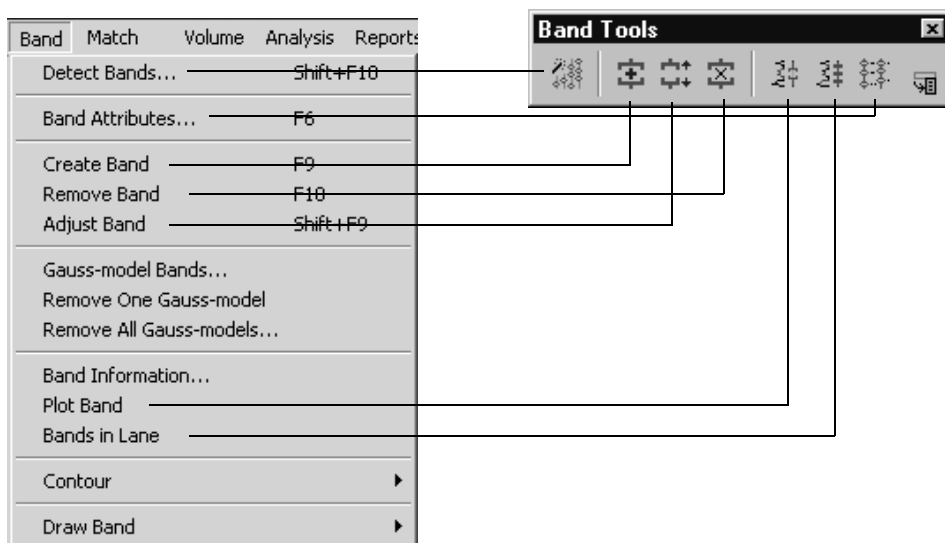


Fig. 5-1. Band menu and Band Tools toolbar.

**Note:** Before detecting bands, you should subtract background from the lanes using the **Lane Background** command (see section 4.2, Lane-Based Background Subtraction).

### 5.1 How Bands Are Identified and Quantified

You can automatically identify all the bands in an image using the **Detect Bands** command (see section 5.2, Band Detection), or you can mark them individually using the **Create Band** command (see section 5.3, Identifying and Editing Individual Bands).

Each identified band is defined by brackets above and below the band. The width of each set of brackets is determined by the lane sampling width (see section 5.2.a, Detection Parameters). The height of the brackets is determined automatically, using a band-finding formula together with parameters that you select.

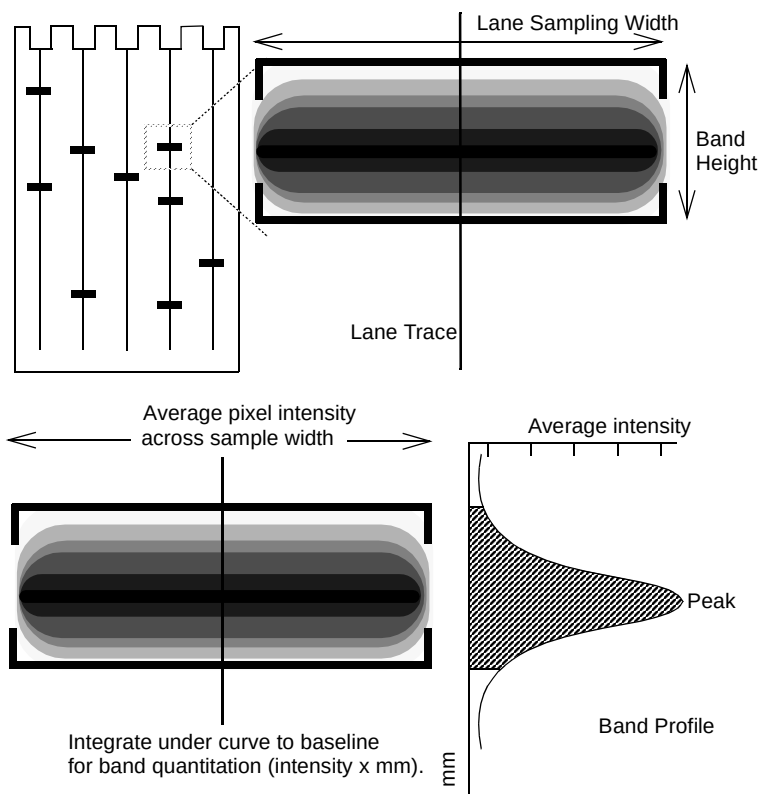


Fig. 5-2. Illustration of bracket quantitation.

When a band is quantitated, the average intensity of each horizontal row of pixels within the brackets is calculated. Next, the number of pixel rows between the top and bottom brackets is determined. Taken together, these result in an intensity profile for the band.

Finally, the area under the profile curve to the baseline is integrated, resulting in units of intensity x millimeters. This is the “trace quantity” of the band.

**Note:** Because the profile of an ideal band conforms to the shape of a Gaussian curve, band profiles can be “fitted” to a Gaussian model. The band quantity can then be quantitated from the area under the Gaussian curve. This is the best to resolve overlapping or closely spaced bands in images. See section 5.7, Gauss-Modeling Bands, for details.

## 5.2 Band Detection

The **Detect Bands** command automatically detects the bands in defined lanes, based on parameters that you select.

**Note:** Before detecting bands, you should subtract background from the lanes using the **Lane Background** command (see section 4.2, Lane-Based Background Subtraction).

Select **Detect Bands** from the *Band* menu or toolbar. The *Detect Bands* dialog will open.

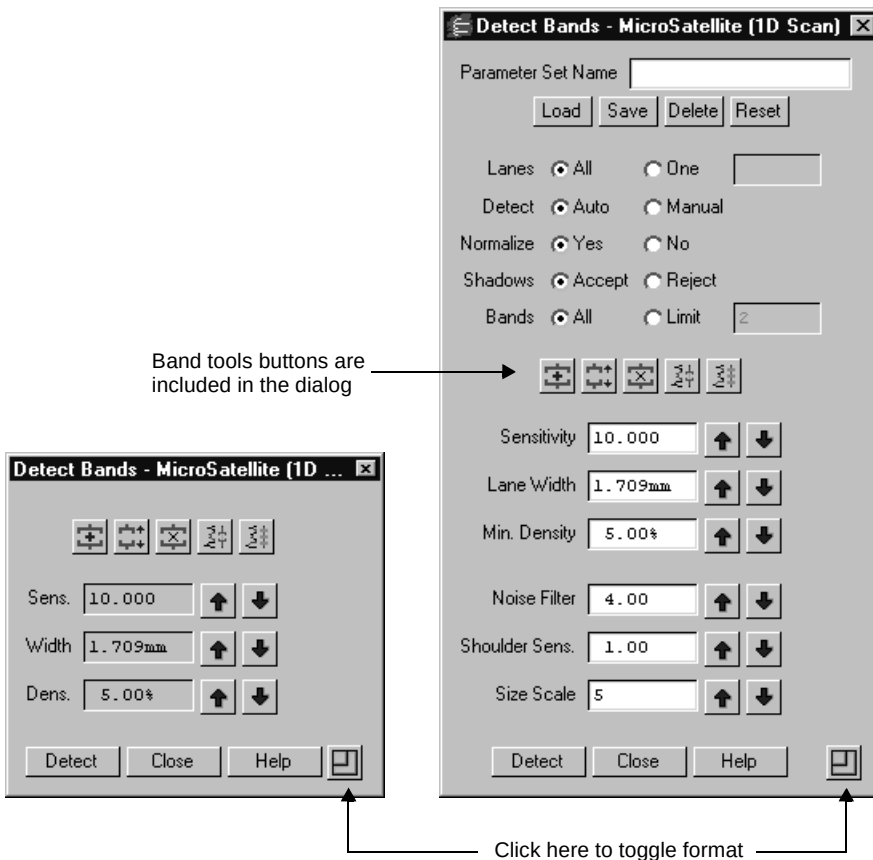


Fig. 5-3. Detect Bands dialog, short and expanded formats, with default values.

When you first open the *Detect Bands* dialog, all bands will be automatically detected based on the default parameters in the dialog, and the bands will appear marked on the image. The bands can be marked by either lines or brackets, depending on the settings in the *Band Attributes* dialog (see section 5.5, *Band Attributes*).

**Note:** If you have already manually identified bands (using **Create Band**, **Adjust Band**, etc.), the **Detect Bands** function will overwrite them. You should use **Detect Bands** first, and then manually add, adjust, or remove bands as needed.



The *Detect Bands* dialog has a short format and an expanded format; toggle between them by clicking the toggle box in the lower right corner.

To change a parameter in the dialog, you can either type in a new value or use the arrows to increase or decrease the setting by 10 percent. Experiment with different settings to find those best suited to your images.

### 5.2.a Detection Parameters

#### Lanes to Detect

If the intensities of the bands vary from lane to lane, you may need to use different detection parameters on different lanes. Specify whether you want to use the detection parameters for all the lanes or a single lane by selecting **All** or **One** next to the *Lanes* prompt. If you choose **One**, type the lane number in the field next to the **One** button.

#### When to Detect

If you select **Auto** next to the *Detect* prompt, band detection will occur immediately each time you change a detection parameter. You will not need to click on the **Detect** button located at the bottom of the form.

To change more than one parameter before detecting, choose **Manual**. With the **Manual** option, you can change parameter settings first, and then apply them by clicking the **Detect** button.

#### Normalization

Normalization is a way to compensate for differences in intensity between lanes. *It does not normalize for band quantitation.*

The intensity of each lane is determined by the darkest band in that lane. For example, suppose that in all but one of the lanes the darkest band has an intensity of 50,000 counts. In the one light lane, the darkest band is only 25,000 counts. With normalization, band detection will be twice as sensitive when processing the light lane, improving the detection of faint bands.

## Quantity One User Guide

---

Select **No** next to the *Normalize* prompt to apply the band detection parameters in the same way to every lane in the gel image. Select **Yes** to normalize for the intensity of the lane.

### Shadow Rejection

“Shadow bands” are common gel artifacts. Shadow bands are spaced at tandem repeat intervals and decrease in intensity as they progress further from a real band. The **Shadows** parameter is designed to limit the detection of shadow bands (see also **Band Limit**).

Select **Reject** to turn on the shadows filter. A band will be detected only if it is darker than the one above it or spaced further than one tandem repeat unit from the previous band. This greatly reduces the number of shadows identified as real bands.

Select **Accept** to turn off the shadows filter.

### Band Limit

If you know that all the lanes in a gel contain a specific number of bands, you can click on the **Limit** button next to the *Bands* prompt and type in the number of bands that you know are present. Only that number of bands will be detected in each lane, reducing the need for later editing.

### Sensitivity

The **Sensitivity** setting determines the minimum signal intensity in the image that will be defined as a band. The higher the sensitivity value, the more bands will be detected.

If the sensitivity is set too high, background noise may be detected as bands. If the setting is too low, real bands may be missed.

The default sensitivity setting is 10.00. If the gel has faint bands (e.g., O.D. < 0.05, counts < 2,000), you may want to increase this value to 20.00.

### Lane Width

The **Lane Width** setting determines the width along the lane lines that are sampled for band detection and quantitation.

When a band is detected, an average intensity for each horizontal row of pixels within the band brackets is calculated. The lane width determines the number of pixels in each row. Study the band lines in the image while you adjust the width. Select a width that is slightly *wider* than the bands in the gel.

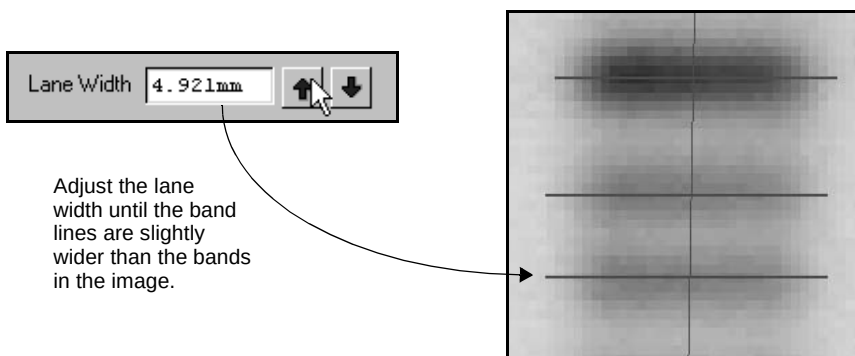


Fig. 5-4. Adjusting the lane width.

You can also change the widths of individual lanes using the **Lane Width** command (see section 4.1.d, Lane Width).

### Minimum Density

When **Normalize** is turned off, the **Min. Density** defines the lowest signal intensity that will be counted as a band.

Before selecting a **Min. Density** value, use the **Plot Lane** command on the *Lane* menu to plot a trace of a lane that includes some faint bands. Then enter a value that is lower than the intensity of the peak of a faint band but is still above the background.

If faint bands are still undetected after adjusting this parameter, you may want to increase the **Sensitivity** setting.

## Quantity One User Guide

---

If **Normalize** is turned on, **Min. Density** changes from an absolute value to a percentage. This percentage is the fraction of the signal intensity of the darkest band in the lane that will be detected as a band. For example, if the darkest band in a lane is 50,000 counts and the **Min. Density** is set to 25,000 counts, when you turn on **Normalize**, the **Min. Density** will switch to 50% (i.e., a band must be at least half as dark as the darkest band in the lane).

### Noise Filter

The **Noise Filter** is used to minimize the number of small fluctuations in the image (i.e., noise) that are called bands while still recognizing larger features (i.e., real bands). This filter becomes especially important at higher **Sensitivity** levels.

The **Noise Filter** value refers to the size of the filter in pixels (e.g., a value of 2.50 equals a filter size of 2.50 x 2.50 pixels). Features smaller than the filter size will not be recognized as bands. Entering a noise filter size of zero turns it off completely. The default value is 4.00.

If band detection detects doublets as single bands, decrease the **Noise Filter** setting and/or increase the **Sensitivity**.

You can also try decreasing the **Size Scale** parameter instead of the **Noise Filter** to improve the detection of closely-spaced bands. However, if you decrease both the **Noise Filter** and the **Size Scale**, the fuzziness around bands may be mistakenly detected as separate bands.

### Shoulder Sensitivity

Normally, band detection tries to distinguish shoulders as separate bands. When looking at a lane trace, these bands appear as flat or gently sloping abutments to darker, better-defined bands (i.e., there is no dip on the trace between the two bands).

Increasing the **Shoulder Sensitivity** will result in more shoulders being detected as bands. Changing this setting to zero will result in no shoulders being recognized as separate bands.

If band detection calls a doublet a single band, check the lane trace to see if there is a dip between the peaks of the two bands. If there is no dip, increasing the **Shoulder Sensitivity** value will help resolve the two bands.

## Size Scale

The **Size Scale** field helps distinguish between trends in signal intensity and random intensity fluctuations. It is the number of pixels in a vertical column that are taken together to determine whether a band is present.

The **Size Scale** parameter is similar to the **Noise Filter** in that it uses the size of objects in the image to determine the nature of those objects. The default **Size Scale** setting of 5 pixels is optimal for most gel images. It can be set to any whole number greater than or equal to 3.

If a gel image has high levels of background noise, a larger **Size Scale** is appropriate. At low noise levels, a smaller value is preferable. You can also increase the **Size Scale** if the gel only has a small number of thick bands scanned at high resolution.

### 5.2.b Parameter Sets

You can save detection parameters for use on similar images. Enter a name for the set in the **Parameter Set Name** field, and click on the **Save** button.

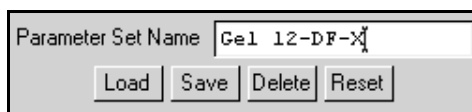


Fig. 5-5. Parameter set controls in the Detect Bands dialog.

To load a saved set of parameters, click on the **Load** button and select the set from the list. To remove a set of parameters, first load it, then click on **Delete**.

### 5.2.c Default Parameters

To return all detection parameters to their default values, click on the **Reset** button at the top of the form.

### 5.3 Identifying and Editing Individual Bands

To manually identify and edit individual bands in an image, use the **Create Band**, **Adjust Band**, and **Remove Band** commands on the *Band* menu and toolbar. (These are also included on the toolbar in the *Detect Bands* dialog.)



Fig. 5-6. Create, Adjust, and Remove Bands buttons.

**Note:** When editing individual bands, it is useful to display the band brackets. Select these in the *Band Attributes* dialog (see section 5.5, Band Attributes). If your bands are very closely spaced, defining them in brackets mode gives you greater control for more precise band definition. It also allows you to define overlapping bands.

#### 5.3.a Identifying Individual Bands

With the bands displayed as brackets, select **Create Band** from the menu or toolbar, then click on either the top or bottom boundary of the band in the gel. An intensity trace of the lane will pop up next to the band.

Drag the cursor until the area of the band that you want to define—represented by the peak on the intensity trace—has been completely enclosed. The area of defined band will be highlighted on the trace.

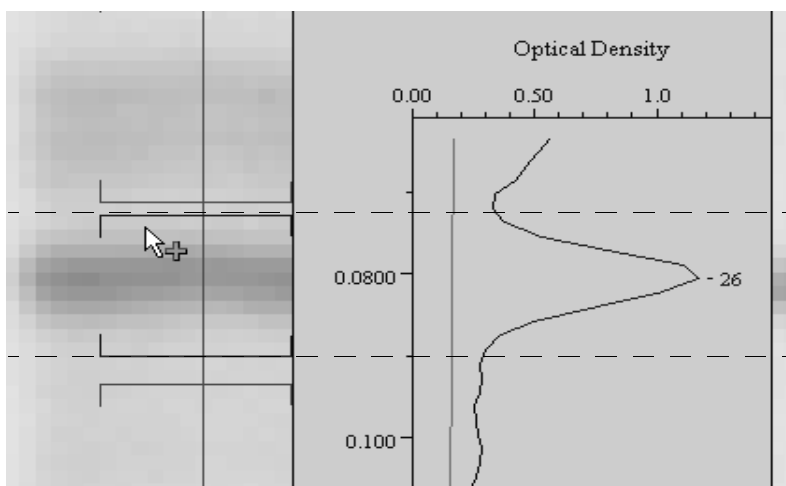


Fig. 5-7. Creating a band with brackets displayed.

When you release the mouse button, brackets will enclose the band on the image.

With the bands displayed as lines, select **Create Band** from the menu or toolbar, then click on the center of the band of interest. A line will appear in the center of the band.

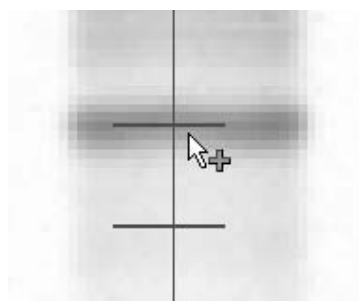


Fig. 5-8. Creating a band with lines displayed.

**Note:** After you have identified several bands, renumber the bands by selecting **Sort and Recalculate** from the *Edit* menu.

### 5.3.b Adjusting Bands

To reposition a band's boundaries, select **Adjust Band** from the menu or toolbar. If the bands are displayed as brackets, drag the upper or lower bracket of the band. If the bands are displayed as lines, drag *near* the upper or lower boundary of the band that you want to adjust.

When you drag the band boundary, a pop-up lane trace will appear, and the area of defined band will be highlighted on the trace. Drag the band boundary to the correct position.

### 5.3.c Deleting Bands

To undetect a band, select **Remove Band** from the menu or toolbar and click on the band. If the bands are displayed as brackets, a trace of the band will be displayed and a pop-up box will ask you to confirm the action. If the bands are displayed as lines, the band line will simply be deleted.

If the bands are displayed as lines, you can delete more than one band at a time. Select **Remove Band**, and drag a box around the bands to be removed.

If you undetect a band, it will no longer be counted as a band, but its intensity will still contribute to the total lane intensity.

**Note:** After you undetect bands, renumber the bands in the image by selecting **Sort and Recalculate** from the *Edit* menu.

## 5.4 Plotting Traces of Bands

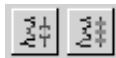


Fig. 5-9. Plot Band and Bands in Lane buttons.

**Plot Band** displays an intensity profile of a band. Select the command from the menu or toolbar and click on the band of interest.



**Bands in Lane** displays an intensity trace of a lane with the defined bands highlighted. Select the command from the menu or toolbar and click on the lane of interest.

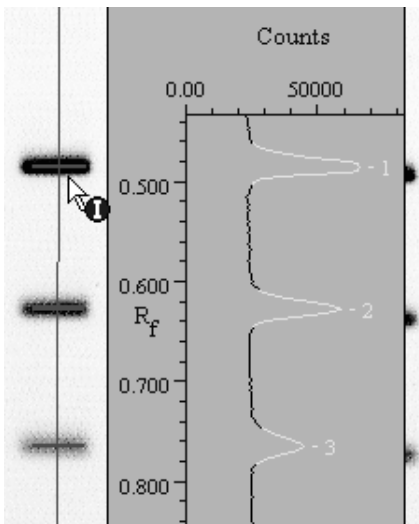


Fig. 5-10. Bands in Lane command.

## 5.5 Band Attributes

You can display different types of information about defined bands. Select **Band Attributes** from the *Band* menu or toolbar to open the dialog.

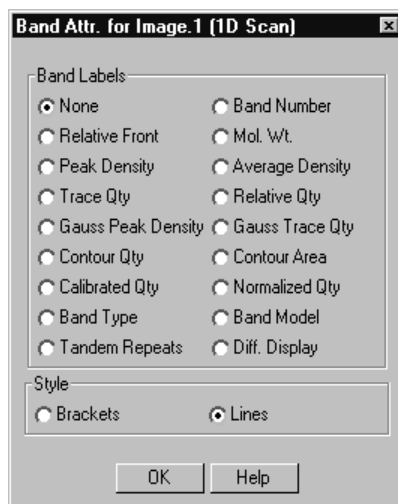


Fig. 5-11. Band Attributes dialog.

At the bottom of the dialog, select how you want to mark defined bands on the image—as **Brackets** or **Lines**. Lines are usually easier to read than brackets in gels with closely packed bands, while brackets are better for displaying and editing the boundaries of bands.

You can also select the band data to display on the image next to each band:

- **None.**
- **Band number**—The sequential number of a band in lane, as counted from the top of the lane.
- **Relative front**—The distance of a band from the top of its lane, divided by the total length of the lane. The lane length can be determined either by measuring a vertical line from the top of the lane to bottom or (if the lane is curved) by measuring along the length of the lane. Set the preferred measuring method in the *Preferences* dialog (see section 2.5.e, Application).

**Note:** Note that normalized Rf is derived from relative front; however, normalized Rf is calculated only for bands that have been modeled using standards or band sets, and can change based on the modeling.

- **Molecular Weight/Isoelectric Point/Base Pairs/other units**—This value is

determined by the type of standards defined for the gel, the band's position in the lane, and any modeling performed on the gel (using band matching or multiple lanes of standards) to compensate for gel distortion or smiling.

- **Peak density**—The intensity value of a band peak.
- **Average density**—The total intensity of the rows of pixels used to generate the profile of a band, divided by the number of rows.
- **Trace qty**—The quantity of a band as measured by the area under its intensity profile curve. Units are intensity x mm.
- **Relative qty**—The quantity of a particular band as measured by its intensity, expressed as a percentage of either the total intensity of all the bands in the lane or the total intensity of the lane (including the areas between bands). The calculation method (**% of Lane** or **% of Bands in Lane**) is set in the *Preferences* dialog (see section 2.5.e, Application).
- **Gauss Peak Density**—The intensity value of a band's Gaussian peak (after Gaussian modeling).
- **Gaussian Trace Quantity**—The quantity of a band as measured by the area under its Gaussian-fitted profile.
- **Contour qty**—The quantity of a band that has been identified using the **Contour** or **Draw Band** tools. It is the sum of the intensities of all the pixels within the band boundary multiplied by the area of each pixel. Units are intensity x mm<sup>2</sup>.
- **Contour area**—The area (in mm<sup>2</sup>) inside the boundary of a band that has been identified using the **Contour** or **Draw Band** tools.
- **Calibrated qty**—The quantity of a band as calculated from the trace quantity and quantity standards. (Note that this is different than quantity determined using volumes.) Units are user-defined.
- **Normalized qty**—The trace quantity of a particular band expressed as a percentage of the quantity of a selected band type that is present in the same lane.
- **Band type**—The band type number of a band that has been matched and placed in a band set.
- **Band model**—Displays the modeling lines across the gel that are generated by band matching, standards, or both. These lines are used to compensate for gel distortion or smiling.
- **Tandem repeats**—The number of repeated base-pair units in a band that has been analyzed using the **VNTR Calculations** function (see section 11.2, Variable

Number Tandem Repeats).

- **Differential Display**—If the band types have been normalized, this displays trends in increasing or decreasing expression of a band type across a gel based on its normalized quantity.

## 5.6 Band Information

The *Band Information* dialog displays information about each defined band. Select **Band Information** from the *Band* menu and click on a band.

An intensity profile of the band's lane will be displayed with the the selected band highlighted, and the dialog will open.

**Band Information for MA00396 (1D Scan)**

|                   |              |                |            |          |         |
|-------------------|--------------|----------------|------------|----------|---------|
| Lane              | 5            | Band Set       |            | Name     | #5      |
| Band              | 17           | Band Type      |            |          |         |
| Rel. Front        | 0.808        |                |            |          |         |
| Mol. Wt.          |              | KDa            |            | Status   | Unknown |
| Peak              | 45.6 CNT     |                | Average    | 31.6 CNT |         |
| Trace             | 152 CNT x mm |                |            |          |         |
| Gauss Model Peak  | 40.6 CNT     |                |            |          |         |
| Gauss Model Trace | 179 CNT x mm |                |            |          |         |
| Contour           | N.A.         |                |            |          |         |
| Relative Qty.     | 3.51%        |                | Norm. Qty. | N.A.     |         |
| Quantity/Units    |              | /              |            | Status   | Unknown |
| Calibration       |              | Not Calibrated |            |          |         |
| OK                |              | Help           |            |          |         |

Fig. 5-12. Band Information dialog.

The lane and band number of the band you clicked on are listed at the top of the dialog. Enter new numbers in these fields to display information for a different band.

The band set and band type will be listed, if known. Other information includes:

- **Relative front**—The distance of the band from the top of its defined lane, divided by the total length of the lane. The lane length can be determined either by measuring a vertical line from the top of the lane to bottom or (if the lane is curved) by measuring along the length of the lane. Set the preferred measuring method in the *Preferences* dialog (see section 2.5.e, Application).

**Note:** Note that normalized Rf is derived from relative front; however, normalized Rf is calculated only for bands that have been modeled using standards or band sets, and can change based on the modeling.

- **Molecular Weight/Isoelectric Point/Base Pairs/other units**—This value is determined by the type of standards defined for the gel, the band's position in the lane, and any modeling performed on the gel (using band matching or multiple lanes of standards) to compensate for gel distortion.
- **Peak density**—The intensity value of the band's peak.
- **Average density**—The total intensity of the rows of pixels used to generate the profile of the band, divided by the number of rows.
- **Trace qty**—The band quantity as measured by the area under its intensity profile curve. Units are intensity x mm.
- **Gauss Peak Density**—The intensity value of the band's Gaussian peak (after Gaussian modeling).
- **Gaussian Trace Quantity**—The band quantity as measured by the area under its Gaussian-fitted profile.
- **Contour qty**—The quantity of a band that has been defined using the **Contour** or **Draw Band** tools. It is the sum of the intensities of all the pixels inside the band boundary multiplied by the area of each pixel. Units are intensity x mm<sup>2</sup>.
- **Relative qty**—The quantity of a band as measured by its intensity, expressed as a percentage of either the total intensity of all the bands in the lane or the total intensity of the lane (including the areas between bands). The calculation method (**% of Lane** or **% of Bands in Lane**) is set in the *Preferences* dialog (see section 2.5.e, Application).
- **Normalized qty**—The trace quantity of the band expressed as a percentage of the quantity of a selected band type that is present in the same lane.

If the quantity of the band is known, you can enter the quantity and units next to the **Quantity/Units** prompt.

To calibrate the band against known quantities, click on the **Calibration** button and select the calibration curve. (See section 6.3.a, Creating and Applying a Set of Quantity Standards, for information on calibration curves.)

### 5.7 Gauss-Modeling Bands

If the bands are closely spaced or overlapping, Gaussian modeling can provide more accurate quantitation than regular band detection.

Gaussian modeling “fits” a Gaussian curve to each band profile and calculates band quantity from the area under the curve. Since the profile of a well-resolved, distinct band conforms to the shape of a Gaussian curve, this creates a band profile that is as close to ideal as possible.

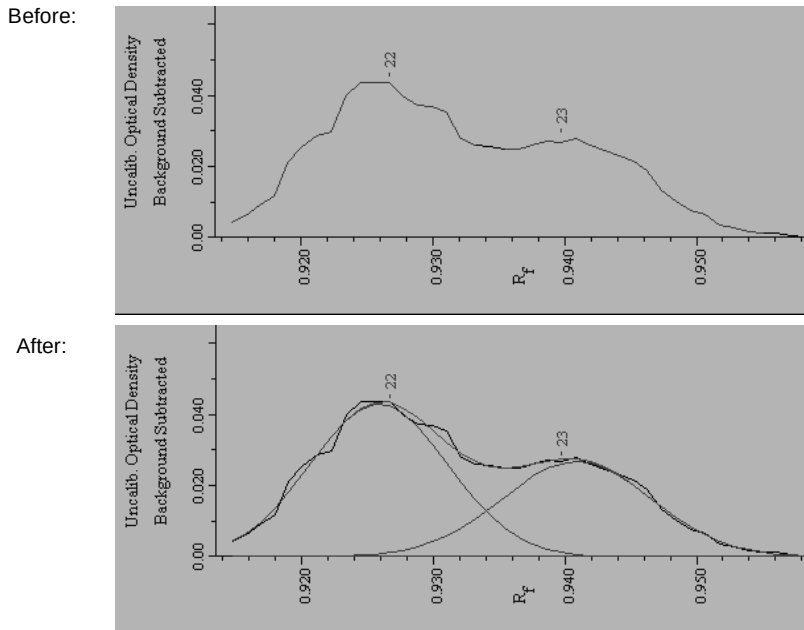


Fig. 5-13. Profiles of two overlapping bands, before (top) and after (bottom) Gaussian modeling. Modeling better resolves the band quantities.

For a band that overlaps with an adjacent band, Gaussian fitting provides the best way to resolve the area that overlaps. (This quantity would be lost with conventional band detection.)

**Note:** Gaussian modeling requires little or no background in lanes. Subtract lane background using the **Lane Background** command (see section 4.2, Lane-Based Background Subtraction) prior to modeling. Also, high-resolution images require significantly more time to model. To reduce image resolution, use the **Reduce File Size** command (see section 2.2.f, Reduce File Size).

To model bands using Gaussian fitting, first detect the bands, then select **Gauss-model Bands** from the *Band* menu.

**Note:** Gaussian modeling will not create bands or eliminate detected bands. It will simply apply a Gaussian curve to the profiles of detected bands.

In the pop-up box, select **All** to model all lanes, or **One** to model a single lane. If you select **One**, type the number of the lane in the field.

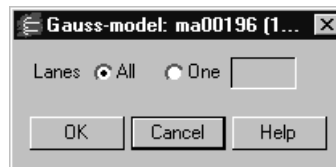


Fig. 5-14. Gauss-model Bands dialog.

Click on **OK**. A status box will display the progress of the modeling.

### Reviewing the Results

Bands that have been Gauss-modeled will appear as normal bands in the image. To view the results of Gaussian fitting, magnify a few bands in a modeled lane using the **Zoom Box** tool, then select **Bands in Lane** or **Plot Band** from the *Band* menu and click on the lane.

**Plot Band** displays the Gaussian curve superimposed on the profile of the selected band. **Bands in Lane** displays the Gaussian profiles of all the bands in the lane, as shown in Fig. 5-13.

## Quantity One User Guide

---

The *Band Information* dialog displays information about the Gaussian peak and trace quantity for modeled bands that you click on. In the dialog, you can compare the Gaussian values to the regular band detection values.

These quantities can also be displayed in the *Lane Report* and *All Lanes Report*.

**Note:** The quantities determined by Gaussian fitting cannot be used to in conjunction with **Quantity Standards** (see section 6.3, Quantity Standards). However, you can use the original trace quantities in calculating **Quantity Standards** after you have Gauss-modeled the bands.

### Adjusting Bands in a Gauss-modeled Lane

If you use any of the individual band commands—**Create Band**, **Delete Band**, **Adjust Band**—in a lane that has been Gauss-modeled, the modeling will be automatically removed from that lane. The Gaussian models in a lane are interdependent, so changing a single band invalidates the modeling. After you have adjusted the bands, you can remodel the lane.

### Incorrect Modeling

The **Gauss-model Bands** command will try to model all the bands in the selected lanes. Carefully review the results of modeling using the **Plot Band**, **Bands in Lane**, and **Band Information** commands as described above.

If the Gaussian curve does not adequately conform to the profile of a band, or if the Gaussian peak and trace quantities differ greatly from the normal peak and trace quantities in the *Band Information* dialog, it may be because there is too much lane background. Use the **Lane Background** command with a smaller rolling disk size to remove more background, then remodel the lane.

If Gaussian modeling does not work well with the bands in the image, simply remove the modeling. To remove the modeling from a particular lane, select **Remove One Gauss-model** from the *Band* menu and click on the lane. To remove the modeling from all lanes, select **Remove All Gauss-models**.

Removing Gaussian modeling will not affect band detection.



## 5.8 Irregularly Shaped Bands in Lanes

If some bands in lanes are irregularly shaped, you can use the contour or drawing features to define them. These functions give you more control over defining bands than either **Detect Bands** or **Create Bands**.

**Note:** These tools are similar to the **Volume Contour Tool** and the **Volume Freehand Tool**, except that they are lane-dependent. To quantify objects without defining lanes, see Chapter 7.

Contoured or hand-drawn bands are quantitated based on the signal intensity of all the pixels within the band boundary, using the following formula:

Quantity = Sum of the intensity of a pixel x pixel size for all the pixels in the boundary.

The intensity of a pixel is multiplied by the area of the pixel. This is done for all the pixels within the contour or drawn boundary. The area of the pixel is determined by the resolution of the image.

The resulting values have units of intensity x mm<sup>2</sup>.

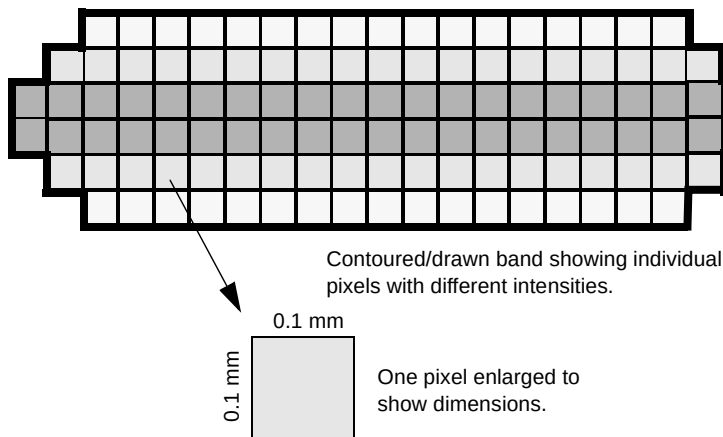


Fig. 5-15. A contoured band scanned at 100 x 100 micrometers.

The commands for contouring and drawing bands are located on the *Contour* submenu and *Draw Band* submenu of the *Band* menu, and on the *Contour Tools* toolbar.

Before using these commands, magnify the image so that the individual pixels in the band are clearly visible. This allows you to position the cursor more accurately.

### 5.8.a Contouring Bands

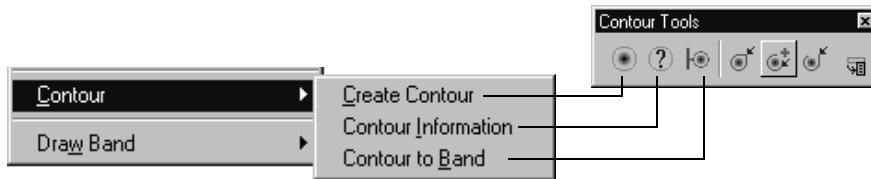


Fig. 5-16. Contour tools.

### Creating Contours

Select **Create Contour** from the *Band > Contour* submenu or toolbar and click on a pixel at the edge of the band. This displays a contour that encloses pixels whose intensity is equal to or greater than that of the pixel at the cursor.

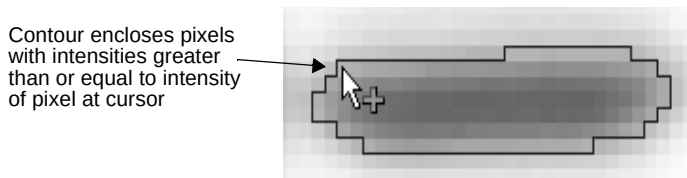


Fig. 5-17. Creating a Contour

If the contour does not encircle the band, reposition the cursor and click again. A new contour will be drawn in place of the old.

## Contour Information

To display information for the selected contour, select **Contour Information**. A pop-up box will show the area, total intensity, and average intensity.

## Converting a Contour into a Band

When you are satisfied with the contour, select **Contour to Band** from the menu or toolbar to redefine the contour as a band and assign it to the nearest lane.

**Note:** Before you can convert a contour into a band, you must define the lane containing the band.

The contour boundary will change color from yellow to red, and a band line will appear on the nearest lane.

**Note:** Gaussian modeling and the **Plot Band** command do not work on contoured bands.

To list areas and quantities of contoured bands in reports, select the **Contour Area** and **Contour Qty** report formatting options. Display this information on the image using the *Band Attributes* dialog.

### 5.8.b Drawing Tools

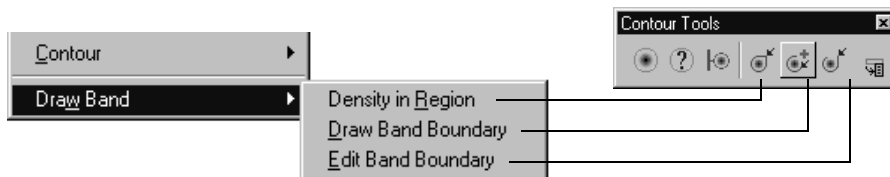


Fig. 5-18. Drawing tools.

## Density in Region

**Density in Region** displays intensity information for any area on an image.

## Quantity One User Guide

---

Select **Density in Region** from the *Band > Draw Band* submenu or *Contour Tools* toolbar, and use the cursor to draw a line around a region of interest.

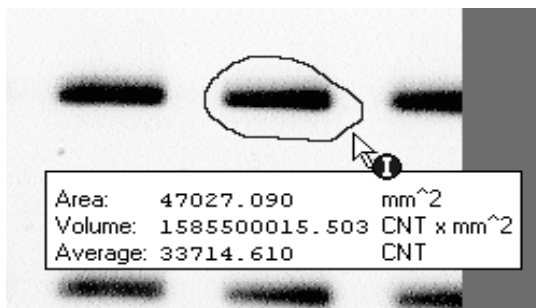


Fig. 5-19. Density in Region tool.

When you close the border, a pop-up box will display information about the enclosed area. For very small regions, magnify the region before using this command.

### Drawing Band Boundaries

**Note:** To use the drawing tools, at least one lane must be defined on the image. Also, magnify the region you want to draw in using **Zoom Box**.

Use **Draw Band Boundary** to draw the boundary of a band manually. Select the command from the *Band > Draw Band* submenu or *Contour* toolbar, and drag the cursor around the region that you want to define as a band. A boundary line will appear.

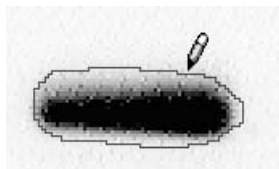


Fig. 5-20. Draw Band Boundary tool.

If you make a mistake and need to retrace part of the band boundary, backtrack with the cursor; the path will be erased and you can redraw it.

**Note:** Be sure to magnify the image before drawing a boundary. If you try to draw a very small boundary, the software will think that you are backtracking and erase the boundary.

When the cursor crosses the line, the color of the line will change to indicate that it is a band boundary, and a band line will appear on the nearest lane.

If you keep drawing, each time the line crosses itself a new band will be created, replacing the old band.

**Note:** Gaussian modeling and the **Plot Band** command do not work on drawn bands.

To list areas and quantities of drawn bands in reports, select the **Contour Area** and **Contour Qty** report formatting options. Display this information on the image using the *Band Attributes* dialog.

### Editing Band Boundaries

To change a drawn band boundary, select **Edit Band Boundary** from the *Band > Draw Band* submenu or *Contour* toolbar and drag the cursor across the previously defined boundary. A line will appear. When you recross the old boundary, the line will change colors and the new boundary will be created.



# 6. Standards and Band Matching

After you have defined the lanes and bands in a gel, you can identify the standard lanes, enter the values of the standards, and determine the values of the experimental bands using those standards. You can also compare sample similarity by matching bands across lanes.

Finally, you can identify bands of known quantity and use these to generate a calibration curve for quantitating unknown bands.

These tools are found on the *Match* menu and *Match Tools* toolbar.

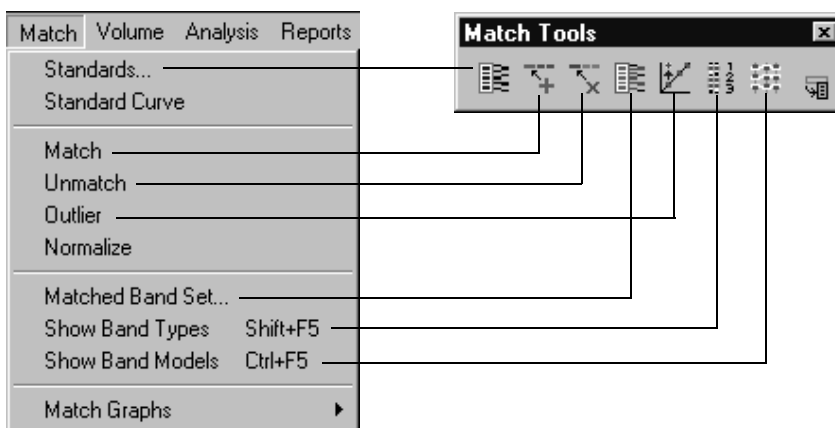


Fig. 6-1. Standards and matching tools.

## 6.1 Standards

If your gel includes lanes of standards, you can enter the values of the standards and automatically calculate the values of the unknown bands in the gel.

## Quantity One User Guide

**Note:** In general, the more standard lanes in a gel, the greater the accuracy of the calculated band values. We recommend a minimum of two standard lanes per gel. Also, the modeling algorithm works best with standard lanes spaced evenly across the gel, as shown in the following figure. Multiple standard lanes also facilitate comparisons of sample similarity.

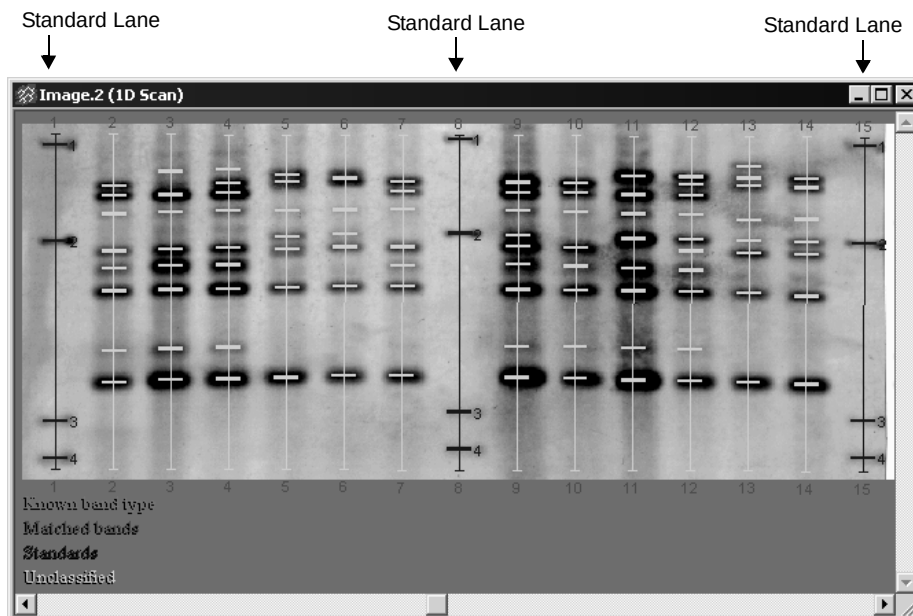


Fig. 6-2. Standard lanes as defined on a gel image.

With the image open and lanes and bands defined, select **Standards** from the *Match* menu or toolbar. A dialog will open, listing any saved standards as well as predefined Bio-Rad standards.





Fig. 6-3. Selecting a set of standards.

Sets of Bio-Rad molecular weight and base pair standards are installed with the software.

### 6.1.a Selecting Predefined Standards

If you are using Bio-Rad or other predefined standards, select them from the pop-up list. The *Standards* dialog will open (see section 6.1.c, Standards Dialog), displaying the values of the standards.

### 6.1.b Creating New Standards

To create a new set of standards, select **New Standards**. A dialog will pop up in which you can specify the units.

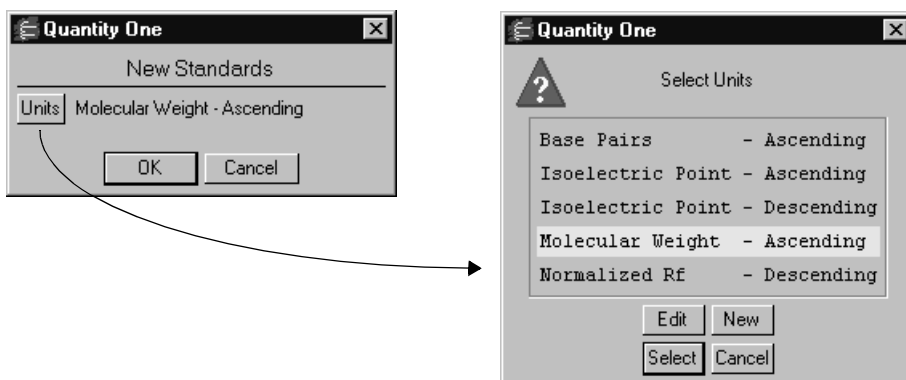


Fig. 6-4. Specifying units.

Click on the **Units** button to specify the units. This opens a dialog in which you can select from a list that includes **Base Pairs**, **Isoelectric Point**, **Molecular Weight**, and **Normalized Rf**.

**Note:** Rf (relative front) expresses the distance a band has traveled down a lane as a fraction of either the total length of the lane or the vertical distance from the top of the lane to the bottom (the calculation method can be specified in the *Preferences* dialog). This provides a generic measure of the positions of bands in lanes. Normalized Rf is derived from relative front, and includes the results of modeling across the gel that comes when multiple lanes of standards are defined on the image. Such modeling is designed to take into account any distortion or smiling across a gel.

To specify a set of units not on the list, click on the **New** button and specify the unit's parameters in the dialog. To edit a set of units, select them and click on **Edit**.



Fig. 6-5. Units dialog.

**Note:** “Ascending” means that bands of higher molecular weight or isoelectric point are at the top of the gel image, and bands of lower molecular weight or isoelectric point are at the bottom of the gel image.

To select a set of units, select them from the list and click on **Select**, then click on **OK**. The *Standards* dialog will open. Here you can enter values for the standards, apply them to the bands on the image, and save them as a set for future use.

### 6.1.c Standards Dialog

The *Standards* dialog contains the values of the standards, and includes tools for applying them to the standard lanes in the image and displaying and adjusting the standards regression curve.

The dialog opens with a default name for the standards. For new standards, this is a generic name (e.g., Standards 1). You can enter a new name in the field.

Proteins - Standard

Standard: Protein

Comment: Pen mark standards.

Type: \$ Standard

Units: Molecular Weight - Ascending

Enzyme: [ ] Primer: [ ]

Category: [ ] Probe: [ ]

Icons: [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

| Type | Name | KDa     |
|------|------|---------|
| ▶ 1  |      | 200.000 |
| ▶ 2  |      | 116.500 |
| ▶ 3  |      | 94.000* |
| ▶ 4  |      | 68.000* |
| ▶ 5  |      | 57.000* |
| ▶ 6  |      | 54.000* |
| ▶ 7  |      | 45.000* |
| ▶ 8  |      | 43.000* |
| ▶ 9  |      | 30.000* |
| ▶ 10 |      | 21.000* |

Buttons: Delete Close Archive Help

Fig. 6-6. Standards dialog.

You can enter additional information next to the **Comment** prompt. The **<category>** buttons and fields can be used to further define the standards.

### Entering Standard Values

If you are creating new standards, enter the values of the standard bands in the table in the middle of the dialog. Predefined standards already have these values entered.

Click here to delete a value and renumber the remaining standards

| Type  | Name | KDa     |
|-------|------|---------|
| ▶ 1 ▶ |      | 6557.00 |
| ▶ 2 ▶ |      | 4361.00 |
| ▶ 3 ▶ |      | 2322.00 |
| ▶ 4 ▶ |      | 2027.00 |
| ▶ N ▶ |      |         |
| ▶ ▶   |      |         |
| ▶ ▶   |      |         |
| ▶ ▶   |      |         |

Standard value

Fig. 6-7. Entering the values of the standards.

The table has three columns, labeled **Type**, **Name**, and the units you previously selected (e.g., KDa, pI, Rf). In the units column, type a value for the first standard band and press the **Enter** key. The cursor will skip to the field below, and you can enter a value for the second standard band. Repeat this process until all the standard values have been entered.

**Note:** The values do not need to be entered sequentially. They will automatically sort themselves in ascending or descending order, depending on how you specified the units.

You can enter a name for each standard band in the **Name** column. This will appear in subsequent reports and printouts.

To remove a standard value, click on the triangle button at the beginning of the row and select **Delete**. The remaining standards will be renumbered.

### Applying Standard Values to Lanes

To apply the values to the standard lanes on the image, click on the **Apply to Lane** button and click on a lane.

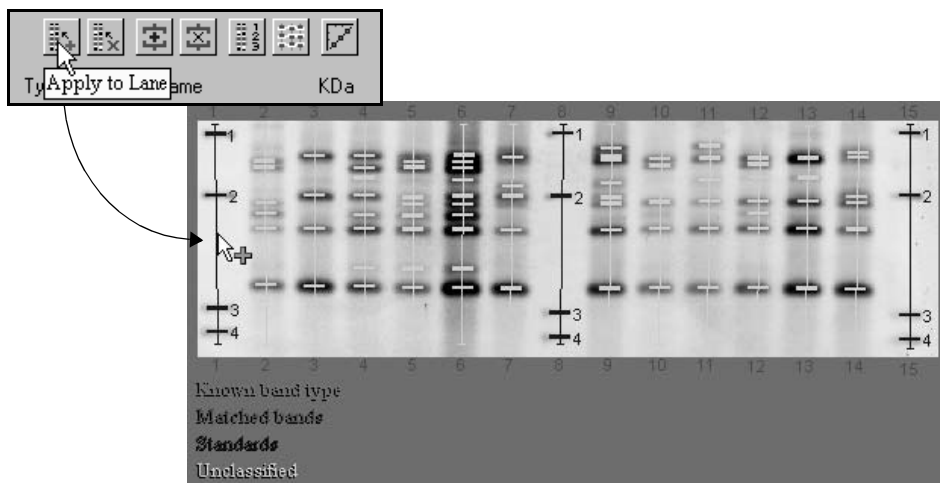


Fig. 6-8. Click on the Apply to Lane button and click on the lanes containing standards.

The values will be applied to the bands in the lane you select. Click on any remaining standard lanes to apply the same values to them.

**Note:** In general, the more evenly spaced the standard lanes, the greater the accuracy of the calculated band values. We recommend a minimum of two standard lanes per gel. Modeling lines that connect the standard bands in different lanes are used to compensate for any smiling or distortion across the gel.

You can also click on the **Arrow** button next to a standard value to apply that value to a particular band in a lane on the image. Click on the button, then click on the standard band. The remaining bands in the lane will be numbered sequentially based on the initial assignment.

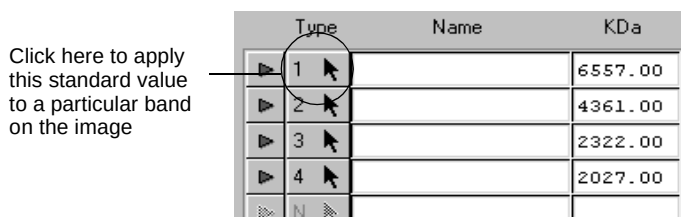


Fig. 6-9. Applying a single standard value.

After you have applied the standard values to a lane, the bands in that lane will change color to blue, indicating that they are now standards.

You are now ready to select the standards regression curve to use for calculating the unknowns.

### Removing Standard Values from Lanes

The **Clear from Lane** command removes all the standard values from the lane(s) you select. Click on the button, then click on the lane or lanes from which you want to delete the standard values.

### Showing the Modeling Lines

Click on the **Show Modeling Lines** button in the **Standards dialog** to display lines across the gel connecting the same standard bands in different lanes. Unknown bands that fall along these lines have the same values as the standards.

To redisplay only the band numbers with no modeling lines, click on the **Show Band Types** button.

### 6.1.d Standards Regression Curve

After you have applied the standard values to the image, you are ready to select the regression model to use to calculate the values of the unknown bands. (Note: You must apply the values to a lane before you can view and adjust the regression curve.)

## Quantity One User Guide

Click on the **Standard Curve** button in the *Standards* dialog, then click on a lane. A graph of the standards regression curve is displayed on the image, and the **Standard Curve Options** dialog is displayed as well.

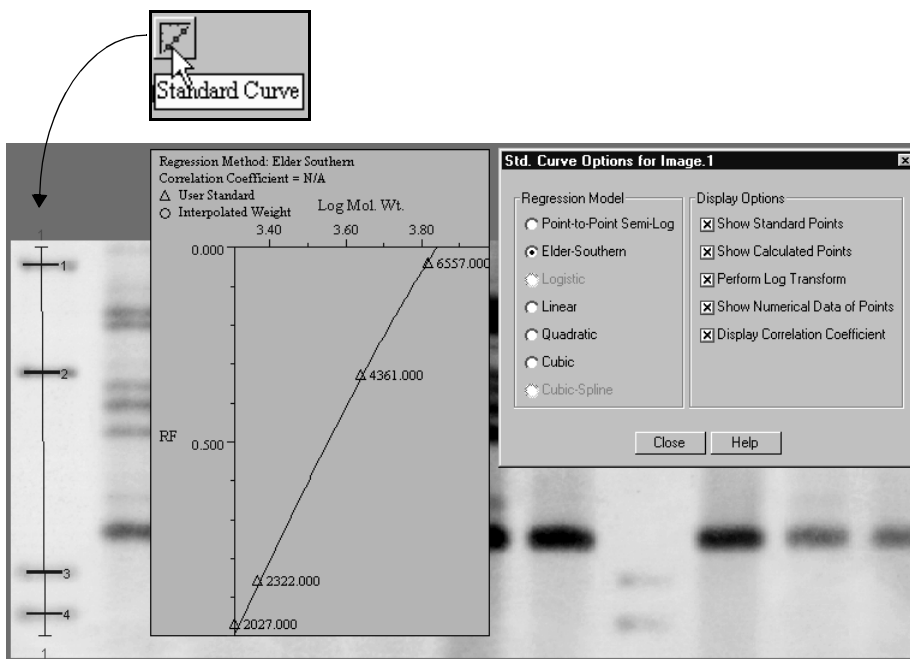


Fig. 6-10. Displaying the standard curve and options dialog.

As you click on different lanes, the curve is displayed for each lane in turn. The X axis is the standard value and the Y axis is the Rf value.

Use the *Std Curve Options* dialog to change the regression model for the curve, as well as various display options.

### Standards Regression Models

Select different regression models in the *Std Curve Options* dialog while you study the standard curve with the standard points displayed. Then choose a curve that best



fits the data points. (The correlation coefficient provides another measure of curve fit.)

**Note:** Note that point-to-point semi-log is the only method available if you perform band matching on the image, because band matching adjusts the positional values of bands in localized areas based on your identification. Point-to-point semi-log is appropriate for this kind of localized variation, whereas the other methods are not. Therefore, you should select point-to-point semi-log if you intend to perform band matching (required for similarity analysis) on the gel.

**Point-to-point semi-log.** This and the Elder-Southern method are especially useful for describing band migration in static-field electrophoresis gels. This is the only method available if you perform band matching on the image (see note above).

**Elder-Southern.** This and the point-to-point semi-log method are especially useful for describing band migration in static-field electrophoresis gels. At least three standard points are required to use this method.

**Linear.** This method of least-squares polynomial fits is useful for modeling pulsed-field electrophoresis gels.

**Quadratic.** At least three standard points are required to use this method of least-squares polynomial fits.

**Cubic.** At least four standard points are required to use this method of least-squares polynomial fits.

**Logistic.** At least five standard points are required to use this method of nonlinear least-squares curve fitting.

**Cubic-Spline.** At least five standard points are required to use this beta-cubic-spline method.

### Display Options

The following options will change how the curve graph is displayed:

**Show Standard Points** displays the standard data points on the graph. The standard points in that lane will be marked on the graph as triangles. Note that known band types will appear marked as standards.

**Show Calculated Points** displays the calculated points on the graph. The calculated points in the lane will be marked on the graph as circles.

**Perform Log Transform** changes the shape of the curve from linear to log. This will not change the calculated values.

**Show Numerical Data of Points** displays the value of each band on the graph next to its corresponding point.

**Display Correlation Coefficient** displays the correlation coefficient for the linear, quadratic, and cubic regression models.

**Note:** The correlation coefficient is a measure of how well the regression model fits the data. It is the square root of the proportion of total variation that can be explained by the regression model. A correlation coefficient of 1.000 would indicate 100 percent certainty of fit.

### 6.1.e Displaying Calculated Values

To view the calculated values of all the bands in the gel image, select **Band Attributes** from the *Band* menu, then select the value to be displayed (molecular weights, base pairs, etc.). Values can be displayed and printed in report format using the lane and match reports on the *Reports* menu.

### 6.1.f Saving, Opening, and Deleting Standards

Standards are saved with the image; you can also save copies of them in a separate archive that is available to all images.

To save the standard values with the image, click on the **Close** button to close the Standards dialog, then save the image. When you open the image again, the standards will be available when you select **Standards**.

To save these standards for use on other images, click on the **Archive** button. The standards will be saved in an archive file separately from the image.

To use a set of archived standards, open any image, then choose **Standards** from the menu or toolbar and select the archived standards. They will be imported into the image.

To delete a set of standards you have created, open them, then click on the **Delete** button at the bottom of the dialog. A pop-up box will ask you to confirm that you want to proceed with the deletion before completing the action. This will delete the standards from both the image and the archive.

To modify a set of standards you have created, open them, make your changes, then save the image and (if desired) archive the new standards.

### Read-Only Standards

You can make archived standards read-only. Read-only standards cannot be deleted from or modified in the archive using the methods described above; they can still be deleted from or modified in the image.

To make standards read-only, insert a tilde character (~) in front of the name of the standards, then click on the **Archive** button. These standards will always be available under that name in the list of standards.

All Bio-Rad standards are read-only.

### Disabling/Deleting the Archive

To delete the archived standards (including Bio-Rad standards), remove the **oneprefs.dbs** database from the **fixed.prm** folder on your hard drive. Under Windows, this folder is located in the Bio-Rad/Program Files/The Discovery Series directory. On the Macintosh, this file is located in The Discovery Series folder in the Preferences folder in the System Folder.

After you remove this file, a new, empty **oneprefs.dbs** database will be automatically created the next time you open an image. You can use this to begin a new archive.

## 6.2 Band Matching

To compare the similarity of samples in a gel (using the phylogenetic tree, similarity matrix, etc.), you must match bands across lanes using the commands on the *Match* menu and toolbar.

## Quantity One User Guide

---

**Note:** If you have run standards on the gel, you should define them before proceeding as described in the previous section. Multiple lanes of standards will facilitate the band matching program; however, they are not required.

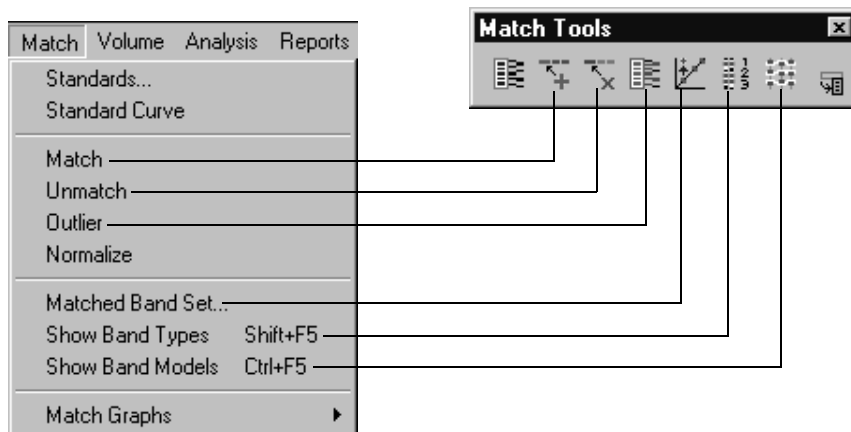


Fig. 6-11. Matching tools.

Select **Match** from the menu or toolbar and click on a representative experimental lane in the gel image. This may be a lane that contains most or all of the bands that you are interested in, and/or a lane in which the bands are particularly well-resolved. Each band in this lane will be designated as a different “band type.”

**Note:** If you have defined standards in the gel, a pop-up box will warn you that the regression model for calculating band values will be restricted to point-to-point semi-log. If you selected a different regression model when defining standards, it will be changed.

The first time you click on an experimental lane with the **Match** command, a pop-up box will prompt you to specify the matching tolerance.



Fig. 6-12. Query box: Apply matching to the whole lane?

**Tolerance** is the minimum spacing that the matching model expects to find between unique bands. It is expressed as a percent of lane height. You can enter a value between 0.2 and 10 percent. If the bands are very close together, enter a tolerance of 2.5 percent or less.

After you select a tolerance, click on **Yes** to automatically match all bands across the lanes. (Click on **No** to match only the specific band you clicked on.)

When you click on **Yes**, the bands in the lane you selected will change to green, indicating that they are known band types that have been identified by you. A band type number will appear next to each band.

The automatic matching mechanism will attempt to match the bands in the other lanes to the known band types. Matched bands are labeled in red, with the number of the band type appearing next to each band. These matched bands are connected by modeling lines.

Yellow bands are bands that the software cannot accurately match. The matching algorithm is deliberately conservative to avoid incorrect labeling, so a number of yellow bands may appear on the image.

Your next step will be to identify the yellow bands as either new band types, or existing band types that could not be automatched.

To summarize:

- **Green bands** are known band types.

- **Red bands** are bands that have been automatically matched with the known band types.
- **Yellow bands** are bands that have not been matched and are unclassified.

### Displaying Band Types and Modeling Lines

To display the band type numbers on the image, select **Show Band Types** from the *Match* menu or toolbar.

Band type modeling lines reveal the path along the gel image that the software uses to match bands of the same type in different lanes. These lines are based on the positions of the known (green) bands and any standards you may have defined.

To display the band type modeling lines, select **Show Band Models** from the *Match* menu or toolbar.

### 6.2.a Editing the Results of Band Matching

After you have matched bands automatically, you can manually indentify the remaining unknown bands as new or existing band types. Use the tools on the *Match* menu and toolbar to manually identify new bands, unmatch bands, or identify bands as outliers from the band set.

**Note:** The modeling lines are designed to give you guidance on identifying new bands.

- To identify an unknown (yellow) band as a new band type, select **Match** and click on the unknown band.
- To match a band to a particular band type, select **Match** and first click on the identified red or green band. Then click on the red or yellow band you want to match. The band will appear green (known) and the modeling line will change to reflect the match.
- If a matched (red) band in a lane is in fact a new band, select **Match**, hold down the **Shift** key, and click on the red band.
- To change a matched band to unknown (yellow), select **Unmatch** and click on that band.
- To remove a green band from the modeling, select **Outlier** and click on that band. An X will appear through the band, indicating that the software is ignoring it when auto-matching bands across the gel image.

Manually match bands until all the bands in the image are identified. The gel should have no yellow (unknown) bands, and the modeling lines should intersect at or near the middle of the bands across the gel. If the modeling lines are not parallel, the lanes may be warped or distorted and thus difficult to compare using the automatic analysis features.

### 6.2.b Band Set Dialog

The *Band Set* dialog contains the values of all the experimental bands in a gel, a tool bar for band matching, and other information about the bands.

To open the dialog, select **Matched Band Set** from the *Match* menu or toolbar and click anywhere on the image.

The dialog will open with a default name for the band set (e.g., Band Set 1). Enter a new name at the top of the dialog, and add any comments or category/attribute information you want to associate with the band set.

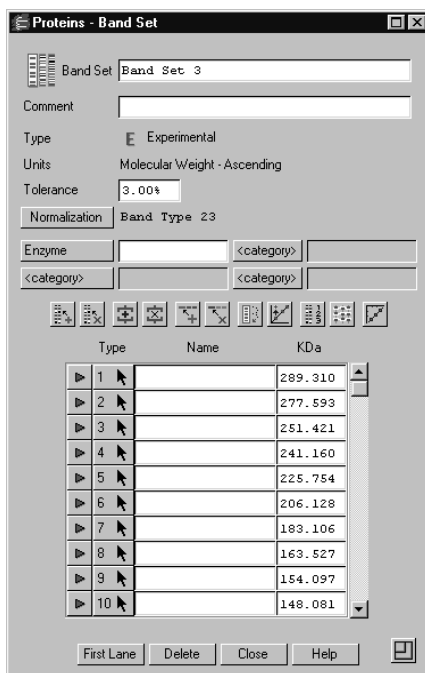


Fig. 6-13. Band Set dialog.

### List of Band Types

The values of the individual bands are listed in the dialog. These values are based on any standards you have defined and the band matching. If you have not defined standards, normalized Rf units are used.



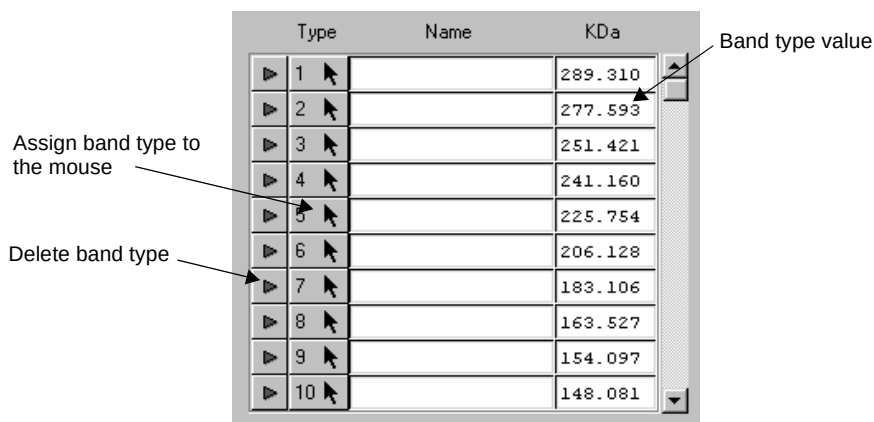


Fig. 6-14. Applying and editing band type values.

If you change any of the values in the list, that will be reflected in the matching on the image.

Click on the numbered **Arrow** button next to a band value and click on a band in the gel to identify that band.

You can enter names for the band types in the **Name** column. These will appear in subsequent reports and print-outs.

To remove a band from the set, click on the triangle icon next to the band value. Confirm the deletion, and the remaining band types will be renumbered.

### Band Set Toolbar

The toolbar in the **Band Set** dialog contains all the commands needed for matching.

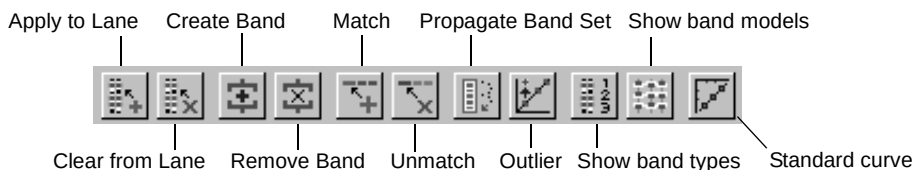


Fig. 6-15. Band set toolbar.

**Apply to Lane** applies the values in the band set to any lane in the gel. Click on the button, then click on the lane. The bands in that lane that can be matched will change to red.

**Clear from Lane** removes the identified values from any lane in the gel. Click on the button, then click on the lane to be cleared. The bands in the lane will change to yellow.

**Create Band** and **Remove Band** are standard band commands that have been included in the toolbar for convenience (see section 5.3.a, Identifying Individual Bands).

**Match** is used to identify a new band. Click on the button, then click on the unknown (yellow) band to identify it and add it to the set.

**Unmatch** removes the identification of a band. Click on the button, then click on the matched band to change it to unknown.

**Propagate Band Set** applies all the values in the set to the bands in a lane, based on a few identified bands in the lane. Click on this button, then click on the lane. The bands in that lane that can be matched will change to green to indicate their known status.

**Outlier** excludes a known (green) band from the band set model. However, the band will still be marked as known.

**Show Band Types** displays all the red, green, and yellow bands on the image, with the band type numbers next to the matched bands.

**Show Band Models** displays the modeling lines across the gel image.

**Standard Curve** displays the *Standards Regression Curve* (see section 6.1.d, Standards Regression Curve). Click on the button, then click on any lane in the image.

### Other Band Set Functions

The units of the bands are displayed in the top half of the dialog, as is the matching tolerance used. Tolerance is the minimum spacing between unique bands that you specified when you created the band set.

The **Normalization** button allows you to pick a specific identified band that appears in all the lanes to normalize the relative quantities of the other bands against (see section 11.1, Differential Display for more information).

Click on the resize button in the lower right corner to reconfigure the dialog to its smaller, palette version, which displays only the tool buttons and band type buttons.

To delete the band set, click on the **Delete** button.

To close the band set, click on **OK**.

Note that the band set is saved when you save the image.

### 6.2.c Tips for Gels Without Standards

If you are not using standards, we recommend that you load at least two lanes per gel with a reference sample containing many if not most of your experimental bands of interest.

Select the **Match** command from the menu or toolbar, and click on this reference sample lane to create a new band set. Then select **Matched Band Set** from the *Match* menu to open the *Band Set* dialog, and apply that band set to other reference sample lanes using the **Propagate Band Set** command.

**Propagate Band Set** is a feature that not only simplifies identifying bands, it allows the software to do some optimizations that will significantly speed up modeling. Choose another reference sample lane that you want to apply the band set to. Start by using the **Match** command to identify one or two bands in the lane, then click on the **Propagate Band Set** button and click on the lane to assign the remaining band types to the lane.

## Quantity One User Guide

---

Once the reference sample lanes have been modeled, identify any unknown bands using the methods outlined in the previous sections.

### 6.2.d Normalizing for Quantity

You can normalize the quantities of the bands in a gel to the quantity of a particular identified band that appears in all lanes. This is useful if you have loaded different amounts of sample in each lane.

**Note:** Quantity normalization is required for calculating Differential Display.

Select **Normalize** from the *Match* menu and click on a matched band that appears in *every lane*. (If the band is not present in a lane, the normalized quantity for that lane will be zero.) The quantity of that band will be set to 100 in each lane, and the quantities of the other bands will be normalized to that band.

You can view the normalized quantities using the *Band Attributes* dialog or in various reports.

### 6.2.e Graphs of Match Data

You can display graphs of different kinds of data associated with the matched bands. The commands for displaying these are located on the *Match > Match Graphs* submenu.

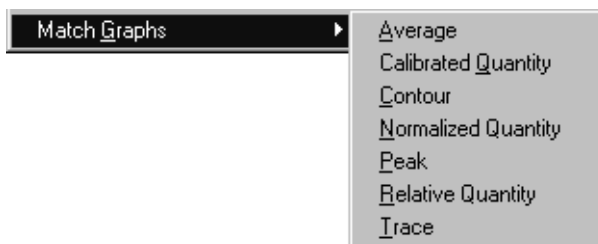


Fig. 6-16. Match Graphs submenu.

From the *Match Graphs* submenu, select a type of graph and click on a matched band. The bands in the matched group will be displayed along with the selected graph.

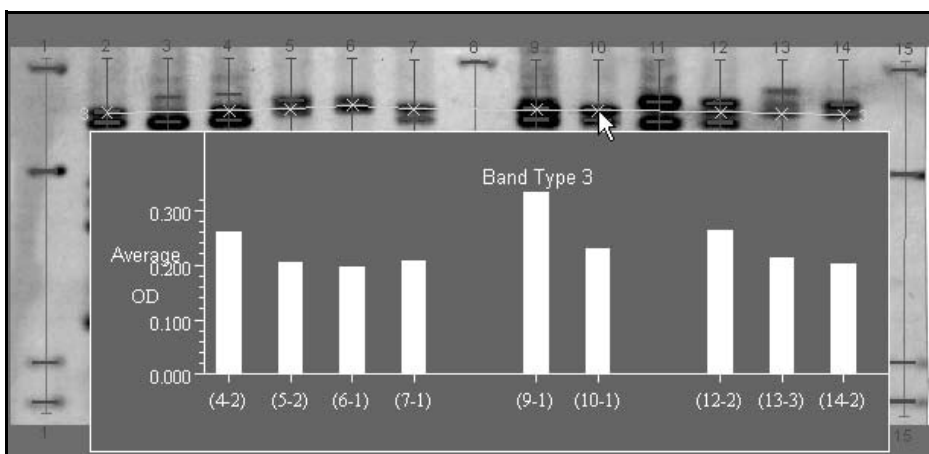


Fig. 6-17. Example of a match graph.

- **Average** displays a histogram of the average densities of the bands in a band type.
- **Calibrated Quantity** displays a histogram of the quantities of the bands in a band type as calculated from a calibration curve (see section 6.3, Quantity Standards).
- **Contour** displays a histogram of the intensity of each contoured band in a band type. This function only works with contoured bands.
- **Normalized Quantity** displays a histogram of the normalized quantities of the bands in a band type. See section 6.2.d, Normalizing for Quantity, for instructions on how to normalize for quantity.
- **Peak** displays a histogram of the peak intensities of the bands in a band type.
- **Relative Quantity** displays a histogram in which each bar represents the quantity of the band in a lane as a percentage of either (1) the total intensity data in the band's lane, or (2) the total intensity of all the bands in the band's lane. The calculation method (**% of Lane** or **% of Bands in Lane**) is set in the *Preferences* dialog.
- **Trace** displays a histogram of the trace quantities of the bands in a band type group.

Each bar of the histogram is labeled on the X axis with the lane number, and (where space permits) the band number. The Y axis is labeled with quantitative values. If the bands in the match group span the entire image window, the histogram will not include bars for the bands in the left-most lanes because of the space required for labeling the axis.

To avoid this problem, decrease the size of the image using **Zoom Out** until there is blank space between the left side of the window and the first band of the match group. Redisplay the histogram. If one or more bars are still not included, continue decreasing the magnification until you can see all the data.

## 6.3 Quantity Standards

From bands of known quantity, you can generate a calibration curve for determining the quantities of all the bands in lanes or cells in lane-based arrays. (To quantitate bands outside of lanes, see Chapter 7.)

Using the *Quantity Standards* dialog, you plot the quantities of the known bands against their intensities to generate a calibration curve. You then apply this curve to unknown bands in the current gel as well as other gels.

To create a calibration curve, the quantities of at least two bands must be known. The greater the number of known bands and the wider the range of their values, the more accurate the calibration curve will be.

**Note:** The band intensities calculated by Gaussian fitting (see section 5.7, Gauss-Modeling Bands) cannot be used in conjunction with **Quantity Standards**. However, you can continue to use the trace intensity (i.e., the area under a band's intensity profile) to calculate **Quantity Standards** after you have Gauss-modeled your bands.

### 6.3.a Creating and Applying a Set of Quantity Standards

Select **Quantity Standards** from the *Analysis* menu. A pop-up box will prompt you to create a new curve or load a saved calibration curve.

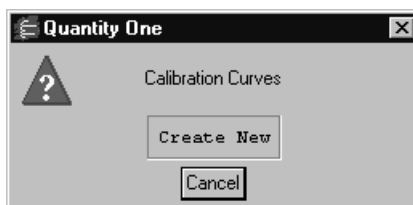


Fig. 6-18. Loading a quantity calibration curve.

Select **Create New** to open a blank *Quantity Standards* dialog.

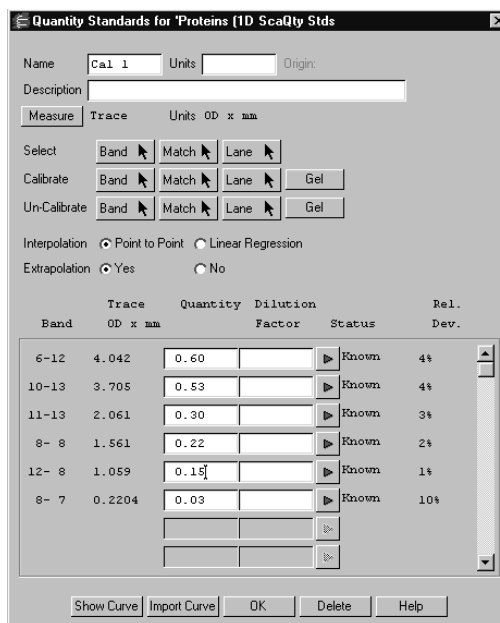


Fig. 6-19. Quantity Standards dialog.

The dialog will open with a default name for the quantity standards (e.g., Cal 1). Enter a new name, specify the quantity value units (e.g.,  $\mu\text{g}$ ), and enter any descriptive information in the appropriate fields at the top of the dialog.

## Quantity One User Guide

---

The calibration curve is generated by plotting the quantities versus the intensities of the known bands. The intensities of the bands can be measured in several different ways. Click on the **Measure** button to display a list.

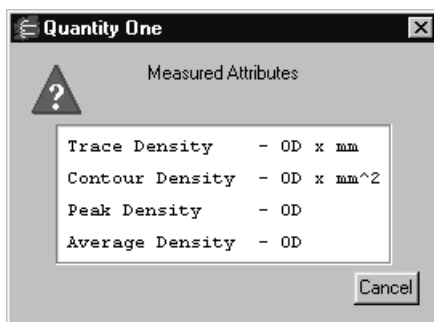


Fig. 6-20. Band attributes to measure for relative quantity.

The options in the *Measure* dialog are:

- **Trace density**—The intensity of a band as measured by the area under its intensity profile curve. Units are intensity x mm.
- **Contour density**—The intensity of a band that has been defined using the **Contour** or **Draw Band** tools (see section 5.8, Irregularly Shaped Bands in Lanes). It is the sum of the intensities of all the pixels inside the band boundary multiplied by the area of each pixel. Units are intensity x mm<sup>2</sup>.
- **Peak density**—The intensity value of a band's peak.
- **Average density**—The total intensity of the rows of pixels used to generate the profile of a band, divided by the number of rows.

### Identifying the Known Bands in the Gel

The dialog includes three buttons that can be used to select the known bands in the gel:

- To select bands one at a time, click on the **Band** button, then click on each band of known quantity. Each band will be highlighted.
- If all of the known bands are the same band type, click on the **Match** button, then click on one of the bands. The matched group of bands will be highlighted.



## Chapter 6. Standards and Band Matching

- If all of the known bands are in one lane, click on the **Lane** button, then click on the lane with the known quantities. The entire lane will be highlighted.



Fig. 6-21. Buttons for selecting bands of known quantity.

In the lower part of the *Quantity Standards* dialog, the lane and band numbers of the selected bands will appear in the **Band** column. The intensity of each band will also be listed.

### Entering the Quantities

Enter the quantity of each known band in the **Quantity** column. When you enter a quantity, the value of the band in the **Status** column will change from **Unknown** to **Known**.

| Band  | Trace   | Quantity | Dilution | Status | Rel. Dev. |
|-------|---------|----------|----------|--------|-----------|
|       | OD x mm |          | Factor   |        |           |
| 6-12  | 4.042   | 0.60     |          | Known  | 2%        |
| 10-13 | 3.705   | 0.53     |          | Known  | 2%        |
| 11-13 | 2.061   | 0.30     |          | Known  | 1%        |
| 8- 8  | 1.561   | 0.22     |          | Known  | 2%        |
| 12- 8 | 1.059   | 0.15     |          | Known  | 0%        |
| 8- 7  | 0.2204  | 0.03     |          | Known  | 11%       |
|       |         |          |          |        |           |
|       |         |          |          |        |           |

Fig. 6-22. Entering the known quantities.

After you have entered a few quantities, the status of the remaining bands in the list may change to **O.R.**, meaning that the remaining bands are out of the current range of values (based on their intensities and what you have already entered).

Alternatively, the software may automatically calculate an unknown quantity (if it is between two known quantities) and enter a value for it; in this case the **Status** column will indicate that the quantity has been calculated (**Calc.**)

In either case, you can type a quantity directly into the **Quantity** column and the status will change to **Known**.

### Relative Deviation of Known Quantities

After three values have been entered, the relative deviations of those bands are automatically calculated and displayed in the **Rel. Dev.** column. The relative deviation is calculated from the known value that you entered and the back-calculated value from the calibration curve.

If the deviation value is too high, you can exclude a band from the calibration curve. Click on the arrow button next to the problem band. In the pop-up box, select **Remove** to remove the band from the *Quantity Standards* dialog. All the information about that band will be deleted.

Alternatively, select **Outlier** from the pop-up list to retain the information about the band in the calibration file but exclude it from the calibration curve.

## 6.3.b Calibration Curve

### Interpolation and Extrapolation

There are two methods for calculating the calibration curve:

- **Point to Point** generates a curve in which each data point is connected directly to the next, regardless of the shape of the resulting curve.
- **Linear Regression** (using the method of Least Squares) generates a smooth curve that is the “best fit” of the values you provided.

Select the preferred option next to **Interpolation** in the dialog.

Next, indicate whether the curve should be extrapolated beyond the highest and lowest known values by selecting **Yes** or **No** next to the **Extrapolation** prompt. Note that values extrapolated from the **Point to Point** curve may be unreliable.

### Displaying the Calibration Curve

To display the calibration curve, click on the **Show Curve** button at the bottom of the *Quantity Standards* dialog. A graph of the known quantities versus measured intensities will be displayed in a separate window.

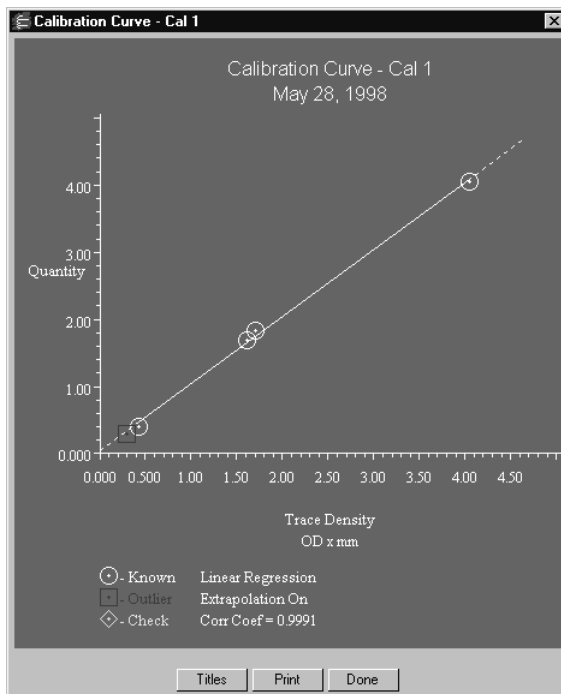


Fig. 6-23. Calibration curve with linear regression and extrapolation.

Known values used to calculate the curve are marked by circles. Values identified as outliers are marked by squares.

- To change the status of a point on the graph, click on it. The status will toggle between **Known** and **Outlier**.
- To display a legend at the bottom of the graph, click on **Titles**.
- To print the graph, click on **Print**.
- To close the graph window, click on **Done**.

### 6.3.c Applying the Calibration Curve

After you have generated a calibration curve, you are ready to calculate the quantities of the unknown bands. Select a button next to **Calibrate** to calculate the quantity of an individual band, match, lane, or the entire gel.



Fig. 6-24. Buttons for calibrating bands of unknown quantity.

If you select **Band**, **Match**, or **Lane**, click on the button and then click on the object in the gel. Click on **Gel** to calculate the entire gel. Bands of calculated quantity are highlighted in the image.

### Unapplying the Calibration Curve

To undo the quantity calculation for a band, match, lane, or gel, use the appropriate button next to **Un-Calibrate**.

### 6.3.d Generating Standard Bands via a Dilution Series

One way to generate bands of known quantity is to start with a stock solution and make several dilutions. Different dilutions can be loaded into different lanes, resulting in a dilution series.

Quantity One will calculate the values in a dilution series if you enter the known quantity and a dilution factor (e.g., for a solution that has been diluted to 10 times the volume, type 1/10 or 0.1).

In the *Quantity Standards* dialog, next to the band from the undiluted stock solution, enter the known quantity.

In the **Dilution Factor** column, type “stock.”

Next to the remaining bands in the series, enter the appropriate dilution factors in the **Dilution Factor** column.

| Band  | Trace   | Quantity |        | Dilution |     | Rel.<br>Dev. |
|-------|---------|----------|--------|----------|-----|--------------|
|       | OD x mm |          | Factor | Status   |     |              |
| 6-12  | 4.042   | 0.60     | Stock  | ▶ Known  | 3%  | ▲<br>▼       |
| 10-13 | 3.705   | 0.48     | .8     | ▶ Known  | 12% |              |
| 11-13 | 2.061   | 0.36     | .6     | ▶ Known  | 13% |              |
| 8- 8  | 1.561   | 0.24     | .4     | ▶ Known  | 3%  |              |
| 12- 8 | 1.059   | 0.18     | .3     | ▶ Known  | 1%  |              |
| 8- 7  | 0.2204  | 0.06     | .1     | ▶ Known  | 9%  |              |

Fig. 6-25. Entering a dilution series.

The quantity of each band will be automatically calculated.

### 6.3.e Importing a Calibration Curve

A calibration curve created for one gel can be applied to other gels.

Make sure that the new gel you want to quantitate and the gel with the existing calibration curve are both open.

Click on the new gel to quantitate. Select **Quantity Standards** from the *Analysis* menu and click on **Create New**, as previously described.

In the new *Quantity Standards* dialog, click on the **Import Curve** button, and select the existing calibration curve from the list. When you make the selection, the values for the curve will be displayed in the new *Quantity Standards* dialog. Each standard value will be labeled **Import** in the **Band** column.

### Checking the Imported Curve

If the quantity of one or more bands in the new image is known, you can verify the accuracy of the imported calibration curve. Go to **Select** in the *Quantity Standards* dialog, click on **Band**, then click on a known band. Its lane number, band number, and intensity will be displayed in one of the standard values fields.

| Band   | Trace   | Quantity | Dilution | Status  | Rel. |  |
|--------|---------|----------|----------|---------|------|--|
|        | OD x mm | Factor   |          |         | Dev. |  |
| Import | 4.042   | 4.04     |          | Known   | 1%   |  |
| Import | 1.702   | 1.82     |          | Known   | 4%   |  |
| Import | 1.610   | 1.68     |          | Known   | 2%   |  |
| Import | 0.4163  | 0.39     |          | Known   | 17%  |  |
| Import | 0.2879  | 0.29     |          | Outlier | 14%  |  |
| 4- 5   | 0.2456  | 0.25     |          | Check   | 16%  |  |

Band used  
to check  
curve →

Fig. 6-26. Checking an imported calibration curve.

When you enter the band's value, its status will change to **Check**, indicating that it is used to verify the accuracy of the calibration curve and is not used in calculating the curve itself.

Click on the **Show Curve** button to display the graph, and note that the **Check** bands are enclosed in diamonds. If the **Check** bands *do not* fall on or very near the calibration curve, we recommend that you *do not use* the imported standards for this gel.

# 7. Volume Tools

You can use the **Volume** tools to quantitate bands, spots, arrays, and other image data.

## What is a Volume?

A volume is the total signal intensity inside a defined boundary drawn on an image. To measure the amount of a particular object (e.g., a band or spot), you draw a volume rectangle, contour, free hand, or circle around the object and compare the intensity data inside the boundary with the data of other objects or a standard using the **Volume Analysis Report** and **Volume Regression Curve** (see section 12.6, Volume Analysis Report).

**Volume** =  $\frac{\text{Sum of the intensities of the pixels within the volume}}{\text{boundary} \times \text{pixel area}}$

**Volume units** = intensity units  $\times \text{mm}^2$

Volumes are similar to band contours (see section 5.8.a, Contouring Bands), except that they are not dependent on lanes and bands.

## 7.1 Creating a Volume

To create a volume, select **Volume Tools** from the main toolbar or the *Edit* menu. These commands are also located on the *Volume* menu.

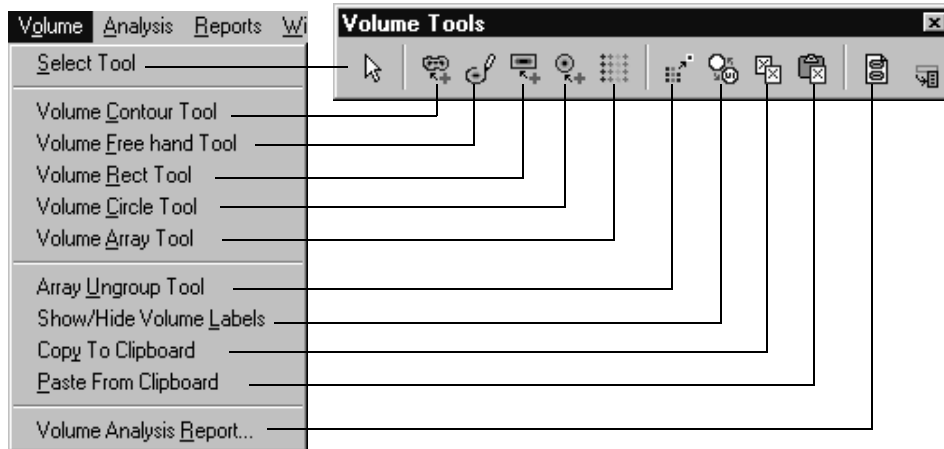


Fig. 7-1. Volume tools.

**Note:** When using any of the following tools, be careful to completely surround the data you want to quantitate. You should also adjust for background intensity (see section 7.6, Volume Background Subtraction). You may want to experiment with several different volumes drawn around the same object before selecting the one that gives you the best quantitation data.

### Volume Contour Tool

Use the **Volume Contour** tool to quickly create a volume boundary that follows the outer edge of the object you want to quantify. To use this tool, first magnify the object, then click on the **Volume Contour** button. Using the tool:

- *Click* on a pixel at the edge of a band or other object to create a contour that encloses pixels of equal or greater intensity.
- *Drag* to create a contour that changes as you move over pixels of different intensity. Drag from inside the object outward until the contour follows the outer edge of the object. When you release the mouse button, the volume is created.



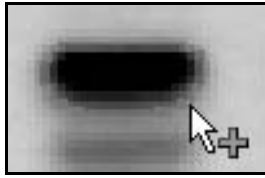


Fig. 7-2. Volume contour.

The contour should completely surround the data you want to quantify.

To edit the contour, position the cursor on the border. The cursor will change to a pencil tool. Drag across the line; a new white line will appear. When you recross the old line, a new contour will be created.

### Volume Free Hand Tool

Use the **Volume Free hand Tool** to manually draw a volume boundary. First magnify the band or other object (you must be able to see the individual pixels). Then click on the **Volume Free hand** button and use the cursor to draw a line around the object. When the line crosses itself, a free hand volume is created.

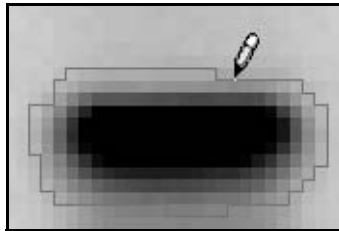


Fig. 7-3. Volume free hand.

If you make a mistake while drawing, backtrack with the mouse. The line you draw should completely surround the data you want to quantify.

To edit the volume, position the cursor on the border and drag across the line; a new white line will appear. When you recross the old line, a new free hand volume will be created.

### Volume Rect Tool

Use the **Volume Rectangle Tool** to create a volume box around an object. Click on the **Volume Rect** button, then drag a box around the object to be quantified. When you release the mouse button, the volume is created.

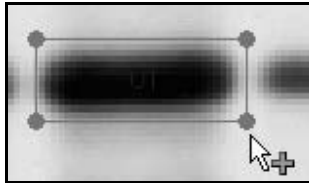


Fig. 7-4. Volume rectangle.

To *resize* the box, click on it to select it, then position the cursor on one of the corner anchor points and drag.

To *rotate* the box, click on it to select it, then hold down the **Shift** key while dragging an anchor point. The volume will pivot around its center. This is useful if the object is lying at an angle—for example, if the gel is smiling.

### Volume Circle Tool

Use the **Volume Circle Tool** to create a circular boundary around an object (such as a spot). To use this tool, click on the **Volume Circle** button, then position the cursor at the center of the object to be quantified and drag outward. As you drag, a circle will appear. When you release the mouse button, the volume is created.

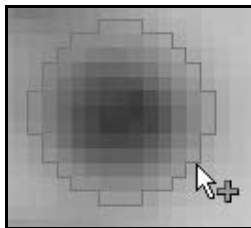


Fig. 7-5. Volume circle.

The volume circle should completely surround the data you want to quantify.

To resize the circle, click on it to select it, then position the cursor on the circle border and drag.

## 7.2 Volume Labels

Volume labels help to identify the type of the volumes you create. As you create each volume, it is given a default Auto label. The default labels are: U1, U2, U3, etc. The “U” stands for unknown, as distinguished from standard (Std) and background (B) volumes. The number indicates the sequence in which the volume was created. To change the volume type, double click the volume to open the Volume Properties dialog box.



Fig. 7-6. Volume Properties dialog box.

At the top of the Volume Properties dialog box is the current label for the selected volume. Volume Type lists the types of volumes: Unknown, Standard, and Background. The type of the current volume is indicated by the selected radio button. Choose a new type by clicking a different radio button. If you choose Standard as the new volume type, you must enter a concentration greater than zero in the Concentration field.

**Note:** If you change a volume's type (e.g., change an unknown to a standard), any subsequent volumes of the original type will be renumbered. For example, if you create volumes U1 and U2, and then designate U1 as a background volume, U2 will be renumbered U1.

### 7.2.a Editing Volume Labels

In the **Volume Properties** dialog box you can choose to give the selected volume a User label as well as modify the style of the Auto labels.

#### Editing User Labels

To create or edit a User label for the selected volume, click **Edit User Label** to open the **Text Overlay Properties** dialog box. In the **Text Overlay Properties** dialog box you can change the, font, size, color, orientation, and background of the selected text. The text field displays the current volume label. To change the text of the label, type a new label in the text field.

**Note:** A change to the text of the label will not cause a renumbering of unedited volume labels. However, if you change the type for a particular volume, this will renumber the subsequent volumes without changing the text or style of the User label.

If you want to remove the User label and return the Auto label for the selected volume, click Use Auto Label. Note that changes do not take effect until you click Done. So if you find you want to keep the User label, click Cancel instead.

#### Editing Auto Labels

To modify the style of the Auto labels, click Edit Auto Labels. Like the Edit User Label tool, this opens the **Text Overlays Properties** dialog box. Use the options in

the **Text Overlays Properties** dialog box to modify the style of the Auto labels. Note that you must first select the text in the text field before making any changes.

**Note:** Editing the Auto labels affects only the style of the labels, not the text or numbering of the Auto labels.

Modifying the style of your Auto labels does not affect the style of any existing User labels. To modify the styles for your User Labels, you must modify each individually. See *Editing User Labels* for further information.

When you first create volumes, they appear labeled. Click **Show/Hide Volume Labels** on the Volume menu or toolbar to hide the labels.

### 7.3 Volume Features

Each new volume you create initially has a green border, which indicates that the volume is selected. If you click elsewhere on the image, the border will change to blue, indicating that the volume is deselected.

To reselect the volume, click on it again. If you move the cursor over the volume, selected or not, the border changes to gold.

After you create a volume, you can view the volume data (area, density, etc.) by selecting the **Volume Analysis Report** from the *Reports* menu.

#### Tips

The volume you draw should completely surround the data you want to quantitate. You should also adjust for background intensity

You may want to experiment with several different volumes drawn around the same object before selecting the one that gives you the best quantitation data.

#### Displaying Volumes

To display previously created volumes after opening an image, select **Volume Tools** from the *Edit* menu or main toolbar.

If you have concealed all overlays using **Hide Overlays**, click on any button in the *Volume* toolbar to display the hidden volumes.

## 7.4 Moving, Copying, and Deleting Volumes

You can move, copy, or delete a single volume or group of volumes within an image. You can also copy and paste volumes between images.

First, select the volume(s). Click on the **Select Tool** button on the *Volume* toolbar. To select a single volume, click on it. To select multiple volumes, either drag a box around them or hold down the **Shift** key while you click on them one at a time. When you drag to select a group of volumes, make sure that you completely surround all the volumes.

Each selected volume will have a green border.

- To *move* the selected volume or volumes, position the cursor over the selection and drag.
- To *copy within an image*, hold down the **Ctrl** key while dragging the selected volume or volumes. The copy will be created and dragged to the new position.
- To *delete* the selected volume or volumes, press the **Delete** key.
- To *copy between images*, click on the **Copy to Clipboard** button on the *Volume* toolbar, then open or select the image you want to copy to and click on the **Paste from Clipboard** button. The copied volume(s) will be pasted into the new image in the same relative position it was copied from.

**Note:** If you are copying to an image with a different pixel size (i.e., resolution), you will receive a message that the placement of the copy may not be exact. Click on **OK** to complete the paste, then position the pasted objects manually.

## 7.5 Volume Standards

You can use volumes of known concentration to calculate the concentrations of unknown volumes.

To classify a particular volume as a standard, double-click on it. This will open the *Volume Properties* dialog.

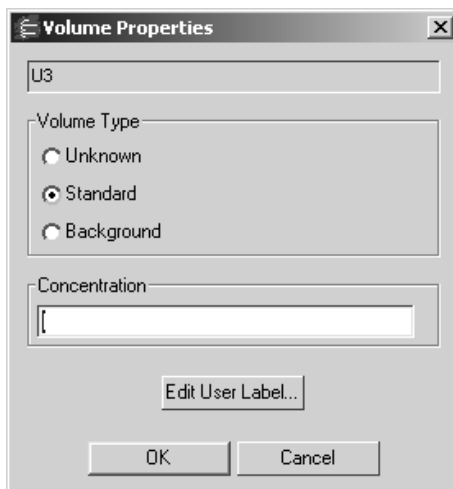


Fig. 7-7. Volume Properties dialog.

Select the **Standard** option button, then enter the concentration in the **Concentration** field. (Do not include units.) Click on **OK** to close the dialog.

Standard volumes have the default names S1, S2, S3, etc., based on their creation sequence. Display or hide volume names using **Show/Hide Volume Labels** command as previously described.

After you have identified two or more standards, you can use the **Volume Regression Curve** (see section 12.7, Volume Regression Curve) under the *Reports* menu to calculate the concentrations of the unknown volumes.

**Note:** Volumes that fall above the greatest standard value, or below the lowest standard value, cannot be guaranteed for accuracy.

To change a standard back to an unknown, double-click on it, then select the **Unknown** button.

# 7.6 Volume Background Subtraction

When you draw a volume, you will probably include some background pixels inside the volume. These background pixels will usually have an intensity value that you do not want to include in the volume quantitation. There are two ways of calculating this background intensity: local and global.

The background subtraction method is selected in the *Volume Report Options* dialog (see section 12.6.a, Volume Report Options).

## Local Background Subtraction

Local background subtraction calculates a separate background intensity for each unknown and standard volume you create. For each volume, the intensities of the pixels in a 1-pixel border around the volume are added together and divided by the total number of border pixels. This gives an average intensity for the background around each volume, which is then subtracted from the intensity of each pixel inside the volume.

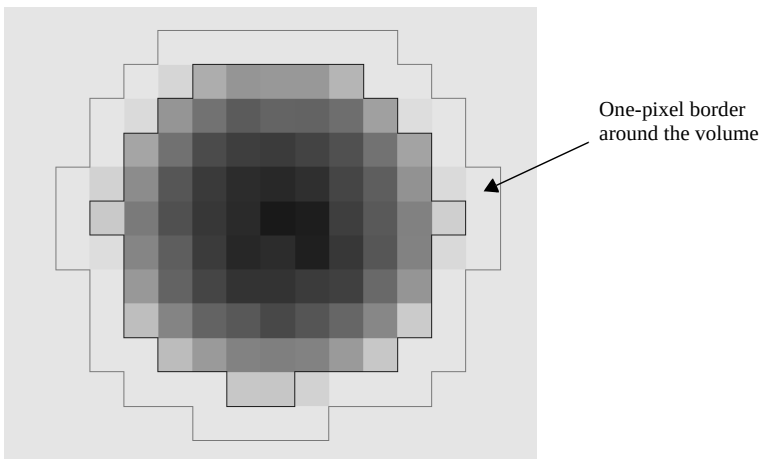


Fig. 7-8. Local background is calculated from a one-pixel border around the volume

Any pixels inside the volume that have the same intensity as the background pixels will be reduced to zero, thereby eliminating them from the quantitation.



### Global Background Subtraction

**Note:** If you select **Global Background Subtraction** in the *Volume Report Options* dialog, but do not define a background volume as outlined below, you will effectively select no background subtraction.

Global background subtraction calculates a single background intensity for the entire gel. This average background intensity is then subtracted from all the volumes in the gel. The steps for calculating global background subtraction are:

1. Create a volume using one of the volume tools in a representative background region of the image (i.e., a region where there is no data and where the average pixel intensity appears to be the same as the background intensity surrounding your data).
2. Double-click on the volume to open the *Volume Properties* dialog, and select the **Background** option button.

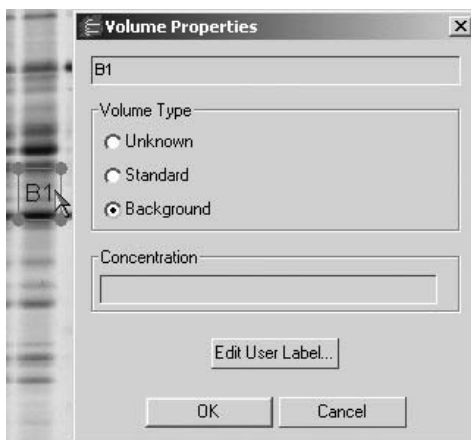


Fig. 7-9. Defining a background volume object.

The average intensity of the pixels in the background volume will be calculated and subtracted from each pixel in all standard and unknown volumes. Any pixels inside the volumes that have the same intensity as the average background will be reduced to zero, thereby eliminating them from the quantitation.

If you create more than one background volume, all the pixels in those background volumes will be used to calculate the average background.

Background volume(s) will have default names B1, B2, etc., based on their creation sequence. You can display/hide volume names using the **Show/Hide Volume Labels** command.

**Note:** If the region you identified as background has a higher average intensity value than a data object, that object will have a negative adjusted volume in the *Volume Analysis Report*. If this happens, select a new background region that has less intensity than the data object.

### Displaying the Results of Background Subtraction

The *Volume Analysis Report* (see section 12.6, Volume Analysis Report) will display both the unadjusted volume and the volume with background subtracted (adjusted volume) of standards and unknowns, so you can see exactly how much intensity was subtracted.

## 7.7 Volume Arrays

The **Volume Array Tool** on the *Volume* menu and toolbar can be used to quantitate dot blots, slot blots, and other arrays.

**Note:** You cannot create a volume array in an image with asymmetric pixels (i.e., different dimensions in x and y). If you want to create a volume array in such an image, select **Reduce File Size** from the *File* menu to change the pixel dimensions of the image (see section 2.2.f, Reduce File Size).

### What Is a Volume Array?

A volume array is a matrix of volume circles or rectangles that can be sized/positioned as a group and overlaid on images of blots, wells, or cells for easy quantitation. The individual cells in the array have the same functionality as standard volumes. You can define cells as background volumes, standards, and/or unknowns, as described in the sections above.

You report the array data as you would standard volumes, using the **Volume Analysis Report**.

### Creating a Volume Array

On the *Volume* menu or toolbar, select the **Volume Array Tool**. This will open the *Build Volume Array* dialog.

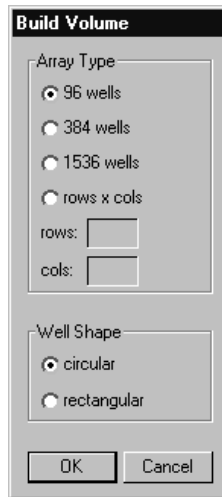


Fig. 7-10. Build Volume Array dialog.

In the dialog, you can select a standard microtiter plate dimension (**96 wells**, **384 wells**, or **1536 wells**) or select **Rows x Cols** and enter the number of rows and columns in the array in the appropriate fields.

Select the shape of the wells/cells (**Circular** or **Rectangular**) and click on **OK**.

The array overlay will be created and displayed on the image.

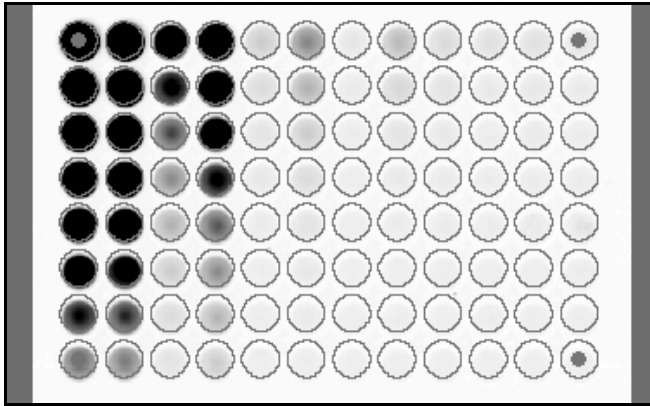


Fig. 7-11. Array overlay.

Like regular volumes, array volumes are initially displayed without labels. To show the labels of the individual wells/cells, click on the **Show/Hide Volume Labels** button on the toolbar. Like regular volumes, array volumes are initially labeled U1, U2, U3, etc.

**Note:** If large volume arrays are slow to display or edit on your computer and the volume labels are showing, try hiding the volume labels using the **Show/Hide Volume Labels** command. This will increase the processing speed considerably.

When you create an array overlay, it is automatically selected (the cells will be displayed with green borders) and the **Select** tool is assigned to the mouse. You can then move the array overlay so that it is properly centered on the image, resize the cells so they fit the blots/wells in the image, and resize the overlay so the four corners fit over the four corners of the array on the image.

To delete the entire array overlay, select it and click on the **Delete** key.

### Moving an Array

To reposition an array overlay, move the cursor over any individual cell until the cursor changes to a multidirectional arrow and the cell border turns yellow. Then hold down the cursor and drag the entire array to a new position.

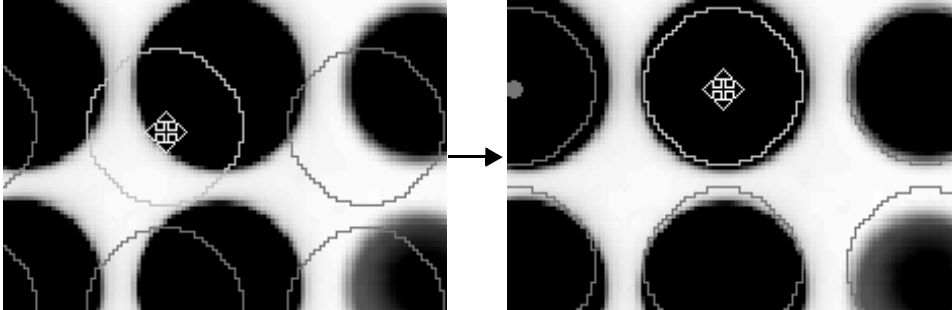


Fig. 7-12. Moving an array.

### Resizing an Array

To resize an array overlay, make sure it is selected, then position the cursor over the dot at the center of one of the corner cells. Green lines will appear connecting the array frame at the four corners.

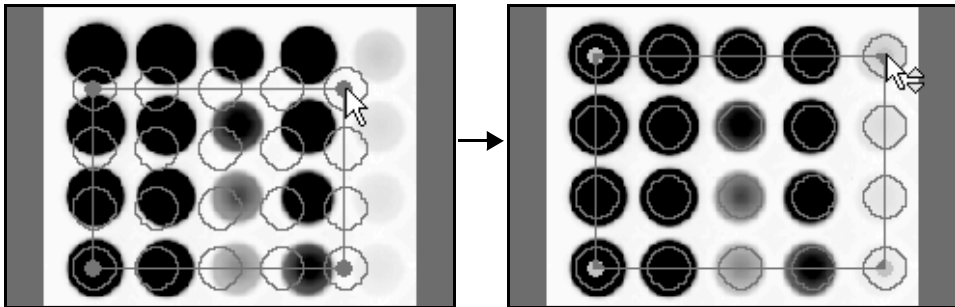


Fig. 7-13. Resizing an array.

Hold down the mouse button and drag the array frame in or out to compress/expand the array.

### Resizing the Array Cells

To resize the individual cells in the array, magnify any individual cell and move the cursor over the cell border (or corner anchor point, in the case of rectangles) until it changes to a cursor with an adjustment symbol. Hold down the mouse button and drag to move the border in or out. All the cells in the array will be resized accordingly.

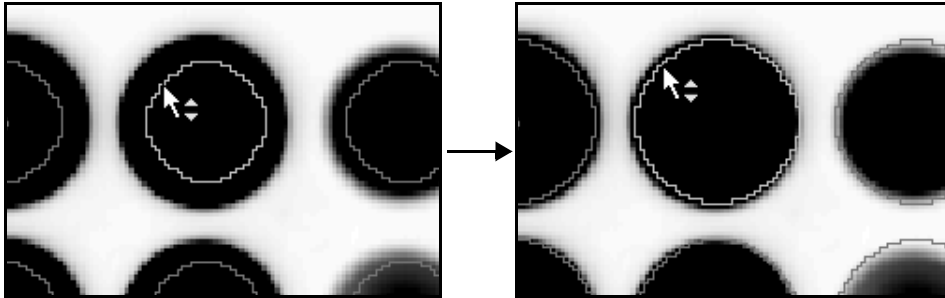


Fig. 7-14. Resizing an array cell.

### Copying an Array

To copy an array within the same image, select it, then hold down the **Ctrl** key while dragging it. The copy will be created and dragged to the new position.

To copy an array between images, select it, then click on the **Copy to Clipboard** button on the *Volume* toolbar. Open or select the image you want to copy to and click on the **Paste from Clipboard** button. The copied array will be pasted into the new image in the same relative position it was copied from.

**Note:** If you are pasting into an image with a different pixel size (i.e., resolution), you will receive a message that the placement of the copy may not be exact. Click on **OK** to complete the paste, then position the pasted array manually.

### Ungrouping an Array

You can ungroup the individual cells in an array, so they behave like normal, stand-alone volumes.

With the array selected, select the **Array Ungroup** command from the menu or toolbar. This command cannot be undone, and you will be prompted to confirm the action.

The ungrouped array will appear deselected (i.e., displayed in blue). You can then move the cells individually, and perform all normal volume operations on the individual cells.





# 8. Colony Counting

You can use Quantity One to automatically count the number of white, blue, or plaque colonies in a Petri dish.

**Note:** For best results, when capturing the image of a Petri dish, the dish should fill the imaging window. Also, images with colonies *should not* have asymmetric pixels. (Asymmetric pixels can be generated by densitometers and the **Reduce File Size** command.) The colony counting function will not work properly on images with asymmetric pixels.

Select **Colony Counting** from the *Analysis* menu to open the *Colony Counting* dialog.

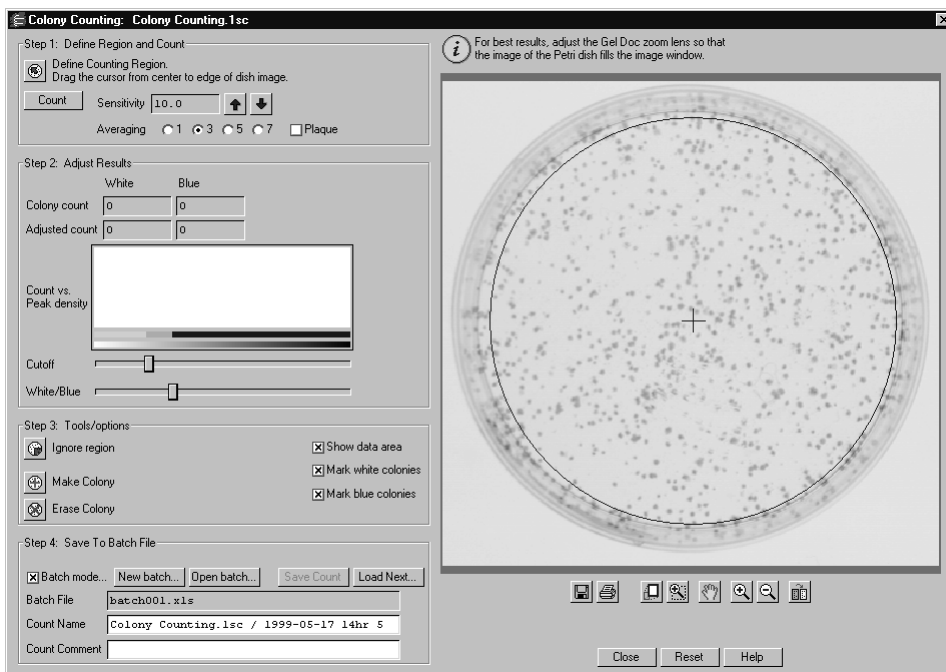


Fig. 8-1. Colony Counting dialog.

The dialog has been arranged from top to bottom to guide you through the procedure.

### 8.1 Defining the Counting Region

First, you must define the region you want to count in the Petri dish image.

Click on the **Define Counting Region** button in the dialog and position the cursor at the center of dish image. Drag the cursor outward. As you drag, a blue circle will expand on the image—this defines the border of the counting region.

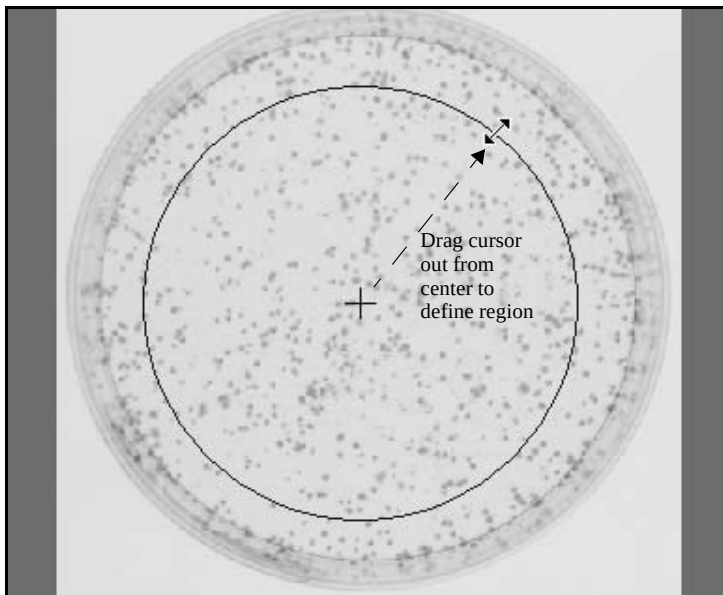


Fig. 8-2. Defining a counting region.

If you make a mistake in defining the counting region, click on the **Reset** button at the bottom of the dialog to start over.

Position the blue border until it is just inside the interior edge of the Petri dish.

**Note:** If the border disappears when you release the mouse button after dragging, check to make sure that the **Show Data Area** checkbox is checked. This checkbox is located at the bottom of the dialog.

If the circle is slightly off-center, you can reposition it by positioning the cursor on the center “target” of the circle. The cursor will change to a multidirectional arrow, and you can drag the entire circle.

To resize the counting region circle, position the cursor on the outer edge of the circle. The cursor will change to a bidirectional arrow and you can drag the border in or out.

## 8.2 Counting the Colonies

After you have positioned the circle, you are ready to detect colonies.

If you are counting plaques, click on the **Plaque** checkbox. (Because plaques appear as clear circles on a darker background, this checkbox must be selected for proper detection.)

Before counting, you may want to adjust the **Sensitivity** and **Averaging** parameters described below.

When you are ready to count, click on the **Count** button.

### Sensitivity

The **Sensitivity** setting determines the minimum signal intensity in the image that will be counted as a colony. (This is based on the slope of the signal’s peak.) The higher the sensitivity, the more colonies will be detected.

If the sensitivity is set too high, background noise will be erroneously detected as colonies. If the setting is too low, real colonies may be missed.

The default sensitivity setting is 10.00. If the image has faint colonies (e.g., O.D. < 0.05, counts < 2,000), you may want to increase this value to 20.00.

### Averaging

**Averaging** is designed to prevent random signal noise (such as salt or pepper) in the image from being detected as colonies. If the image is noisy, you should select the highest value that still results in good separation of colonies (default = 3).

A low averaging value may result in noise being detected as colonies. A high averaging value may result in two closely spaced colonies being counted as one.

## 8.3 Displaying the Results

When you click on **Count**, the number of detected colonies will appear in the **Results** section of the dialog in the **White** column.

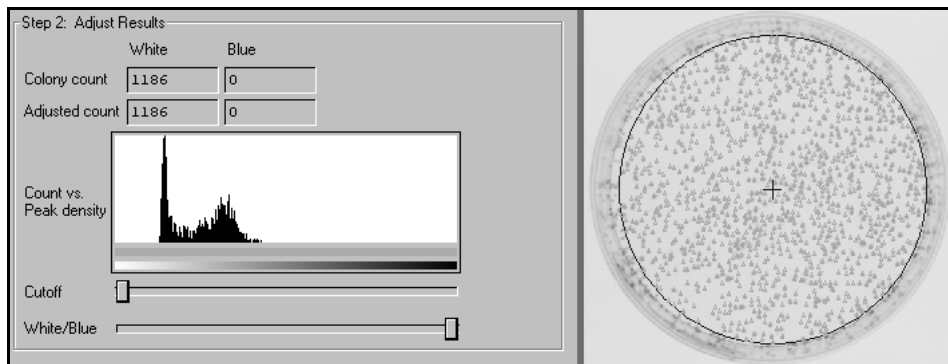


Fig. 8-3. Example of a dish with white colonies.

The colonies will also appear marked as gold triangles on the image itself.

**Note:** If the colonies are not marked on the image, make sure that the **Mark White Colonies** checkbox at the bottom of the dialog is checked.

A text box on the image will indicate how many colonies were detected on the image.

### Doing a Recount

To recount using different parameters, change the **Sensitivity** and/or **Averaging** settings. This will erase the old count. Then click on **Count** again to recount.

Redrawing the counting region circle or clicking on the **Reset** button will also erase the count.

## 8.4 Making and Erasing Individual Colonies

If automatic colony detection has missed or erroneously detected some colonies, you can manually mark or unmark them directly on the image using the buttons under **Tools/Options**.

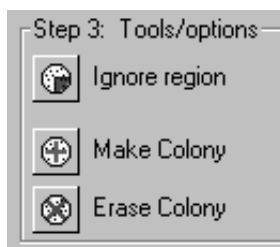


Fig. 8-4. Colony counting tools.

To mark a colony, click on the **Make Colony** button, then click on the spot on the image that you want to identify as a colony.

To unmark a colony, click on the **Erase Colony** button, then click on the colony on the image that you want to unmark.

The colony count will change accordingly.

### 8.5 Using the Histogram to Distinguish Colonies

The histogram in the *Colony Counting* dialog is a graphical representation of the signal data in the image. You can use the histogram and associated sliders to reduce the number of incorrectly identified colonies and/or distinguish between white and blue colonies in the image.

#### Colonies Versus Background Noise

If there is a clear peak on the left end of the colony counting histogram, it is probably due to background intensity or noise in the image. (For information on subtracting background from entire images, see section 3.10, Whole-Image Background Subtraction; for information on filtering noise from images, see section 3.11, Filtering Images.)

If background is being detected as colonies, you can use the histogram and the **Cutoff** slider to correct this.

Drag the **Cutoff** slider to the right until it is centered on the right edge of the background peak.

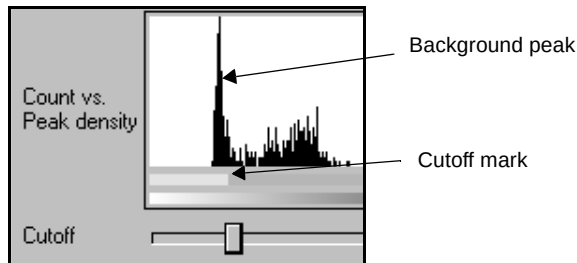


Fig. 8-5. Using the Cutoff slider.

The yellow portion of the bar beneath the histogram marks the range of image data has been designated as background noise, and is not being considered for colony counting purposes. The gold portion of the bar marks white colony data range.

The colony count displayed in the dialog and on the image should decrease. On the image, you should also see the incorrectly identified colonies disappear as you drag the slider.

### White and Blue Colonies

If you know you have white and blue colonies in the image, and there are two clear peaks on the histogram to the right of the background peak, you can use the histogram to distinguish between these types of colonies.

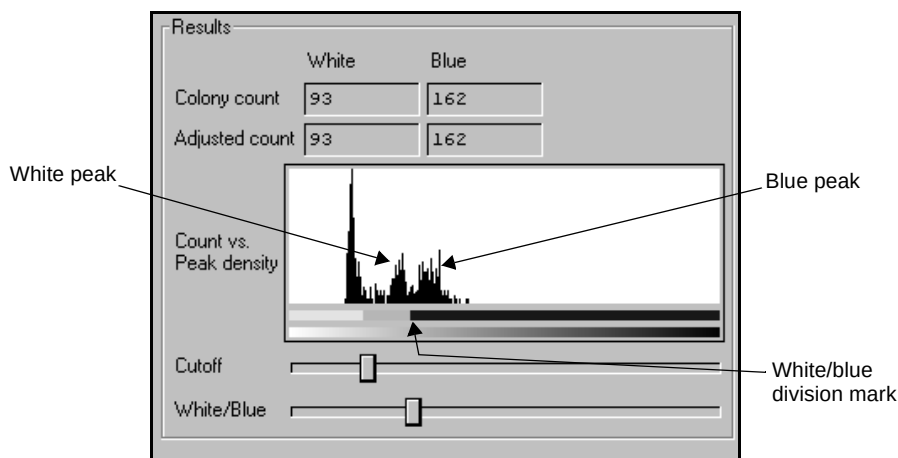


Fig. 8-6. Using the White/Blue slider.

Drag the **White/Blue** slider to the left until it is positioned between the two peaks. The white colony data range is indicated by gold on the bar beneath the histogram, and the blue colony data range is marked with blue.

As you drag the slider, the numbers of white and blue colonies will change in the dialog and in the text box on the image. Also on the image, you should see the marked white colonies (gold triangles) change to blue colonies (blue squares).

**Note:** If the blue colonies are not marked on the image, check to make sure that the **Mark Blue Colonies** checkbox at the bottom of the dialog is checked.

### 8.6 Ignoring a Region of the Dish

If a particular region of your Petri dish is damaged and you do not want to consider the colonies (if any) that appear there in the final count, you can exclude that region of the dish from the calculations.

Click on the **Ignore Region** button, then position the cursor on one edge of the region you want to ignore. Drag the cursor on the image, defining the full region you want to ignore.

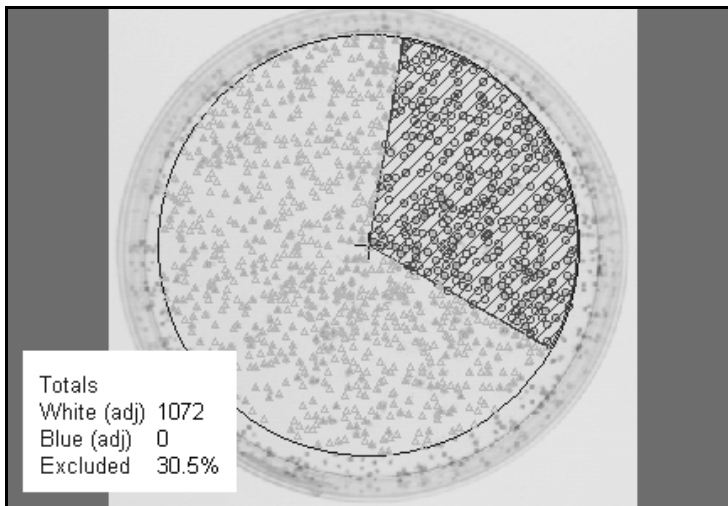


Fig. 8-7. Marking a region to ignore.

As you drag, you will create a “pie slice” marked with red cross-hatching. Any colonies in this region will not be considered in the final count.

When you have defined the region, release the mouse button. If you want to change the size of the ignored region, position the cursor on the edge of the pie slice near the rim of the blue circle. The cursor will change to a bidirectional arrow, and you can drag the edge of the pie slice.



### Colony Count and Adjusted Count

After you have defined a region to ignore, two different counts will appear in the dialog: the colony count and the adjusted count.

The **Colony Count** is the number of colonies that appear in the defined circle minus those in the ignored region.

The **Adjusted Count** is an estimate of the total colony count in the Petri dish; it uses the known colonies to extrapolate the number of colonies that might have appeared in the ignored region if it had not been damaged. The adjusted count is calculated based on the area of the ignored region and the density distribution of colonies in the rest of the circle.

## 8.7 Saving/Resetting the Count

A colony count can be saved to the image and/or a separate spreadsheet file.

### Saving to the Image

Any count you perform is automatically stored with the image. To save the count with the image, exit the *Colony Counting* dialog by clicking on the **Close** button, and use the **Save** commands under the *File* menu to save the image.

To view the count data again, simply open the image and open the *Colony Counting* dialog.

To save a count or multiple counts to a spreadsheet file, see the following section.

### Resetting the Count

Click on the **Reset** button to clear the *Colony Counting* dialog and any changes you have made to the image. This command cannot be undone.

### 8.8 Saving to a Spreadsheet

The **Batch File** controls allow you to export colony data from an image or multiple images to a Microsoft® Excel spreadsheet for review and comparison. To activate these controls, click on the **Batch Mode** checkbox.

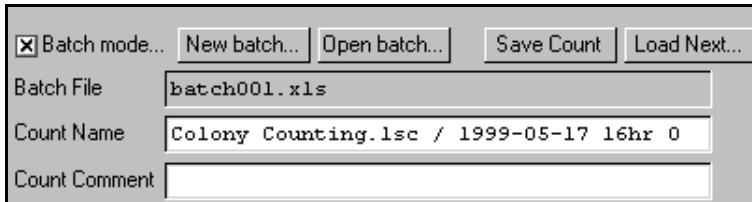


Fig. 8-8. Batch Mode controls.

#### Creating/Opening a Batch File

To create a new batch file, click on the **New Batch** button. This will open a dialog in which you can specify the name and location of the spreadsheet you want to create.

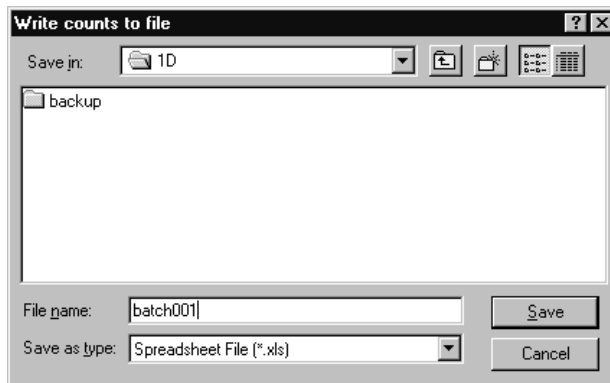


Fig. 8-9. Creating a batch file.

When you click on **Save**, the new batch file name will be displayed in the *Colony Counting* dialog.

To open an existing batch file, click on the **Open Batch** button. This will open a similar dialog. Select the Excel file you want to open from the appropriate directory.

### Naming/Saving a Count

Enter a name for the count you want to save in the **Count Name** field, or use the default name (the file name plus a time stamp). Enter any comments in the **Count Comment** field. This data will be included in the spreadsheet.

To save the currently displayed count to the batch file, click on the **Save Count** button. The number of colonies, as well as associated count settings, will be added to the spreadsheet. After you have saved the current count, the **Save Count** button will become deactivated. If you adjust the count in any way, the button will become active again and you can add the adjusted count to the spreadsheet.

### Loading Another Image

After you have saved the count(s) for the current image, you can open another dish image by clicking on the **Load Next** button. This will open a standard *Open* dialog from which you can select the image.

The new image will be loaded into the *Colony Counting* dialog.

**Note:** The image will only be loaded into the *Colony Counting* dialog; it will not open in a separate image window in Quantity One.

After you have saved the count(s) for the new image to the batch file, you can either load another image using the **Load Next** command or click on **Close** to close the *Colony Counting* dialog.



# 9 ReadyAgarose™ 96 Plus

ReadyAgarose 96 plus gels are pre-cast agarose gels that accommodate 96 samples and standards. These high throughput gels allow loading samples from 96 well microtiter plates. The gels are 4 and 12 multi-channel pipet compatible, and the tip-to-tip spacing of the multi-channel pipetors results in adjacent samples from microtiter plates being loaded in alternating wells on the ReadyAgarose 96 plus gel.

The ReadyAgarose 96 plus feature of Quantity One allows you to manipulate the gel image to rearrange the lanes from samples run on the ReadyAgarose 96 plus gels and put them in a 96 well microtiter plate format, thus simplifying sample tracking for analysis. This allows easy comparison and analysis of gel data based on the microtiter plate format.

When acquiring the gel image, zoom in such that there is minimum blank space around the gel. This is extremely important with low resolution imagers. Once the gel image is acquired, the ReadyAgarose 96 plus feature uses a step-by-step approach to analyze the image. The First step is to align the gel image. Quantity One then places lane overlays over each lane. You can then make slight adjustments to the size and position of the lane overlays. The final step allows you to determine the lane format. There are three lane formats from which to choose, 8 x 14, 4 x 28 (default), or 2 x 56 format (includes standard lanes).

## 9.1 ReadyAgarose Wizard

Once you have acquired your image, select ReadyAgarose 96 plus from the Analysis menu. This opens the ReadyAgarose 96 plus Wizard.

### **Step 1 - Positioning the Image**

The first panel of the wizard displays your image with four red crosshairs.

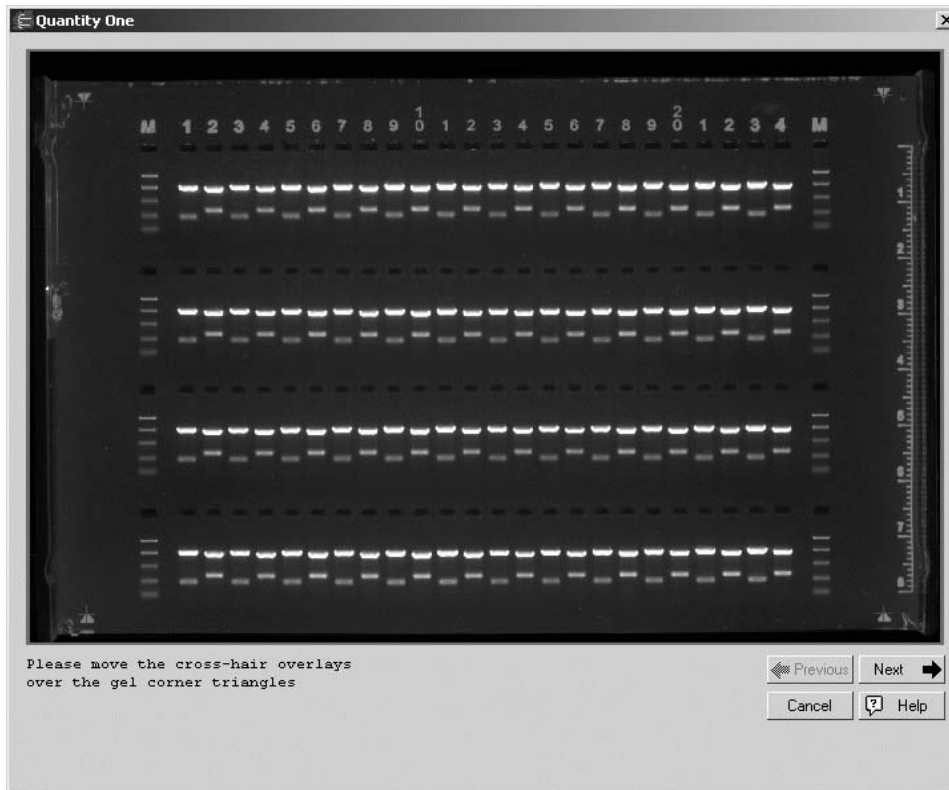


Fig. 9-1. Positioning the Image

Click and drag each crosshair to place them over the four triangles visible in the gel holder. The ideal position is at the tip of the triangle. When you have correctly positioned the crosshairs, click the Next button.

### **Step 2 - Resizing, Repositioning, and Spacing the lanes**

In the next panel, the image displays 96 lanes based on the position of the crosshairs in the previous panel.

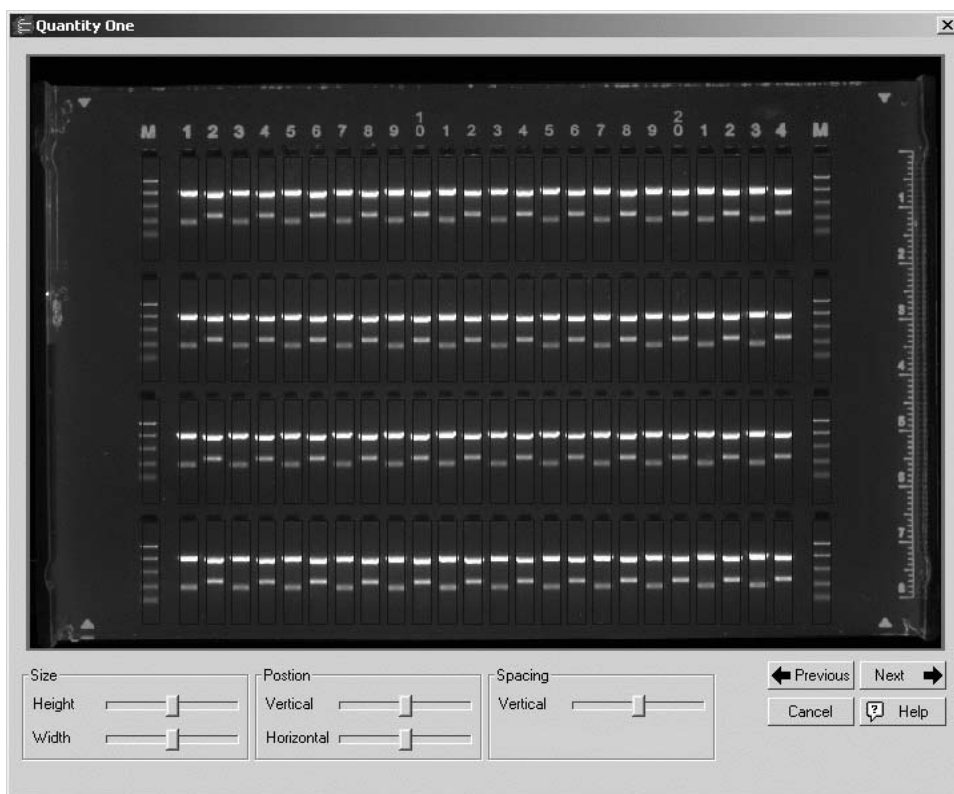


Fig. 9-2. Resizing and Repositioning the Lanes

Use the slider bars at the bottom to change position, resize the lanes, or modify the vertical spacing of the lanes. Note that the sliders move, resize, or space all the lanes at once. When you are satisfied with the location of the lanes, click Next.

### Step 3 - Arranging the Lane View

This panel places the 96 lanes in a new view. The default view is 4 x 28.

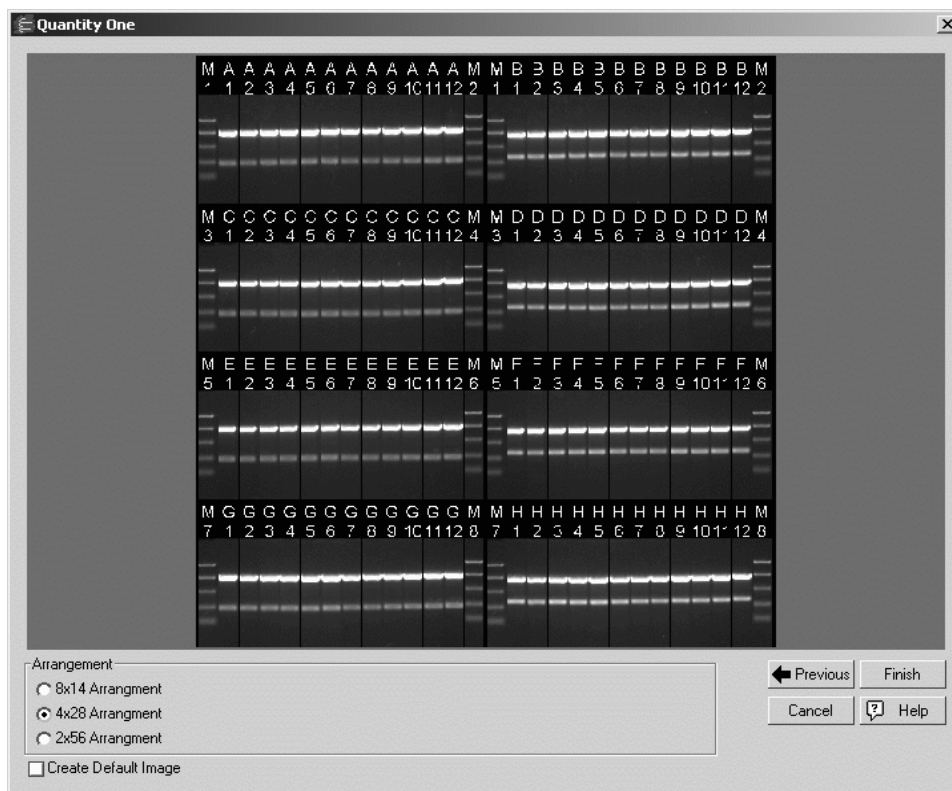


Fig. 9-3. Arranging the Lanes

To arrange the lanes in a different view, select the appropriate view at the bottom of the panel. If you would like Quantity One to create an additional default image with the lanes in their original orientation, check the box labeled, "Create Default Image". When you are satisfied with the gel arrangement, click Finish to create your new image.



# 10. EXQuest Spot Cutter

The EXQuest Spot Cutter is a precision instrument that accurately locates and excises protein spots from 1D gels or blots, and loads them into microplates for down-stream processing and analysis.



Fig. 10-1. EXQuest Spot Cutter

The following sections describe how to calibrate the EXQuest Spot Cutter and excise spots using Manual Excision or lane/Volume Excision.

### 10.1 Calibration and setup

Before you can use the spot cutter for the first time, you must first calibrate the instrument. The calibration wizard helps you to quickly calibrate the camera as well as set the plate and wash station positions. The calibration wizard functions the same whether you are using the gel tip or the membrane tip.

To open the calibration wizard, go to the Devices tab of the Preferences dialog box, make sure you have selected the EXQuest Spot Cutter, and click Setup Spot Cutter.

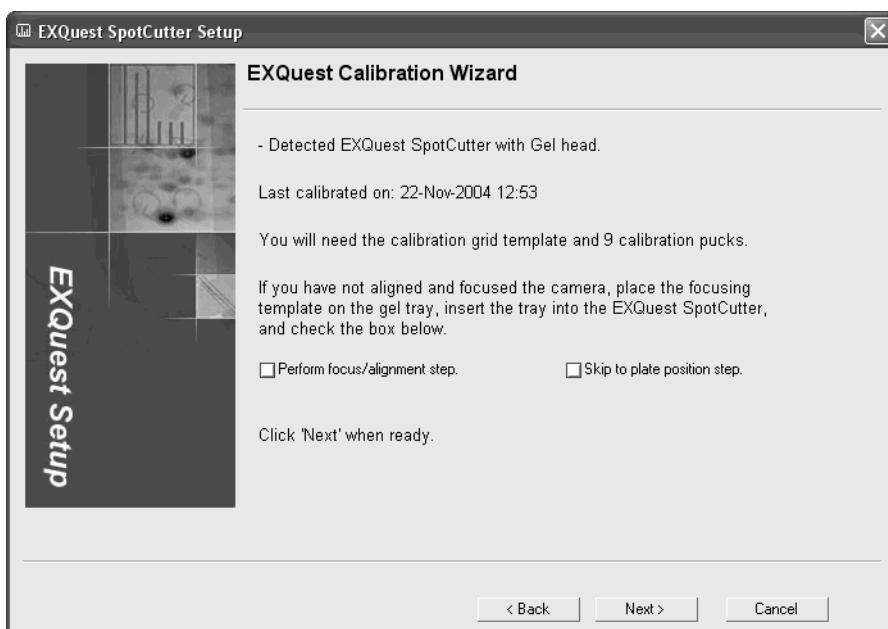


Fig. 10-2. EXQuest Spot Cutter wizard

#### 10.1.a System Calibration

The wizard guides you step-by-step through the system calibration. System calibration synchronizes the coordinates of the camera and the coordinates of the

cutter. Follow the directions on each panel then click Next to continue. If you find you made an error, or need to repeat a step for any reason, click Back.

**Note:** Clicking Back requires you to re-perform any completed steps.

### Focusing the Camera

If this is the first time calibrating the spot cutter, or you have replaced the camera, you need to focus the camera. Check the box labeled Perform Focus/Alignment step, then click Next.

If you do not need to focus the camera, clear the Perform Focus/Alignment step checkbox. You can also skip system calibration and go directly to plate alignment provided system calibration was performed earlier and is still valid.

### Flat Field Calibration

In the Flat Field Calibration step, the spot cutter needs to create an image of the cutting stage. Make sure the cutting stage is clean and dry before proceeding.

### Lens Calibration

In the Lens Calibration step, the spot cutter needs to create an image of the grid placed on the cutting stage. Place the calibration grid on the stage before proceeding.

### System Position Calibration

This step requires the placement of pucks on the stage in a three-by-three grid. As the gel tip positions above each puck, it is necessary to center the puck beneath the tip. Once placed, click lower tip. This places the puck exactly where it needs to be. Once all pucks are placed properly, the camera takes an image of the pucks.

**Auto placement** - If you have previously performed calibration, you can have the spot cutter automatically place the pucks for you. Simply place the pucks on the cutting stage. Check the box labelled Auto place pucks. When you click Next, the spot cutter takes an image of the pucks on the stage, then automatically moves the pucks to their proper positions. When the pucks are positioned, the spot cutter creates the system position image.

### 10.1.b Calibrating the Plate and Wash Station Positions

The second part of the calibration wizard focuses on plate position and wash station position. Due to minor imperfections in the construction of the spot cutter, it is necessary to fine tune the position of the microtiter plate positions and the wash station.

#### Plate Position Calibration

To calibrate plate position, select a plate type from the list, place the plate in the right, rear plate position, and click Next. The spot cutter moves the tip to the location above the A1 well.

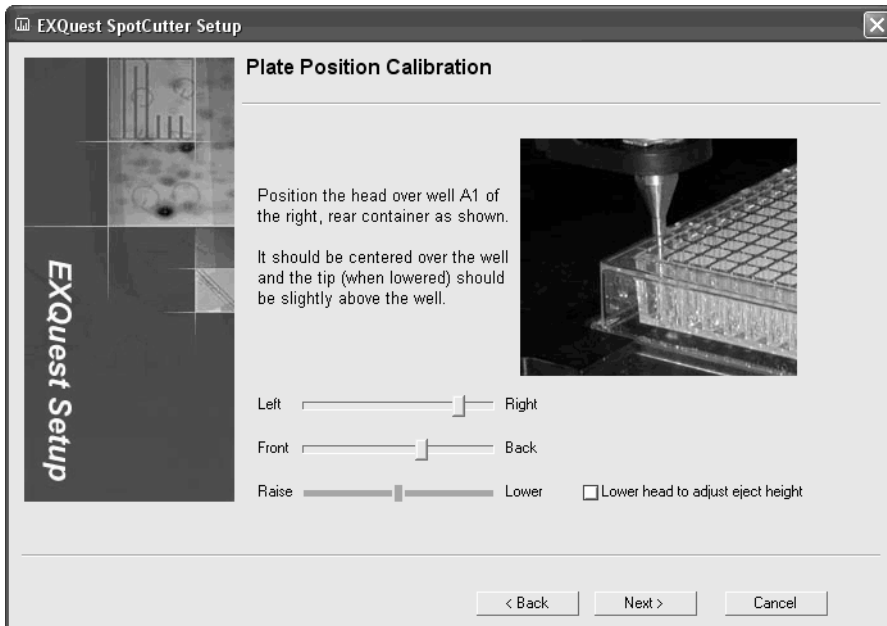


Fig. 10-3. Plate position for a 384 well plate

The sliders allow you to adjust the tip by moving it along X and Y coordinates. Click Lower to lower the tip into the well. This gives you a clear idea of where the tip is in relation to the center of the well.

The tip height slider adjusts the height of the tip above the well when ejecting material into the well. This should be slightly above the lip.

After calibrating one type, you have the option to return to the plate selection panel to calibrate another type. If you anticipate using only one type of plate, you can skip calibrating the other types. If at a later time you find you need to use another plate type, you can return to this step without performing system calibration.

### Wash Station Calibration

In the Wash Station Position step the tip moves to the location above the Wash station. If you are doing Membrane tip calibration, the tip moves to just above Wash Station #1.

The sliders allow you to adjust the tip by moving it along X and Y coordinates. Click Lower to lower the tip. This gives you a clear idea of where the tip is in relation to the center of the wash station.

The tip height slider adjusts the height of the tip in relation to the wash station. The tip should be low enough to effectively wash the tip.

### Calibration Summary

The Calibration summary describes all changes made during calibration. PDQuest generates a log that will be updated each time the spot cutter is calibrated. To view the log, click View Log.

**Note:** If this is the first time calibrating the spot cutter, the log is unavailable.

Click Finish to close the wizard. You are now ready to cut spots using the EXQuest Spot Cutter.

### 10.2 Manual Excision

Manual Excision is used for basic spot cutting operations. If you want to excise entire lanes or create volumes to excise, use Lane/Volume Excision.

Before using this tool, make sure the spot cutter is connected, is switched on, and has warmed up, then select Manual Excision from the Excision menu. The software will connect with the spot cutter platform and camera, and the Manual Excision dialog box will open.



Fig. 10-4. Manual Excision

In the event that Quantity One cannot locate the spot cutter, a warning dialog box displays. Check your connections and click Retry. You also have the option of running Manual Excision in simulation mode.

**Note:** See the spot cutter hardware manual for instructions on installing the platform, camera, and associated hardware and software.

The Manual Excision tool is designed to guide you through the spot cutting procedure. Click on the link for more information about each step.

- **Acquire Image.** In this step, you capture an image of your gel or membrane using the spot cutter camera.
- **Specify the Cut Run Options.** These options can be adjusted for each cut run.
- **Specify Cuts.** In this step, you identify the spots in the image that you want to cut.
- **Performing the Cut Run.** In this step, you perform the cut run. When the run is complete, you can acquire a confirmation image.

### 10.2.a Acquire the Image

Before acquiring an image, select the light option appropriate for your gel/membrane/stain type:

- White Trans (gels and PVDF membranes).
- UV EPI (fluorescent gels).

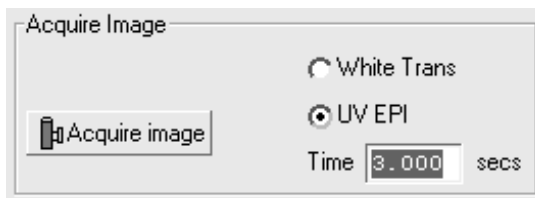


Fig. 10-5. Acquire Image.

If UV light is selected, specify an exposure time in seconds in the field. Also, make sure that the UV lamp switch on the front of the cutter is on.

**Note:** For Sypro Ruby stains, an exposure time of 20-30 seconds is recommended.

When you have selected a light option, click Acquire Image.

**Note:** For UV exposures, there will be a 15-second delay while the lamps heat up before the exposure begins.

The camera will take a single image of the gel or membrane on the platform, then a background image to adjust for image background. When the exposure is complete, the image will appear in the window.



To adjust the brightness and contrast of the image, open the Transform window by clicking on the button to the right of the image window.

### 10.2.b Specify Options

Under specify Options, you can make changes to various settings.

#### Cut

On the Cut tab you can choose to make multiple cut when possible, select the specific cutting tip size and the type of material you are cutting.

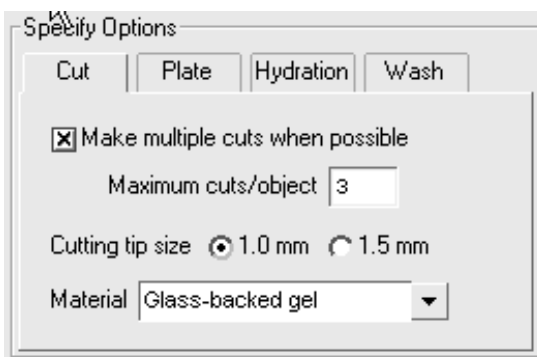


Fig. 10-6. Cut options.

If you want to make multiple cuts, check the box labelled Make multiple cuts when possible. Then enter the maximum number of cuts per object.

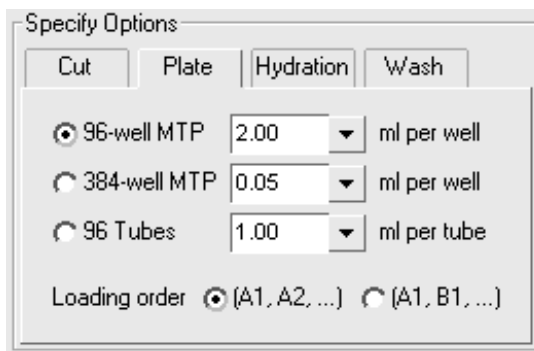
Next, select the cutting tip size you are using. Quantity One uses this to determine when you have reached the maximum capacity for wells on the plate.

Finally, select the material you are cutting from the drop down list.



## Plate

On the plate tab, select the plate type you are using for this cut run, then select the volume of each well. PDQuest uses this setting in conjunction with your cut settings to determine whether you will exceed the well capacity.



The image shows a software dialog box titled "Specify Options". It has four tabs: "Cut", "Plate", "Hydration", and "Wash". The "Plate" tab is selected. Inside the "Plate" tab, there are three radio button options for plate types: "96-well MTP" (selected), "384-well MTP", and "96 Tubes". Each option has a corresponding volume input field and unit: "2.00 ml per well" for 96-well MTP, "0.05 ml per well" for 384-well MTP, and "1.00 ml per tube" for 96 Tubes. At the bottom, there is a "Loading order" section with two radio button options: "(A1, A2, ...)" (selected) and "(A1, B1, ...)".

Fig. 10-7. Plate options.

Load order determines the order in which wells will be filled on the plate.

## Hydrate

To keep gels from drying out during long cut runs, you can have the EXQuest spot cutter automatically hydrate your gel(s) during the cut run. Check the hydrate box then enter the time, in minutes between hydration runs.

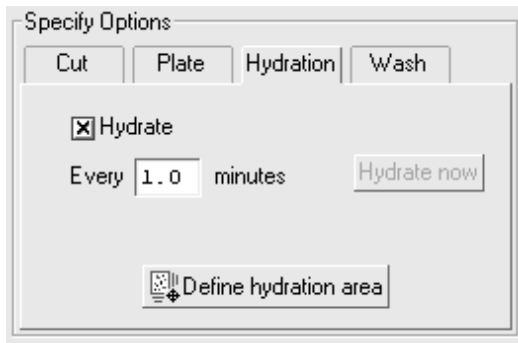


Fig. 10-8. Hydrate option.

To determine the area you want to hydrate, click Define hydration area, then click and drag on the image to create a box. To redraw the hydration area, click and drag again.

**Note:** You must define a hydration area for hydration to function.

### Wash

When using the EXQuest with the gel tip, EXQuest will flush the tip after each cut is made. Enter the wash volume to flush through the cutter tip.

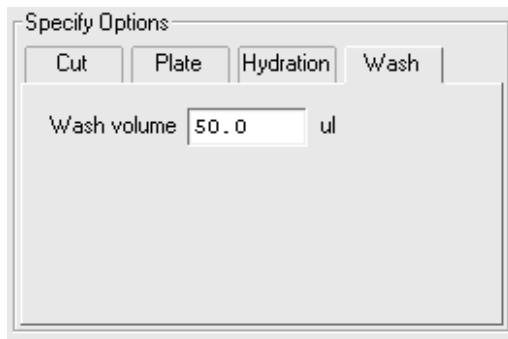


Fig. 10-9. Wash option.

If you are using the membrane tip, you cannot adjust the wash volume as the tip is washed in the three membrane wash stations.

## 10.2.c Specify Cuts

In Specify Cuts, click Add/move cut then click on spots in the image to identify which spots to cut. To make multiple cuts of large spots (if selected on the Cut tab of the Specify Options section), click and drag over the spot in the image.



Fig. 10-10. Specify cuts.

Use the Zoom Box tool on the right of the Basic Excision Tool to magnify each spot. When clicking on spots in the image, select the Snap to Peak checkbox to automatically center the cut circle on the spot peak (calculated from the spot intensity in the image).

To remove a cut, click Remove cut and click on a cut selection in the image.

## Run Info

The Run Info section contains information regarding the current cut run.

**Cuts** - indicates the total number of cuts in the run including multiple cuts.

**Wells** - indicates the number of wells to be used.

**Plates** - indicates the number of plates needed for the run.

**Min. well volume** - indicates the minimum well capacity to avoid overflow based on the number of multiple cuts selected and cutter tip size. If the well volume for your

selected plate is less than the Min. well volume, then the overflow warning flashes and you will not be able to start a cut run.

**Note:** Minimum well volume and the overflow warning only display if you are using the EXQuest Spot Cutter with a gel head.

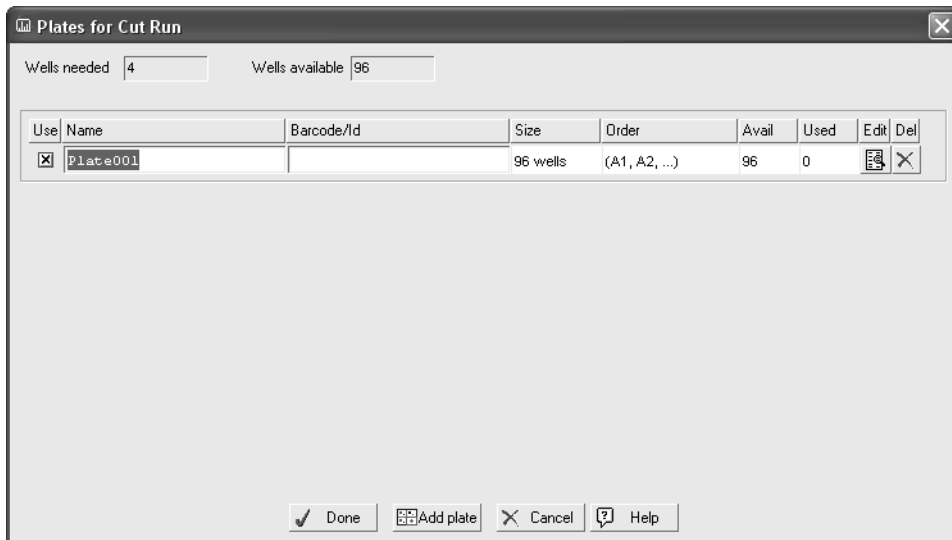
### 10.2.d Performing the Cut Run

After you have identified all the cuts you want to make and selected the appropriate options, you are ready to begin the cut run.

Click the Begin/Resume button.

#### Plate Selection

The Plate selection dialog box lists all the plates available for the cut run. The table shows the plate name, barcode/ID, plate size (96 well, 384 well, 96 tube), wells available on the plate, and used wells. If this is a new cut run, the number of plates listed is determined by the number of wells required for the cut run.



The dialog box titled "Plates for Cut Run" features a close button (X) in the top right corner. It contains two input fields: "Wells needed" with the value "4" and "Wells available" with the value "96". Below these is a table with the following columns: Use, Name, Barcode/Id, Size, Order, Avail, Used, Edit, and Del. The table contains one row with the following data: Use (checked checkbox), Name (Plate001), Barcode/Id (empty), Size (96 wells), Order ((A1, A2, ...)), Avail (96), Used (0), Edit (pencil icon), and Del (X icon). At the bottom of the dialog are four buttons: Done (with a checkmark icon), Add plate (with a plus icon), Cancel (with an X icon), and Help (with a question mark icon).

| Use                                 | Name     | Barcode/Id | Size     | Order         | Avail | Used | Edit | Del |
|-------------------------------------|----------|------------|----------|---------------|-------|------|------|-----|
| <input checked="" type="checkbox"/> | Plate001 |            | 96 wells | (A1, A2, ...) | 96    | 0    |      |     |

Fig. 10-11. Select plates.

In the Plate selection dialog box you can change the name of a plate and enter a barcode/ID directly in the table. The barcode helps to properly place the plates in the spot cutter.

To add a plate click Add plate. To remove a plate from the list, click Delete.

If the Use checkbox is cleared, the plate will be skipped in the cut run even if it has wells available.

To skip specific wells in a plate to reserve them for such things as standards, click Edit in the table. This opens a plate diagram dialog box.

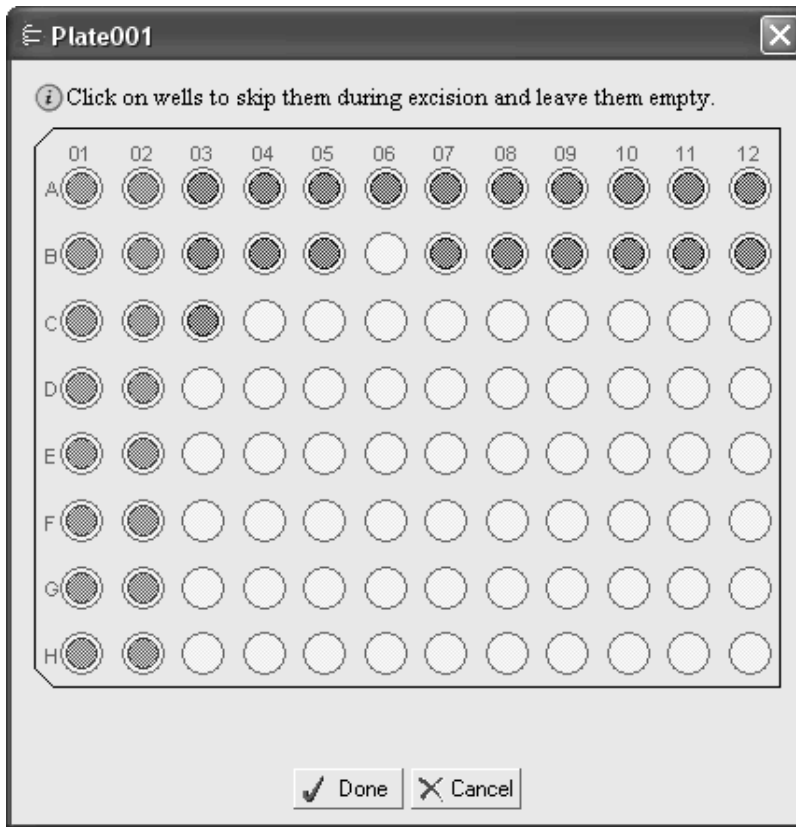


Fig. 10-12. Mark wells for skipping.

Click on a well to mark it for skipping in the cut run. Red wells are reserved wells, while blue wells are wells with material in them from a previous run.

**Note:** You cannot change the state of wells that have material in them.

When you have finished making your plate selections, click Done.

The plate wizard will direct you as to where to place the plate in the spot cutter.

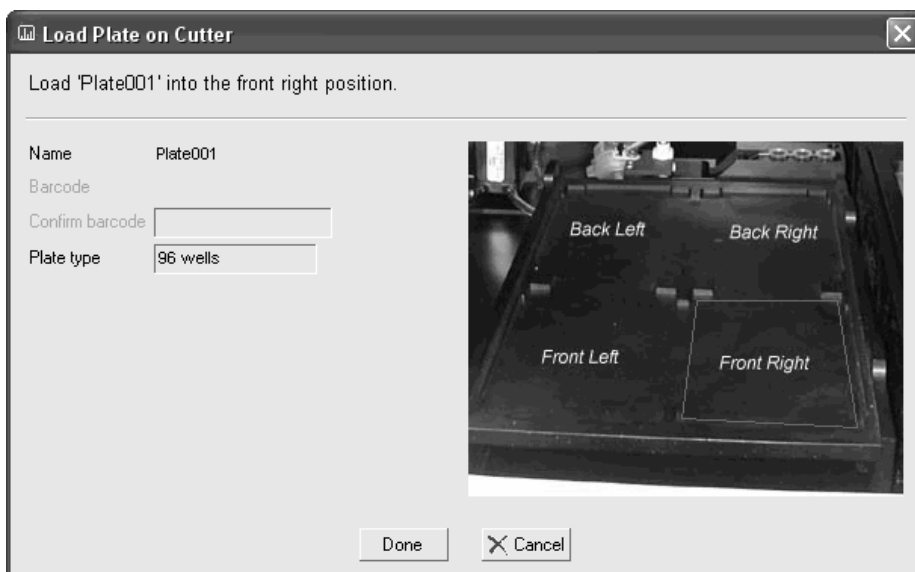


Fig. 10-13. Plate loading in the EXQuest Spot Cutter

As each cut is being made, it will be highlighted in yellow on the screen. After each cut has been made, the cut circle appears in white on the screen. Cuts that have not yet been made are circled in green.

## Run Info

The Run Info section contains information regarding the current cut run.

**Cuts** - indicates the total number of cuts in the run including multiple cuts.

**Wells** - indicates the number of wells to be used.

**Plates** - indicates the number of plates needed for the run.

**Min. well volume** - indicates the minimum well capacity to avoid overflow based on the number of multiple cuts selected and cutter tip size. If the well volume for your

## Quantity One User Guide

---

selected plate is less than the Min. well volume, then the overflow warning flashes and you will not be able to start a cut run.

**Note:** Minimum well volume only displays if you are using the EXQuest Spot Cutter with a gel head.

### Ending the Cut Run

Click Pause to pause the cut run. You will be prompted to confirm the pause. Note that the cutter will pause only after the most recent cut has been deposited in a well. Any remaining cut requests will remain selected in the image, and you can click Begin/Resume to complete the remainder of the cuts.

When all cuts are complete, the cutting tip will return to the home position.

### Confirming Cuts

When the cut run is completed, you have the option of taking a confirmation image. If you click Yes, PDQuest acquires a temporary image of the gel with the cut spots identified by plate and well number.

The specification for the EXQuest Spot Cutter spot pick-up is greater than 99% effective at picking up a spot that has been cut from a gel. However, there may be some spots that have not been picked up, and these can be re-cut. The efficiency of pick up on the second cut is again >99%, so the spot is reliably picked up the second time.

Use the zoom tools to confirm all spots were cut. If there are spots that need to be re-cut, click Add/move cut, and then right click the spot in the Manual Excision window.

When you have identified all the spots that need to be re-cut, click Begin/Resume.

## 10.2.e Other Options

### Prime Pump (Gel head only)

The spot cutter should be primed: when it is first installed; anytime the water bottle is filled, when the cutting head is changed; and for the first cut run of the day. Click



Prime pump until no large air bubbles are visible in the tubing. For an empty system this usually takes 3 prime cycles.

## Save

To save the image taken by the spot cutter, click the Save button in the Manual Excision Tool window. You will be prompted to enter a name and specify a location for the image before saving.

## Import

If you have a cut list saved to a text file, you can import it into the Manual Excision Tool. This is useful if you have a large number of cuts for similar gels.

**Note:** You can only import a cut list to an image that does not have any cuts added. However, after you import cuts, you can add additional cuts to the image.

Click Import to open the Import Excision Coordinate List dialog box.



Fig. 10-14. Importing a cut list

If the first line of your file is header information, select Skip first line of file. Next enter which column the coordinates are located in the file. For example, if your file

lists the spot number in column one, the X coordinates in column two and Y in three, then you would enter two for column X and three for column Y. Columns must be separated by tabs or by columns.

Click the browse button to locate the file you want to import.

Units indicates the measurements of the coordinates, and Origin determines starting point for the measurements.

The image in the Manual Excision Tool displays the placement of the spots as orange circles before you complete the import. If the cuts are out of position, you can click and drag all cuts at once to better position them. If you need to move cuts individually, you can do so after import by clicking Add cut, then clicking and dragging the cut you want to move.

When you are satisfied with the placement of the cuts, click Done to complete the import.

### Display Options

Select the Well check box (below the Manual Excision image window) to display the well number associated with each cut request.

Select the Position checkbox to display the position of the cut (in millimeters) on the gel.

## 10.3 Lane/Volume Excision

Lane/Volume Excision functions much like Manual excision, except that you define what to cut based on defined lanes or volumes.

Before using this tool, make sure the spot cutter is connected, is switched on, and has warmed up, then select Lane/Volume Excision from the Excision menu. The software will connect with the spot cutter platform and camera, and the Manual Excision dialog box will open.

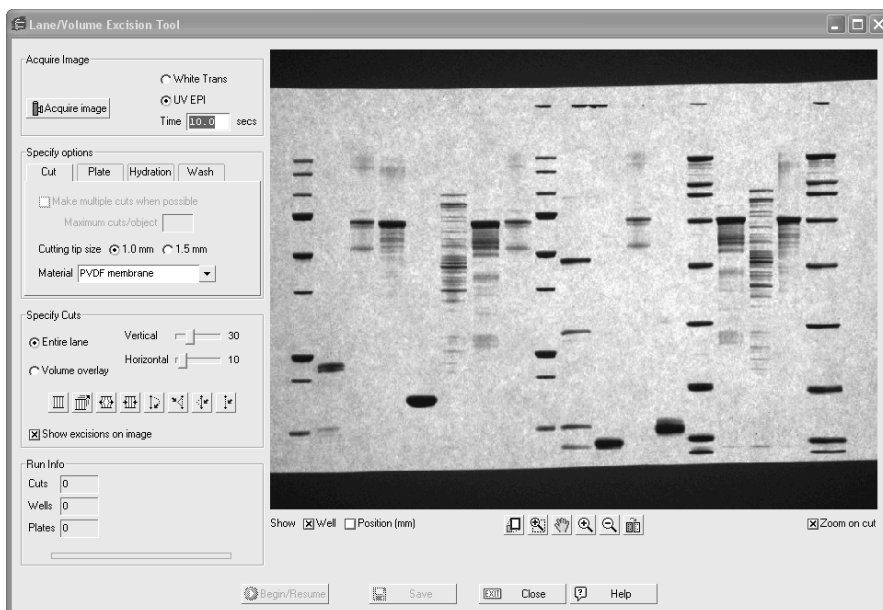


Fig. 10-15. Lan/Volume Excision

In the event that Quantity One cannot locate the spot cutter, a warning dialog box displays. Check your connections and click Retry. You also have the option of running Lane/Volume Excision in simulation mode.

**Note:** See the spot cutter hardware manual for instructions on installing the platform, camera, and associated hardware and software.

The Manual Excision tool is designed to guide you through the spot cutting procedure. Click on the link for more information about each step.

- **Acquire Image.** In this step, you capture an image of your gel or membrane using the spot cutter camera.
- **Specify the Cut Run Options.** These options can be adjusted for each cut run.
- **Specify Cuts.** Select whether to cut lanes or volumes.
- **Performing the Cut Run.** In this step, you perform the cut run. When the run is complete, you can acquire a confirmation image.

### 10.3.a Acquire the Image

Before acquiring an image, select the light option appropriate for your gel/membrane/stain type:

- White Trans (gels and PVDF membranes).
- UV EPI (fluorescent gels).

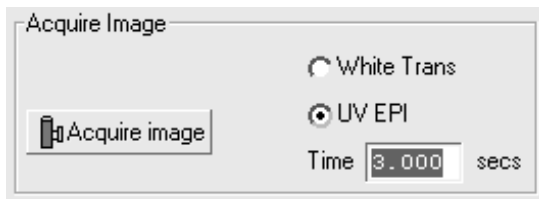


Fig. 10-16. Acquire Image.

If UV light is selected, specify an exposure time in seconds in the field. Also, make sure that the UV lamp switch on the front of the cutter is on.

**Note:** For Sypro Ruby stains, an exposure time of 8-10 seconds is recommended.

When you have selected a light option, click Acquire Image.

**Note:** For UV exposures, there will be a short delay while the lamps heat up before the exposure begins.

The camera will take a single image of the gel or membrane on the platform, then a background image to adjust for image background. When the exposure is complete, the image will appear in the window.



To adjust the brightness and contrast of the image, open the Transform window by clicking on the button to the right of the image window.

### 10.3.b Specify Options

Under specify Options, you can make changes to various settings.

## Cut

On the Cut tab you can choose to make multiple cut when possible, select the specific cutting tip size and the type of material you are cutting.

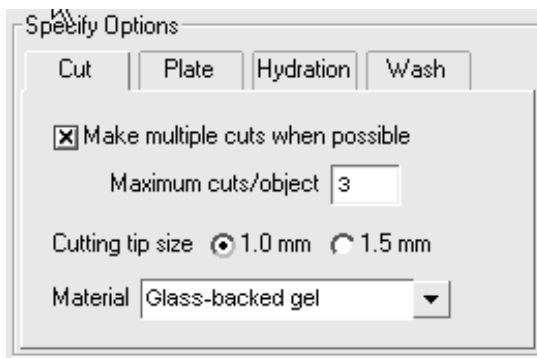


Fig. 10-17. Cut options.

If you want to make multiple cuts, check the box labelled Make multiple cuts when possible. Then enter the maximum number of cuts per object.

**Note:** Make multiple cuts when possible is inactive when using Lane Excision.

Next, select the cutting tip size you are using. Quantity One uses this to determine when you have reached the maximum capacity for wells on the plate.

Finally, select the material you are cutting from the drop down list.

## Plate

On the plate tab, select the plate type you are using for this cut run, then select the volume of each well. Quantity One uses this setting in conjunction with your cut settings to determine whether you will exceed the well capacity.

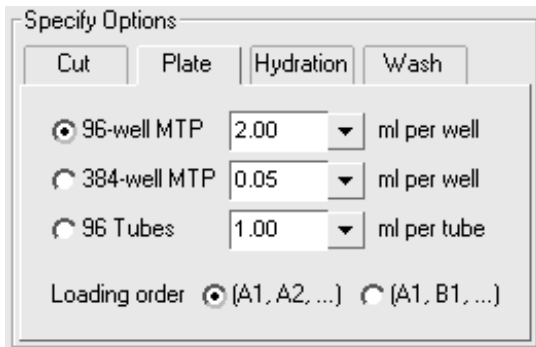


Fig. 10-18. Plate options.

Load order determines the order in which wells will be filled on the plate.

### Hydrate

To keep gels from drying out during long cut runs, you can have the EXQuest spot cutter automatically hydrate your gel(s) during the cut run. Check the hydrate box then enter the time, in minutes between hydration runs.

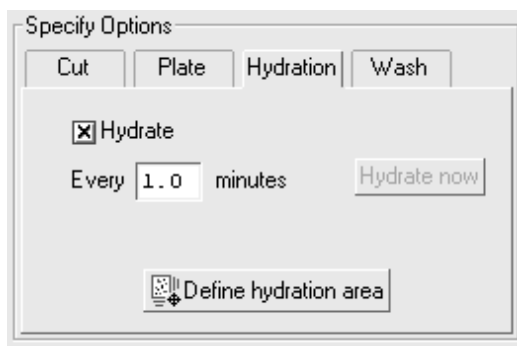


Fig. 10-19. Hydrate option.

To determine which area you want to hydrate, click Define hydration area, then click and drag on the image to create a box. To redraw the hydration area, click and drag again. If you choose not to define a hydration area, then the EXQuest Spot Cutter will hydrate the entire cutter stage.

### Wash

Enter the wash volume to flush through the cutter tip.

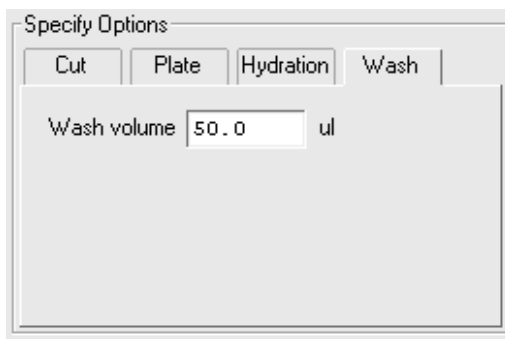


Fig. 10-20. Wash option.

### 10.3.c Specify Cuts

Under Specify Cuts, determine whether to use Lane Excision or Volume Excision.

#### Lane Excision

If you select Lane Excision, use the Lane tools to define lanes in the image. See Section 4.1, Defining Lanes, for information using the Lane tools.

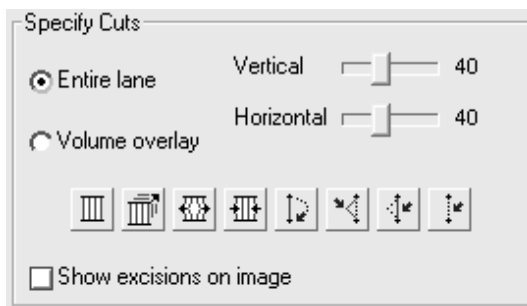
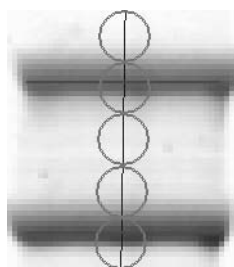


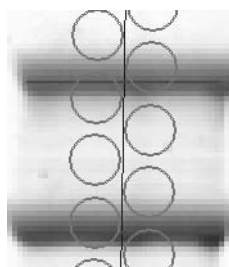
Fig. 10-21. Specify Cuts using Lane excision

Because of the circular shape of the cutting tip, it is necessary to overlap the cuts to some degree to reliably obtain a sample of everything in the lane.

Use the vertical slider to adjust the percent of overlap. Zero will give a single column of cuts and 100 will stagger the spots at approximately 45 degrees. Use the horizontal slider to stagger the cuts so that a complete gel piece is obtained with each cut. The lower the number, the closer the cuts will be. At zero, the cut edges overlap.



Vertical = 100  
Horizontal = 0



Vertical = 20  
Horizontal = 50

Fig. 10-22. Use the sliders to adjust cut overlap.

Note that when using lane excision, each individual cut will be placed in a separate well.



## Volume Excision

If you want to excise individual bands from a gel or blot, select Volume excision under Specify cuts.

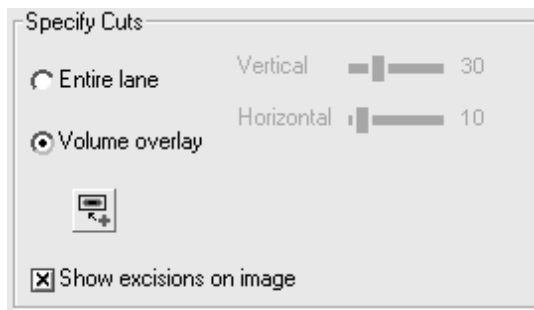


Fig. 10-23. Specify cuts by volume

To create a volume to excise, select the Volume rectangle tool then click and drag on the image to make the volume. If Make multiple cuts when possible is selected on the Cut options tab, the volume will contain up to the maximum number of cuts indicated.

**Note:** If you select multiple cuts for volumes, all cuts for each volume will be placed in the same well.

## Run Info

The Run Info section contains information regarding the current cut run.

**Cuts** - indicates the total number of cuts in the run including multiple cuts.

**Wells** - indicates the number of wells to be used.

**Plates** - indicates the number of plates needed for the run.

**Min. well volume** - indicates the minimum well capacity to avoid overflow based on the number of multiple cuts selected and cutter tip size. If the well volume for your selected plate is less than the Min. well volume, then the overflow warning flashes and you will not be able to start a cut run.

### 10.3.d Performing the Cut Run

After you have identified all the cuts you want to make and selected the appropriate options, you are ready to begin the cut run. Close the door of the spot cutter before proceeding.

Click the Begin/Resume button.

#### Plate Selection

The Plate selection dialog box lists all the plates available for the cut run. The table shows the plate name, barcode/ID, plate size (96 well, 384 well, 96 tube), wells available on the plate, and used wells. If this is a new cut run, the number of plates listed is determined by the number of wells required for the cut run.

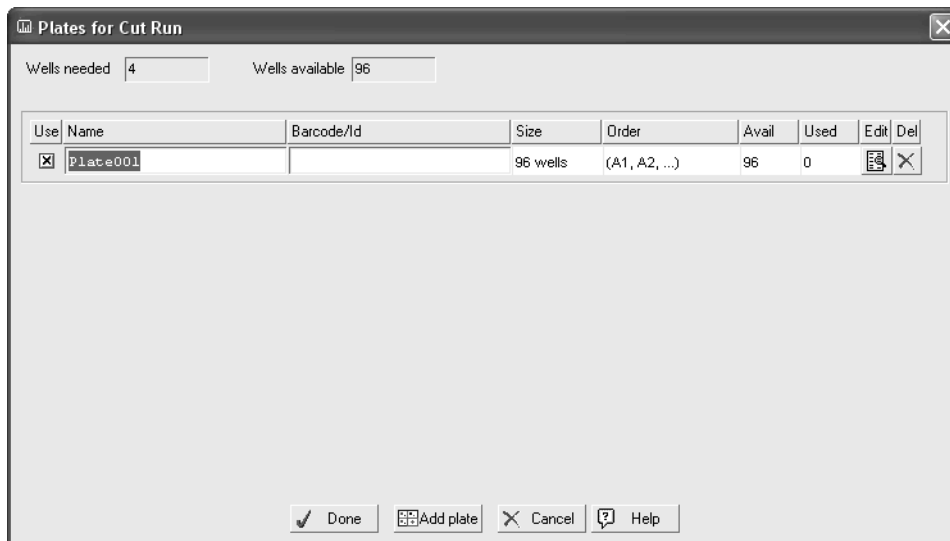


Fig. 10-24. Select plates.

In the Plate selection dialog box you can change the name of a plate and enter a barcode/ID directly in the table. The barcode helps to properly place the plates in the spot cutter.

To add a plate click Add a plate. To remove a plate from the list, click Delete.

If the Use checkbox is cleared, the plate will be skipped in the cut run even if it has wells available.

To skip specific wells in a plate to reserve them for such things as standards, click the icon under Edit in the table. This opens a dialog box displaying the plate.

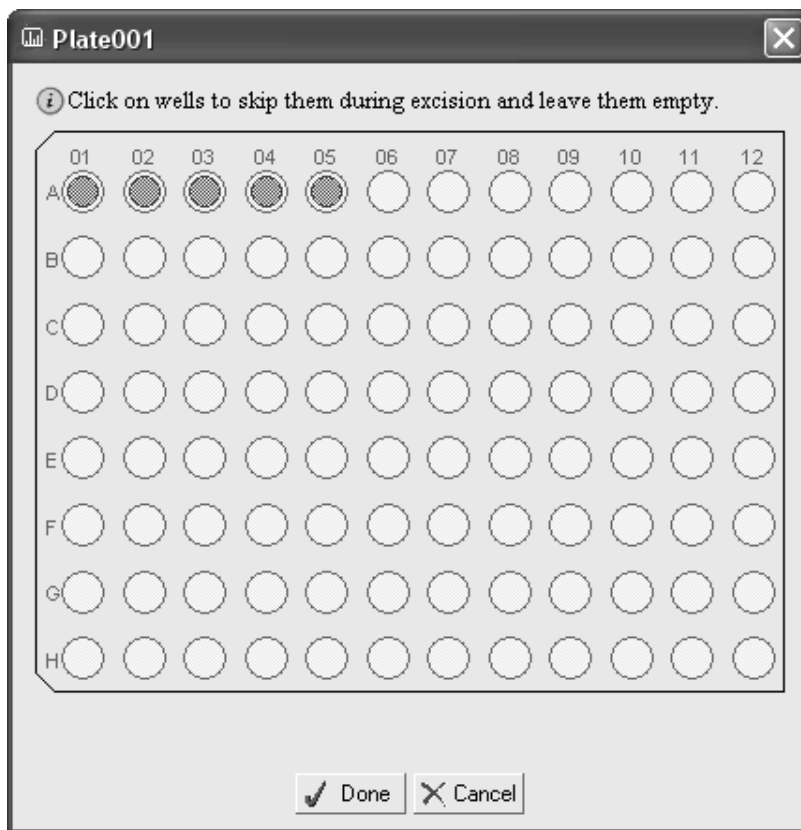


Fig. 10-25. Mark wells for skipping.

Click on a well to mark it for skipping in the cut run. If you are continuing an unfinished cut run, some of the wells may already be marked as reserved.

## Quantity One User Guide

**Note:** Reserved wells from unfinished cut runs may have material in them. Be sure there is no material in the well before you mark it as unreserved.

When you have finished making your plate selections, click Done.

The plate wizard will direct you as to where to place the plate in the spot cutter.

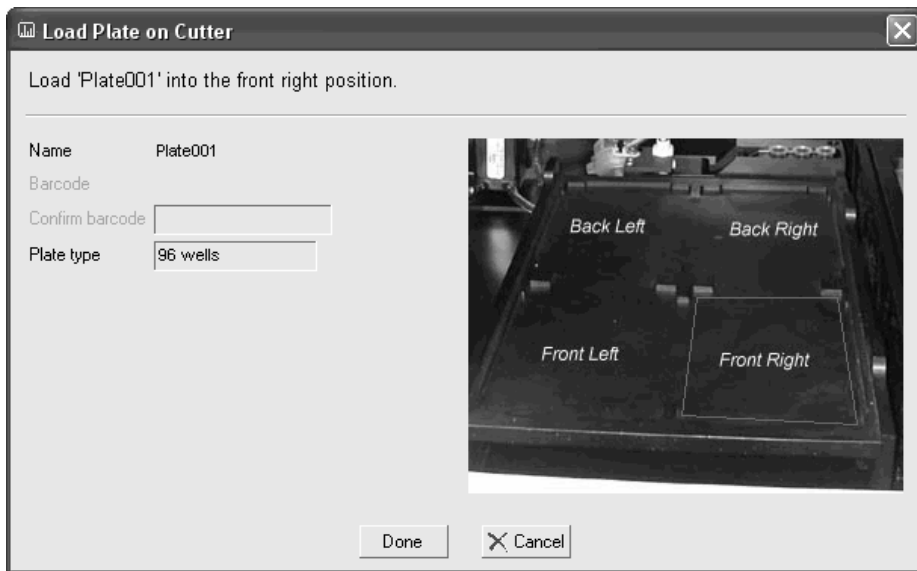


Fig. 10-26. Plate loading in the EXQuest Spot Cutter

As each cut is being made, it will be highlighted in yellow on the screen. After each cut has been made, the cut circle appears in white on the screen. Cuts that have not yet been made are circled in green.

### 10.3.e Other Options

#### Prime Pump

The spot cutter should be primed: when it is first installed; anytime the water bottle is filled, when the cutting head is changed; and for the first cut run of the day. Click

Prime pump until no large air bubbles are visible in the tubing. For an empty system this usually takes 3 prime cycles.

### Save

To save the image taken by the spot cutter, click the Save button in the Manual Excision Tool window. You will be prompted to enter a name and specify a location for the image before saving.

### Display Options

Select the Well Number check box (below the excision image window) to display the well number associated with each cut request.

Select the Cut Position checkbox to display the position of the cut (in millimeters) on the gel.



# 11. Differential Display and VNTRs

Quantity One has tools for studying gene expression using differential display analysis and counting VNTRs or other repeated elements in gels.

These functions are located on the *Analysis* menu.

## 11.1 Differential Display

Differential display is a popular technique using mRNA and PCR amplification to identify genes that are differentially expressed between and among cell types.<sup>1</sup> The mRNA in a cell is amplified using RT-PCR and resolved on a DNA sequencing gel, where it can be compared with mRNA from related cell types to determine differential gene expression.

The **Differential Display** tool in Quantity One facilitates the side-by-side comparison of the bands in a sequencing gel by highlighting the up or down regulation of bands across lanes.

**Note:** You must match the bands in the gel image before using this function. See Chapter 6 for more information.

### 11.1.a Normalization

Before using the **Differential Display** function, you must normalize the bands in the gel for quantity across all the lanes. With the gel open and matched, select **Normalize** from the *Match* menu and click on a distinct, well-resolved band that appears in every lane of the gel. The quantity of that band in each lane will be set to 100, and the quantities of the other bands in the lane will be normalized to that band. (If the band is not present in a lane, the normalized quantity for that lane is zero.)

---

1. Liang, P., and Pardee, A. (1992) *Science* 257: 967-970.

See section 6.2.d, Normalizing for Quantity, for more information on normalization.

### 11.1.b Differential Display Searches

After you have normalized the gel, select **Differential Display** from the *Analysis* menu. A dialog will open in which you can specify certain parameters.

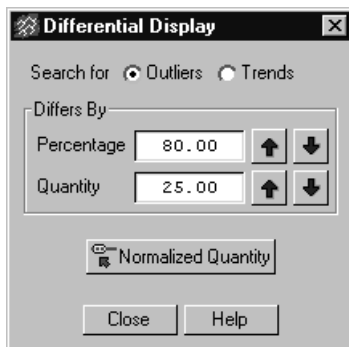


Fig. 11-1. The Differential Display dialog box.

You can perform two types of searches on bands across a differential display gel: an **Outliers** search or a **Trends** search. Both are determined by the **Percentage** and **Quantity** values that you specify in the **Differs By** field.

### Outliers

Select the **Outliers** button to highlight bands with a normalized quantity that differs from the mean normalized quantity for that band type by the specified percentage and/or the specified quantity. An outlier is defined as a band with normalized quantity ( $n_q$ ) that satisfies one of the following criteria:

1.  $n_q > \text{ratio} \times \text{mean}$  and  
 $n_q > \text{mean} + \text{quantity}$
2.  $n_q < (1/\text{ratio}) \times \text{mean}$  and  
 $n_q < \text{mean} - \text{difference}$

Where:  $\text{ratio} = 1.0 + (\text{percentage}/100)$



### Trends

Select the **Trend** button to search for increasing or decreasing levels of gene expression across samples, represented by trends in normalized quantity across lanes. With **Trend** selected, a linear regression of ( $n_q$ ) versus lane number is computed for each band type. The leftmost and rightmost lanes containing that band type are determined. The normalized quantities for these lanes are calculated from the regression model. If:

$$\begin{aligned} \text{abs}(\text{leftmost\_}n_q - \text{rightmost\_}n_q) &\geq \text{difference and} \\ \text{Max}(\text{leftmost\_}n_q, \text{rightmost\_}n_q) / \text{Min}(\text{leftmost\_}n_q, \text{rightmost\_}n_q) &\geq \text{ratio} \end{aligned}$$

then the band type is flagged as a trend.

### Displaying Results

Bands identified as outliers or belonging to a trend are highlighted with a **white line** in the image.

If a lane is missing a band assigned to a band type that is flagged as an outlier or a trend, the expected location of the band will be highlighted by a **white box**.

Use the **Normalized Quantity** button in the *Differential Display* dialog to display a histogram of the normalized quantities and the mean quantity for a band type. Click on the button, then click on the band. A graph will display the normalized quantities of that band across the entire gel.

## 11.2 Variable Number Tandem Repeats

If your experiments involve the use of microsatellites, VNTRs, or other repeated elements, you can calculate the number of times a repeated element occurs in a band.

**Note:** You must define the **Standards** in the gel before you can use this function (see section 6.1, Standards).

To calculate the VNTRs in the bands in a gel, select **VNTR Calculations** from the *Analysis* menu. (You can also select the **VNTR Quick Guide** from the *Help* menu for guidance on analyzing gels with VNTRs.)

## Quantity One User Guide

---

The *Tandem Repeat Calculations* dialog box will open.

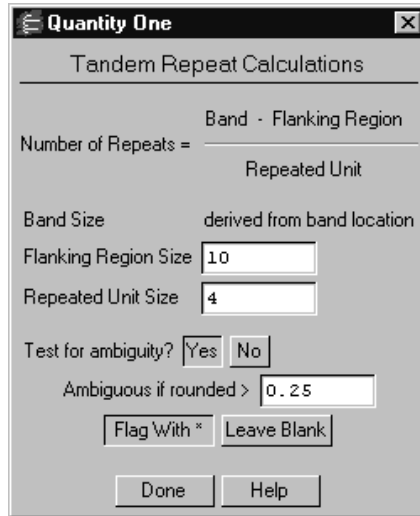


Fig. 11-2. Tandem Repeat Calculations dialog box.

At the top of the dialog box is the equation used to calculate the number of times an element is repeated in a band.

The band size is determined by the position of the band on the gel image.

In the **Flanking Region Size** field, enter the size (in base pairs or other repeated units) of the part of the fragment that does not include any repeated elements. This would include primer length and the length of any sequences that fall between the end of the primer region and the beginning of the stretch of repeats.

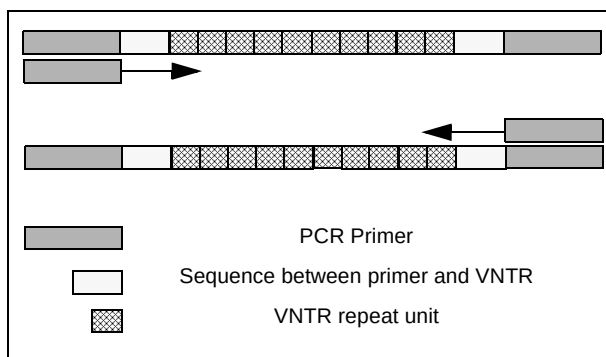


Fig. 11-3. Diagram of components of a DNA fragment.

In the **Repeated Unit Size** field, enter the size in base pairs of the repeated element (e.g., if you are working with (CA) repeats, enter “2”).

### Testing for Ambiguity

The number of repeats calculated by the software may not be a whole number, due to the limitations of exact band size determination. Since the number of times an element is repeated must be a whole number, the calculated value is rounded to the nearest whole number.

In the dialog box, you have the option to “flag” those numbers

Select **Yes** after **Test for ambiguity?** to flag values that deviate from the nearest whole number by a certain fraction. These values may warrant further review. In the **Ambiguous if rounded >** field, specify how much the calculated value must deviate from the nearest whole number for it to be flagged.

Finally, specify how to identify the ambiguous values using the **Flag with \*** button or the **Leave Blank** button.

When you have entered all the information in the dialog box, click on the **Done** button. The number of tandem repeats will be displayed next to the bands. (Note that if no standards have been defined on the gel, no numbers will be displayed.)

If the numbers have been concealed (e.g., by the **Hide Overlays** command), you can redisplay them by selecting **VNTR Display**.



# 12.Reports

Quantity One can display and print a variety of analysis reports. You can format the reports to include different kinds of data.

The available reports are listed under the *Reports* menu.



Fig. 12-1. Reports Menu.

## 12.1 Report Window

Many of the reports share the same basic report window.

Click here to close the report window

The screenshot shows a window titled "Proteins - Report". Inside, it displays a "Detail Report by Lane" for "Proteins" dated "November 28, 2001". The report is for "Lane 3", "Name: cerebellum", and "Band Set: Band Set 3". It contains a table with 8 columns: Band #, Band Type #, Relative Front, Mol. Wt. KDa, Peak OD, Average OD, Trace OD x mm, and Relative Qty. The table lists 6 bands. Below the table, it states "Molecular Weight Calculation Method: Point to Point" and "= Known x Extrapolated". The bottom right corner of the report area says "Screen Page 17 of 136". At the bottom of the window is a toolbar with icons for printing, scrolling, and exporting. Arrows point from labels below to these icons: "Print One Page" to the first print icon, "Print All Pages" to the second print icon, "Controls for scrolling through screen pages" to the navigation icons, "Export" to the spreadsheet icon, and "Reformat" to the document icon. An arrow from the text "Click here to close the report window" points to the close button (X) in the window's title bar.

| Band # | Band Type # | Relative Front | Mol. Wt. KDa | Peak OD | Average OD | Trace OD x mm | Relative Qty |
|--------|-------------|----------------|--------------|---------|------------|---------------|--------------|
| 3 - 1  | 1           | 0.005          | 291.70 x     | 0.53    | 0.47       | 0.395         | 1.0          |
| 3 - 2  | 2           | 0.014          | 279.88       | 0.48    | 0.26       | 0.476         | 1.2          |
| 3 - 3  | 5           | 0.058          | 227.52       | 0.78    | 0.42       | 1.060         | 2.7          |
| 3 - 4  | 6           | 0.079          | 205.92       | 0.58    | 0.33       | 0.832         | 2.1          |
| 3 - 5  | 7           | 0.103          | 183.11 =     | 0.14    | 0.11       | 0.146         | 0.4          |
| 3 - 6  | 8           | 0.126          | 163.79       | 0.33    | 0.20       | 0.412         | 1.1          |

Molecular Weight Calculation Method: Point to Point  
= Known x Extrapolated

Screen Page 17 of 136

Print One Page   Print All Pages   Controls for scrolling through screen pages   Export   Reformat

Fig. 12-2. Example of a report window.

The standard report window has buttons for printing the report, scrolling through the screen pages of the report, and exporting the report to a spreadsheet application. Some report windows also have a **Reformat** button for changing the data display.

To close a report window, click on the **Close** box in the upper right corner of the window.

## Scrolling

If a report has multiple pages, the scroll buttons in the report window become active. Use them to scroll to the next page, previous page, first page, or last page of the report. You can also enter a specific page number in the field to skip to that page.

## Printing

Print the report using the **Print One Page** or **Print All Pages** commands in the report window.

Click on either of these buttons to open a smaller version of the standard print dialog (described in the next chapter).

Windows version:



Macintosh version:



Fig. 12-3. Print Report dialog.

**Print One Page** prints only the current screen page. **Print All Pages** prints all the pages in the report.

**Note:** If you select **Print All Pages**, the number of pages printed may be less than the number of screen pages listed in the report window. This is because the print command reformats the data to make maximum use of the paper size.

### Exporting Report Data

You can export the report data to a spreadsheet application for further computation and analysis.

Click on the **Export** button to open the *Export Report* dialog.



Fig. 12-4. Export Report dialog.

Exported data can be separated by commas or tabs, depending on the requirements of your spreadsheet application.

You can save the data to a text file or to the clipboard by selecting the appropriate option button. If you select **File**, when you click on **OK** a dialog will open in which you can save the file.

### Reformatting Report Data

Click on the **Reformat** button to open the options dialog for the report and select different display options. This button is not available in all reports.



## 12.2 Lane and Match Reports

There are four different lane and match reports: **Lane Report**, **All Lanes Report**, **Match Report**, and **All Matches Report**.

**Lane Report** generates a report on any you lane you select. First select the report from the *Reports* menu, then click on the particular lane.

**All Lanes Report** generates a report on all the lanes in the current gel image.

**Match Report** generates a report on a matched band that you select. First select the report from the menu, then click on a matched band. This report requires band matching (see section 6.2, Band Matching).

**All Matches Report** generates a report on all the matched bands in the gel . This report requires band matching (see section 6.2, Band Matching).

When you select any of these reports, the *Report Options* dialog will open, where you can specify the type and formatting of the data.

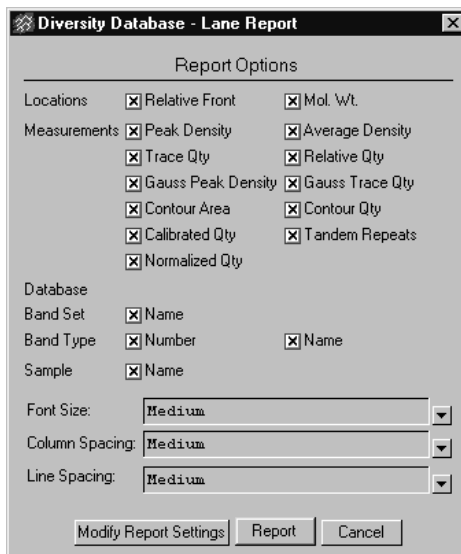


Fig. 12-5. Report Options dialog.

Select the data to display using the checkboxes.

Select the **Font Size**, **Column Spacing**, and **Line Spacing** settings to be used in the report by clicking on the button next to each field and selecting from the list of options.

To save the report options, click on the **Modify Report Settings** button and enter a name for the report settings in the field.

To load or delete previously saved settings, click on the button next to the **Settings to Load or Delete** field and select from the list of saved settings. Then click on **Load** or **Delete**.

To display the report, click on the **Report** button. The report will be displayed in a standard report window (see section 12.1, Report Window).

## 12.3 Band Types Report

The **Band Types Report** shows the presence or absence of specific bands in specific lanes of a gel. The report window displays a schematic representation of all the bands and lanes in the gel, with the presence or absence of bands shown as specified in the options dialog.

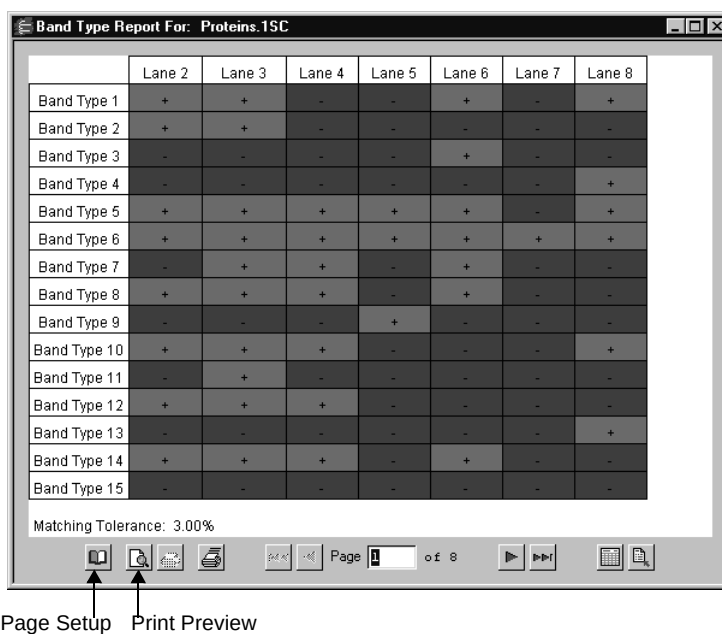


Fig. 12-6. Band Type Report.

**Note:** This report requires band matching (see section 6.2, Band Matching).

Select the report from the *Reports* menu, and select the display options in the *Band Types Report Options* dialog.

To save the report options, click on the **Settings** button in the options dialog and enter a name for the settings in the field. To load or delete previously saved settings, click on the button next to the **Settings to Load or Delete** field and select from the list of saved settings. Then click on **Load** or **Delete**.

To display the report, click on the **Report** button.

This report window includes two additional options: **Page Setup** and **Print Preview**.

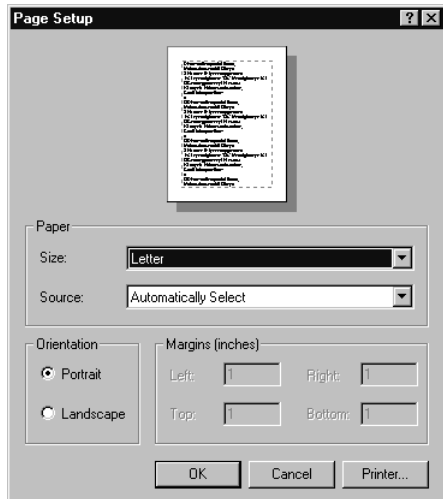


Fig. 12-7. Page Setup dialog.

The **Page Setup** dialog includes controls for selecting the paper size and orientation, paper tray, page margins, and printer. The **Print Preview** dialog displays the report as it will be printed.

## 12.4 1-D Analysis Report

The **1-D Analysis Report** displays all the advanced analysis data (including band types, normalized quantities, amount of sample loaded, etc.) for all the lanes in the gel. The lanes will also be ranked in similarity to the lane you initially select to generate the report.

**Note:** This report requires band matching (see section 6.2, Band Matching).

Select **1-D Analysis Report** from the *Reports* menu, then click on any experimental lane in the gel. A dialog will pop up, in which you can select the report data to display.

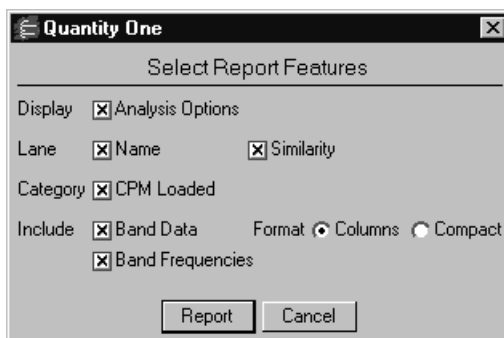


Fig. 12-8. 1-D Analysis Report options.

To display the selected data, click on the **Report** button. The report will be displayed in a standard report window (see section 12.1, Report Window).

## 12.5 Similarity Comparison Reports

Quantity One has three reports for comparing the similarity of lane-based samples in a gel: **Compare Lane Images**, **Phylogenetic Tree**, and **Similarity Matrix**.

**Note:** These reports require band matching (see section 6.2, Band Matching).

### Comparison Options

Before opening any of these reports, select **Comparison Options** from the *Reports* menu to specify some similarity settings.

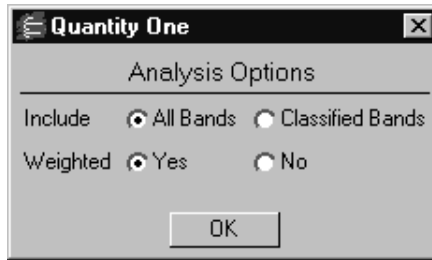


Fig. 12-9. Comparison Options dialog.

In the *Comparison Options* dialog, select **All Bands** to include every band in the sample similarity comparison, or selected **Classified Bands** to include only matched bands.

Next to **Weighted**, select **Yes** to use both band position *and* intensity when comparing sample similarity. Select **No** to use only band position when comparing samples. The next section describes how weighting is used in the similarity comparison equation.

### Comparison Method

The method for computing similarity in Quantity One is the Dice Coefficient. The formula for the Dice Coefficient is:

$$sim = 200 \times \frac{\sum_{i=1}^B \min(s_i, t_i)}{\sum_{i=1}^B (s_i + t_i)}$$
$$dist = 100 - sim$$

where S and T are vectors representing two lanes in the same band set that are being compared.

To compute similarity, a vector is constructed that represents the bands identified in the lane. The vector depends on the comparison options (see above) selected. If the

search was done on classified (matched) bands only, then the vector  $S$  contains  $B$  elements ( $S = (s_1, s_2, s_3 \dots s_B)$ ), where  $B$  is the number of band types in the lane's band set. The values for  $s_1, s_2, s_3 \dots s_B$  have the following values:

### Weighting Off Search

$s_i = 1$  if the  $i$ 'th band type is found in the lane.

$s_i = 0$  if it is not found.

### Weighting On Search

If the band set has a normalizing band type, then:

$s_i =$  The normalized density of the band assigned to the  $i$ 'th band type.

$s_i = 0$  if the lane does not have a band assigned to the  $i$ 'th type.

Otherwise:

$s_i =$  The quantity of the band assigned to the  $i$ 'th band type.

$s_i = 0$  if the lane does not have a band assigned to the  $i$ 'th type.

## 12.5.a Compare Lane Images

The **Compare Lane Images** report displays the lanes in decreasing order of similarity to a lane that you select.

Select **Compare Lane Images** from the *Reports* menu, then click on a reference lane. Select the report features in the pop-up box.

:

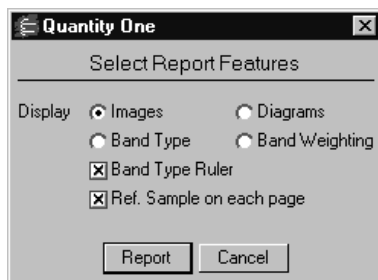


Fig. 12-10. Compare Lane Images options.

If you select the **Images** option, actual images of the lanes will be displayed in the report. If you select **Diagrams**, schematic representations of the lanes will be displayed.

To display the selected data, click on the **Report** button.



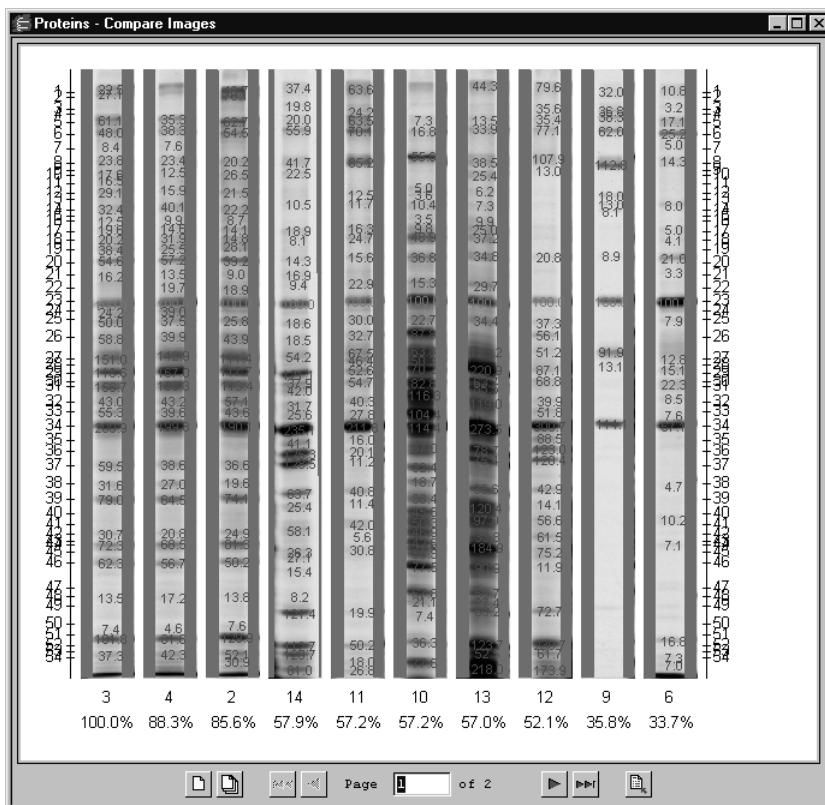


Fig. 12-11. Compare Lane Images report.

The *Compare Lane Images* report is displayed in a standard report window (see section 12.1, Report Window). Change the report options by clicking on the **Reformat** button.

The **Print Report** dialog for this report contains special fields for entering a report title.

### 12.5.b Phylogenetic Tree

Phylogenetic trees are schematic representations of sample similarity. To compare the similarity of samples in a phylogenetic tree format, select **Phylogenetic Tree** from the *Reports* menu. (You can also select **Phylogenetic Tree Quick Guide** under the *Help* menu to guide you through the process of generating a phylogenetic tree.)

In the pop-up box, select a clustering method for creating the tree. See the following pages for information about the different methods.

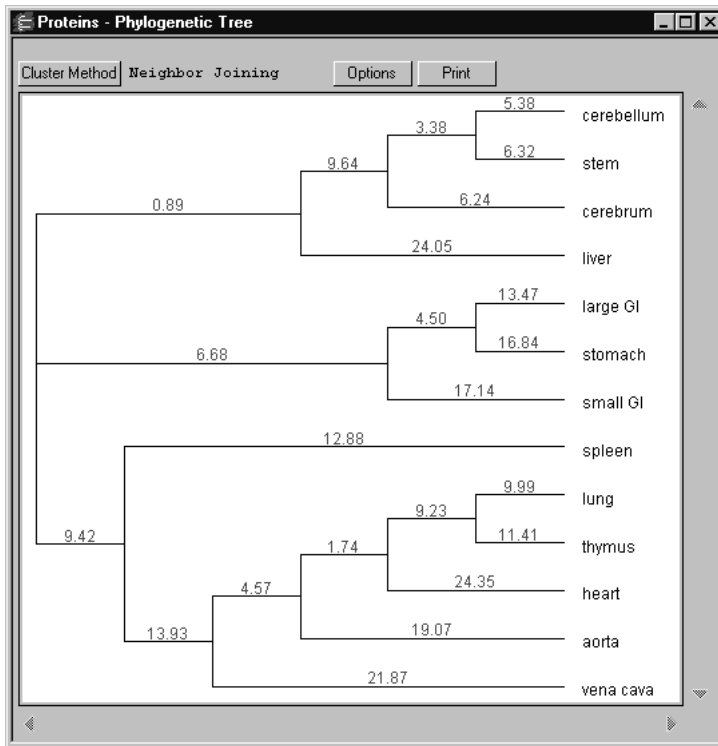


Fig. 12-12. Phylogenetic Tree.

**Note:** We have found that Ward's method, UPGAMA, and WPGAMA give the most plausible clusters and are affected the least by samples that are outliers.

The report window for a phylogenetic tree is slightly different than the standard report window. The tree appears in a window with scroll bars for moving up and down and left and right.

To print the tree, click on the **Print** button.

Click on the **Cluster Method** button to select a different clustering method for displaying the tree. This will automatically recalculate and redisplay the tree.

### Neighbor Joining

This type of phylogenetic tree is computed is based on minimizing the total branch length at each stage of clustering. The method also finds branch lengths between nodes. The approximate distance between any two samples in this tree can be found by adding the branch lengths that connect the samples.

### Other Methods<sup>1, 2</sup>

The following methods are based on the algorithm below:

1. Begin with n clusters—one cluster for each sample.
2. Compute the similarity matrix for the samples.
3. Convert the similarity matrix into a distance matrix d using the appropriate distance formula.
4. Join the two clusters with the minimum distance into one cluster. Compute the similarity value for this cluster.
5. Recompute the distance matrix d using the cluster that was formed in Step 4.

Steps 4 and 5 are repeated until there is only one cluster. The difference between the methods below is based on the definition of minimum distance in Step 4, and on the

- 
1. For a complete explanation of the calculations and assumptions used to generate these dendrograms, please refer to Sneath and Sokal. *Numerical Taxonomy*, San Francisco: W. H. Freeman & Company, 1973.
  2. Vogt and Nagel, *Clinical Chemistry* **38** (2): 182-198, (1992).

## Quantity One User Guide

---

method of computing the new distance matrix in step 5. In the discussion that follows, let:

$p, q$  be indices indicating two clusters that are to be joined into a single cluster.

$k$  be the index of the cluster formed by joining clusters  $p$  and  $q$ .

$i$  be the index of any remaining clusters other than cluster  $p, q$ , or  $k$ .

$n_p$  the number of samples in the  $p$ 'th cluster.

$n_q$  the number of clusters in the  $q$ 'th cluster.

$n$  the number of clusters in the  $k$ 'th cluster formed by joining the  $p$ 'th and  $q$ 'th cluster.

$$n = n_p + n_q$$

$d_{pq}$  the distance between cluster  $p$  and cluster  $q$ .

### *Single Linkage*

This is also called Nearest Neighbor or Minimum Method.

$$d_{ki} = \min(d_{pi}, d_{qi})$$

### *Complete Linkage*

This is also called the Furthest Neighbor or Maximum Method.

$$d_{ki} = \max(d_{pi}, d_{qi})$$

Single and Complete linkage are good algorithms for indicating outlier clusters.

### *UPGAMA*

Unweighted pair group method using arithmetic averages. This is also called Weighted Average Linkage.

$$d_{ki} = \left(\frac{n_p}{n}\right) \cdot d_{pi} + \left(\frac{n_q}{n}\right) \cdot d_{qi}$$

### *WPGAMA*

Weighted pair group method using arithmetic averages. This is also called Average Linkage.

$$d_{ki} = 0.5 \cdot d_{pi} + 0.5 \cdot d_{qi}$$

WPGAMA is a special case of UPGAMA that favors the most recent member clusters in forming new clusters.

### *Centroid*

$$d_{ki} = \left(\frac{n_p}{n}\right) \cdot d_{pi} + \left(\frac{n_q}{n}\right) \cdot d_{qi} - \frac{(n_p \cdot n_q)}{n^2} d_{pq}$$

### *Median*

$$d_{ki} = 0.5 \cdot d_{pi} + 0.5 \cdot d_{qi} - 0.25 \cdot d_{pq}$$

Centroid and Median are similar to UPGAMA and WPGAMA, respectively, but the distance formula contains an additional third term. Centroid and Median methods are not monatomic hierarchical clustering algorithms. In other words, the similarity value between cluster k and any other cluster may be greater than the similarity between cluster p and cluster q. This condition occurs if the centroids (or medians) of the different clusters have approximately the same distance as the distances between the samples that make up the cluster.

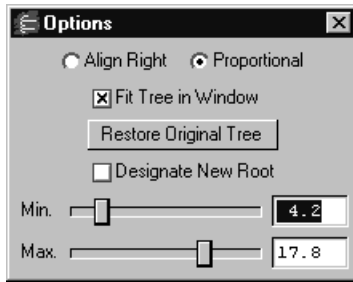
### *Ward's*

This method attempts to minimize the information by describing a set of N samples using a fewer number of clusters.

$$d_{ki} = \frac{n_p + n_i}{n + n_i} d_{pi} + \frac{n_q + n_i}{n + n_i} d_{qi} - \frac{n_i}{n + n_i} d_{pq}$$

### Options

Click on the **Options** button in the *Phylogenetic Tree* window to open the *Options* dialog.



Neighbor Joining Only



All Other Methods

Fig. 12-13. Two forms of the Options dialog.

The box on the left has controls that are specific for the Neighbor Joining method. The box on the right is applicable to the other tree methods. Both boxes share certain features.

Each allows you to choose between aligning the tree to the right or proportionally aligning it. Right alignment uses a fixed branch length for displaying the distance between nodes; proportional alignment differs for Neighbor Joining and other tree modes.

With the Neighbor Joining method, proportional alignment shows branch length sizes proportional to the min/max values set in the *Options* dialog. Other methods plot the nodes at locations determined by their similarity values.

The **Fit Tree in Window** option scales the tree so that the entire tree will fit onto a single printed page. With this turned off, the entire tree will appear with the correct distances between nodes preserved, and the printed tree can be tiled across multiple sheets of paper.

The Neighbor Joining *Options* dialog, click on the **Designate New Root** button to designate a new root node (the node at the very top of the tree), enabling you to look at the relationships between lanes from a different perspective. This will highlight all the nodes in the tree. Click on the node that you want to serve as the new root and the

dendrogram will be recalculated. To return to the original tree, click on **Restore Original Tree**.

The **Min.** and **Max.** sliders are active only for proportionally aligned Neighbor Joining trees. Use them to adjust the minimum and maximum distances between nodes on branches to highlight specific regions of data. Distances below the **Min.** or above the **Max.** values will be collapsed to fit in the window.

For non-Nighbor Joining trees, use the **Clusters** option to define the number of clusters (0 to 18) in the tree. The separate clusters will be identified in the tree by letters.

### 12.5.c Similarity Matrix

The **Similarity Matrix** report compares the similarity of all the lanes to one another. If there are N lanes in the gel, then the similarity matrix is an N by N matrix that is computed using the Dice Coefficient as described in section 12.4, 1-D Analysis Report.

The matrix has the following properties:

- The diagonal elements always have values of 100. This is because a lane is always 100 percent similar to itself.
- The matrix is symmetrical ( $M_{ij} = M_{ji}$ ).

Select **Similarity Matrix** from the *Reports* menu to open this report.

Population 1 - Band Frequencies

Band Frequency Report

Population 1 of Database.1

October 7, 1998

Bandset: expt Units: Molecular Weight

Normalization: Band Type 13

Samples: 24

| Band # | Name | KDa     | Band Frequency |            |
|--------|------|---------|----------------|------------|
|        |      |         | Unweighted %   | Weighted % |
| 1      |      | 5945.99 | 16.7           | 5.3        |
| 2      |      | 5687.71 | 50.0           | 22.7       |
| 3      |      | 5477.17 | 79.2           | 39.6       |
| 4      |      | 5243.58 | 58.3           | 29.7       |
| 5      |      | 4838.65 | 70.8           | 16.6       |
| 6      |      | 4640.99 | 8.3            | 1.3        |
| 7      |      | 4520.08 | 0.0            | 0.0        |
| 8      |      | 4352.12 | 58.3           | 22.6       |
| 9      |      | 4179.56 | 83.3           | 37.6       |
| 10     |      | 3943.85 | 54.2           | 27.1       |
| 11     |      | 3628.10 | 100.0          | 66.7       |
| 12     |      | 2950.30 | 45.8           | 12.7       |
| 13     |      | 2655.65 | 100.0          | 100.0      |

Screen Page 1 of 1

Fig. 12-14. Similarity Matrix report.

The *Similarity Matrix* report is displayed in a standard report window (see section 12.1, Report Window).

## 12.6 Volume Analysis Report

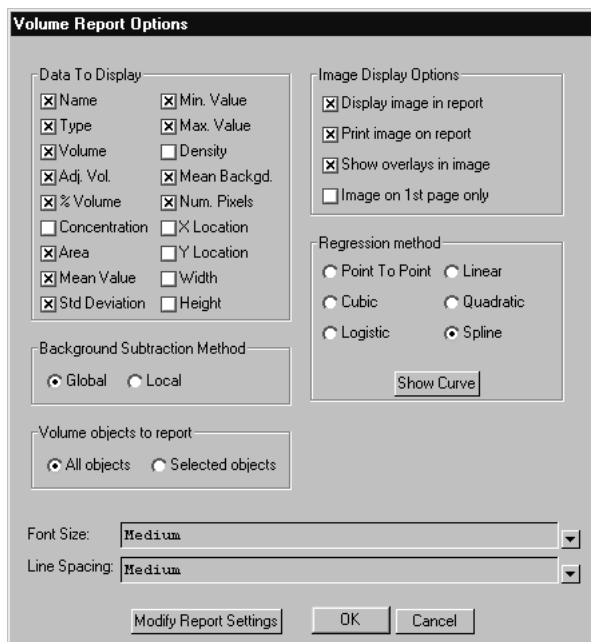
The **Volume Analysis Report** displays volume data.

Select **Volume Analysis Report** from the *Reports* menu or *Volume* toolbar. The *Volume Report Options* dialog will open, in which you can specify the information that will appear in the report.



When you click on **OK**, the report will be displayed in a standard report window (see section 12.1, Report Window).

### 12.6.a Volume Report Options



The dialog box is titled "Volume Report Options". It contains several sections for configuring the report:

- Data To Display:** A list of checkboxes for data to include in the report. The checked items are: Name, Type, Volume, Adj. Vol., % Volume, Area, Mean Value, Std Deviation, Min. Value, Max. Value, Mean Backgd., Num. Pixels, and Y Location. The unchecked items are: Density, Concentration, X Location, Width, and Height.
- Image Display Options:** A list of checkboxes for image display. The checked items are: Display image in report, Print image on report, and Show overlays in image. The unchecked item is: Image on 1st page only.
- Regression method:** A group of radio buttons for selecting the regression method. The selected method is "Spline". Other options are: Point To Point, Linear, Cubic, and Quadratic. A "Show Curve" button is located below the radio buttons.
- Background Subtraction Method:** A group of radio buttons for selecting the background subtraction method. The selected method is "Global". The other option is "Local".
- Volume objects to report:** A group of radio buttons for selecting the volume objects to report. The selected option is "All objects". The other option is "Selected objects".
- Font Size:** A dropdown menu set to "Medium".
- Line Spacing:** A dropdown menu set to "Medium".
- Buttons:** "Modify Report Settings", "OK", and "Cancel".

Fig. 12-15. Volume Report Options dialog.

The volume report options are described below:

- **Name**—The name that is automatically assigned to the volume based on its type (U=Unknown, Std=Standard, B=Background) and order in which it was created.
- **Type**—Unknown, standard, or background.
- **Volume**—Sum of the intensities of the pixels inside the volume boundary x area of a single pixel (in mm<sup>2</sup>).
- **Adj. Vol.**—Volume minus the background volume; if there is no background volume, this is simply the volume.

## Quantity One User Guide

---

- **% Volume**—The volume expressed as a percentage of all the volumes in the image.
- **Concentration**—The quantity as calculated from the standards and the regression method. If you have not defined standards, this is not calculated.
- **Area**—The total area of the volume box you have drawn in  $\text{mm}^2$ .
- **Mean Value**—The mean intensity of the pixels inside the volume boundary.
- **Std. Deviation**—The standard deviation from the mean intensity.
- **Min. Value**—The value of the lowest intensity pixel in the volume.
- **Max. Value**—The value of the highest intensity pixel in the volume.
- **Density**—The total intensity of all the pixels in the volume divided by the area of the volume.
- **Mean Background**—The mean intensity of the pixels in the background volume.
- **Num. Pixels**—The number of pixels inside the volume.
- **X location**—The distance in mm from the left edge of the image to the center of the volume.
- **Y location**—The distance in mm from the top edge of the image to the center of the volume.
- **Width**—The width of the volume in mm.
- **Height**—The height of the volume in mm.

Specify the preferred **Background Subtraction Method (Global or Local)**.

**Note:** If you select **Global** and have not defined a background volume, no background subtraction will be performed on the image.

The **Image Display Options** affect how the image is displayed and/or printed on the report.

You can choose whether to report on all volume objects (**All objects**) or only those objects you have selected (**Selected objects**)

Select the regression method for calculating the **Volume Regression Curve**. To display the curve, click on the **Show Curve** button.

Select the **Font Size** and **Line Spacing** settings to be used in the report by clicking on the button next to each field and selecting from the list of options.

### Saving the Report Options

To save the report options, click on the **Modify Report Settings** button at the bottom of the dialog. In the pop-up dialog, enter a name for the report settings in the field.

To load or delete previously saved settings, click on the button next to the **Settings to Load or Delete** field and select from the list of saved settings. Then click on **Load** or **Delete**.

## 12.7 Volume Regression Curve

If you have defined at least two standard volumes on the image, you can display a regression curve for calculating the quantities of the unknowns.

**Note:** Volumes that fall above the greatest standard value, or below the lowest standard value, cannot be guaranteed for accuracy.

Select **Volume Regression Curve** from the *Reports* menu.

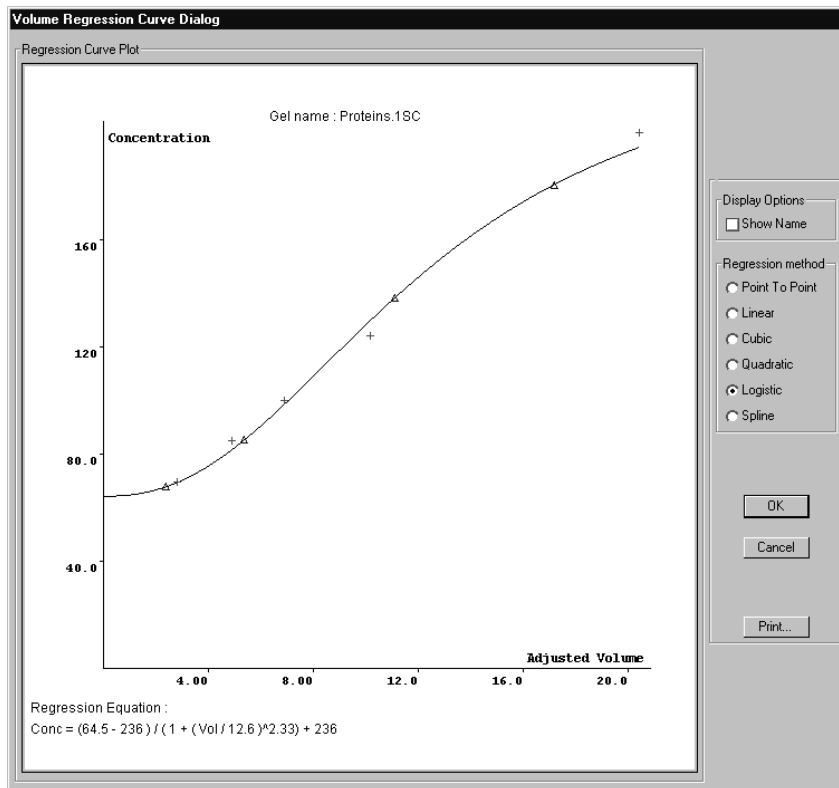


Fig. 12-16. Volume Regression Curve window.

On the curve, each standard volume is marked by a red X and each unknown is marked by a blue triangle.

The X axis is the adjusted volume and the Y axis is the concentration, based on the standards you have marked on the image.

To display the numbers and names of the volumes, click on the **Show Name** checkbox.

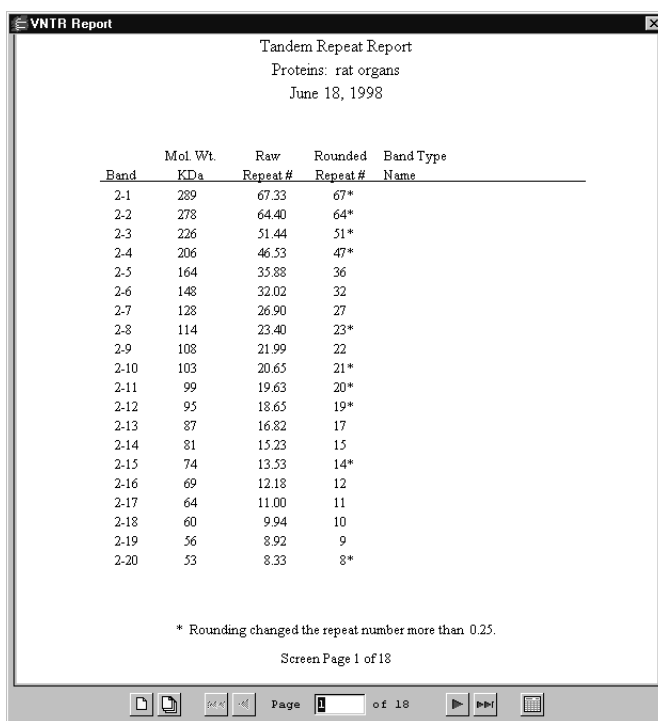
Select the preferred regression method from the list of option buttons. The regression equation for the selected method is displayed in the lower left of the window.

To print the curve, click on the **Print** button. To close the window, click on the **OK** button.

## 12.8 VNTR Report

The **VNTR Report** displays the calculated VNTR data (see section 11.2, Variable Number Tandem Repeats).

Select **VNTR Report** from the *Reports* menu. A pop-up box will ask if you want to display the gel image in the report.



The screenshot shows a window titled "VNTR Report" with a subtitle "Tandem Repeat Report". Below the subtitle, it says "Proteins: rat organs" and "June 18, 1998". The main content is a table with five columns: "Band", "Mol. Wt. KDa", "Raw Repeat #", "Rounded Repeat #", and "Band Type". The table lists 20 bands (2-1 to 2-20) with their corresponding molecular weights, raw repeat numbers, rounded repeat numbers, and band types. At the bottom of the table, there is a note: "\* Rounding changed the repeat number more than 0.25." and "Screen Page 1 of 18". The window has a standard toolbar at the bottom with icons for print, save, and navigation.

| Band | Mol. Wt. KDa | Raw Repeat # | Rounded Repeat # | Band Type |
|------|--------------|--------------|------------------|-----------|
| 2-1  | 289          | 67.33        | 67*              |           |
| 2-2  | 278          | 64.40        | 64*              |           |
| 2-3  | 226          | 51.44        | 51*              |           |
| 2-4  | 206          | 46.53        | 47*              |           |
| 2-5  | 164          | 35.88        | 36               |           |
| 2-6  | 148          | 32.02        | 32               |           |
| 2-7  | 128          | 26.90        | 27               |           |
| 2-8  | 114          | 23.40        | 23*              |           |
| 2-9  | 108          | 21.99        | 22               |           |
| 2-10 | 103          | 20.65        | 21*              |           |
| 2-11 | 99           | 19.63        | 20*              |           |
| 2-12 | 95           | 18.65        | 19*              |           |
| 2-13 | 87           | 16.82        | 17               |           |
| 2-14 | 81           | 15.23        | 15               |           |
| 2-15 | 74           | 13.53        | 14*              |           |
| 2-16 | 69           | 12.18        | 12               |           |
| 2-17 | 64           | 11.00        | 11               |           |
| 2-18 | 60           | 9.94         | 10               |           |
| 2-19 | 56           | 8.92         | 9                |           |
| 2-20 | 53           | 8.33         | 8*               |           |

\* Rounding changed the repeat number more than 0.25.

Screen Page 1 of 18

Fig. 12-17. Example of a VNTR Report.

## Quantity One User Guide

---

The report is displayed in a standard report window (see section 12.1, Report Window).

The report displays the molecular weight, the raw repeat number, and the rounded repeat number for each band in the gel image.

The raw repeat number is the number of repeats calculated using the information that you provided in the *Tandem Repeat Calculation* dialog, and is likely to include fractional values. The rounded repeat number is the raw repeat number rounded to the nearest whole number.

An asterisk (\*) will appear next to some of the rounded repeat numbers if you selected **Test for Ambiguity** and **Flag with \*** in the *Tandem Repeat Calculations* dialog. The asterisk will appear next to numbers that vary from the raw repeat number by more than the selected ambiguity value.

# 13. Printing and Exporting

The commands for printing and exporting images are located on the *File* menu.

Reports are printed from within the individual report windows.

## 13.1 Print Image

Print Image prints a copy of the active image window and any image overlays that are displayed. The Print Image command opens the standard print dialog box for your operating system. When you are satisfied with all the print parameters, click Print to send the image to the printer.

## 13.2 Page Setup

Specify the printer settings using the **Page Setup** command on the *File > Print* submenu. This will open the standard *Page Setup* dialog box for Windows or Macintosh, in which you can select the paper size, page orientation, printer, etc.

## 13.3 Print Settings

Use the Print Settings command to configure what information to include and how the image appears each time you print an image. To open Print Settings, click Print Settings in the File>Print submenu. Any changes made to the Print Settings take effect immediately. You can also leave the Print Settings dialog box open to quickly change the settings for images as you print.

**Note:** You can have multiple Print Settings dialog boxes open, one for each open image, with different settings. However, If you use the print command (file>Print>Print Image or CTRL+P) for an image that does not have the Print Settings dialog box open, the print format will be based on the last Print Settings dialog box where changes were made.

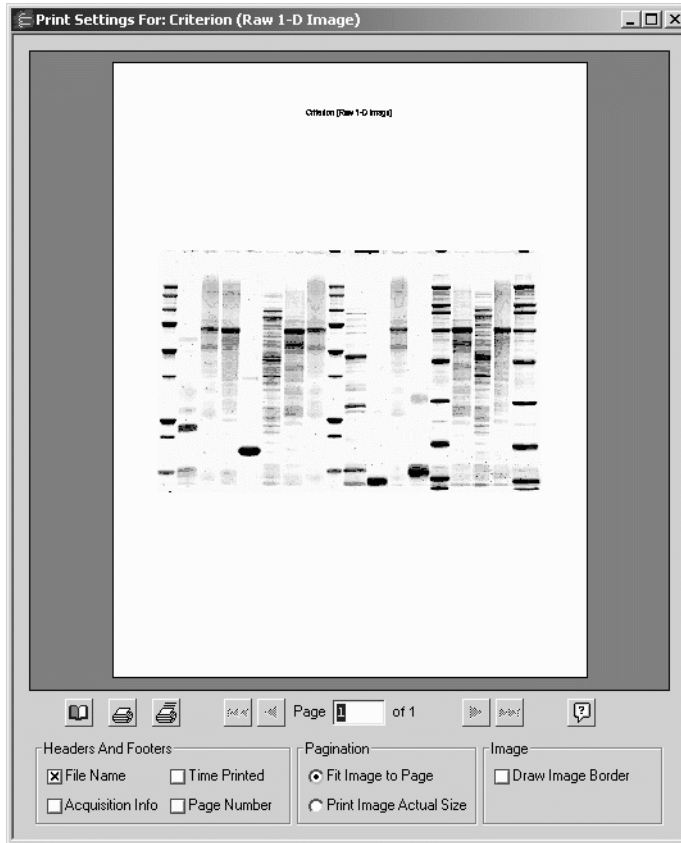


Fig. 13-1.

Under the Headers and Footers settings, select what information you want to include in your prints.

**File name** - Select File name add the name of the image file to the header of the print.\



**Acquisition Info** - If Acquisition Info is selected, the date the image was acquired and the data range of the scan is included in the footer information.

**Time printed** - To add the date and time the image was printed, select Time printed.

**Page number** - Select Page number to add the page number to the footer information.

The Pagination settings determine allow you to determine whether to print your images actual size or fit the images to the page.

**Fit Image to Page** - This fits the current view of the image to a single page based on the layout of the image.

**Print Actual Size** - Select Print actual size to print your images based on the scan size of the gels.

**Note:** If you are using the Gel Doc EQ, ChemiDoc EQ, Fluor-S, or Fluor-S MAX, you must specify the correct image area size when capturing your images to ensure accurate 1:1 printing. You can specify the image area size in the acquisition window for the instrument. See the chapter on each imaging device for more information.

Under Image you can add a border to the image area of the prints. Adding a border is useful for determining the edges of faint images.

Click the Page Setup button to open the Page Setup dialog box. Note that, although you can print directly from the Print Settings dialog box by clicking either Print or Print Current Page, the print settings affect each image printed using the Print Image command.

**Note:** The Print Settings dialog box attempts to accurately display the image exactly how it will print. However, in some cases, such as images with large pixel dimensions set to Print Actual Size, small differences may be noticed between the preview window and the actual print.

### 13.4 Image Report

The **Image Report** displays the image and basic image information, including the image dimensions, pixel size, date of scan, type of imager, etc.

## Quantity One User Guide

To print an image report for a particular gel, select **Image Report** from the *File > Print* submenu. This will open a smaller version of the standard printer dialog box.

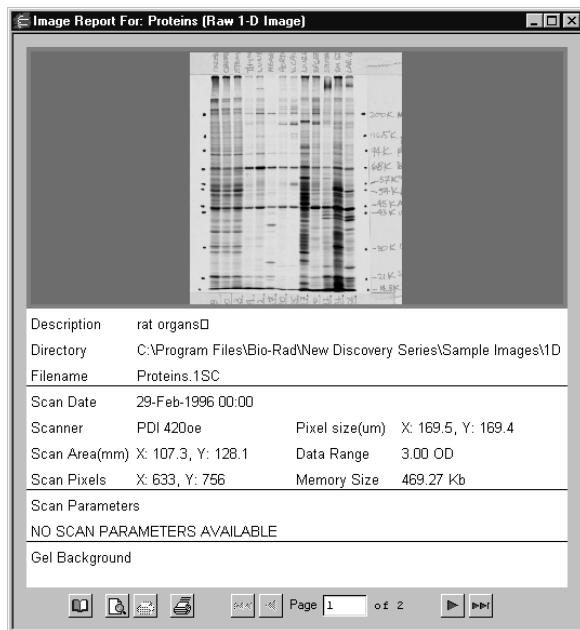


Fig. 13-2. Image Report.

**Note:** TIFF images do not contain all the tagged information that would normally be included in an image file (for example, imaging device, scan date, image color, etc.). For this reason, the **Image Report** may list this information as “Unknown” for imported TIFF files.

## 13.5 Video Print

Select **Video Print** from the *File > Print* submenu to print images and reports to a video printer.

**Note:** Video printing requires installation of the video board and cable that come with the Gel Doc EQ and ChemiDoc EQ systems. The video board and cable can also be ordered separately.

### Settings for the Mitsubishi P90W/P91W Video Printer

There are three settings for the Mitsubishi P90W/P91W video printer. Set **Contrast** to zero, **Brightness** to zero, and **Gamma** to 5.

The dip switches should stay in the orientation in which they are shipped: Pin 1 is up (on), and Pins 2–10 are down (off).

## 13.6 Export to TIFF Image

You can export images in TIFF format for analysis and publishing using other applications. Select **Export to TIFF Image** from the *File* menu to open a dialog box in which you can specify the export parameters.



Fig. 13-3. Export to TIFF Image dialog box.

There are two basic export options:

- Export the **raw image data** for further analysis
- Export the **displayed view** of the image for publishing.

If you choose to export the displayed view, you can export with or without the image overlays.

### 13.6.a Analysis Export Mode

To export a TIFF image for further analysis, select the **Export raw data** button. The publishing controls in the dialog box will become inactive.

### 13.6.b Publishing Export Mode

To export a TIFF image that looks like the image as it is currently displayed on the screen, select one of the options under **Publishing**.

#### Export View Including Overlays

Select this option to include overlays with the exported image. The other controls in the dialog box will become inactive.

#### Export View Excluding Overlays

Select this option to *not* include overlays with the exported image. This is the only mode available if you are exporting images from the *Multi-channel Viewer* (see section 3.4, Multi-Channel Viewer).

With this option selected, specify a resolution for the TIFF image by selecting **72 dpi** (typical computer screen resolution), **150 dpi**, **300 dpi** (standard printing resolution), **Same As Scan**, or any resolution that you enter in the **Specify** field (up to the resolution of the scan).

Under **Transform**, you can specify a **Linear** transform or preserve the **Current View** (which is automatically log transformed).

Images with a higher **Bit Depth** (e.g., **16 bit** or **24 bit RGB**) can be compressed to a lower bit depth (e.g., **8 bit**) during export.

**Note:** TIFF images are automatically exported from the *Multi-channel Viewer* in **24 bit RGB** mode to preserve the colors displayed in the viewer.

Finally, if you are only displaying part of the image due to magnification or repositioning, you can preserve the **Current View** or export the **Entire Image**.

### 13.6.c Exporting the Image

The size of the pixels in the image and the file size of the image are listed at the bottom of the dialog. When you are ready to export, click on the **Export** button.

In the **Save As** dialog box, the default file name will have a .tif extension, and the file type will indicate that this is a TIFF image.

### 13.7 Exporting a JPEG Image

The **Export to JPEG Image** command allows you to export an image as a JPEG image. The **Export to JPEG Image** is primarily for exporting your images for use in reports, presentations, etc. rather than a TIFF image because a JPEG image is considerably smaller in size than a TIFF image. Unlike TIFF images, JPEG images cannot be opened by TDS applications.

To export an image as a JPEG, open the **File** menu and click **Export to JPEG Image**. This opens the **Export to JPEG Image** dialog box.

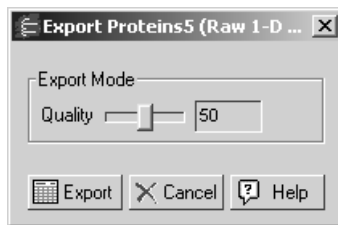


Fig. 13-4. Exporting to JPEG

In the **Export to JPEG** dialog box, under **Export mode**, use the slider to adjust the image quality. The higher the quality, the larger the file size.

Click **Export** to complete the operation. The default file name and location are based on the original image. You can change the file name or select a different folder to save in. Click **Save** to complete the export.

# Appendix A

## Gel Doc EQ



Fig. A-1. Gel Doc EQ.

Before you can begin acquiring images, the Gel Doc EQ system must be properly installed and connected with the host computer. See the Gel Doc EQ hardware manual for installation, startup, and operating instructions.

## Quantity One User Guide

---

To use the Gel Doc EQ, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install the main software application.

Make sure that your Gel Doc EQ camera is turned on. If the camera is not turned on, the Gel Doc EQ acquisition window will open and an image capture error will be displayed.

### Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

## A.1 Gel Doc EQ Acquisition Window

To acquire images using the Gel Doc EQ, go to the File menu and select Gel Doc EQ.... The acquisition window for the instrument will open, displaying a control panel and a video display window.



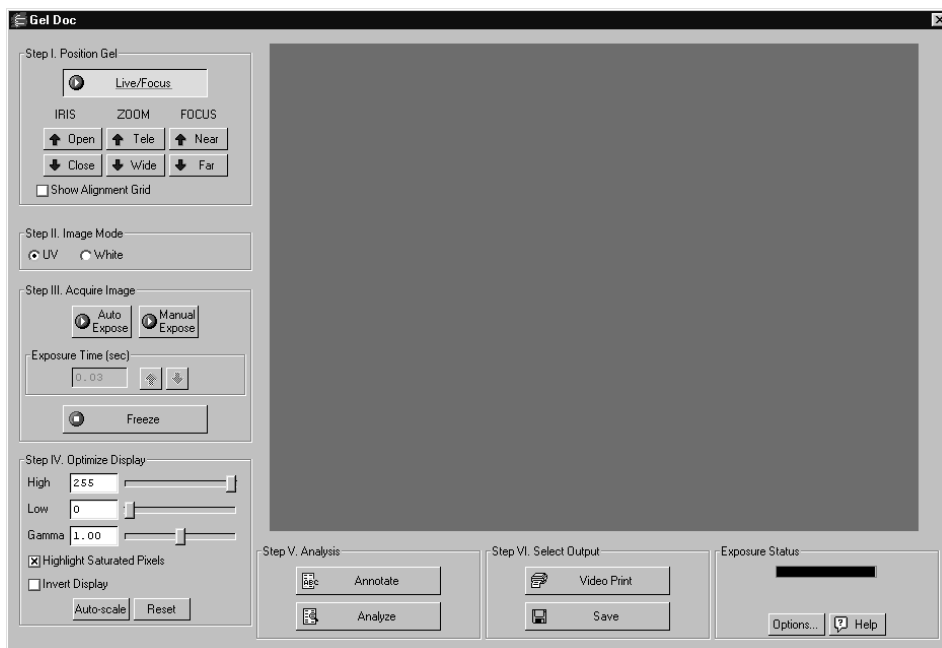


Fig. A-2. Gel Doc EQ acquisition window

The Gel Doc EQ video display window will open in “live” mode, giving you a live video display of your sample. If no image is visible, make sure the camera is on, check the cable connections, make sure the iris on the camera is not closed, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are six basic steps to acquiring an image using the Gel Doc EQ:

1. Position and focus the gel or other object to be imaged.
2. Select Mode.
3. Acquire the image.

4. Optimize the display.
5. Analysis.
6. Select the output.

## A.2 Step I. Position Gel

The Gel Doc EQ window will open in “live” mode, giving you a live video display of your sample. In this mode, the Live/Focus button will appear selected, and frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.

You can use live mode to zoom, focus, and adjust the aperture on the camera, while positioning the sample within the area.

**Note:** Newer versions of the Gel Doc EQ feature a motorized zoom lens that can be controlled directly from the acquisition window using the Iris, Zoom, and Focus arrow buttons. Click on the Up/Down buttons while viewing your sample in the window to adjust the lens. These buttons will not be visible if you are connected to older versions of the Gel Doc EQ without the motorized zoom lens.



Fig. A-3. Newer Gel Doc EQs feature camera control buttons in the acquisition window.

You can also select the Show Alignment Grid checkbox to facilitate positioning.

**Note:** After positioning your sample, you should check the Imaging Area dimensions under Options (see section A.9, Options) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are

focusing on, you can place a ruler in the Gel Doc EQ cabinet so that it is visible in the image.

### A.3 Step II. Select Image Mode

The Image Mode option buttons allow you to set the type and scale of your data.

#### UV

Select this mode for fluorescent samples. With this mode selected, the data will be measured in linear intensity units.

#### White Light

Select this mode for reflective and transmissive samples. With this mode selected, the data will be measured in uncalibrated optical density (uOD) units.

### A.4 Step III. Acquire Image

For many white light applications, you can skip this step and save and print images directly from Live/Focus mode.

For UV light or faint samples, you can take an automatic exposure based on the number of saturated pixels in the image or you can enter a specific exposure time.

**Note:** “Exposure” refers to the integration of the image on the camera CCD over a period of time. The effect is analogous to exposing photographic film to light over a period of time.

#### Auto Expose

Auto Expose will take an exposure whose time length is determined by the number of saturated pixels in the image. This is useful if you are uncertain of the optimal exposure length.

## Quantity One User Guide

---

**Note:** If you know the approximate exposure time you want ( $\pm 3$  seconds), you can skip this step and go directly to Manual Expose.

Click on the Auto Expose button to cancel Live/Focus mode and begin an automatic exposure. The Auto Expose button will appear selected throughout the exposure.

During the auto exposure, the image is continuously integrated on the camera CCD until it reaches a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.15 percent. See section A.9, Options.)

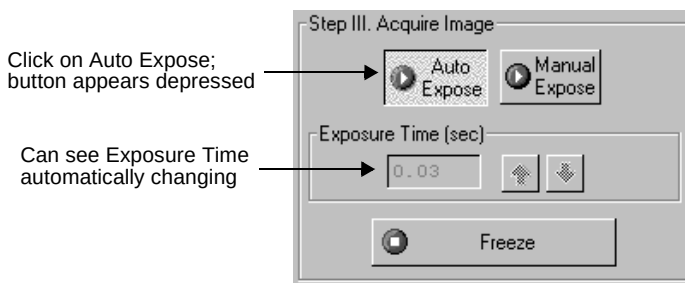


Fig. A-4. Auto Expose.

Once an image has reached the specified percent of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, the exposure time appears active in the Exposure Time field, and Manual Expose is activated.

**Note:** If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

### Manual Expose

If you know the approximate exposure time you want, you can click on the Manual Expose button. Manual Expose is automatically activated after Auto Expose has deactivated.

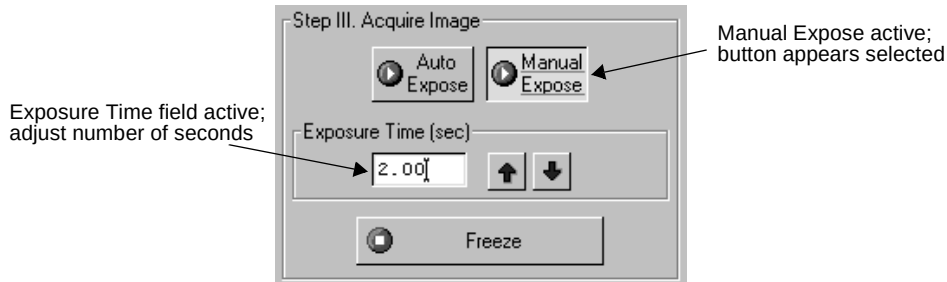


Fig. A-5. Manual Expose.

With Manual Expose activated, you can adjust the exposure time directly by changing the number of seconds in the Exposure Time field. Type in a number or use the arrow buttons next to the field.

When the specified exposure time is reached, the last captured image will be displayed in the Gel Doc EQ image window. The camera continues to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

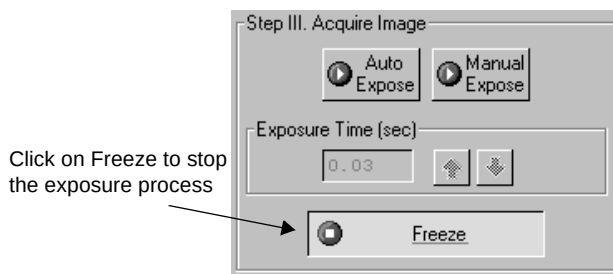


Fig. A-6. Freezing the exposure.

**Note:** Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Video Print, Image Mode, Display controls, etc.).

### A.5 Step IV. Optimize Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a video printer. Adjusting these controls will automatically freeze the video display and allow you to alter the image within the Gel Doc EQ window.

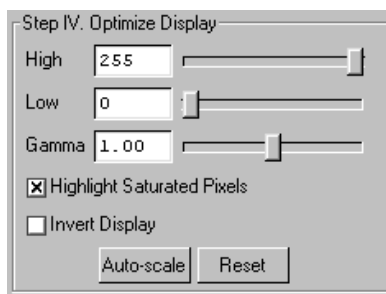


Fig. A-7. Display controls.

These controls are similar to those in the Transform dialog box.

**Note:** The Display controls will only change the appearance of the image. They will not change the underlying data.

#### High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. In white light mode, dragging the High slider handle to the left will make weak signals appear darker. In UV mode, dragging the High slider handle to the left will make weak signals appear brighter. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

### Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

### Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

### Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa. This will only affect how the image is displayed on the screen, not the actual image data.

### Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

### Reset

Reset will return the image to its original, unmodified appearance.

## A.6 Step V. Analysis

The Analysis step of the Gel Doc EQ acquisition window allows you to add annotations and analyze the newly acquired image.

### Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Text Overlay toolbar will also pop up to allow you to annotate your image.

The image will not be saved until you select Save or Save As from the File menu.

### Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

You can then analyze the image using the other features in the main application.

The image will not be saved until you select Save or Save As from the File menu.

## A.7 Step VI. Select Output

In Select Output you can select Video Print and Save as your output options.

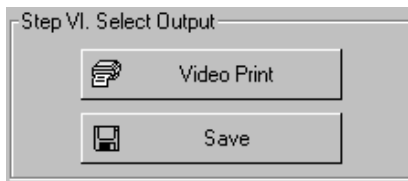


Fig. A-8. Output options.

### Video Print

Clicking on Video Print will automatically send the currently displayed frame (either live or integrated) to a video printer. You can add information about your image to the



bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See section A.9, Options.)

### Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

You can also export your image as a TIFF image for viewing with other applications.

## A.8 Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

If the exposure time is less than 1 second, the status bar will not refresh itself for each exposure; it will remain at 100 percent.

## A.9 Options

Click on the Options button to open the Options dialog box. Here you can specify certain settings for your Gel Doc EQ system.

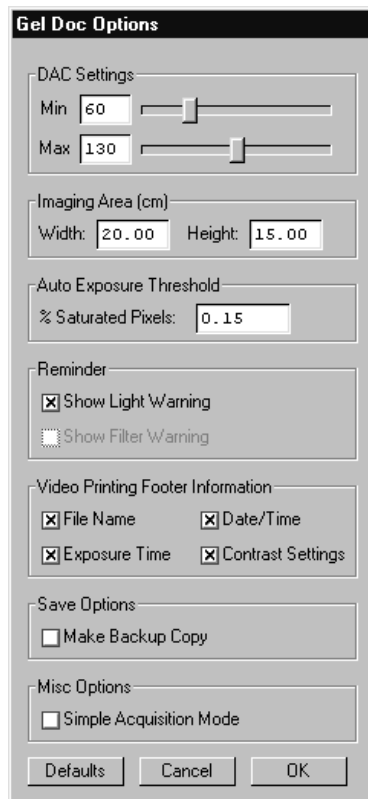


Fig. A-9. Available options in the Gel Doc EQ acquisition window.

Click on OK to implement any changes you make in this box. Clicking on Defaults restores the settings to the factory defaults.

### DAC Settings

**Note:** The default DAC settings are highly recommended and should be changed with caution.

These sliders may be used to adjust the minimum and maximum voltage settings of your video capture board. The minimum slider defines the pixel value that will appear

as white in the image, while the maximum slider defines the pixel value that will appear as black. The slider scale is 0–255, with the defaults set to 60 minimum and 130 maximum.

### Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turn determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

**Note:** Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

### Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.15 percent.

### Reminder

When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the Gel Doc EQ acquisition window or when your system is “idle” for more than 5 minutes.

### Video Printing Footer Information

The checkboxes in this group allow you to specify the information that will appear at the bottom of your video printer printouts.

### Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

## Quantity One User Guide

---

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Backup images are identified by "Backup of" followed by the original image file name.

**Note:** Backup files are read only files. If you attempt to save a backup file, you will be prompted to give it a new name. This protects the backup file and preserves it from any changes.

### Simple Acquisition Mode

The Simple Acquisition mode option allows you to reconfigure the Gel Doc EQ acquisition window to simple mode. In simple mode, the acquisition window contains the same steps as the advanced acquisition window with the exception of Step IV, Optimize display, and Step V, Analysis. Manual exposure is also disabled in simple mode.

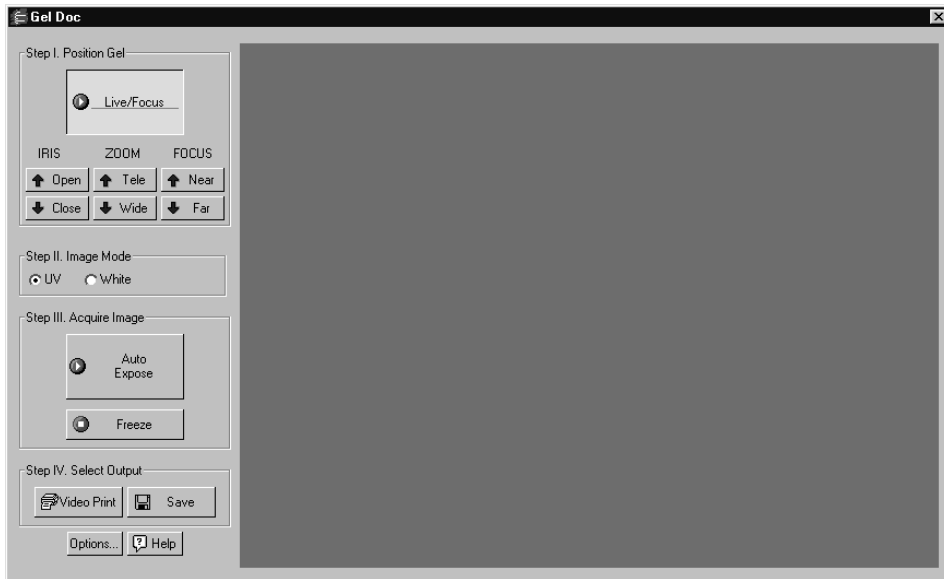


Fig.A-10. Simple Acquisition Window

To change the acquisition window to Simple Acquisition Mode, check the box marked Simple Acquisition Mode. The change will take effect the next time you open the acquisition window.



# Appendix B

## Gel Doc XR



Fig. B-1. Gel Doc XR.

Before you can begin acquiring images, the Gel Doc XR system must be properly installed and connected with the host computer. See the Gel Doc XR hardware manual for installation, startup, and operating instructions.

To use the Gel Doc XR, be sure the system is properly installed. Installation instructions can be found in the Gel Doc XR Hardware Installation Manual.

Make sure that your Gel Doc XR system is powered on. If not, the Gel Doc XR acquisition window will not open and an error will be displayed stating the Gel Doc XR camera could not be found.

### Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

## B.1 Gel Doc XR Acquisition Window

To acquire images using the Gel Doc XR, go to the File menu and select Gel Doc XR.... The acquisition window for the instrument will open, displaying a control panel and a video display window.



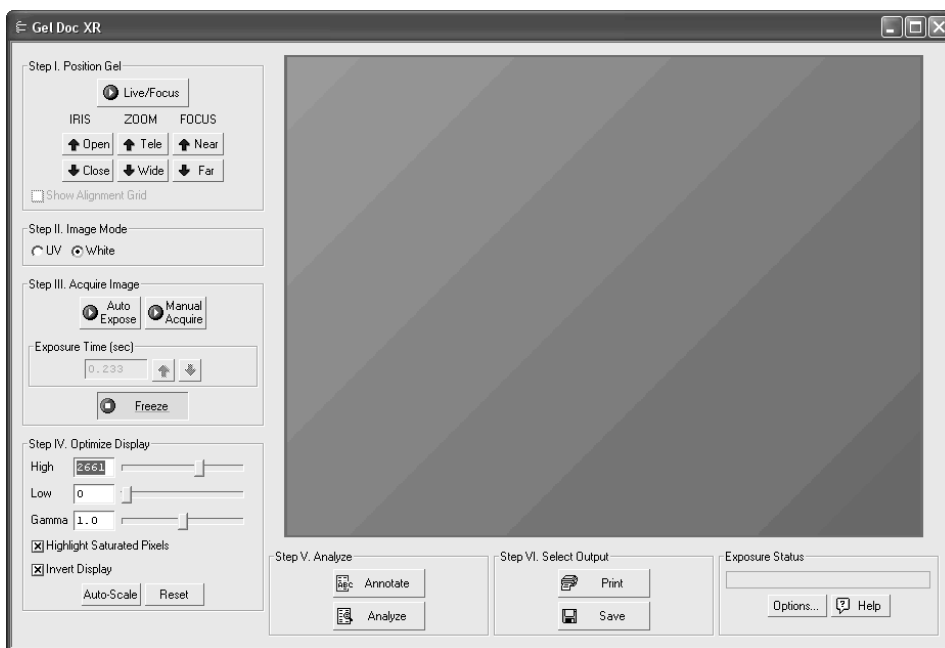


Fig. B-2. Gel Doc XR acquisition window

The Gel Doc XR display window will open in “live” mode, giving you a live display of your sample. If no image is visible, make sure the camera is on, check the cable connections, make sure the iris on the camera is not closed, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are six basic steps to acquiring an image using the Gel Doc XR:

1. Position and focus the gel or other object to be imaged.
2. Select an illumination mode.
3. Acquire the image.

4. Optimize the display.
5. Analyze the image.
6. Select the output.

## B.2 Step I. Position Gel

The Gel Doc XR window will open in “live” mode, giving you a live display of your sample. In this mode, the Live/Focus button will appear selected, and frames will be captured and displayed.

You can use live mode to zoom, focus, and adjust the aperture on the camera, while positioning the sample within the area.

**Note:** The Gel Doc XR features a motorized zoom lens that can be controlled directly from the acquisition window using the Iris, Zoom, and Focus arrow buttons. Click on the Up/Down buttons while viewing your sample in the window to adjust the lens.

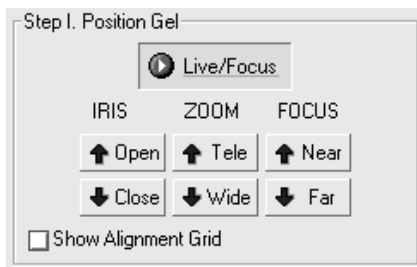


Fig. B-3. Camera control buttons in the acquisition window.

You can also select the Show Alignment Grid checkbox to facilitate positioning.

**Note:** After positioning your sample, you should check the Imaging Area dimensions under Options (see section B.9, Options) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are focusing on, you can place a ruler in the Gel Doc XR cabinet so that it is visible in the image.

## B.3 Step II. Select Image Mode

The Image Mode option buttons allow you to set the type and scale of your data.

### UV

Select this mode for fluorescent samples. With this mode selected, the data will be measured in linear intensity units.

### White Light

Select this mode for reflective and transmissive samples. With this mode selected, the data will be inverted so that darker areas will be of higher intensity levels.

## B.4 Step III. Acquire Image

For many white light applications, you can skip this step and save and print images directly from Live/Focus mode.

For UV light or faint samples, you can take an automatic exposure based on the number of saturated pixels in the image or you can enter a specific exposure time.

**Note:** “Exposure” refers to the integration of the image on the camera CCD over a period of time. The effect is analogous to exposing photographic film to light over a period of time.

### Auto Expose

Auto Expose will take an exposure whose time length is determined by the number of saturated pixels in the image. This is useful if you are uncertain of the optimal exposure length.

**Note:** If you know the approximate exposure time you want, you can skip this step and go directly to Manual Expose.

Click on the Auto Expose button to cancel Live/Focus mode and begin an automatic exposure. The Auto Expose button will appear selected throughout the exposure.

## Quantity One User Guide

---

During the auto exposure, the image is continuously integrated on the camera CCD until it reaches a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.15 percent. See section B.9, Options.)

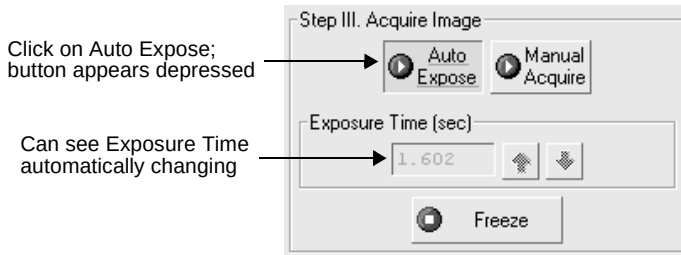


Fig. B-4. Auto Expose.

Once an image has reached the specified percent of saturated pixels, it is captured and displayed in the display window, Auto Expose is automatically deactivated, the exposure time appears active in the Exposure Time field, and Manual Expose is activated.

**Note:** If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

### Manual Expose

If you know the approximate exposure time you want, you can click on the Manual Expose button. Manual Expose is automatically activated after Auto Expose has deactivated.

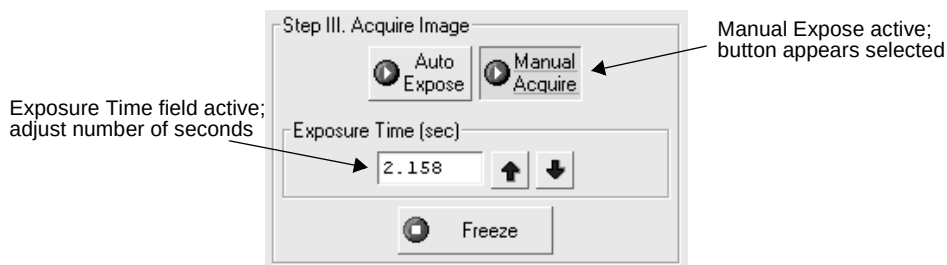


Fig. B-5. Manual Expose.

With Manual Expose activated, use the arrow buttons next to the field to adjust the exposure. You can also type in the new exposure time then press ENTER for the change to take effect.

When the specified exposure time is reached, the last captured image will be displayed in the Gel Doc XR image window. The camera continues to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

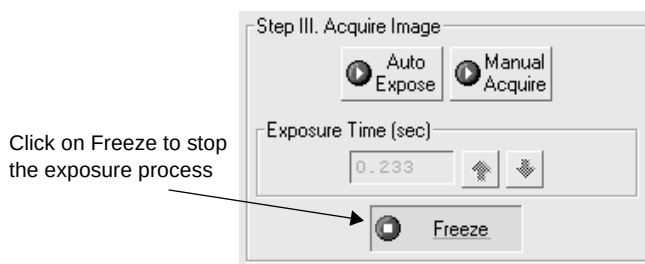


Fig. B-6. Freezing the exposure.

**Note:** Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Print, Image Mode, Display controls, etc.).

### B.5 Step IV. Optimize Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a printer. Adjusting these controls will automatically freeze the display and allow you to alter the image within the Gel Doc XR window.

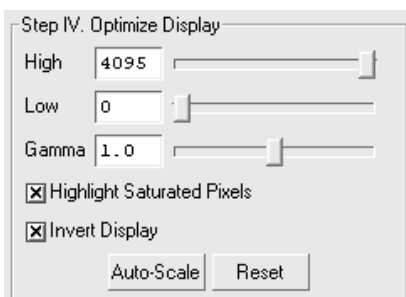


Fig. B-7. Display controls.

These controls are similar to those in the Transform dialog box.

**Note:** The Display controls will only change the appearance of the image. They will not change the underlying data.

#### High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. In white light mode, dragging the High slider handle to the left will make weak signals appear darker. In UV mode, dragging the High slider handle to the left will make weak signals appear brighter. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

### Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

### Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

### Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa. This will only affect how the image is displayed on the screen, not the actual image data.

### Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

### Reset

Reset will return the image to its original, unmodified appearance.

## B.6 Step V. Analysis

The Analysis step of the Gel Doc XR acquisition window allows you to add annotations and analyze the newly acquired image.

### Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Text Overlay toolbar will also pop up to allow you to annotate your image.

The image will not be saved until you select Save or Save As from the File menu.

### Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

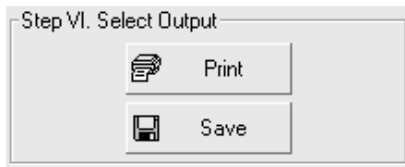
You can then analyze the image using the other features in the main application.

The image will not be saved until you select Save or Save As from the File menu.

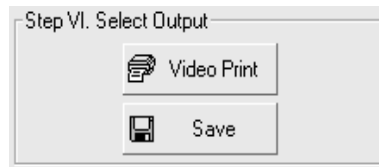
You can export your image as a TIFF image for analysis with other applications. You can also export your image as a JPEG image for use in reports and presentations.

## B.7 Step VI. Select Output

In Select Output you can Print and Save as your output options.



Digital or Desktop Printer Connected



Video Printer Connected

Fig. B-8. Output options.



### Print

If you are printing to a desktop or digital printer, clicking Print sends the currently displayed frame (either live or integrated) to the printer.

If you have a video printer connected to your computer, Video Print will be displayed in the Select Output window.

You can add information about your image to the bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See section B.9, Options.)

### Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

## B.8 Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

## B.9 Options

Click on the Options button to open the Options dialog box. Here you can specify certain settings for your Gel Doc XR system.

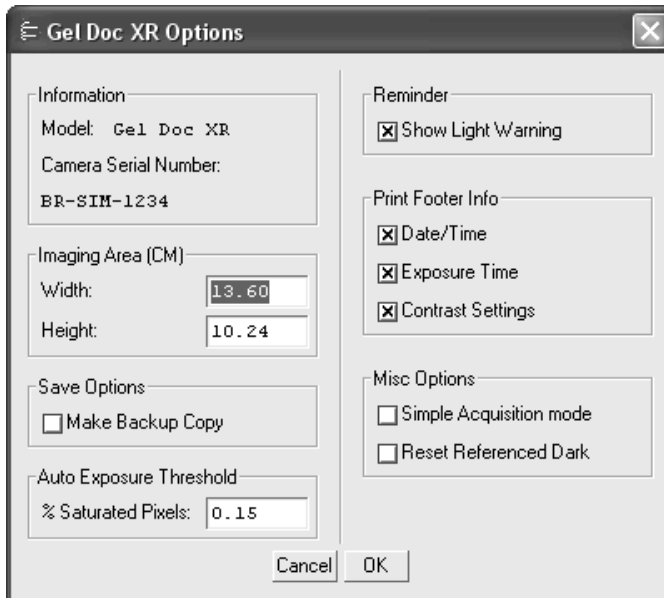


Fig. B-9. Available options in the Gel Doc XR acquisition window.

Click OK to implement any changes you make in this box.

### Information

This field describes the instrument model and the camera serial number.

### Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turns determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

**Note:** Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

### Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy check box.

With this check box selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Backup images are identified by "Backup of" followed by the original image file name.

**Note:** Backup files are read only files. If you attempt to save a backup file, you will be prompted to give it a new name. This protects the backup file and preserves it from any changes.

### Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.15 percent.

### Reminder

When this check box is selected, the software will warn you to turn off your transilluminator light when you exit the Gel Doc XR acquisition window or when your system is “idle” for more than 5 minutes.

### Print Footer Information

The check boxes in this group allow you to specify the information that will appear at the bottom of your printouts.

### Simple Acquisition Mode

The Simple Acquisition mode option allows you to reconfigure the Gel Doc XR acquisition window to simple mode. In simple mode, the acquisition window contains the same steps as the advanced acquisition window with the exception of Step IV,

Optimize display, and Step V, Analysis. Manual exposure is also disabled in simple mode.



Fig.B-10. Simple Acquisition Window

To change the acquisition window to Simple Acquisition Mode, check the box marked Simple Acquisition Mode. The change will take effect the next time you open the acquisition window.

### Reset Reference Dark

The first time you open the Gel Doc XR acquisition window, a reference dark image file is generated and saved to your hard drive. The purpose of this file is to reduce dark current noise generated from the CCD. Dark current noise is typical of all CCDs and is a result of the accumulation of charge in the absence of light.

To reset the reference dark image, check the Reset Reference Dark checkbox. The next time you open the Gel Doc XR acquisition window, the software will acquire a new dark reference image.

# Appendix C

## ChemiDoc EQ



Fig.C-1. ChemiDoc EQ.

Before you can begin acquiring images, the ChemiDoc EQ system must be properly installed and connected with the host computer. See the ChemiDoc EQ hardware manual for installation, startup, and operating instructions.

To use the ChemiDoc EQ, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install the main software application.

Make sure that your ChemiDoc EQ camera is turned on. If the camera is not turned on, the ChemiDoc EQ acquisition window will open and an image capture error will be displayed.

### Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

## C.1 ChemiDoc EQ Acquisition Window

To acquire images using the ChemiDoc EQ, go to the File menu and select ChemiDoc EQ.... The acquisition window for the instrument will open, displaying a control panel and a video display window.

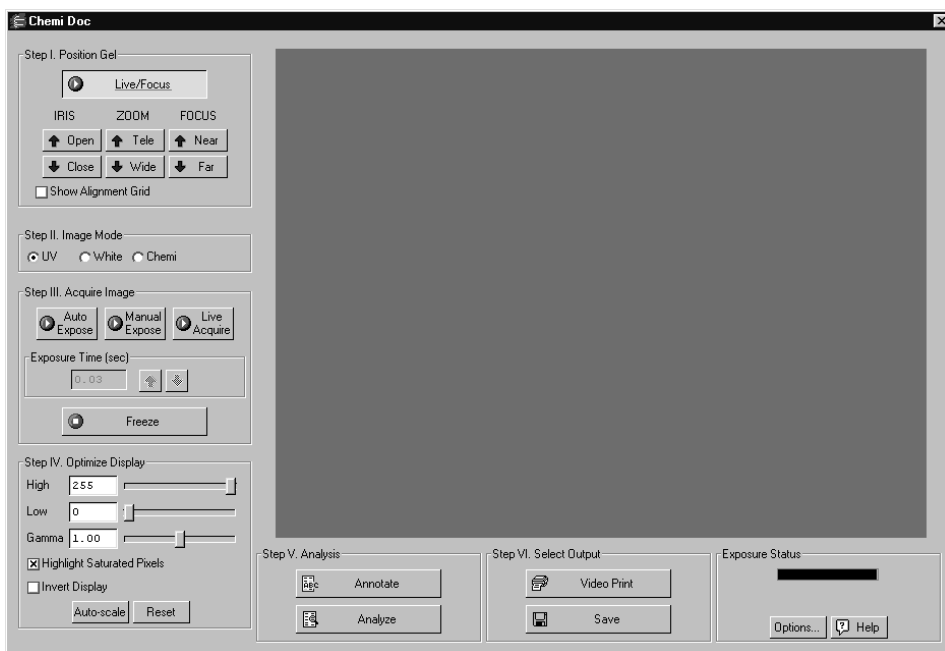


Fig.C-2. ChemiDoc EQ acquisition window

The ChemiDoc EQ video display window will open in “live” mode, giving you a live video display of your sample. If no image is visible, make sure the camera is on, check the cable connections, make sure the iris on the camera is not closed, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are six basic steps to acquiring an image using the ChemiDoc EQ:

1. Position and focus the gel or other object to be imaged.
2. Select Mode.
3. Acquire the image.

4. Optimize the display.
5. Analysis.
6. Select the output.

## C.2 Step I. Position Gel

The ChemiDoc EQ window will open in “live” mode, giving you a live video display of your sample. In this mode, the Live/Focus button will appear selected, and frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.

You can use live mode to zoom, focus, and adjust the aperture on the camera, while positioning the sample within the area.

**Note:** Newer versions of the ChemiDoc EQ feature a motorized zoom lens that can be controlled directly from the acquisition window using the Iris, Zoom, and Focus arrow buttons. Click on the Up/Down buttons while viewing your sample in the window to adjust the lens. These buttons will not be visible if you are connected to older versions of the ChemiDoc EQ without the motorized zoom lens.



Fig.C-3. Live/Focus button and camera control buttons.

You can also select the Show Alignment Grid checkbox to facilitate positioning.

**Note:** After positioning your sample, you should check the Imaging Area dimensions under Options (see section C.9, Options) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are



focusing on, you can place a ruler in the ChemiDoc EQ cabinet so that it is visible in the image.

### C.3 Step II. Image Mode

The Image Mode option buttons change the type and scale of the data, as well as the behavior of the ChemiDoc EQ when acquiring an image.

#### UV

Select this mode for fluorescent samples. With this mode selected, the data will be measured in linear intensity units.

#### White Light

Select this mode for reflective and transmissive samples. With this mode selected, the data will be measured in uncalibrated optical density (uOD) units.

#### Chemi

This mode is designed for chemiluminescent samples. With this mode selected, the data is measured in linear intensity units; however, the data is inverted, so that samples will appear dark on a light background.

Also, Chemi mode changes the behavior of the Auto and Manual Expose functions, as described above.

### C.4 Step III. Acquire Image

For many white light applications, you can skip this step and save and print images directly from Live/Focus mode.

For UV light, chemiluminescent applications, or faint samples, the ChemiDoc EQ control panel has several features for creating image exposures. You can take an automatic exposure based on the number of saturated pixels in the image, you can

## Quantity One User Guide

---

enter a specific exposure time, or you can take a series of exposures and select the best one.

**Note:** “Exposure” refers to the integration of the image on the camera CCD over a period of time. The effect is analogous to exposing photographic film to light over a period of time.

### Auto Expose

Use Auto Expose if you want to take a single exposure but are uncertain of the optimal exposure time.

**Note:** If you know the approximate exposure time you want ( $\pm 3$  seconds), you can skip this step and go directly to Manual Expose.

Click on the Auto Expose button to cancel Live/Focus mode and begin an automatic exposure. The Auto Expose button will appear selected throughout the exposure.

During the auto exposure, the image is continuously integrated on the camera CCD until it reaches a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.15 percent. See section C.9, Options.)

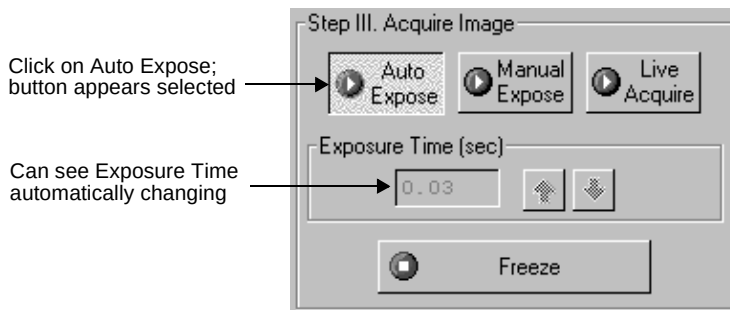


Fig.C-4. Selecting Auto Expose.

Once an image has reached the specified percentage of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, and the exposure time appears active in the Exposure Time field.

At this point, if you are in UV or White image mode, Manual Expose will be automatically activated. If you are in Chemi mode, the Freeze button will be automatically activated.

**Note:** If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most non-chemi applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

### Manual Expose

If you know the approximate exposure time you want, you can click on the Manual Expose button. In UV or White image mode, Manual Expose is automatically activated after Auto Expose is complete.

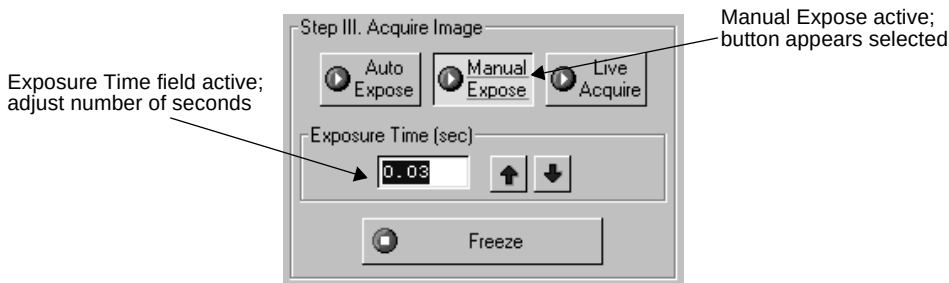


Fig.C-5. Setting a manual exposure.

With Manual Expose activated, you can adjust the exposure time directly by changing the number of seconds in the Exposure Time field. Type in a number or use the arrow buttons next to the field.

In UV or White image mode, when the specified exposure time is reached, the last captured image will be displayed in the ChemiDoc EQ image window. The camera will continue to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

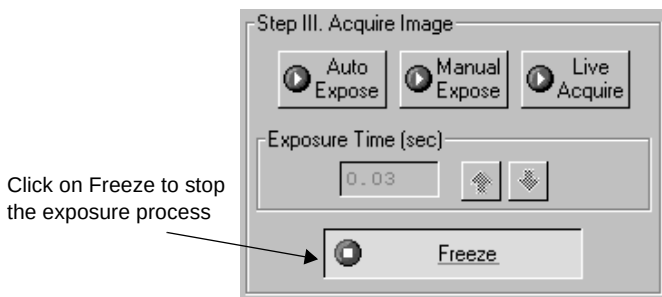


Fig.C-6. Freezing the manual exposure.

In Chemi mode, Manual Expose will expose an image over the specified exposure time and then stop automatically.

**Note:** Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Video Print, Image Mode, Display controls, etc.).

### Live Acquire

Live Acquire mode allows you to specify an interval over which a series of progressively longer exposures are taken. All exposures are then displayed on the screen, and you can choose the one with the best image.

Click on the Live Acquire button. A settings dialog box will open in which you can specify the total exposure time, starting exposure time, and number of exposures.

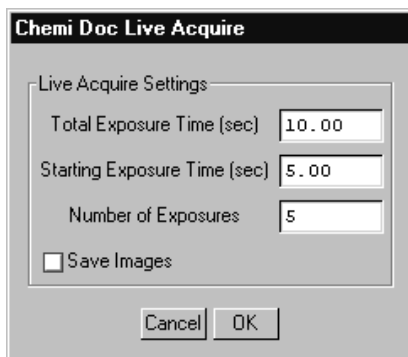


Fig.C-7. Live Acquire settings.

**Note:** You should specify no more than 10 exposures in the Live Acquire Settings dialog, to avoid excessive build-up of image background in later exposures. The fewer the exposures, the less background will be added to the image. See the Release Notes for additional instructions on reducing background in images captured using Live Acquire.

Select the Save Images checkbox if you want to automatically save each exposure as it is taken.

Click on OK in the settings dialog to begin taking exposures. If you selected Save Images, a Save dialog box will open in which you can specify the base file name and location of the exposure files. When you click on Save, the exposures will be taken.

The specified number of exposures will be taken at equal intervals between the starting exposure time and total exposure time. When each exposure is complete, an image window containing that exposure will open behind the ChemiDoc EQ window. When the full exposure time has lapsed, all the image windows will move in front of the ChemiDoc EQ window.

Note that the first exposure will have the base file name (the default base file name is the computer user name and a time stamp). Each subsequent exposure will have a version number (v2, v3, v4, etc.) appended to the base file name. The highest version number will be the final exposure. If you did not elect to auto-save the exposures as they were created, then each image will be unsaved.

To stop the Live Acquire, click on the Freeze button or adjust any of the subsequent controls (e.g., Video Print, Image Mode, Display controls, etc.).

**Note:** Exposures captured before freezing will be displayed in image windows.

Study the different images and select the best exposure(s) to keep. You can then proceed to the next step.

## C.5 Step IV. Optimize Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a video printer. Adjusting these controls will automatically freeze the video display and allow you to alter the image within the ChemiDoc EQ window.

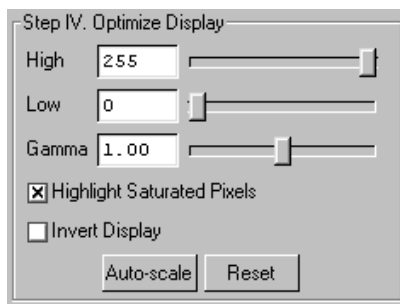


Fig.C-8. Display controls.

These controls are similar to those in the Transform dialog box.

**Note:** The Display controls will only change the appearance of the image. They will not change the underlying data.

### High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. In white light mode and chemi mode, dragging the High slider handle to the left will make weak signals appear darker. In

UV mode, dragging the High slider handle to the left will make weak signals appear brighter. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

### Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

### Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

### Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa. This will only affect how the image is displayed on the screen, not the actual image data.

### Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

### Reset

Reset will return the image to its original, unmodified appearance.

### C.6 Step V. Analysis

The Analysis step of the ChemiDoc EQ acquisition window allows you to add annotations and analyze the newly acquired image.

#### Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Text Overlay toolbar will also pop up to allow you to annotate your image.

The image will not be saved until you select Save or Save As from the File menu.

#### Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

You can then analyze the image using the other features in the main application.

The image will not be saved until you select Save or Save As from the File menu.

### C.7 Step VI. Select Output

The ChemiDoc EQ window has several output options.

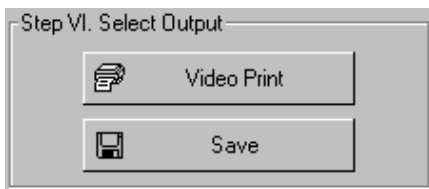


Fig.C-9. Output options.



### Video Print

Clicking on Video Print will automatically send the currently displayed frame (either live or integrated) to a video printer. You can add information about your image to the bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See section C.9, Options.)

### Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

You can also export your image as a TIFF image for viewing with other applications.

## C.8 Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

If the exposure time is less than 1 second, the status bar will not refresh itself for each exposure; it will remain at 100 percent.

## C.9 Options

Click on the Options button to open the Options dialog box. Here you can specify certain settings for your ChemiDoc EQ system.



Fig.C-10. Available options in the ChemiDoc EQ acquisition window.

Click on OK to implement any changes you make in this box. Clicking on Defaults restores the settings to the factory defaults.

### DAC Settings

**Note:** The default DAC settings are highly recommended and should be changed with caution.

These sliders may be used to adjust the minimum and maximum voltage settings of your video capture board. The minimum slider defines the pixel value that will appear as white in the image, while the maximum slider defines the pixel value that will appear as black. The slider scale is 0–255, with the defaults set to 60 minimum and 130 maximum.

### Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turn determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

**Note:** Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

### Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.15 percent.

### Reminder

When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the ChemiDoc EQ acquisition window or when your system is “idle” for more than 5 minutes.

**Note:** If you are performing experiments that are longer than 5 minutes (e.g., chemiluminescence), this should be deselected.

### Video Printing Footer Information

The checkboxes in this group allow you to specify the information that will appear at the bottom of your video printer printouts.

### Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Backup images are identified by "Backup of" followed by the original image file name.

**Note:** Backup files are read only files. If you attempt to save a backup file, you will be prompted to give it a new name. This protects the backup file and preserves it from any changes.

### Simple Acquisition Mode

The Simple Acquisition mode option allows you to reconfigure the ChemiDoc EQ acquisition window to simple mode. In simple mode, the acquisition window contains the same steps as the advanced acquisition window with the exception of Step IV, Optimize display, and Step V, Analysis. Manual exposure is also disabled in simple mode.

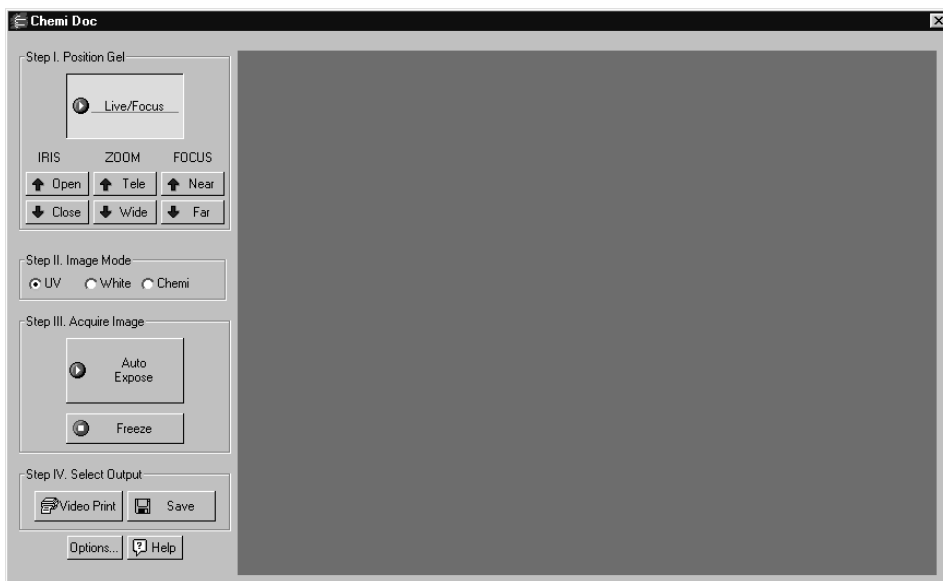


Fig.C-11. Simple Acquisition Window

To change the acquisition window to Simple Acquisition Mode, check the box marked Simple Acquisition Mode. The change will take effect the next time you open the acquisition window.



# Appendix D

## ChemiDoc XRS



Fig.D-1. ChemiDoc XRS.

Before you can begin acquiring images, the ChemiDoc XRS system must be properly installed and connected with the host computer. See the ChemiDoc XRS hardware manual for installation, startup, and operating instructions.

To use the ChemiDoc XRS, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install the main software application.

Make sure that your ChemiDoc XRS camera is turned on. If the camera is not turned on, the ChemiDoc XRS acquisition window will not open.

### Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

## D.1 ChemiDoc XRS Acquisition Window

To acquire images using the ChemiDoc XRS, go to the File menu and select ChemiDoc XRS.... The acquisition window for the instrument will open, displaying a control panel and a video display window.



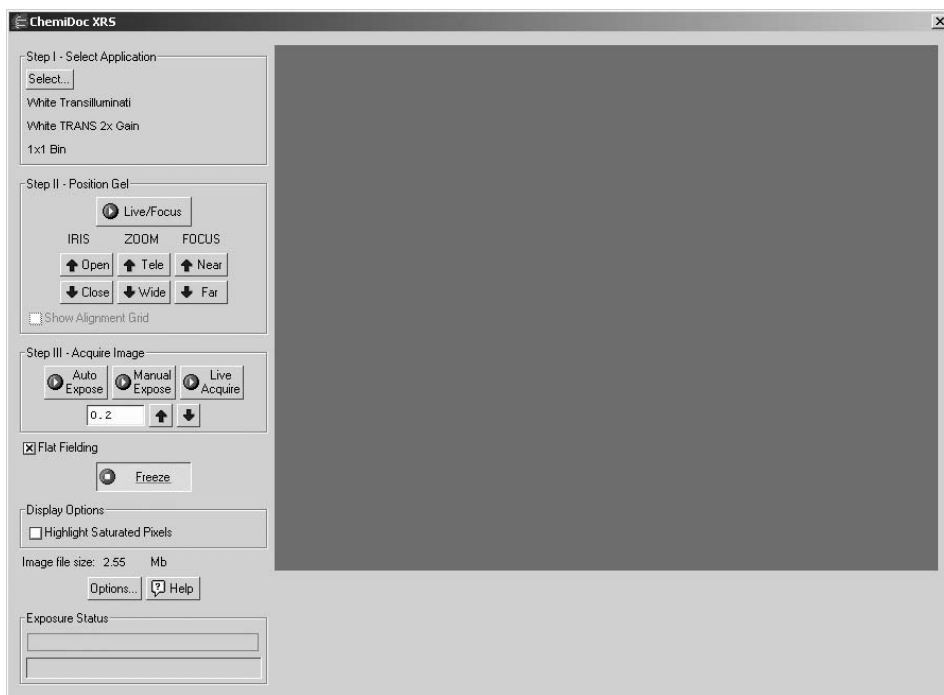


Fig.D-2. ChemiDoc XRS acquisition window

When the ChemiDoc XRS window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to acquiring an image using the ChemiDoc XRS:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Acquire the image.

# D.2 Step I. Select Application

To set the appropriate parameters for the type of object you are imaging, click on the Select button under Select Application. You have the option of selecting a UV, White, or Chemiluminescence application. You can also select a custom setting. Once you select an application, the name of the application and its settings appear in the Select application step.

## D.2.a. Custom Applications

The Bio-Rad installed applications have pre-set gain and bin settings. If the settings for the available applications do not meet your needs, you have the option of creating a custom application, which allows you to set your own binning settings.

To use a custom application, click Select and choose custom. Next, select a custom application from the list, or select Create to create a new application. This opens the custom dialog.

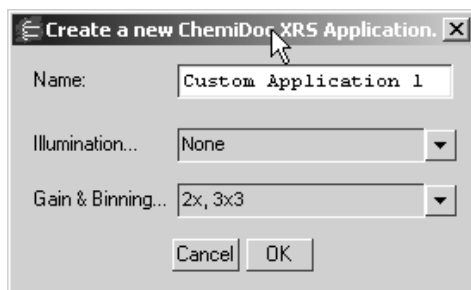


Fig.4-3. Creating a custom ChemiDoc XRS application.

Enter a name for the application. From the Illumination drop down list, select an illumination mode. You can select UV, white or no illumination. Next select Gain and Binning settings from the drop down list.

A higher Binning setting (2x2, 3x3) provides optimal sensitivity for low-light applications such as chemiluminescence. In this mode, the pixels in the camera are “binned” (e.g., four pixels are combined into one) to increase the amount of signal per pixel without increasing noise. Note that combining the pixels results in a reduction in the resolution of the image.

Click OK to save and add the new custom application to the custom list. The new application is automatically selected as your application.

### D.3 Step II. Position Gel

Click the Live/Focus button, and frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.

You can use live mode to zoom, focus, and adjust the aperture on the camera, while positioning the sample within the area.

**Note:** ChemiDoc XRS features a motorized zoom lens that can be controlled directly from the acquisition window using the Iris, Zoom, and Focus arrow buttons. Click on the Up/Down buttons while viewing your sample in the window to adjust the lens.



Fig.D-4. Live/Focus button and camera control buttons.

You can also select the Show Alignment Grid checkbox to facilitate positioning.

**Note:** After positioning your sample, you should check the Imaging Area dimensions under Options (see D.5, Options) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are focusing on, you can place a ruler in the ChemiDoc XRS hood so that it is visible in the image.

### D.4 Step III. Acquire Image

The ChemiDoc XRS control panel has several features for creating image exposures. You can take an automatic exposure based on the number of saturated pixels in the image, you can enter a specific exposure time, or you can take a series of exposures and select the best one.

**Note:** “Exposure” refers to the integration of the image on the camera CCD over a period of time. The effect is analogous to exposing photographic film to light over a period of time.

#### Auto Expose

Use Auto Expose if you want to take a single exposure but are uncertain of the optimal exposure time.

**Note:** If you know the approximate exposure time you want ( $\pm 3$  seconds), you can skip this step and go directly to Manual Expose.

Click on the Auto Expose button to cancel Live/Focus mode and begin an automatic exposure. The Auto Expose button will appear selected throughout the exposure.

During the auto exposure, the image is continuously integrated on the camera CCD until it reaches a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.75 percent. See D.5, Options.)

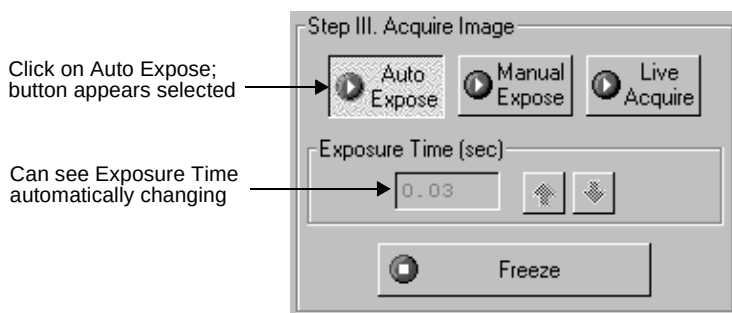


Fig.D-5. Selecting Auto Expose.

Once an image has reached the specified percentage of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, and the exposure time appears active in the Exposure Time field.

**Note:** If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most non-chemiluminescent applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

### Manual Expose

If you know the approximate exposure time you want, you can click on the Manual Expose button. Manual Expose is automatically activated after Auto Expose is complete.

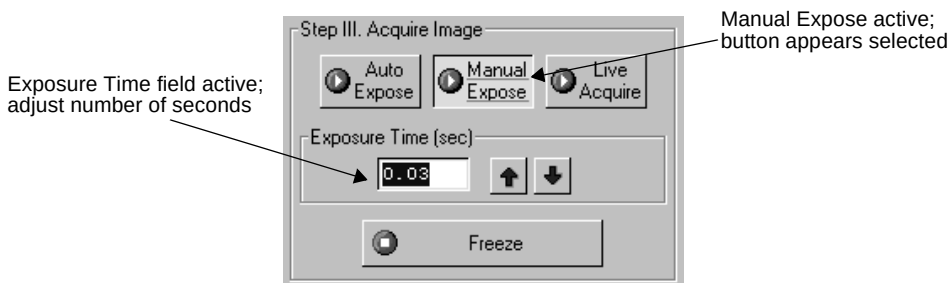


Fig.D-6. Setting a manual exposure.

With Manual Expose activated, you can adjust the exposure time directly by changing the number of seconds in the Exposure Time field. Type in a number or use the arrow buttons next to the field.

When the specified exposure time is reached, the last captured image will be displayed in the ChemiDoc XRS image window.

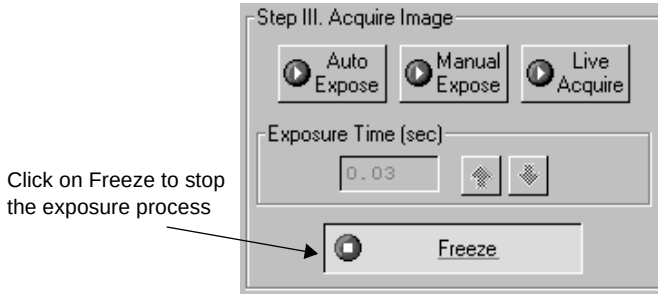


Fig.D-7. Freezing the manual exposure.

### Live Acquire

Live Acquire mode allows you to specify an interval over which a series of progressively longer exposures are taken. All exposures are then displayed on the screen, and you can choose the one with the best image.

Click on the Live Acquire button. A settings dialog box will open in which you can specify the total exposure time, starting exposure time, and number of exposures.

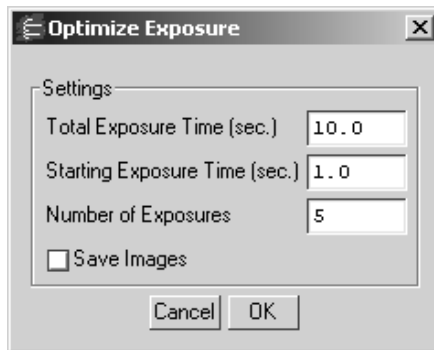


Fig.D-8. Live Acquire settings.

**Note:** You should specify no more than 10 exposures in the Live Acquire Settings dialog, to avoid excessive build-up of image background in later exposures. The

fewer the exposures, the less background will be added to the image. See the Release Notes for additional instructions on reducing background in images captured using Live Acquire.

Select the Save Images checkbox if you want to automatically save each exposure as it is taken.

Click on OK in the settings dialog to begin taking exposures. If you selected Save Images, a Save dialog box will open in which you can specify the base file name and location of the exposure files. When you click on Save, the exposures will be taken.

The specified number of exposures will be taken at equal intervals between the starting exposure time and total exposure time. When each exposure is complete, an image window containing that exposure will open in front of the ChemiDoc XRS window.

Note that the first exposure will have the base file name (the default base file name is the computer user name and a time stamp). Each subsequent exposure will have a version number (v2, v3, v4, etc.) appended to the base file name. The highest version number will be the final exposure. If you did not elect to auto-save the exposures as they were created, then each image will be unsaved.

To stop the Live Acquire, click on the Freeze button.

**Note:** Exposures captured before freezing will be displayed in image windows.

Study the different images and select the best exposure(s) to keep.

### D.4.a. Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

### Illumination Flat Fielding

For UV or white light transillumination applications, you should use the appropriate reference plate to ensure a uniform intensity in the image. This will compensate for normal variations in image pixel intensity that occur with a transilluminating light source.

To enable this feature, select the Flat Fielding checkbox.

*UV Illumination Flat Fielding:* When you first select the Flat fielding checkbox, and then acquire an image using the UV transilluminator, you will be prompted to remove your sample and place the fluorescent reference plate on the ChemiDoc XRS sample stage and turn on the UV transilluminator (see the ChemiDoc XRS User Manual).

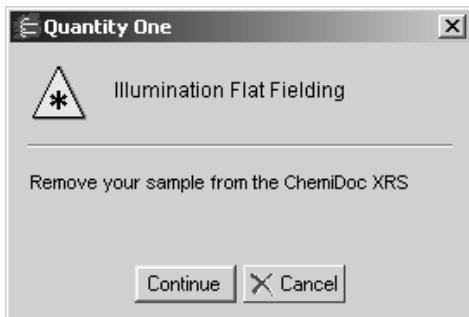


Fig.D-9. Illumination flat fielding reminder

A reference image of the plate will be acquired and saved on your computer hard drive. The reference image will be applied to the sample image to generate a flat field corrected exposure.

*White Light Illumination Flat Fielding:* When you first select the checkbox, and then acquire an image using the white light conversion screen or the white light transilluminator, you will be prompted to remove your sample, turn on the white light source, and collect an exposure of the white light screen/transilluminator (see the ChemiDoc XRS User Manual). A reference image of the screen/transilluminator will be acquired and saved on your computer hard drive. The reference image will be applied to the sample image to generate a flat field corrected exposure.

For subsequent UV or white light trans exposures, you will be prompted to either use the appropriate saved flat field image or acquire a new one.



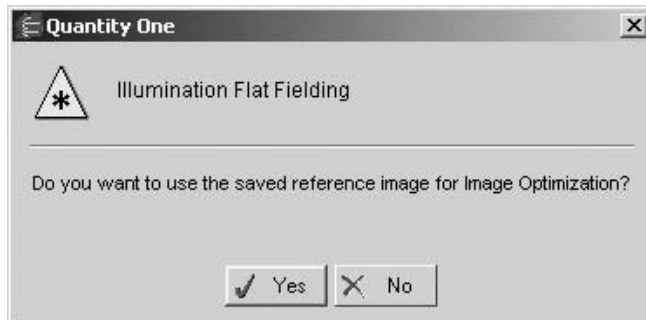


Fig.D-10. Using the saved reference image.

**Note:** Flat fielding is unavailable for White EPI illumination and chemiluminescence applications.

## D.5 Options

Click on the Options button to open the Options dialog box. Here you can specify certain settings for your ChemiDoc XRS system.

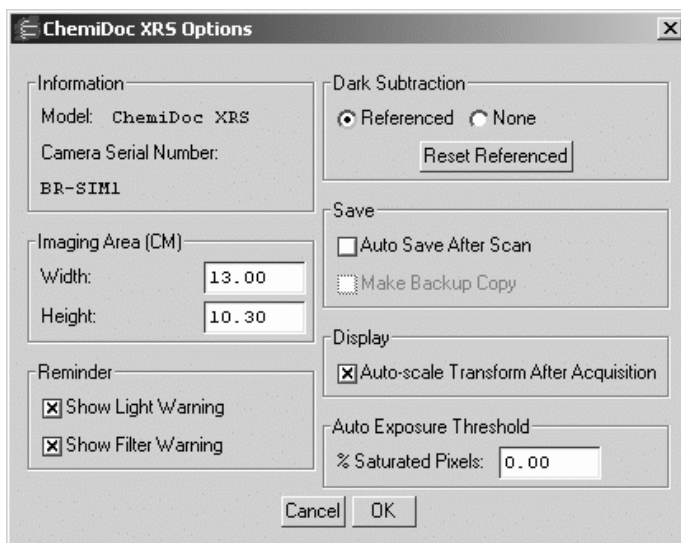


Fig.D-11. Available options in the ChemiDoc XRS acquisition window.

Click on OK to implement any changes you make in this box.

### D.5.a. Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turn determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

**Note:** Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

### D.5.b. Reminder

**Show Light Warning** - When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the ChemiDoc XRS acquisition window or when your system is "idle" for more than 5 minutes.

**Note:** If you are performing experiments that are longer than 5 minutes (e.g., chemiluminescence), this should be deselected.

**Show Filter Warning** - When this checkbox is selected, the software will warn you to move the filter slider either to chemi or the UV/White position.

### D.5.c. Dark Subtraction

At the time of installation a reference dark image file was generated and saved to your hard drive. The purpose of this file is to reduce dark current noise generated from the CCD. Dark current noise is typical of all CCDs and is a result of the accumulation of charge in the absence of light

The default time of the reference dark image is ten minutes. The default reference file is optimized for ChemiDoc XRS applications. If you select to change the reference dark image, ensure that the time of the reference dark image is of equal or greater duration than your typical sample image. When the sample image time is longer than the reference dark image time the ability to detect very faint samples may be compromised.

To reset the reference dark image time, select options in the ChemiDoc XRS acquisition window. Select Reset Referenced.

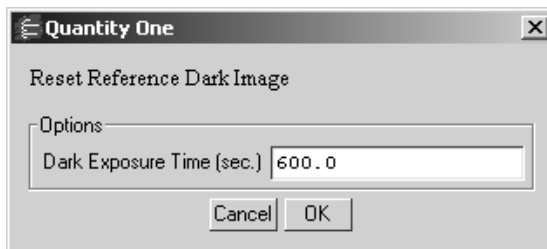


Fig.D-12. Reset Reference Dark pop-up box

Enter the desired time in seconds. Select OK. You will be prompted to close and re-open the acquisition window. Place the lens cap on the camera and close the door of the hood (See ChemiDoc XRS instruction manual). Select OK at the prompt and new reference dark images of the entered time interval will be taken for each of the five possible combinations of camera gain and bin settings. These will be retained as the reference dark images until the time is reset again.

### D.5.d. Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Backup images are identified by "Backup of" followed by the original image file name.

**Note:** Backup files are read only files. If you attempt to save a backup file, you will be prompted to give it a new name. This protects the backup file and preserves it from any changes.

### D.5.e. Auto-scale Transform

Auto-scale Transform after Acquisition allows the user the option of having the image automatically perform the Auto-Scale transform function upon completion of image acquisition. This eliminates the need to transform when the acquisition time was too short or the iris not opened enough.

To enable the auto-scale transform function, check the box labeled, Auto-Scale Transform after Acquisition in the Options dialog.

### D.5.f. Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.75 percent.

## D.6 Other Features

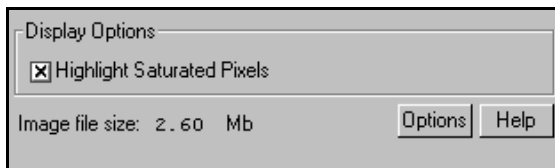


Fig.D-13. Other ChemiDoc XRS acquisition window features.

### Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

### File Size of Images

Image File Size shows the size of the image file you are about to create. This size is determined by the bin setting used for the selected application.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)



# Appendix E

## GS-800 Imaging Densitometer



Fig.E-1. GS-800 Imaging Densitometer

Before you can begin scanning images with the GS-800 Imaging Densitometer<sup>®</sup>, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

### PC Only: A Note About SCSI Cards

The GS-800 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-800, you must have a SCSI card installed in your PC. If

you have a PC with a Windows 98 or Windows ME operating system, you may also need to load the SCSI and WinASPI drivers that came with the card.

### Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the Ctrl key and select the name of the device from the File menu. The title of the GS-800 acquisition window will indicate that it is simulated.

**Note:** There is no simulated calibration for densitometers.

## E.1 GS-800 Acquisition Window

To begin acquiring images, go to the File menu and select GS-800. The acquisition window for the densitometer will open, displaying a control panel and a scanning window.



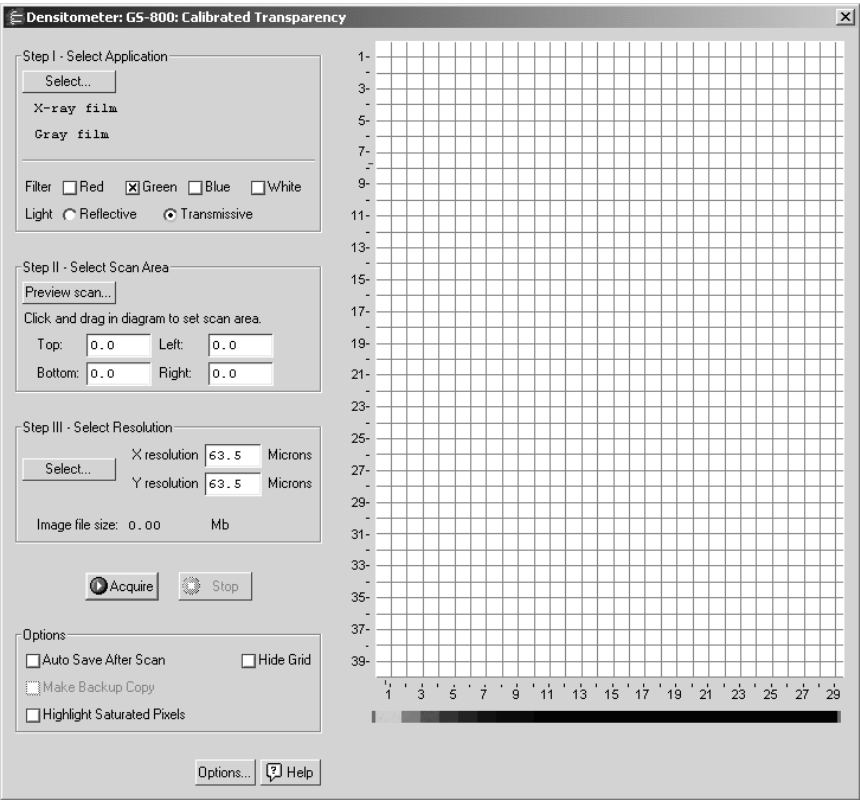


Fig. E-2. GS-800 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1-40 top to bottom and 1-30 left to right if the light source is reflective, and 1-40 top to bottom and 1-29 left to right if the light source is transmissive.

To hide the gridlines, select the Hide Grid checkbox under Options.

Below the main scanning window is the calibration strip window. Every time the densitometer calibrates, an image of the calibration strip will appear in this window.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are five basic steps to scanning an image using the GS-800:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Calibrate the instrument. (This is automatic, after you enter the step tablet values before you first scan after installation.)
5. Acquire the image.

## E.2 Step I. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

### Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the **Select** button under *Step I. Select Application*.

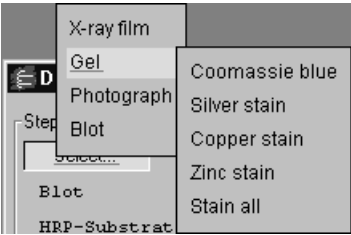


Fig. E-3. Example of the application tree in the GS-800 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the **Esc** key.

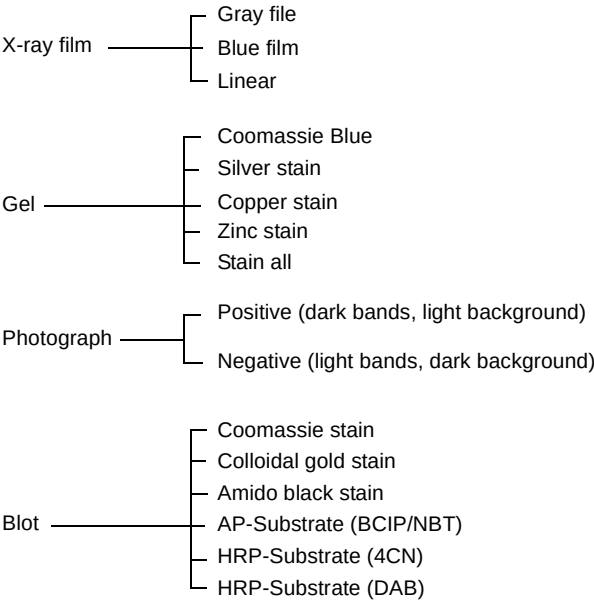


Fig. E-4. Applications available in the GS-800 acquisition window.

### Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.



Fig. E-5. GS-800 Custom Application controls.

Next to *Filter*, click on either the **Red**, **Green**, **Blue**, or **White** checkbox, or a combination of two of the first three (**Red-Green**, **Green-Blue**, **Red-Blue**).

| Filter Color | Wavelength | Application Examples                                      |
|--------------|------------|-----------------------------------------------------------|
| Red          | 595–750 nm | Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue |
| Green        | 520–570 nm | Coomassie R-250, Basic Fuchsin                            |
| Blue         | 400–530 nm | Crocein Scarlet                                           |
| White        | 400–750 nm | Silver Stains, Copper Stains, Film                        |

Fig. E-6. Examples of filter colors and applications.

Next to *Light*, select the appropriate light source.

- Select **Reflective** for scanning opaque mediums such as dried gels on filter paper, arrays, or photographs. The scanning dimensions in this mode are 30 cm x 40 cm.
- Select **Transmissive** for scanning transparent mediums such as films, gels, or slides. The scanning dimensions in this mode are 29 cm x 40 cm.

## E.3 Step II. Select Scan Area

### Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

### Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

**Note:** The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To reposition the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To resize the box, position your cursor on a box side and drag. The side you have selected will move.
- To redo the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

### E.4 Step III. Select Resolution

To select from a list of possible scanning resolutions, click on the Select button under Step III. Select Resolution. This will open the Select Scan Resolution dialog box.

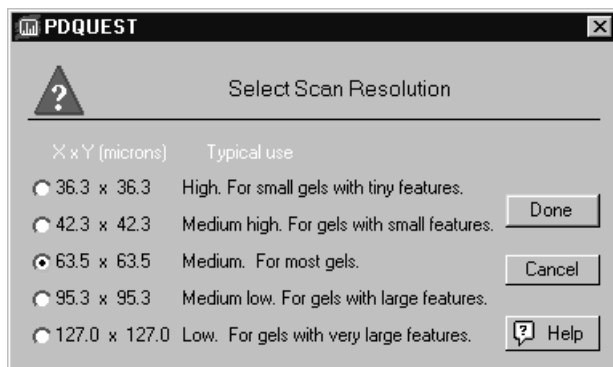


Fig. E-7. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

#### Specifying Your Own Resolution

If you select Oversample in the Options dialog box, you can specify your own resolution within the range of 32–169 micrometers. With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

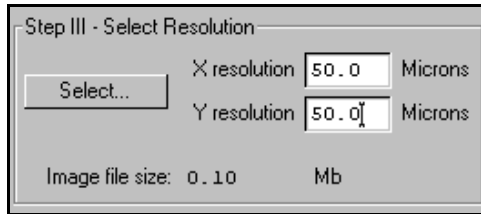


Fig. E-8. Entering a custom resolution (with Oversample selected).

### Image File Size

The size of the scan file for the selected resolution is listed next to Image File Size. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

## E.5 Calibration

The GS-800 automatically calibrates the densities of scanned images to ensure accuracy and reproducibility of results. The GS-800 has built-in step tablets for both transmissive and reflective scanning.

- The transmissive step tablet calibrates transmissive scans from 0 to 3.0 O.D.
- The reflective step tablet targets reflective scans from 0 to 2.0 O.D.

The first time you use the GS-800, you must select some settings to ensure accurate calibration. Click on the Options button in the main acquisition window. This will open the Densitometer Options dialog box.

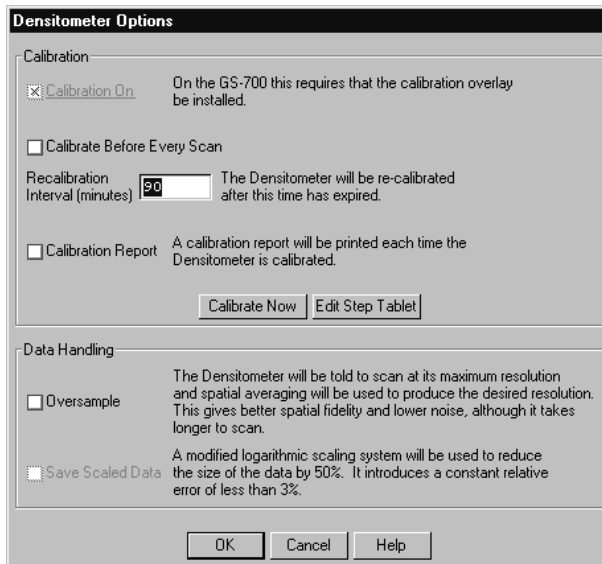


Fig. E-9. Densitometer Options dialog box.

Note that calibration is always on for the GS-800.

### E.5.a Step Tablet Values

The built-in transmissive and reflective step tablets on your densitometer have specific density values. Before scanning for the first time, you must enter the values for your transmissive step tablet into the software. You can use the default values in the software for the reflective step tablet.

#### Transmissive Step Tablet

Before you scan in transmissive mode for the first time, you must specify the values for your transmissive step tablet.

First, make sure that the Transmissive checkbox is selected in the main acquisition window, then click on the Options button. In the Densitometer Options dialog, click on the Edit Step Tablet button. The Step Tablet Values for GS-800 Transmissive dialog box will open.



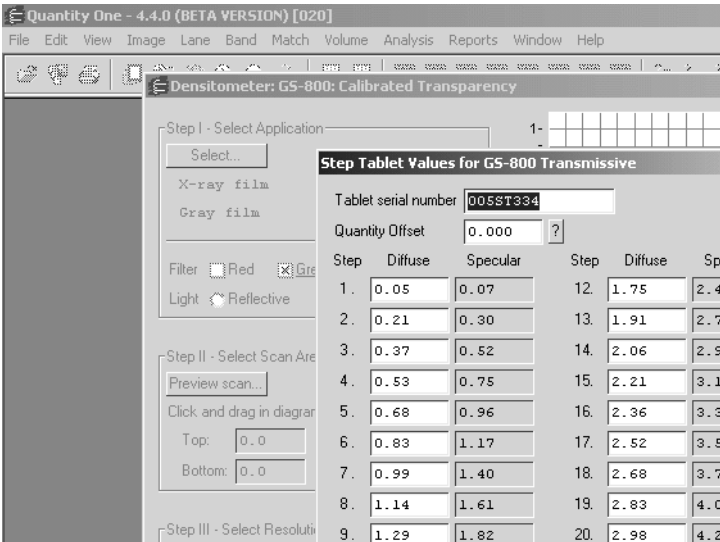


Fig. E-10. Step Tablet dialog box.

Attached to the outside of your GS-800, you will find a copy of the manufacturer’s printout of the diffuse density values for each segment of the transmissive step tablet, as well as a serial number for the tablet. These exact density values must be entered into the software to associate a correct density value with each step on the step tablet.

**Note:** Scanning in transmissive mode with incorrect step tablet values can cause significant errors in the reported densities of your scans.

First, type the serial number for the tablet into the Tablet Serial Number field.

The Quantity Offset field does not apply in the GS-800 in transmissive mode. This value should remain at zero.

Next, enter the step tablet values into the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet. The density values do not need to be reentered each time you calibrate.

When you are finished entering the transmissive step tablet values, click on OK.

### Reflective Step Tablet

For the reflective step tablet, it is recommended that you use the default target values in the software. Although you can edit these values based on your own measurements and testing, the default values have been preset based on careful testing and review.

To view or edit the default reflective values, make sure that the **Reflective** checkbox is selected in the main acquisition window, then click on the **Options** button. In the **Densitometer Options** dialog, click on the **Edit Step Tablet** button. The **Step Tablet Values for GS-800 Reflective** dialog box will open.

When you are finished viewing/editing the reflective step tablet values, click on **OK**.

### Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular/diffuse O.D. ratio. This conversion does not affect quantitation.

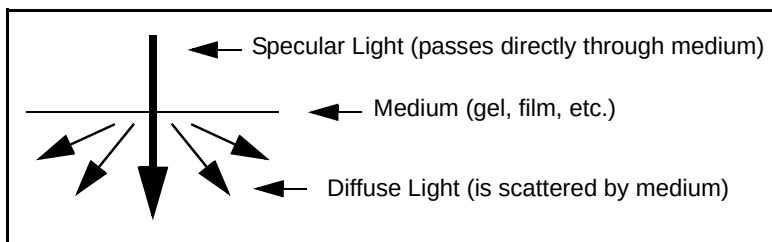


Fig. E-11. Specular and diffuse density

Diffuse density values are converted to specular optical density units according to the following formula:

$$\text{Specular OD} = 1.4 \cdot \text{Diffuse OD}$$

## E.5.b Other Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the **Calibrate Now** button (in the *Densitometer Options* dialog box).

You can also specify how often you want the GS-800 to automatically recalibrate. Either click on the **Calibrate Before Every Scan** checkbox or specify a period between automatic recalibrations (in minutes) in the *Recalibration Interval* field.

**Note:** The scanner will automatically recalibrate each time you change your filter or your reflective/transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

To print out a calibration report each time the densitometer calibrates, click on the **Calibration Report** checkbox.

## E.6 Acquire the Image

**Note:** Before scanning in transmissive mode, make sure the white balance region of the scanning area is not covered or obstructed in any way.

To begin to scan, click on the **Acquire** button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the **Stop** button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can analyze and save it.

**Note:** The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected **Auto Save After Scan**, the file will not be saved until you perform a **Save** or **Save As** operation.

# E.7 Other Options

## Oversample

This feature allows you to scan at the maximum resolution of the scanner and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the **Options** button in the acquisition window and select the **Oversample** checkbox in the options dialog.

With oversampling on, you can specify your own resolution within the range of the densitometer by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

## Auto Save After Scan

To automatically save any scan you create, select the **Auto Save After Scan** checkbox.

With this checkbox selected, when you click on **Acquire**, a **Save As** dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the **Save** button.

## Make Backup Copy

If you have selected **Auto Save After Scan**, you can also automatically create a backup copy of any scan you create.

Click on the **Make Backup Copy** checkbox. With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Backup images are identified by "Backup of" followed by the original image file name.

**Note:** Backup files are read only files. If you attempt to save a backup file, you will be prompted to give it a new name. This protects the backup file and preserves it from any changes.

### Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the **Transform** command on the main toolbar or *Image* menu.

### Hide Grid

To hide the gridlines in the scanning area window, click on the **Hide Grid** checkbox.



# Appendix F

## Personal Molecular Imager FX



Fig.F-1. Personal Molecular Imager FX

Before you can begin acquiring images using the Personal Molecular Imager<sup>®</sup> FX, the instrument must be properly installed and connected with the host computer. See the Personal FX hardware manual for installation, startup, and operating instructions.

**Note:** The Personal FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

### PC Only: A Note About SCSI Cards

The Personal FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Personal FX, you must have a SCSI card installed in your PC. If you have a PC with a Windows 98 or Windows ME operating system, you may also need to load the SCSI and WinASPI drivers that came with the card.

### Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

## F.1 Personal FX Acquisition Window

To acquire images using the Personal FX, go to the File menu and select Personal FX.... The acquisition window for the imager will open, displaying a control panel and the scanning area window.



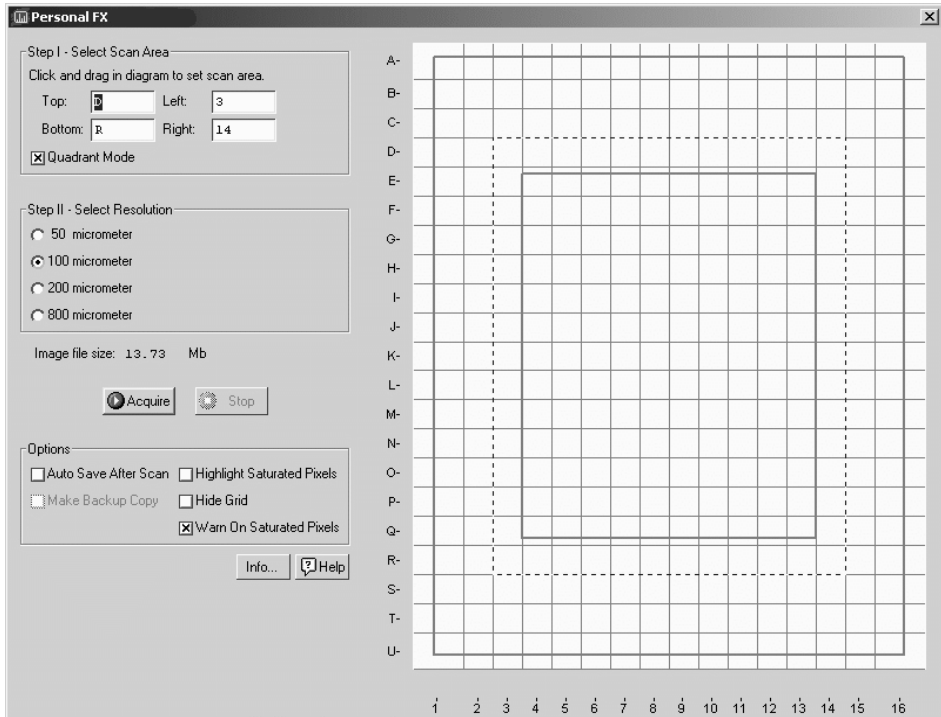


Fig.F-2. Personal FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Personal FX:

1. Select the scan area

2. Select the resolution
3. Acquire the image

### F.2 Step I. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

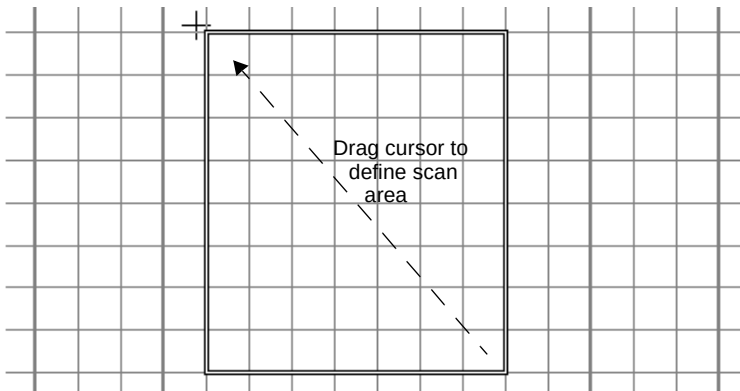


Fig.F-3. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

### F.3 Step II. Select Resolution

The Personal FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

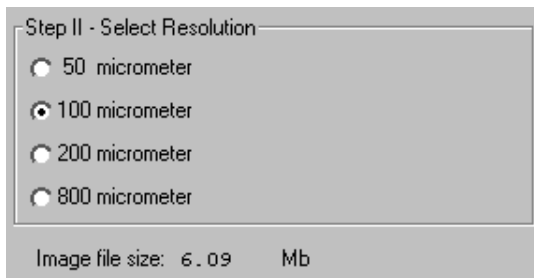


Fig.F-4. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution should be used for typical gels and arrays.
- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

### File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

## F.4 Acquire the Image

Once you have selected your scan area and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

**Note:** If the image you are scanning has more than 10 saturated pixels, you will receive a warning message.

### Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image.

You can then save and analyze the image using the standard menu and toolbar functions.

## F.5 Options

### Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

**Note:** In PDQuest, this option is preselected and cannot be turned off. All images must be automatically saved when acquired.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

### Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Backup images are identified by "Backup of" followed by the original image file name.

**Note:** Backup files are read only files. If you attempt to save a backup file, you will be prompted to give it a new name. This protects the backup file and preserves it from any changes.

### Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

### Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

### Warn on Saturated Pixels

The Warn on Saturated Pixels option is checked by default. Uncheck this box to disable this warning.

# Appendix G

## Molecular Imager FX Family (FX Pro, FX Pro Plus and Molecular FX)



Fig. G-1. Molecular Imager FX

Before you can begin acquiring images using any of the Molecular Imager FX family of products, the particular instrument must be properly installed and connected with the host computer. See the corresponding FX hardware manual for installation, startup, and operating instructions.

## Quantity One User Guide

---

**Note:** The FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

### PC Only: A Note About SCSI Cards

The FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the FX, you must have a SCSI card installed in your PC. If you have a PC with a Windows 98 or Windows ME operating system, you may also need to load the SCSI and WinASPI drivers that came with the card.

### Simulation Mode

Any of the imaging device acquisition windows can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, hold down the CTRL key and select the name of the device from the File menu. The title of the acquisition window will indicate that it is simulated.

## G.1 FX Acquisition Window

To acquire images using the FX, go to the File menu and select FX.... The acquisition window for the imager will open, displaying the control panel for the imager and the scanning area window.



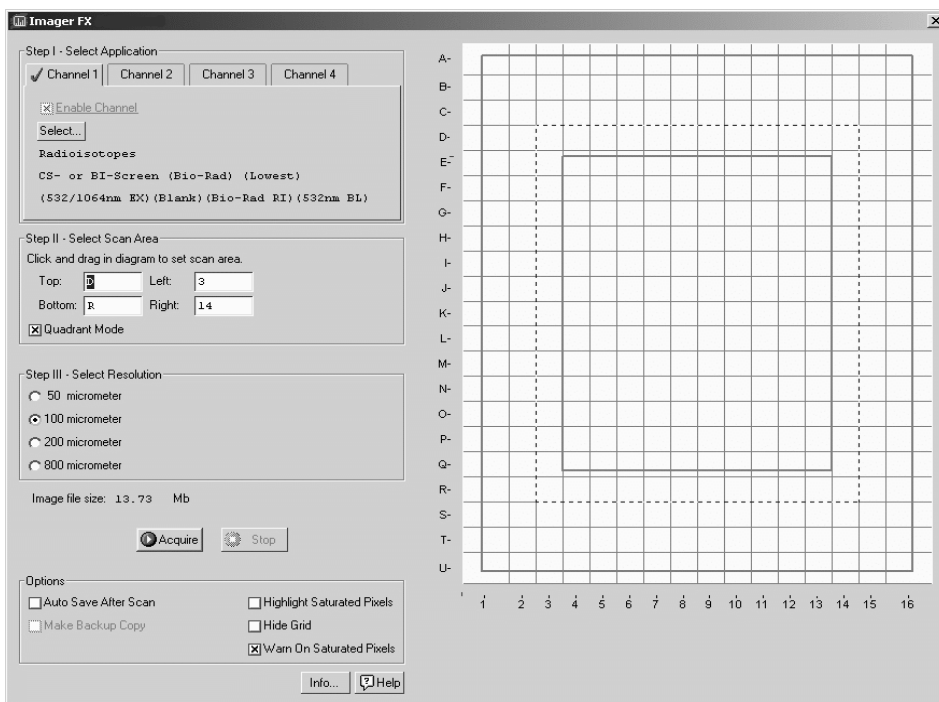


Fig. G-2. FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the FX:

1. Select the application(s).
2. Select the scan area.

3. Select the resolution.
4. Acquire the image.

## G.2 Step 1. Select Application

In Step 1, you select the appropriate filters and other scanning parameters for the type of gel, blot, plate, or other object that you are imaging. Each set of parameters you select is called an “application.”

The FX now supports multi-channel sequential scanning. This allows you to automatically scan the same object using up to four different applications (e.g., to detect different types of stains on the same gel).

First you select a channel, then you select the application under that channel.

### G.2.a Selecting a Channel

The four channels are accessed using the tabs under Step 1. Channel 1 is always enabled—that is, the FX will always scan using the application settings selected under Channel 1 first.

To enable any of the remaining channels, click on a channel tab, then select the Enable Channel checkbox and select the application for that channel as described in the following section.

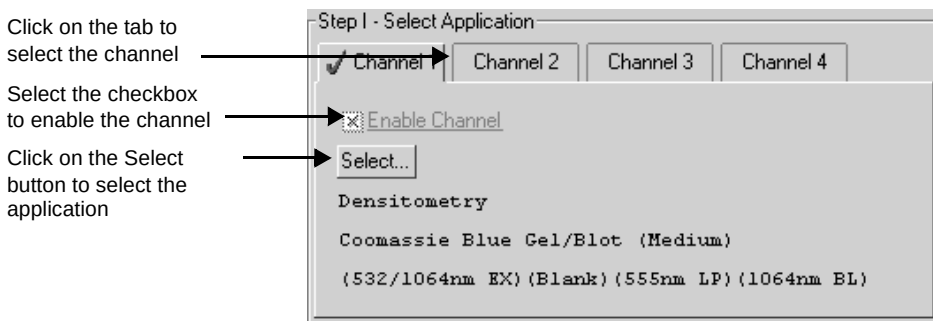


Fig. G-3. Enabling Channel 2.

Enabled channels have a green check mark on their tabs.

You do not need to enable Channels 2–4 in sequence. For example, you can set up your four most common applications using the different channels, but only enable Channel 4 for a particular gel. Channel 1 would be scanned first, Channel 4 second.

Enabled channels are scanned sequentially. The separate scans are displayed and saved as separate images. The total scanning time depends on the number and type of enabled applications.

If you have selected Auto Save After Scan, the image created using Channel 1 will have the base file name, and images created using subsequent channels will have the base file name plus a version number. (v. 2, v. 3, v. 4). Note that the image version number does not necessarily correspond to the channel number. For example, if you scanned an image using only Channels 1 and 4, the image created using Channel 4 will be saved as version 2 (v. 2).

## G.2.b Selecting an Application

To select an application (i.e., the appropriate filter and other parameters for the type of object you are imaging), click on the Select button under a channel tab.

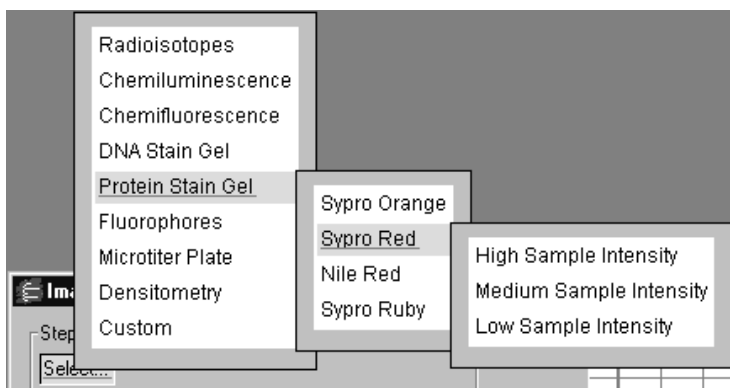


Fig. G-4. Example of an application tree: Ethidium Bromide gel.

## Standard Applications

The standard applications and associated settings are listed in a tree that expands from left to right. When you select a standard application, the software automatically selects the appropriate filter(s) in the FX for that particular application. Some applications are inactive in the FX Pro and FX Pro Plus systems.

### Standard FX Applications

| Category          | Application                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------|
| Radioisotopes     | CS- or BI-Screen (Bio-Rad) <sup>1</sup><br>K-Screen (Kodak) <sup>2</sup><br>Fuji-Screen <sup>2</sup> |
| Chemiluminescence | Chemi-Screen (Bio-Rad) <sup>1</sup>                                                                  |
| Chemifluorescence | ECL-Plus<br>Attophos                                                                                 |
| DNA Stain Gel     | Ethidium Bromide<br>Sybr Green I & II<br>Sybr Gold                                                   |

Standard FX Applications

|                   |                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| Protein Stain Gel | Sypro Orange<br>Sypro Red<br>Nile Red<br>Sypro Ruby                                                            |
| Fluorophores      | Alexa 488<br>Alexa 532<br>Alexa 546<br>FITC<br>FAM<br>CY3<br>HEX<br>R6G<br>Texas Red                           |
| Microtiter Plate  | DNA (Sybr Green I)<br>Protein (Nano Orange)<br>ssDNA (Oligreen)<br>DNA (Picogreen)<br>B-Gal (FDG)<br>GUS (FDG) |
| Densitometry      | Coomassie Blue Gel/Blot<br>Copper Stain Gel/Blot<br>Silver Stain Gel/Blot<br>X-Ray Film (Grey Type)            |

<sup>1</sup>Not supported in the FX Pro and FX Pro Plus systems.

<sup>2</sup>Not supported in the FX Pro systems.

First select your general application, next select the particular stain or medium you are using, and finally (if appropriate) select the intensity of your samples.

**Note:** Some applications require an external laser. If you choose one of these without having an external laser attached, you will receive a warning.

To exit the tree without selecting, press the ESC key.

Your selection and settings will be displayed below the Select button.

### Sample Intensity

Many FX applications require that you select a sample intensity (High, Medium, or Low) from the application tree. This is simply a rough estimate of how much sample is visible in your gel or other object.

If you are unsure of the level of intensity of your sample, you can always select a level, capture an image, then adjust the level and capture another image.

For example, if you select Low Sample Intensity and the resulting image has too many saturated pixels, you will receive a warning message. Simply change the setting to Medium Sample Intensity and rescan. If you select High Sample Intensity and the resulting image is too faint, select Medium or Low and rescan.

### Custom Applications

If your application is not listed, if you want to use user-installed filters, or if you want to use an external laser, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

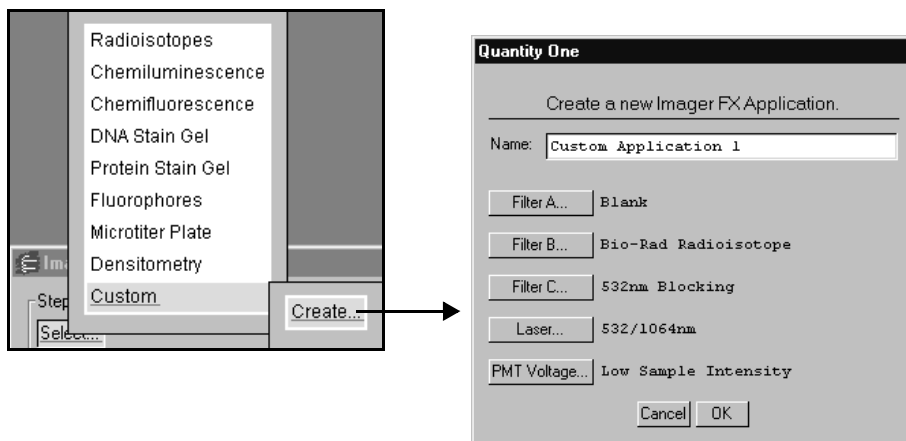


Fig. G-5. Creating a custom application.

To select a filter (including user-defined) or filter combination, click on the buttons for Filters A, B, and C, and make your choice from each pop-up list.

**Note:** Filter wheel C is not available in the FX Pro or the FX Pro Plus systems.

**Note:** The user-defined filters (User1, User2, etc.) cannot be renamed in the pop-up list, so be sure to remember which filter you insert into each position in the FX (i.e. 690 nm filter, 605 nm BPfilter).

To use an external laser, click on the Laser button and select it from the pop-up list (i.e. 488 nm or 635 nm laser). Otherwise, use the default internal laser (532/1064nm).

**Note:** The dual laser (532/1064 nm) is not available in the FX Pro or the FX Pro Plus systems.

Click on the PMT Voltage button to select a standard voltage for your custom application or create a custom PMT voltage.

To select a custom voltage, click on the Custom option. In the dialog box, adjust the slider to select a PMT voltage as a percentage of the maximum.

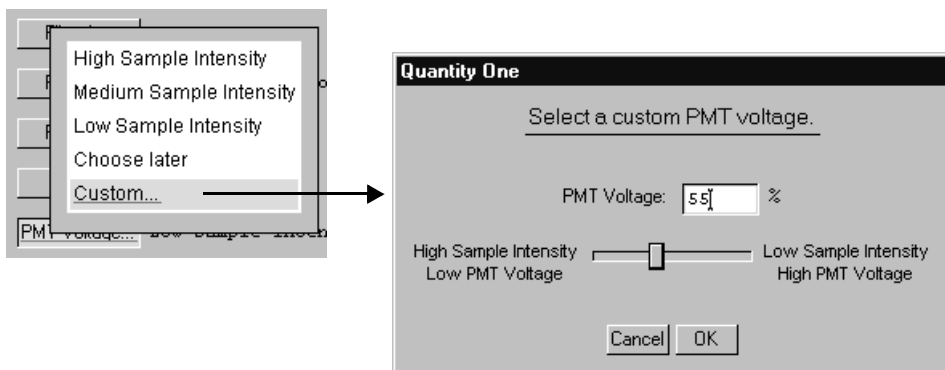


Fig. G-6. Selecting a custom PMT voltage.

**Note:** For voltages above 80% of maximum, you will receive a warning message that the high voltage could damage the PMT.

If you select Choose Later from the list of PMT voltages, the choices of sample intensity will be displayed when you select your custom application.

Finally, enter a name for your application in the Name field and click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

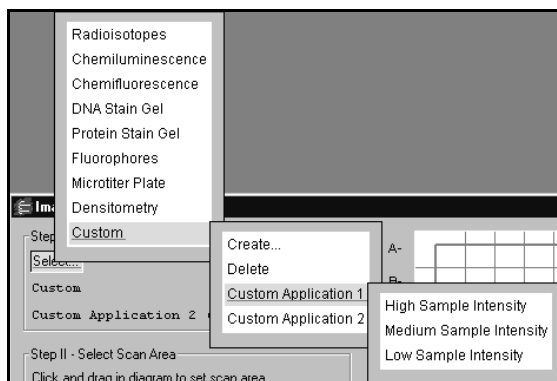


Fig. G-7. Selecting a custom application.

You can edit a custom application by selecting **Custom**, **Edit**, and the name of the application. You can also use this feature to create a new custom application from an existing one.

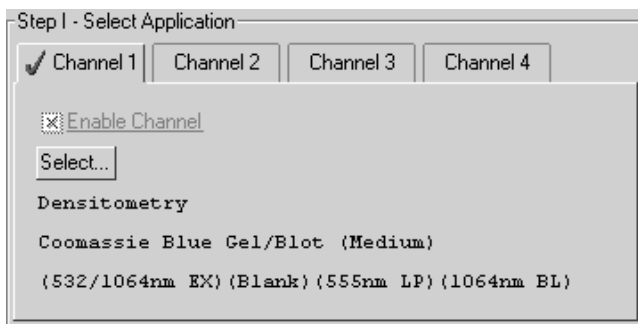


Fig. G-8. Application selection and settings.



Once you select an application, the application name and settings appear below the Select button.

## G.3 Step II. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

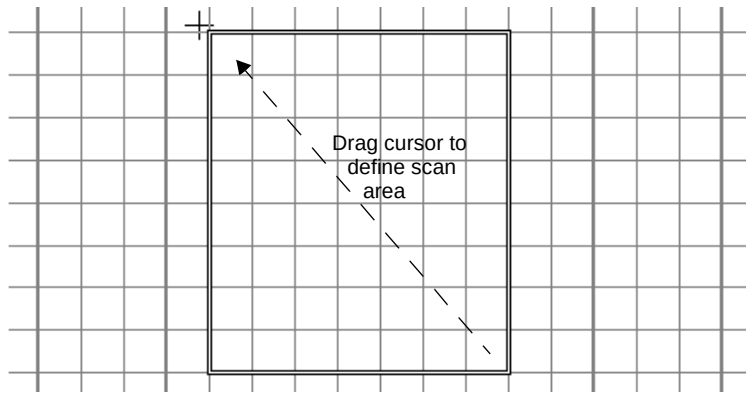


Fig. G-9. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

## G.4 Step III. Select Resolution

The FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

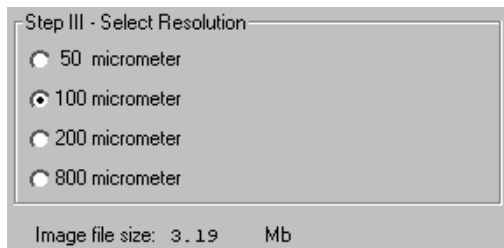


Fig. G-10. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution is useful for typical gels and arrays.
- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

## File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

## G.5 Acquire the Image

Once you have selected your application, scan area, and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

**Note:** If the image you are scanning has more than 10 saturated pixels, you will receive a warning message. If this happens, you can go back and select a higher sample intensity in the application tree.

## Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image.

You can then save and analyze the image using the standard menu and toolbar functions.

# G.6 Options

## Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

If you are scanning using multiple channels, the image created using Channel 1 will be saved using the base file name, and images created using subsequent channels will have the base file name plus a version number. (v. 2, v. 3, v. 4). Note that the image version number does not necessarily correspond to the channel number. For example, if you scanned an image using only Channels 1 and 4, the image created using Channel 4 will still be saved as version 2 (v. 2).

## Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Backup images are identified by "Backup of" followed by the original image file name.

**Note:** Backup files are read only files. If you attempt to save a backup file, you will be prompted to give it a new name. This protects the backup file and preserves it from any changes.

## Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Image > Transform command.

### Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

### Warn on Saturated Pixels

The Warn on Saturated Pixels option is checked by default. Uncheck this box to disable this warning.



# Appendix H

## VersaDoc



Fig.H-1. VersaDoc.

Before you can acquire images using the VersaDoc Imaging System, you need to install the Roper Scientific interface adapter and its associated device drivers in your computer. The driver installation procedures for Windows operating systems include pre-installation and use of the Windows Hardware Wizards.

**Note:** *Important!* Please read and follow the driver installation procedures contained in the README documents before you install the VersaDoc camera interface adapter. You can find the README documents in the VersaDoc Drivers folders on The Discovery Series (TDS) CD.

## Quantity One User Guide

---

The CD also contains Roper Scientific camera interface driver installation instructions for Macintosh computers.

See the VersaDoc hardware manual for further installation, startup, and operating instructions for the VersaDoc Imaging System.

### H.1 VersaDoc Acquisition Window

To acquire images using the VersaDoc, go to the File menu and select VersaDoc.... The acquisition window for the instrument will open. The acquisition window opens, displaying a control panel and an image display window.

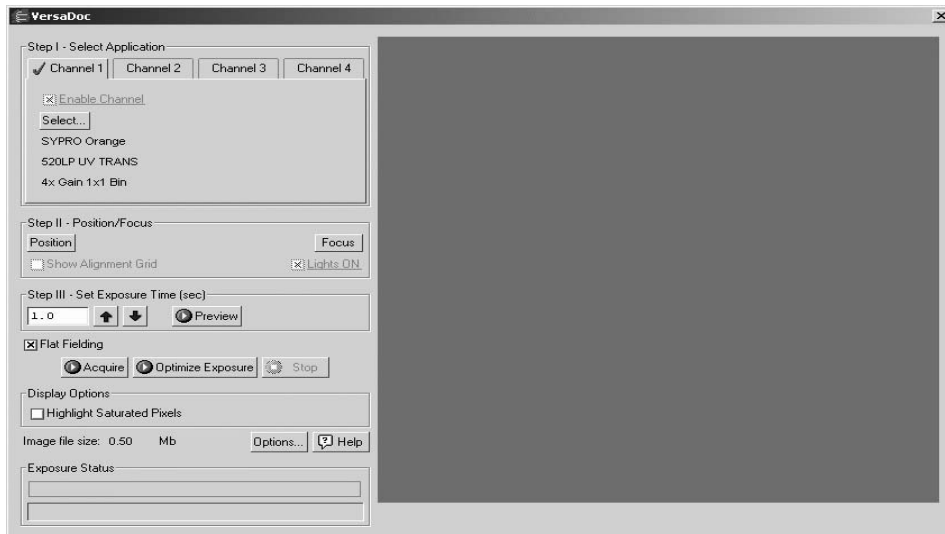


Fig.H-2. VersaDoc acquisition window.

When the VersaDoc window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the VersaDoc:



1. Select the application.
2. Position and focus the object to be imaged.
3. Set the exposure time.
4. Acquire the image.

## H.2 Step 1. Select Application

In Step 1, you select the appropriate filters and other imaging parameters for the type of gel, blot, plate, or other object that you are imaging. Each set of parameters you select is called an application.

The VersaDoc supports multichannel sequential imaging. This allows you to automatically image the same object using up to four different applications (e.g., to detect different types of stains on the same gel).

First you select a channel, then you select the application under that channel.

### H.2.a. Selecting a Channel

**Note:** Illumination Flat Fielding will be disabled if you use multiple channels.

The four channels are accessed using the tabs under Step 1. Channel 1 is always enabled—that is, the VersaDoc will always capture an image using the application settings selected under Channel 1 first.

To enable any of the remaining channels, click on a channel tab, then select the Enable Channel checkbox and select the application for that channel as described below.

Enabled channels have a green check mark on their tabs.

**Note:** If you use the Optimize Exposure feature, only the application under the selected channel tab will be imaged; other channels, even if enabled, will be ignored.

You do not need to enable Channels 2–4 in sequence. For example, you can set up your four most common applications using the different channels, but only enable Channel 4 for a particular gel. Channel 1 would be scanned first, Channel 4 second.

## Quantity One User Guide

---

Enabled channels are imaged sequentially. The separate scans are displayed and saved as separate images. The total imaging time depends on the number and type of enabled applications.

If you have selected Auto Save After Scan, the image created using Channel 1 will have the base file name, and images created using subsequent channels will have the base file name plus a version number. (v. 2, v. 3, v. 4). Note that the image version number does not necessarily correspond to the channel number. For example, if you captured an image using only Channels 1 and 4, the image created using Channel 4 will be saved as version 2 (v. 2).

### H.2.b. Selecting an Application

To select an application (i.e., the appropriate filter and illumination source for the type of object you are imaging), click on the Select button under a channel tab.

### Standard Applications

The standard applications and associated settings are listed in a tree that expands from left to right. When you select an application, the software automatically sets the appropriate filter in the VersaDoc for that particular application.

First select your general application, then select the particular stain or medium you are using. When you select the stain or medium, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none), light type (UV, white, or none), and light source (Trans, Epi, or neither) in the VersaDoc for that particular application.

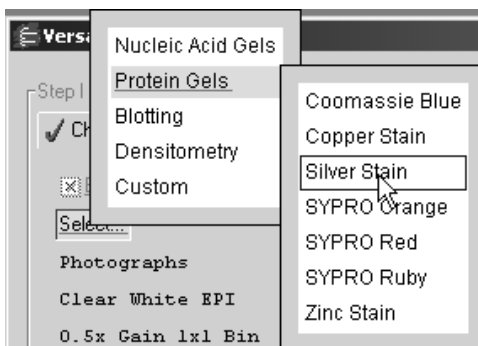


Fig.H-3. The application tree in the VersaDoc acquisition window.

**Note:** If you select an application that requires trans white illumination, you will need to place the *white light conversion screen* on the sample stage area.

To exit the tree without selecting, press the Esc key or click outside the dialog box.

Your selection will be displayed below the Select button.

### Custom Applications

If your application is not listed, if you want to use a user-installed filter, or if you want to set the gain and bin settings manually, you can create and save your own custom applications.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

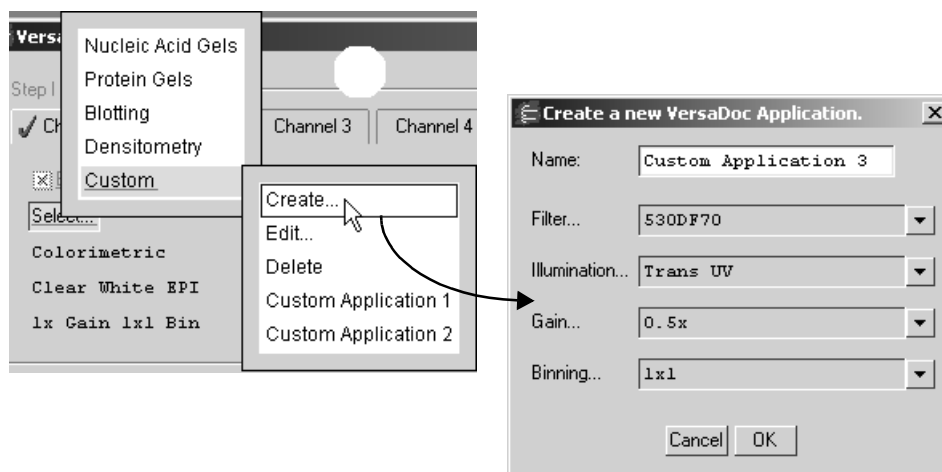


Fig.H-4. Creating a new custom application.

Enter a name for your application in the Name field.

Next, click on the buttons next to the Filter, Illumination, Gain and Binning fields to change these settings.

**Note:** Note that if you select trans white illumination, you will need to place the *white light conversion screen* on the sample stage.

A higher Binning setting (2x2, 3x3) provides optimal sensitivity for low-light applications such as chemiluminescence. In this mode, the pixels in the camera are “binned” (e.g., four pixels are combined into one) to increase the amount of signal per pixel without increasing noise. Note that combining the pixels results in a reduction in the resolution of the image.

Selecting a higher Gain (4x) provides higher sensitivity without reduced resolution; however, noise will also increase. This is useful for faint signals (bright spots will saturate). Selecting a lower gain (0.5x) is useful for brighter images that tend to saturate. A gain setting of 1x provides the greatest dynamic range.

Click OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created.

You can edit a custom application by selecting Custom, Edit, and the name of the application. You can also use this feature to create a new custom application from an existing one.

You can delete a custom application by selecting Custom, Delete, and the name of the application.

### H.3 Step II. Position/Focus

Before positioning and focusing the image, select the application that you will be using under Step I.

Note that the optimal position and focus will be different for a filtered image versus an unfiltered one. (All standard applications except Chemiluminescence use a filter.)

#### Position

After you have selected your application, you are ready to center your gel or other object within the camera frame. To do so, click on the Position button. The VersaDoc will begin capturing a “live” image and updating it every second.

With the Position button selected, study the image in the acquisition window while you position your object in the center of the sample stage. If you have a zoom lens on the camera, you can adjust the magnification while you position. (See the VersaDoc User Manual for details on positioning.)

While you are positioning, you can select the Show Alignment Grid checkbox to display a target grid overlay on the image.

When you click on Position, the light inside the camera box automatically switches on. To turn the light off while positioning, deselect the Light On checkbox.

When you are finished positioning, click on the Stop button.

#### Focus

**Note:** Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the

desired setting. See Table 1, “Recommended Exposure Times and Lenses,” on page 9.

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while adjusting the focus on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.) (See the VersaDoc User Manual for details on focusing.)

When you click on Focus, the light inside the camera box automatically switches on. To turn the light off while positioning, deselect the Light On checkbox.

When you are finished focusing, click on the Stop button.

## H.4 Step III. Set Exposure Time

The “exposure time” is the period of time an image is integrated on the CCD. The effect is analogous to exposing photographic film to light.

### Setting an Exposure Time

Different applications have different optimal exposure times. If you are imaging using multiple channels, you can select a different exposure time for each channel. First select the channel using the tabs, then select the appropriate exposure time for that channel.

See Table 1, “Recommended Exposure Times and Lenses,” on page 9.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.



Fig.H-5. Selecting an exposure time.

The following table provides recommended exposure times for various applications

**Table 1: Recommended Exposure Times and Lenses**

| Sample                                                                                           | Recommended Exposure | Lens & Filter | Accessories Used                       |
|--------------------------------------------------------------------------------------------------|----------------------|---------------|----------------------------------------|
| Fluorescent Stain Gel                                                                            | 3–30 sec.            | Zoom/IR       | None                                   |
| Fluorescence End-Label Gel                                                                       | 30 sec.–5 min.       | Zoom/IR       | None                                   |
| Fluorescent Blot                                                                                 | 0.5–5 sec.           | Zoom/IR       | Sample/Chemi Tray (if sample is small) |
| Chemifluorescent Blot                                                                            | 0.5–5 sec.           | Zoom/IR       | None                                   |
| Colorimetric Gel                                                                                 | 0.1–1 sec.           | Zoom/IR       | White light conversion screen          |
| Colorimetric Blot                                                                                | 0.1–1 sec.           | Zoom/IR       | Sample/Chemi Tray (if sample is small) |
| X-ray film                                                                                       | 0.1–1 sec.           | Zoom/IR       | White light conversion screen          |
| Weak Chemiluminescence <sup>1</sup>                                                              | 5–10 min.            | 50 mm         | Sample/Chemi Tray (if sample is small) |
| Strong Chemiluminescence <sup>1</sup>                                                            | 10 sec.–2 min.       | 50 mm         | Sample/Chemi Tray (if sample is small) |
| <sup>1</sup> For chemi applications, the 50mm lens is recommended. Always remove the 660 filter. |                      |               |                                        |

**Note:** For most applications, you can select an exposure time, capture an image, study it, and then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For chemiluminescent samples, which degrade over time and emit low levels of light, select a high exposure time initially or use the Optimize Exposure command described on page 12.

### Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button to create a preview exposure and display it in the acquisition window.

**Note:** The camera on the VersaDoc must be at the correct operating temperature before capturing images. The temperature adjustment can take several minutes after the camera is turned on, depending on your VersaDoc model and the ambient room temperature. See your hardware manual for details.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image. The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

## H.5 Acquire the Image

You can acquire a single exposure for each channel based on the time selected in Step III (Set Exposure Time), or take a series of exposures for a particular channel over a specified interval (Optimize Exposure).

### Illumination Flat Fielding

For applications using the UV or white light transillumination, you should use the appropriate reference plate to ensure a uniform intensity in the image. This will compensate for normal variations in image pixel intensity that occur with a transilluminating light source.



To enable this feature, select the Flat Fielding checkbox.

**Note:** Flat fielding is only available in single channel, single exposure mode. If you are using multiple channels, or take multiple exposures using the Optimize Exposure feature described below, Flat Fielding will be disabled.

*UV Illumination Flat Fielding:* When you first select the Flat fielding checkbox, and then acquire an image using the UV transilluminator, you will be prompted to remove your sample and place the fluorescent reference plate on the VersaDoc sample stage (see the VersaDoc User Manual). A reference image of the plate will be acquired and saved on your computer hard drive. The reference image will be applied to the sample image to generate a Flat Field corrected image.

*White Illumination Flat Fielding:* When you first select the checkbox, and then acquire an image using the white light conversion screen, you will be prompted to remove your sample and collect an exposure of the conversion screen. A reference image of the screen will be acquired and saved on your computer hard drive. The reference image will be applied to the sample image to generate a Flat Field corrected image.

For subsequent UV or white light trans exposures, you will be prompted to either use the appropriate saved Flat Fielding image or acquire a new one. Any changes in light source, filter, or lens setting will require the acquisition of a new Flat Fielding reference image.

### Acquire

Click on the Acquire button to capture a single image for each enabled channel. An exposure will be taken for each enabled channel based on the time selected for that channel in Step III.

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the acquisition window.

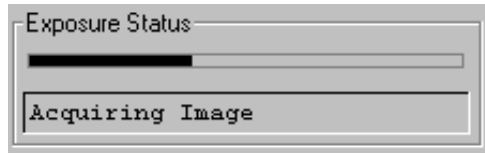


Fig.H-6. Exposure Status bar when acquiring an image.

Depending on which dark subtraction type you have selected under Options (see section H.6, Options), a dark image may be acquired immediately following image acquisition.

If you want to stop an acquisition that is in progress, click on the Stop button. The current acquisition will be terminated. If multiple channels are selected, you must click on the Stop button once per channel to stop all acquisitions.

After an image has been acquired, a separate window will pop up containing the new image. You can then analyze the image using the analysis functions.

### Optimize Exposure

Optimize Exposure allows you to specify an interval over which a series of progressively longer exposures are taken. All exposures are then displayed on the screen, and you can choose the one that provides the best image.

**Note:** Multiple exposures will be taken for only the selected multichannel tab. Other channels, even if enabled, will not be used. Illumination Flat Fielding will be disabled if you are using Optimize Exposure.

Click on the Optimize Exposure button. A settings dialog box will open in which you can specify the total exposure time, starting exposure time, and number of exposures. (The specified number of exposures will be taken at regular intervals between the starting exposure time and the total exposure time.)

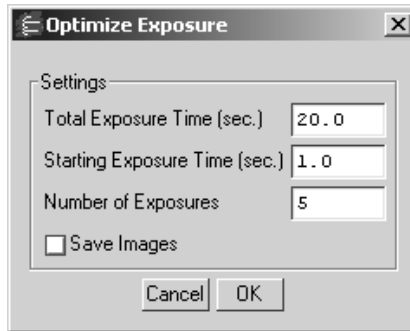


Fig.H-7. Optimize Exposure dialog.

**Note:** You should specify no more than 10 exposures in the Optimize Exposure dialog, to avoid excessive build up of image background in later exposures. The fewer the exposures, the less background will be added to the image. See the Release Notes for additional instructions on reducing background in images captured using Optimize Exposure.

Select the Save Images checkbox if you want to automatically save each exposure as it is taken.

Click on OK in the settings dialog to begin taking exposures. If you selected Save Images, a Save dialog box will open in which you can specify the base file name and location of the exposure files. When you click on Save, the exposures will be taken.

The specified number of exposures will be taken at equal intervals between the starting exposure time and total exposure time. The exposure status bar will show the progress of each exposure.

Depending on which dark subtraction type you have selected under Options, a dark image may be acquired immediately following each exposure.

When each exposure is complete, an image window containing that exposure will open in front of the VersaDoc window. Subsequent exposures are tiled in front of the VersaDoc window.

Note that the first exposure will have the base file name; the default base file name is the computer user name and a time stamp. Each subsequent exposure will have a version number (v2, v3, v4, etc.) appended to the base file name. (If you are using the

default base file name, the time stamp may change in the course of the series of exposures; in this case, the base file name will change and the version numbering will reset for subsequent exposures.)

The highest version number will be the final exposure. If you did not elect to auto-save the exposures as they were created, then each image will be unsaved.

To stop the image acquisitions, click on the Stop button.

**Note:** Exposures captured before stopping will be displayed in image windows.

Study the different images and select the best exposure(s) to keep.

## H.6 Options

Click on the Options button to open the Options dialog box.

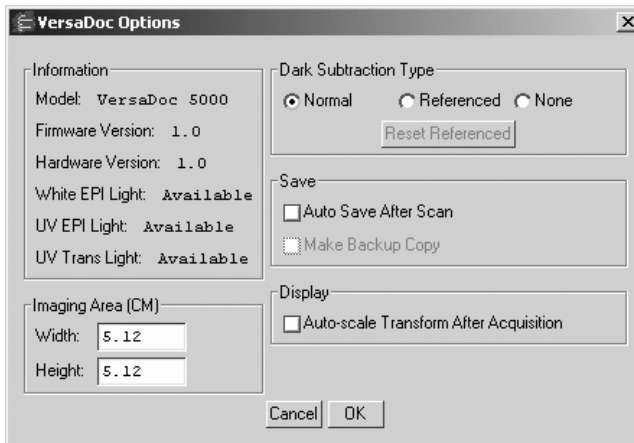


Fig.H-8. Options dialog box.

### H.6.a. Dark Subtraction Type

All CCD cameras accumulate electrons that produce a signal that is indistinguishable from light. This “dark current” adds to the noise in your images, particularly for long

exposures. In most cases, you will want to subtract this dark current from your images.

The settings for subtracting the dark current are in the Options dialog box. Click on the Options button in the acquisition window to open this dialog.

### Normal

The Normal option button selects the default dark subtraction type. In this mode, after you acquire an image, a “dark” image of the same exposure length will be taken, and this will be subtracted from your image.

The progress of the dark exposure will be displayed in the Exposure Status bar following the regular image exposure.

In Normal mode, a dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark exposure will be taken.

### Reference

If you do not want to perform a dark exposure with each acquisition, you can take a “reference” dark exposure that will be saved and subtracted from all subsequent acquisitions. Click on the Referenced button to activate this feature.

The first time you acquire an image after selecting this option, the VersaDoc will take a dark exposure that will be saved and used to subtract the dark current from all subsequent acquisitions.

**Note:** The VersaDoc 1000 will take a 60-second reference dark exposure; the VersaDoc 3000 and 5000 will take a 180-second reference dark exposure.

For image exposures that are longer or shorter than the reference dark, the reference dark will be scaled accordingly and then subtracted. It is recommended that the reference dark exposure time be equal to or greater than the sample exposure time. You can change the default reference dark exposure time using the Reset Reference button (see below).

If you deselect the Referenced button and then reselect it, the old reference dark exposure will still be available.

Separate reference dark exposures will be taken for images that have different levels of binning or gain. Once you have created a reference dark for each level of binning or gain, the appropriate reference dark will be used according to the settings of your selected application.

### Reset Reference

If you would like a reference dark with an exposure time that more closely matches that of your typical scans, click on the Reset Reference button.

A pop-up box will prompt you to enter a new dark exposure time in seconds.

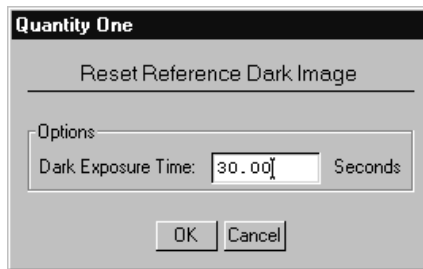


Fig.H-9. Reset Reference Dark pop-up box.

Click on OK to implement your change. The new reference dark will be created when you acquire your next image.

Because of the high sensitivity of the CCD, fluctuations in background radiation and/or temperature in the room can affect the level of dark count. If you feel that radiation/temperature conditions have changed in the room since your last reference dark was created, use the Reset Reference button to delete your old reference and create a new one under current conditions.

### None

If you do not want to perform dark subtraction, select None. No dark exposure will be acquired or subtracted.

### H.6.b. Save

#### Auto Save After Scan

To automatically save any image you acquire using the Acquire button, click on the Auto Save After Scan checkbox in the Options dialog box.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

#### Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan, then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Backup images are identified by "Backup of" followed by the original image file name.

**Note:** Backup files are read only files. If you attempt to save a backup file, you will be prompted to give it a new name. This protects the backup file and preserves it from any changes.

### H.6.c. Imaging Area Size

The imaging area is the area of the sample (in centimeters) that is captured by the camera and displayed in the scan window. To specify the size of this area, enter a dimension in the appropriate field under Imaging Area in the Options dialog box. Click on the Options button in the acquisition window to open this dialog.

When you change one imaging area dimension, the other will change to maintain the aspect ratio of the camera lens.

The imaging area will change depending on your zoom factor. For example, if you have zoomed in on a area that is 4.5 x 3.5 cm, then you would enter 4.5 for the width (3.5 for the height would be calculated automatically).

**Note:** Your imaging area settings must be correct if you want to do 1:1 printing. They must also be correct if you want to compare the quantities of objects (e.g., using the Volume Tools) in different images.

The imaging area dimensions also determine the size of the pixels in your image (i.e., resolution). A smaller imaging area will result in a higher resolution.

### H.6.d. Auto-scale Transform

Auto-scale Transform after Acquisition allows the user the option of having the image automatically perform the Auto-Scale transform function upon completion of image acquisition. This eliminates the need to transform an image or re-scan an image when the acquisition time was too short or the iris not opened enough.

To enable the auto-scale transform function, check the box labeled, Auto-Scale Transform after Acquisition in the Options dialog.

## H.7 Other Features

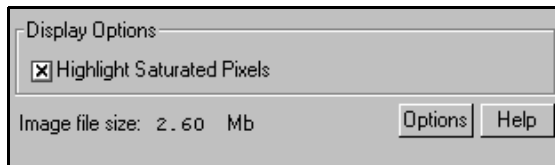


Fig.H-10. Other VersaDoc acquisition window features.

### Highlight Saturated Pixels

When this box is checked, any saturated pixels in the image will appear highlighted in red in the scan window and in the pop-up image window. To view/hide saturated pixels in the pop-up image window, use the Transform command.



### File Size of Images

Image File Size shows the size of the image file you are about to create. This size is determined by the resolution of the camera and any binning you perform when capturing the image.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image.

Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.



# Appendix I

## Other Features

The following features are available in Quantity One, but have more utility in its more powerful companion application, Diversity Database. You can examine your gel images in Quantity One and then database them using Diversity Database.

### I.1 Categories and Attributes

User-defined <category> buttons are available in the Standards dialog box, Matched Band Set dialog box, and Gel Layout dialog box. They allow you to categorize the characteristics of your particular gel or any related gel to which you might apply the same set of standards.

To define a new category, click on one of the <category> buttons. A dialog will pop up in which you can select from a list of categories or create a new one.



Fig.I-1. Category pop-up box.

To create a new category, click on the Edit button. This will open another pop-up box in which you can enter the name of your new category.

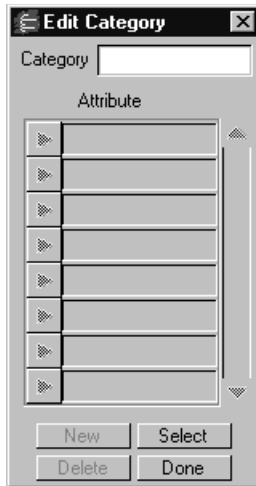


Fig.I-2. Edit Category dialog box.

Type the name of the new category (e.g., “Color”) next to the Category prompt and list attributes of that category in the Attribute fields (e.g., “Red,” “Green,” “Blue,” etc.). The form will automatically sort your attributes alphabetically within the Attribute fields. Categories and attributes can be defined for any characteristics of your gel that would be useful to sort by. Typical categories might be “Enzyme,” “Primer,” “Probe,” “Type,” “Who,” etc.

Once you’ve created a category and attributes, you can use them in the dialog box. Once again, click on a <category> button.

Select the category to be applied from the available categories on the list (or select <none>). Click on the Attribute button to specify an attribute. Click OK to apply your selection to the Standards box.

## I.2 Gel Layout

In Diversity Database, you can use the Gel Layout dialog box to compare samples across multiple gel images. In Quantity One, you can use it to enter general information about your image.

To open the Gel Layout form, select Gel Layout from the Edit menu.

| Lane # | Name       | CPM Loaded | <category> |
|--------|------------|------------|------------|
| S 1    | Protein    |            |            |
| E 2    | cerebrum   | 18,268     |            |
| E 3    | cerebellum | 13,740     |            |
| E 4    | stem       | 19,463     |            |
| E 5    | thymus     | 5,939      |            |
| E 6    | lung       | 8,012      |            |
| E 7    | heart      | 4,017      |            |
| E 8    | aorta      | 7,085      |            |

Fig.I-3. Gel Layout dialog box.

The following information about your gel can be specified at the top of the Gel Layout form:

- Category and attribute information (see above) for the whole gel.
- A brief description of the gel.
- Gel run date.
- Number of lanes.
- Lane width.

## Quantity One User Guide

---

- Lane-based background subtraction and disk size.

Each lane in your gel has its own line on the Gel Layout form. The following information can be specified for a lane.

- Band set type (standard or experimental).
- Lane number.
- Band set name.
- Sample name.
- Category/attributes information (see above) for individual lanes.

The pop-up buttons on the left side of the form offer a number of choices pertaining to the individual lanes on your gel. Clicking on one will open the Lane Choices list.

Lane Report will open a customizable lane report form that can be printed or exported.

Assign Band Set allows you to select the specific band set that is to be applied to that lane.

Unassign Band Set allows you to remove the band set that is currently applied to that lane.

View Band Set displays the band set form for the band set that is currently applied to that lane.

# Appendix J

## 21 CFR Part 11

### J.1 Introduction

Effective August 20, 1997 the United States Food and Drug Administration (FDA) released Part 11 “Electronic Records; Electronic Signatures” of title 21 of the Code of Federal Regulations. This rule states the conditions under which the FDA considers electronic signatures and electronic records to be trustworthy, reliable, and equivalent to traditional handwritten signatures. In this manner it defines the conditions under which an organization must operate to meet its record keeping and record submission requirements when using electronic signatures and records rather than handwritten records and signatures.

#### J.1.a. CFR and The Discovery Series

The purchase of a separate license allows the use of the 21 CFR Part 11 System Security features of Quantity One. Quantity One utilizes these security features in concert with the operating system to assist the laboratory, research institution, etc. in becoming CFR compliant. These features include audit trail, electronic signature, and secure user login.

**Note:** Bio-Rad makes no claim that Quantity One is CFR compliant in and of itself, nor does it guarantee compliance for the user. The organization must establish policies and standard operating procedures that work in conjunction with the tools provided by Bio-Rad to ensure compliance with 21 CFR Part 11.

#### Audit Trail

Section 11.10 (e) of 21 CFR Part 11 requires the use of “secure, computer-generated, time-stamped audit trails to independently record the date and time” of activities within the system. In System Security mode, Quantity One automatically records any events that affect data and analysis in the Audit Report. For instance, such things as

## Quantity One User Guide

---

cropping and band detection are recorded in the Audit Report, while transform will not because it does not affect the data.

Although there are software controls to detect modification of data, protection of the data from deletion must happen at the level of the operating system. As a result, data must be stored only on an NTFS partition. Access to directories where data is stored should be granted only to those who need to access that data. Ideally, access for a single user will prevent the possibility of two users accessing the same file. Also, a regular backup procedure should be in place.

### Electronic Signature

Section 11.3 (b) (7) of 21 CFR Part 11 defines an electronic signature as “a computer data compilation of any symbol or series of symbols executed, adopted, or authorized by an individual to be the legally binding equivalent of the individual's handwritten signature.” Quantity One allows the user to “sign” images thereby locking the files against future changes. If a user changes a “signed” file, and attempts to save the file, the file automatically receives a new version number, and the signed tag is removed from the window title.

### Secure User Login

Section 11.3 (b)(4) defines a closed system as “an environment in which system access is controlled by the persons who are responsible for the content of electronic records that are on the system.” Quantity One utilizes the Microsoft Windows 2000 and XP Professional security model as the secure user login.

## J.2 Using The Discovery Series in CFR Mode

### Access Levels

Before CFR mode can be utilized, the system must be correctly configured by the network administrator. There are four levels of access to Quantity One in CFR Mode:

Guest - A user with guest access has the ability to open and view files, but a guest cannot change data, nor can a guest save a file.



Tech - The tech level user has the ability to acquire images, transform images, crop, flip, rotate, use the text and line tools, perform basic analysis, print, and export. The tech level user also has the ability to save and sign files.

User - At this level, all functions are available to the user. However, the user cannot change the CFR settings.

Administrator - The administrator level has all the capabilities of the full user as well as the ability to make changes to the CFR settings.

### CFR Image Files

For files to be opened in CFR mode, the files must have been created on a CFR system, either through acquisition or importing a TIFF file. Any files created on a non-CFR system cannot be opened. To open a file created on a non-CFR system, open the file on a non-CFR system and export it to TIFF. You can then import it in CFR mode. To ensure accuracy, remember to select the analysis option as your export mode when you export your non-CFR images.

**Note:** Due to operating system limitations, it is possible to have image files open in two or more locations over a network simultaneously. To safeguard your image files keep them in a protected folder or on your local machine.

### J.2.a Setting Security Preferences

Setting the Security preferences for CFR mode requires the user have administrator privileges. To locate the Security Preferences, select Preferences from the Edit menu and click the Security tab.

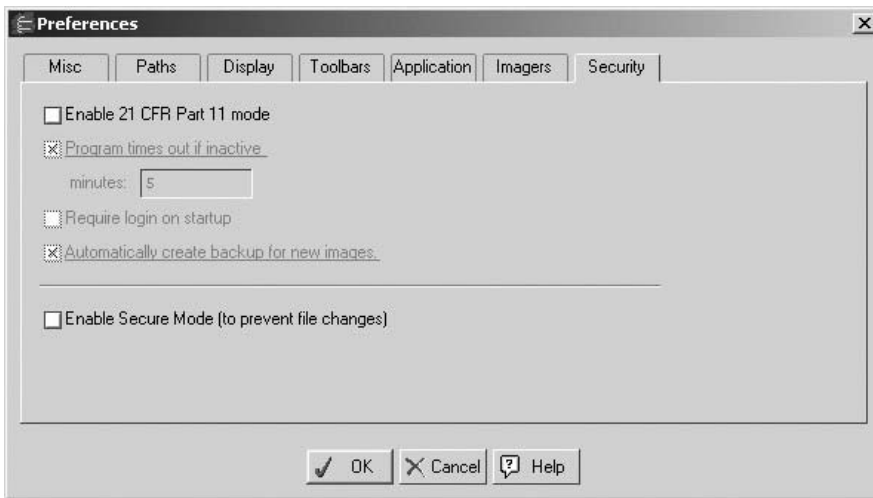


Fig. J-1. Security Preferences

The following options are available on the Security tab of the Preferences dialog box:

**Enable 21 CFR Part 11 mode** - If this box is unchecked, all other options are inactive. To activate the other options and CFR mode check this box.

**Program times out if inactive** - When this box is checked, you must enter your password to resume using Quantity One. The minutes field indicates the amount of time that must elapse before the application times out.

**Require login on startup** - If this box is checked, you must enter your password when Quantity One starts. This password is the same password as the one used for logging on to Windows. This is an added security measure to prevent unauthorized users from starting Quantity One on a machine where they are not logged in.

**Automatically create backup for new images** - This option, when selected, creates a backup file when you save a newly acquired or imported image. Locate the backup file in the same folder as the original. The backup file is identified by “backup of” preceding the filename of the original.

**Enable Secure Mode** - This option is not associated with CFR. For further information on Secure Mode, See Section 2.5, Preferences.

### J.2.b Secure User Login

Quantity One utilizes the Windows security model. When you open Quantity One, Quantity One determines the user name by the ID of the user currently logged on to that system. If you have Require Login on startup checked on the Security tab of the Preferences dialog box, then you must enter your Windows password to start Quantity One.

To view the current user information, select Show User Information from the CFR menu. The message displays the username, full name, and group membership for the currently logged in user.

**Note:** Although Windows XP allows you to switch users without logging off, you must first close Quantity One before switching users. Windows 2000 does not allow the ability to switch users.

### J.2.c Audit Report

The Audit Report included with CFR mode tracks all changes to actual data of images as well as any events associated with analysis.

| Audit Report        |                |                  |                                                 |
|---------------------|----------------|------------------|-------------------------------------------------|
| Time                | User           | Description      | Detail                                          |
| 2003-08-04/09:27:38 | Matthew Halsey | Open...          | Converted from TIFF                             |
|                     |                |                  | Original file name: Proteins.tif                |
| 2003-08-04/09:27:48 | Matthew Halsey | Save As...       | Old file name: Proteins.1sc                     |
|                     |                |                  | New file name: CFR-Proteins.1sc                 |
| 2003-08-04/09:27:48 | Matthew Halsey | Save             | Created backup file: Backup of CFR-Proteins.1sc |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 1                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 2                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 3                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 4                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 5                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 6                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 7                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 8                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 9                          |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 10                         |
| 2003-08-04/09:27:53 | Matthew Halsey | New Lane Created | Initial Lane Number: 11                         |

Fig. J-2. Audit Report

Each entry in the Audit report contains the date and time of the event as well as the name of the user who performed the event. The Audit Report also contains a description of the event with details, the file name, and the version of Quantity One used when the event occurred. The comment column allows you to enter comments about particular events. To enter a comment, click the cell under the comments heading for the event to which you want to add the comment. This opens the comment editor dialog box. Enter your comment and click Done.

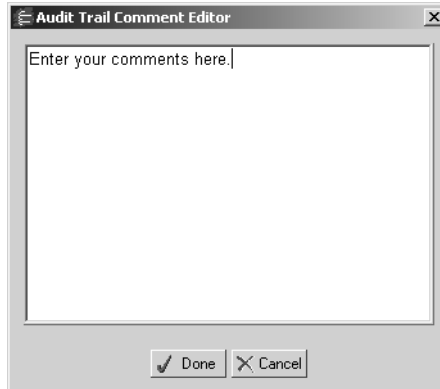


Fig. J-3. Comment Editor

### Audit Report Options

The Audit Report Options dialog box allows you to customize the Audit Report view. Click the Options button to open the Audit Report Options Dialog box.

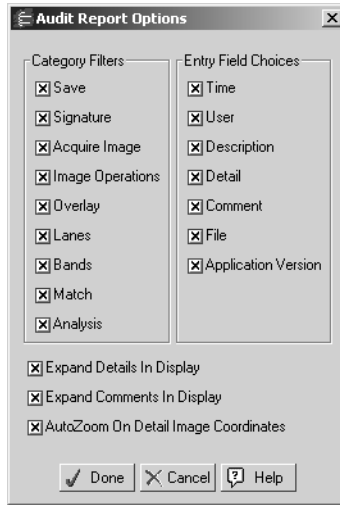


Fig. J-4. Audit Report Options

The left hand column of the Audit Report dialog box lists the category filters. If a box is checked, all events of that type display in the Audit Report. To hide a category in the report viewer, clear the check box next to the corresponding category.

The right hand column lists the columns the report viewer displays. To hide a column, clear the check box next to the corresponding column name.

**Note:** These options only affect what is displayed in the report viewer, not what is recorded. As previously stated, all changes to image data and events associated with analysis are recorded.

If the Expand Details in Display option is checked, then the all details for each event are displayed in the viewer. Otherwise, only the first row of each event is displayed. Similarly, the Expand comments in Display functions the same way.

The AutoZoom on Detail Image Coordinates option is a convenient way to pinpoint the location an event took place in the image. If the box is checked, click the coordinates in the report viewer and the image will automatically zoom to the location of the event. Note that this tool is only available when a specific location has been recorded in the report.

### Printing and Exporting the Audit Report

The Audit Report Viewer allows you to print or export the report for use in a spreadsheet application.

To print the report, click either the Print Current Page button or the Print button. Print Current Page is only available in print preview mode and allows you to print the currently viewed page. The print all pages button opens the Print dialog box where you can either print all pages or specify a print range. Click Page Setup to change the print settings.

To export the report click the Export button. This opens the Export Options dialog box where you can determine the delimiter (tab or comma) and the export destination (save as a file or copy to the clipboard).

### J.2.d Electronic Signature

As was previously stated, Section 11.3 (b) (7) of 21 CFR Part 11 defines an electronic signature as “a computer data compilation of any symbol or series of symbols executed, adopted, or authorized by an individual to be the legally binding equivalent of the individual's handwritten signature.” Quantity One allows the user to “sign” an image thereby locking the file against future changes. If a user changes a “signed” file, and attempts to save the file, the file automatically receives a new version number, and the signed tag is removed from the window title.

To “sign” a file in Quantity One, with the file open select Sign from the CFR menu. A modified version of the Comments dialog box opens. Enter any comments you may have, and click Sign. This opens the Login User dialog box, which requires you to enter your user name and password for signature validation.



Fig. J-5. Electronic Signature dialog box

Once you have “signed” your file, the image window title bar indicates the file has been signed, and the event is recorded in the Audit report.

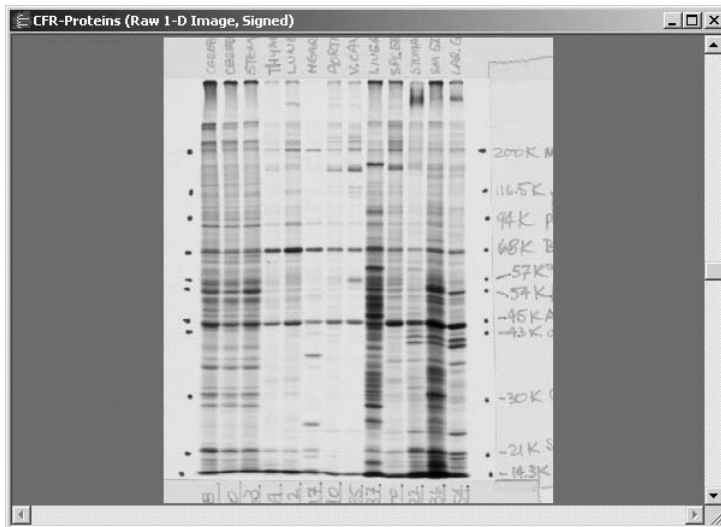


Fig. J-6. A signed Image

If your policies require signature verification by more than one person, such as a lead researcher, a signed file can be signed again without changing the file version provided the file has not been modified between signatures.



# Index

## Numerics

3D Viewer ..... 3-10

## A

Annotations. See Text Overlays.

Arrays

    Lane-based ..... 4-16

    Volume ..... 7-12

Arrow keys ..... 3-3

Automation Manager ..... 3-42

Average density ..... 5-15

## B

Background subtraction

    Entire image ..... 3-29

    Lane-based ..... 4-9

    Volume-based ..... 7-10

Band data, displaying ..... 5-13

Band Set form

    List of band types ..... 6-18

    Toolbar ..... 6-19

Band sets ..... 6-17

Band types

    Defining ..... 6-14

    Listing ..... 6-18

## Quantity One User Guide

---

|                                                |      |
|------------------------------------------------|------|
| Band Types report .....                        | 12-6 |
| Bands                                          |      |
| Adjusting .....                                | 5-12 |
| Attributes, displaying .....                   | 5-13 |
| Band information, displaying .....             | 5-16 |
| Brackets or lines preference .....             | 2-18 |
| Contouring .....                               | 5-22 |
| Deleting .....                                 | 5-12 |
| Detecting automatically .....                  | 5-3  |
| Detection parameters .....                     | 5-5  |
| Detection parameters, saving and loading ..... | 5-9  |
| Displaying as brackets/lines .....             | 5-13 |
| Doublets, detecting .....                      | 5-8  |
| Drawing .....                                  | 5-24 |
| Gaussian modeling .....                        | 5-18 |
| Identification and quantitation .....          | 5-2  |
| Identifying individually .....                 | 5-10 |
| Irregular shapes .....                         | 5-21 |
| Normalized for quantity .....                  | 6-22 |
| Numbering .....                                | 5-14 |
| Plotting traces .....                          | 5-12 |
| Shadow bands, rejecting .....                  | 5-6  |
| Bio-Rad Technical Service .....                | 1-14 |

## C

|                                 |            |
|---------------------------------|------------|
| Calibrated quantity .....       | 5-15       |
| Categories and attributes ..... | I-1        |
| Centering an image .....        | 3-4        |
| Chemi Doc                       |            |
| Acquiring the image .....       | C-5, D-6   |
| Adjusting the display .....     | C-11       |
| Annotating images .....         | A-10, C-12 |
| Auto Expose .....               | C-6, D-6   |
| Auto exposure settings .....    | C-15, D-14 |
| Backup images .....             | C-16, D-14 |

|                                      |               |
|--------------------------------------|---------------|
| Chemi mode .....                     | C-5           |
| DAC settings .....                   | C-14          |
| Exposure Status bar .....            | C-13, D-9     |
| Exposure time .....                  | A-7, C-7, D-7 |
| Freeze .....                         | C-8           |
| Image mode .....                     | C-5           |
| Imaging area .....                   | C-15, D-12    |
| Invert display .....                 | C-11          |
| Live Acquire .....                   | C-8, D-8      |
| Live image display .....             | C-4           |
| Manual Expose .....                  | C-7, D-7      |
| Options .....                        | C-13, D-11    |
| Positioning the sample .....         | C-4, D-5      |
| Saturated pixels, highlighting ..... | C-11          |
| Saving images .....                  | C-13          |
| Simulation mode .....                | C-2, D-2      |
| UV mode .....                        | C-5           |
| Video card .....                     | C-2, D-2      |
| Video display window .....           | C-3           |
| Video printing .....                 | C-13          |
| Video printing, footer info .....    | C-15          |
| White light mode .....               | C-5           |
| Clear Analysis .....                 | 3-42          |
| Closing images .....                 | 2-10          |
| Colony Counting                      |               |
| Adjusted count .....                 | 8-9           |
| Averaging .....                      | 8-4           |
| Background noise .....               | 8-6           |
| Batch files .....                    | 8-10          |
| Defining counting region .....       | 8-2           |
| Displaying results .....             | 8-4           |
| Histogram .....                      | 8-6           |
| Ignoring a region .....              | 8-8           |
| Image specifications .....           | 8-1           |
| Making/erasing colonies .....        | 8-5           |
| Plaques .....                        | 8-3           |
| Saving a count with the image .....  | 8-9           |
| Saving multiple counts .....         | 8-10          |
| Saving to a spreadsheet .....        | 8-10          |

## Quantity One User Guide

---

|                                  |       |
|----------------------------------|-------|
| Sensitivity .....                | 8-3   |
| White and blue colonies .....    | 8-7   |
| Colors, setting .....            | 3-14  |
| Compare Lane Images .....        | 12-11 |
| Compare Lanes .....              | 4-13  |
| Comparison reports .....         | 12-9  |
| Computer requirements            |       |
| Macintosh .....                  | 1-6   |
| PC .....                         | 1-5   |
| Context menu .....               | 2-5   |
| Contour area .....               | 5-15  |
| Contour quantity .....           | 5-15  |
| Contours. See Bands, Contouring. |       |
| Contrast. See Transform.         |       |
| Crop                             |       |
| Saving crop settings .....       | 3-26  |
| Cropping .....                   | 3-24  |
| Custom .....                     | D-4   |
| Custom Rotation .....            | 3-27  |

## D

|                                                   |       |
|---------------------------------------------------|-------|
| Default Automation .....                          | 3-45  |
| Density in Region .....                           | 5-23  |
| Density Tools .....                               | 3-5   |
| Density at Cursor .....                           | 3-5   |
| Density in Box .....                              | 3-5   |
| Plot Cross-section .....                          | 3-6   |
| Plot Density Distribution .....                   | 3-5   |
| Plot Vertical Trace .....                         | 3-6   |
| Detect Bands. See Bands, Detecting automatically. |       |
| Dice Coefficient .....                            | 12-10 |
| Differential Display .....                        | 11-1  |
| DOS file names, parsing .....                     | 2-16  |
| Downloading from Internet                         |       |
| Windows .....                                     | 1-13  |

---

## E

|                             |       |
|-----------------------------|-------|
| Editing .....               | 7-6   |
| Exit .....                  | 2-15  |
| Exporting TIFF files .....  | 13-5  |
| EXQuest .....               | 10-1  |
| Calibration and setup ..... | 10-2  |
| Lane/Volume Excision .....  | 10-18 |
| Manual Excision .....       | 10-6  |

## F

|                                      |      |
|--------------------------------------|------|
| File locations in Windows .....      | 1-8  |
| Filter Wizard .....                  | 3-34 |
| Filtering image noise .....          | 3-33 |
| Flipping images .....                | 3-26 |
| Fluor-S MAX MultiImager              |      |
| File size of images .....            | H-19 |
| Fluor-S MultiImager                  |      |
| File size of images .....            | D-15 |
| Saturated pixels, highlighting ..... | D-15 |
| FX, see Molecular Imager FX          |      |

## G

|                                  |            |
|----------------------------------|------------|
| Gaussian modeling of bands ..... | 5-18       |
| Gaussian Peak Density .....      | 5-15       |
| Gaussian Trace Quantity .....    | 5-15       |
| Gel Doc                          |            |
| Acquiring the image .....        | A-5        |
| Auto expose .....                | A-5        |
| Auto exposure settings .....     | A-13, B-13 |
| Backup images .....              | A-13, B-13 |

|                                      |            |
|--------------------------------------|------------|
| DAC settings .....                   | A-12       |
| Display settings .....               | A-8        |
| Exposure Status bar .....            | A-11, B-11 |
| Freeze .....                         | A-7        |
| Image mode .....                     | A-5        |
| Imaging area .....                   | A-13, B-12 |
| Invert display .....                 | A-9        |
| Live image display .....             | A-4        |
| Manual expose .....                  | A-6        |
| Options .....                        | A-11, B-11 |
| Positioning the sample .....         | A-4        |
| Saturated pixels, highlighting ..... | A-9, B-9   |
| Saving images .....                  | A-11, B-11 |
| Simulation mode .....                | A-2, B-2   |
| UV mode .....                        | A-5        |
| Video card .....                     | A-2        |
| Video display window .....           | A-3        |
| Video printing, footer info .....    | A-13, B-13 |
| White light mode .....               | A-5        |
| Gel Doc XR                           |            |
| Acquiring the image .....            | B-5        |
| Annotating images .....              | B-10       |
| Auto expose .....                    | B-5        |
| Display settings .....               | B-8        |
| Freeze .....                         | B-7        |
| Image mode .....                     | B-5        |
| Invert display .....                 | B-9        |
| Live image display .....             | B-4        |
| Manual expose .....                  | B-6        |
| Positioning the sample .....         | B-4        |
| UV mode .....                        | B-5        |
| White light mode .....               | B-5        |
| Gel Doc XR display window .....      | B-3        |
| Gel Layout form .....                | I-2        |
| Grab .....                           | 3-3        |
| GS-800 Imaging Densitometer          |            |
| Applications, selecting .....        | E-4        |
| Calibration .....                    | E-9        |
| Calibration settings .....           | E-13       |

|                                           |      |
|-------------------------------------------|------|
| Calibration step tablets .....            | E-10 |
| Filters and light source, selecting ..... | E-6  |
| Options .....                             | E-14 |
| Oversampling .....                        | E-14 |
| Preview scan .....                        | E-7  |
| Resolution, selecting .....               | E-8  |
| Saturated pixels, highlighting .....      | E-15 |
| Scan area, selecting .....                | E-7  |
| Scanning an image .....                   | E-13 |
| Scanning window .....                     | E-3  |
| SCSI card .....                           | E-1  |
| Simulation mode .....                     | E-2  |

## I

|                                             |      |
|---------------------------------------------|------|
| Image information .....                     | 2-10 |
| Image Report .....                          | 13-3 |
| Image size, changing .....                  | 2-11 |
| Image Stack Tool .....                      | 3-13 |
| Images                                      |      |
| Background subtraction .....                | 3-29 |
| Colors .....                                | 3-14 |
| Contrast .....                              | 3-17 |
| Cropping .....                              | 3-24 |
| Filtering noise from .....                  | 3-33 |
| Inverting data .....                        | 3-39 |
| Magnifying .....                            | 3-1  |
| Positioning .....                           | 3-1  |
| Rotating and flipping .....                 | 3-26 |
| Stacking .....                              | 3-13 |
| Text overlays .....                         | 3-39 |
| Viewing in multiple channels .....          | 3-7  |
| Imaging devices, listing on File menu ..... | 2-21 |
| Imitate Zoom .....                          | 3-4  |
| Importing TIFF files .....                  | 2-9  |
| Installation                                |      |

|                     |      |
|---------------------|------|
| Windows .....       | 1-7  |
| Invert              |      |
| Image data .....    | 3-39 |
| Image display ..... | 3-23 |

## K

|                         |     |
|-------------------------|-----|
| Keyboard commands ..... | 2-6 |
|-------------------------|-----|

## L

|                                 |       |
|---------------------------------|-------|
| Lane frame                      |       |
| Adjusting .....                 | 4-4   |
| Creating .....                  | 4-2   |
| Lane/Volume Excision .....      | 10-18 |
| Lane-based Arrays               |       |
| Adjusting .....                 | 4-18  |
| Cell height .....               | 4-18  |
| Cell width .....                | 4-18  |
| Lanes                           |       |
| Adjusting .....                 | 4-7   |
| Comparing .....                 | 4-13  |
| Defining .....                  | 4-6   |
| Deleting .....                  | 4-7   |
| Framing automatically .....     | 4-2   |
| Framing manually .....          | 4-3   |
| Profiling .....                 | 4-8   |
| Reports .....                   | 12-5  |
| Unadjusting .....               | 4-7   |
| Width of all lanes .....        | 5-7   |
| Width of individual lanes ..... | 4-8   |
| License .....                   | 1-9   |
| Registration .....              | 1-10  |
| Line tool .....                 | 3-40  |



---

# M

|                                                  |      |
|--------------------------------------------------|------|
| Macintosh .....                                  | 1-8  |
| Macintosh, memory assigned to Quantity One ..... | 1-6  |
| Magnifying images .....                          | 3-1  |
| Manual Excision .....                            | 10-6 |
| Match graphs .....                               | 6-22 |
| Matching                                         |      |
| Displaying band types .....                      | 6-16 |
| Displaying modeling lines .....                  | 6-16 |
| Manually matching bands .....                    | 6-16 |
| Match commands .....                             | 6-16 |
| Match graphs .....                               | 6-22 |
| Matched band sets .....                          | 6-17 |
| Reports .....                                    | 12-5 |
| Maximizing the application window .....          | 2-17 |
| Memory allowance .....                           | 2-16 |
| Menus .....                                      | 2-1  |
| Modeling lines                                   |      |
| Displaying .....                                 | 5-15 |
| Molecular Imager FX                              |      |
| Acquiring the image .....                        | G-13 |
| Control panel .....                              | G-2  |
| File size of images .....                        | G-13 |
| Options .....                                    | G-14 |
| Saturated pixels, highlighting .....             | G-14 |
| Saving the image .....                           | G-13 |
| Scanning window .....                            | G-3  |
| Selecting an application .....                   | G-4  |
| Selecting resolution .....                       | G-12 |
| Selecting scan area .....                        | G-11 |
| Simulation mode .....                            | G-2  |
| Multi-channel viewer .....                       | 3-7  |

### N

|                           |            |
|---------------------------|------------|
| Normalized quantity ..... | 5-15, 6-22 |
| Normalized Rf .....       | 6-4        |

### O

|                                    |     |
|------------------------------------|-----|
| Opening images .....               | 2-7 |
| Optimizing images. See Transform.  |     |
| Overlays, showing and hiding ..... | 3-6 |

### P

|                                      |       |
|--------------------------------------|-------|
| Page Setup .....                     | 13-1  |
| Password, entering .....             | 1-12  |
| PCR gel analysis .....               | 11-1  |
| Peak density .....                   | 5-15  |
| Personal Molecular Imager FX         |       |
| Acquiring an image .....             | F-6   |
| Control panel .....                  | F-2   |
| File size of images .....            | F-6   |
| Options .....                        | F-7   |
| Saturated pixels, highlighting ..... | F-7   |
| Saving the image .....               | F-6   |
| Scanning window .....                | F-3   |
| Selecting resolution .....           | F-5   |
| Selecting scan area .....            | F-4   |
| Simulation mode .....                | F-2   |
| Phylogenetic Trees                   |       |
| Clustering methods .....             | 12-15 |
| Display options .....                | 12-18 |
| Displaying .....                     | 12-14 |
| Preferences                          |       |

|                                    |      |
|------------------------------------|------|
| Display .....                      | 2-17 |
| File paths .....                   | 2-17 |
| General .....                      | 2-15 |
| Imagers .....                      | 2-21 |
| Relative Front Calculation .....   | 2-20 |
| Relative Percent Calculation ..... | 2-20 |
| Toolbars .....                     | 2-19 |
| Print Image .....                  | 13-1 |
| Printing                           |      |
| Image Report .....                 | 13-3 |
| Video print .....                  | 13-4 |
| Propagate Band Set .....           | 6-21 |

## Q

|                                         |                 |
|-----------------------------------------|-----------------|
| Quantity .....                          | 1-14            |
| Calibrated .....                        | 5-15            |
| Contour .....                           | 5-15            |
| Normalized .....                        | 5-15            |
| Relative .....                          | 5-15, 5-17      |
| Trace .....                             | 5-3, 5-15, 6-26 |
| Quantity Standards                      |                 |
| Applying to bands .....                 | 6-30            |
| Calibration curve .....                 | 6-28            |
| Checking imported values .....          | 6-31            |
| Creating .....                          | 6-24            |
| Definition .....                        | 6-24            |
| Dilution series .....                   | 6-30            |
| Entering standards .....                | 6-27            |
| Importing .....                         | 6-31            |
| Relative deviation .....                | 6-28            |
| Selecting bands of known quantity ..... | 6-26            |
| Quick Guides .....                      | 2-4             |

### R

|                              |            |
|------------------------------|------------|
| ReadyAgarose 96 Plus .....   | 9-1        |
| ReadyAgarose Wizard .....    | 9-1        |
| Reduce File Size .....       | 2-12       |
| Registration                 |            |
| By fax/e-mail .....          | 1-12       |
| By Internet .....            | 1-11       |
| Registration form .....      | 1-10       |
| Relative Front .....         | 2-20, 5-14 |
| Relative quantity .....      | 5-15, 5-17 |
| Reports                      |            |
| 1-D Analysis .....           | 12-8       |
| Band Types .....             | 12-6       |
| Compare Lane Images .....    | 12-11      |
| Exporting .....              | 12-4       |
| Lanes .....                  | 12-5       |
| Matches .....                | 12-5       |
| Phylogenetic Tree .....      | 12-14      |
| Printing .....               | 12-3       |
| Report window features ..... | 12-1       |
| Similarity Matrix .....      | 12-19      |
| VNTR .....                   | 12-25      |
| Volume Analysis .....        | 12-20      |
| Revert to Saved .....        | 2-10       |
| Rotating images .....        | 3-26       |

### S

|                                                   |      |
|---------------------------------------------------|------|
| Sample images .....                               | 2-9  |
| Saturated pixels, highlighting .....              | 3-23 |
| See also individual imaging devices               |      |
| Saving images .....                               | 2-10 |
| Show/Hide Volume Labels .....                     | 7-7  |
| Similarity of lane-based samples, comparing ..... | 12-9 |

|                                                  |      |
|--------------------------------------------------|------|
| Simulation mode, see individual imaging devices  |      |
| Size of images, changing .....                   | 2-11 |
| Software license .....                           | 1-9  |
| Sort and Recalculate .....                       | 3-42 |
| Standards                                        |      |
| Applying .....                                   | 6-7  |
| Archive .....                                    | 6-12 |
| Archive, disabling/deleting .....                | 6-13 |
| Bio-Rad .....                                    | 6-1  |
| Calculated values, viewing .....                 | 6-12 |
| Creating .....                                   | 6-3  |
| Deleting .....                                   | 6-12 |
| Dialog .....                                     | 6-5  |
| Entering values .....                            | 6-6  |
| Modeling lines .....                             | 6-9  |
| Opening existing .....                           | 6-12 |
| Predefined .....                                 | 6-1  |
| Read-only .....                                  | 6-13 |
| Regression curve .....                           | 6-9  |
| Regression curve display options .....           | 6-11 |
| Regression models .....                          | 6-10 |
| Removing standards from lanes .....              | 6-9  |
| Saving .....                                     | 6-12 |
| Status boxes .....                               | 2-2  |
| Step .....                                       | D-4  |
| Subtract Background. See Background subtraction. |      |

## T

|                                     |      |
|-------------------------------------|------|
| Tandem repeats .....                | 5-15 |
| Tandem repeats, calculating .....   | 11-3 |
| Technical Service, contacting ..... | 1-14 |
| Text Overlays                       |      |
| Creating .....                      | 3-39 |
| Editing .....                       | 3-41 |
| Line tool .....                     | 3-40 |

## Quantity One User Guide

---

|                                  |                 |
|----------------------------------|-----------------|
| Viewing .....                    | 3-41            |
| TIFF files .....                 |                 |
| Exporting .....                  | 13-5            |
| Importing .....                  | 2-9             |
| Tile commands .....              | 3-4             |
| Tool Help .....                  | 2-2             |
| Toolbars .....                   |                 |
| Main .....                       | 2-2             |
| Secondary .....                  | 2-3             |
| Trace quantity .....             | 5-3, 5-15, 6-26 |
| Transform .....                  | 3-17            |
| Auto-scale .....                 | 3-20            |
| Controls .....                   | 3-20            |
| Gamma slider .....               | 3-22            |
| High/Low Sliders .....           | 3-20            |
| Highlight Saturated Pixels ..... | 3-23            |
| Histogram .....                  | 3-20            |
| Invert Display .....             | 3-23            |

## U

|                                |      |
|--------------------------------|------|
| UNIX file names, parsing ..... | 2-17 |
| Users, multiple .....          | 2-23 |

## V

|                           |            |
|---------------------------|------------|
| VersaDoc .....            |            |
| Acquire .....             | H-11       |
| Acquiring the image ..... | H-10       |
| Binning .....             | D-4, H-6   |
| Control panel .....       | H-2        |
| Custom settings .....     | H-5        |
| Dark subtraction .....    | D-13, H-14 |
| Exposure time .....       | H-8        |

|                                           |       |
|-------------------------------------------|-------|
| Exposure times, recommended .....         | H-9   |
| Gain .....                                | H-6   |
| Illumination Flat Fielding .....          | H-10  |
| Imaging area size .....                   | H-17  |
| Optimize exposure .....                   | H-12  |
| Optimize uniformity .....                 | D-9   |
| Options .....                             | H-14  |
| Positioning .....                         | H-7   |
| Preview scan .....                        | H-10  |
| Saturated pixels, highlighting .....      | H-18  |
| Save options .....                        | H-17  |
| Selecting an application .....            | H-3   |
| Video printing .....                      | 13-4  |
| View Entire Image .....                   | 3-4   |
| Viewing images in multiple channels ..... | 3-7   |
| VNTR Report .....                         | 12-25 |
| VNTRs, calculating .....                  | 11-3  |
| Volumes                                   |       |
| Arrays .....                              | 7-12  |
| Background subtraction .....              | 7-10  |
| Copying between images .....              | 7-8   |
| Copying within an image .....             | 7-8   |
| Creating .....                            | 7-1   |
| Deleting .....                            | 7-8   |
| Displaying .....                          | 7-7   |
| Labeling .....                            | 7-5   |
| Moving .....                              | 7-8   |
| Regression curve .....                    | 12-23 |
| Report .....                              | 12-20 |
| Showing/hiding labels .....               | 7-7   |
| Standards .....                           | 7-8   |
| Unknown .....                             | 7-5   |

## W

Windows

## Quantity One User Guide

---

|                              |     |
|------------------------------|-----|
| Tiling .....                 | 3-4 |
| Windows file locations ..... | 1-8 |

## Z

|                                                 |      |
|-------------------------------------------------|------|
| Zoom Box .....                                  | 3-2  |
| Zoom commands, imitating in other windows ..... | 2-18 |
| Zoom In/Zoom Out .....                          | 3-3  |
| Zoom tools                                      |      |
| Changing behavior .....                         | 2-17 |



## **The Discovery Series™ Software Software License Agreement**

READ THIS! THIS IS A LEGAL AGREEMENT BETWEEN YOU, THE CUSTOMER, AND BIO-RAD LABORATORIES, INC. ("BIO-RAD") CONCERNING THE ENCLOSED COMPUTER PROGRAM ("THE PROGRAM"). YOU SHOULD REVIEW THE FOLLOWING TERMS AND CONDITIONS OF THIS AGREEMENT CAREFULLY BEFORE OPENING THIS PACKAGE. BY OPENING THIS PACKAGE YOU INDICATE YOUR ACCEPTANCE OF SUCH TERMS AND CONDITIONS. IF YOU DO NOT AGREE WITH ANY OF THEM, YOU SHOULD PROMPTLY RETURN THE PACKAGE UNOPENED, YOUR MONEY WILL BE REFUNDED.

### **LICENSE**

Pursuant to the license granted to you by this Agreement, you may (a) use the Program on a single computer; (b) copy the Program into any computer readable or printed form for back-up or modification purposes in support of your use of the Program on the single computer (certain programs, however, may include mechanisms to limit or inhibit copying; they are marked "copy protected"); (c) modify the Program and/or merge it into another program for your use on the single computer, and (d) transfer the Program and license to another party if the other party agrees to accept the terms and conditions of this Agreement.

Any portion of the Program merged into another program shall continue to be subject to the terms and conditions of this Agreement. If you must transfer the Program, you must at the same time either transfer all copies whether in printed or computer readable form to the same part or destroy any copies not transferred, including without limitations all modifications and portions of the Program contained or merged into other programs. You must also reproduce and include the copyright notice on any copy, modification or portion merged into another program.

YOU MAY NOT USE, COPY, MODIFY OR TRANSFER THE PROGRAM OR ANY COPY, MODIFICATION, OR MERGED PORTION, IN WHOLE OR IN PART, EXCEPT AS EXPRESSLY PROVIDED FOR IN THIS AGREEMENT. If you transfer possession or any copy modification, or merged portion of the Program to another party, your license is automatically terminated.

### **TERM**

The license is effective until terminated. You may terminate it at any other time by destroying the Program together with all copies, modifications and merged portions in any form. It shall also terminate upon conditions set forth elsewhere in this Agreement or if you fail to comply with any term or condition of this Agreement. Upon such termination, you must destroy the Program together with all copies, modifications, and merged portions in any form.

## LIMITED WARRANTY AND REMEDIES

BIO-RAD warrants the media on which the Program is furnished to be free from defects in materials and workmanship under normal use for a period of ninety (90) days from the date of delivery to you as evidenced by a copy of your receipt. This warranty is limited to you and is not transferable.

During the 90 day warranty period, BIO-RAD shall (a) replace any media item not meeting the foregoing warranty and which is returned to BIO-RAD or an authorized BIO-RAD dealer ("Authorized Dealer") with a copy of your receipt; or (b) if BIO-RAD or the Authorized Dealer is unable to deliver a replacement media item which is free of defects in materials or workmanship, you may terminate this Agreement by returning the Program, and you money shall be refunded.

The following warranty does not extend to any media item which has been damaged as a result of accident, misuse, abuse, or as a result of service or modification by anyone other than BIO-RAD or an Authorized Dealer.

EXCEPT AS EXPRESSLY SET FORTH ABOVE, THE PROGRAM IS PROVIDED "AS IS" AND NO OTHER WARRANTIES EITHER EXPRESSED OR IMPLIED, ARE MADE WITH RESPECT TO THIS PROGRAM, INCLUDING, WITHOUT LIMITATION THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. BIO-RAD WITHOUT LIMITATIONS EXPRESSLY DISCLAIMS ALL WARRANTIES NOT STATED HEREIN. YOU ASSUME THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM SHOULD THE PROGRAM PROVE DEFECTIVE. YOU AND NOT BIO-RAD OR AN AUTHORIZED DEALER ASSUME THE ENTIRE COST OF NECESSARY SERVICING REPAIR OR CORRECTION. SOME STATES DO NOT ALLOW THE EXCLUSION OF IMPLIED WARRANTIES, SO THE ABOVE EXCLUSION MAY NOT APPLY TO YOU. THIS WARRANTY GIVES YOU SPECIFIC LEGAL RIGHTS, AND YOU MAY ALSO HAVE OTHER RIGHTS WHICH VARY FROM STATE TO STATE.

BIO-RAD DOES NOT WARRANT THAT THE FUNCTIONS CONTAINED IN THE PROGRAM WILL MEET YOUR REQUIREMENTS OR THAT THE OPERATION OF THE PROGRAM WILL BE UNINTERRUPTED OR ERROR-FREE. YOU ASSUME RESPONSIBILITY FOR THE SELECTION OF THE PROGRAM TO ACHIEVE YOUR INTENDED RESULTS, AND FOR THE INSTALLATION, USE AND RESULTS OBTAINED FROM THE PROGRAM.

YOUR SOLE REMEDIES AND BIO-RAD'S ENTIRE LIABILITY ARE SET FORTH ABOVE, IN NO EVENT WILL BIO-RAD BE LIABLE TO YOU OR ANY OTHER PERSON FOR ANY DAMAGES, INCLUDING WITHOUT LIMITATION ANY INCIDENTAL OR CONSEQUENTIAL DAMAGES, EXPENSES, LOST PROFITS, LOST SAVINGS OR OTHER DAMAGES ARISING OUT OF THE USE OR INABILITY TO USE

SUCH PROGRAM EVEN IF BIO-RAD OR AN AUTHORIZED DEALER HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES, OR FOR ANY CLAIM BY ANY OTHER PARTY.

SOME STATES DO NOT ALLOW THE LIMITATION OR EXCLUSION OF INCIDENTAL OR CONSEQUENTIAL DAMAGES FOR CONSUMER PRODUCTS, SO THE ABOVE LIMITATIONS OR EXCLUSIONS MAY NOT APPLY TO YOU.

#### GENERAL

You may not sublicense, assign, or transfer the license or the Program except as expressly provided in this Agreement. Any attempt to otherwise sublicense, assign or transfer any of the rights, duties or obligations hereunder is null and void and not merely voidable.

If you have any questions concerning this Agreement, including warranty service, you should contact BIO-RAD LABORATORIES, INC., 1000 Alfred Nobel Drive, Hercules, CA 94547.

This Agreement shall be governed by the laws of the state of California as applicable to contracts between residents of California which are entered into and performed entirely within California.

YOU HAVE ACKNOWLEDGED THAT YOU HAVE READ THIS AGREEMENT, UNDERSTAND IT, AND AGREE TO BE BOUND BY ITS TERMS AND CONDITIONS. YOU FURTHER AGREE THAT IT IS THE COMPLETE AND EXCLUSIVE STATEMENT OF THE AGREEMENT BETWEEN US WHICH SUPERCEDES ANY PROPOSAL OR PRIOR AGREEMENT, ORAL OR WRITTEN, AND ANY OTHER COMMUNICATIONS BETWEEN US RELATING TO THE SUBJECT MATTER OF THIS AGREEMENT.

